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(54) **DIDEHYDRO-3'-DEOXY-4'-ETHYNYLTHYMIDINES AND RELATED COMPOUNDS AND THEIR USE IN TREATING MEDICAL CONDITIONS**

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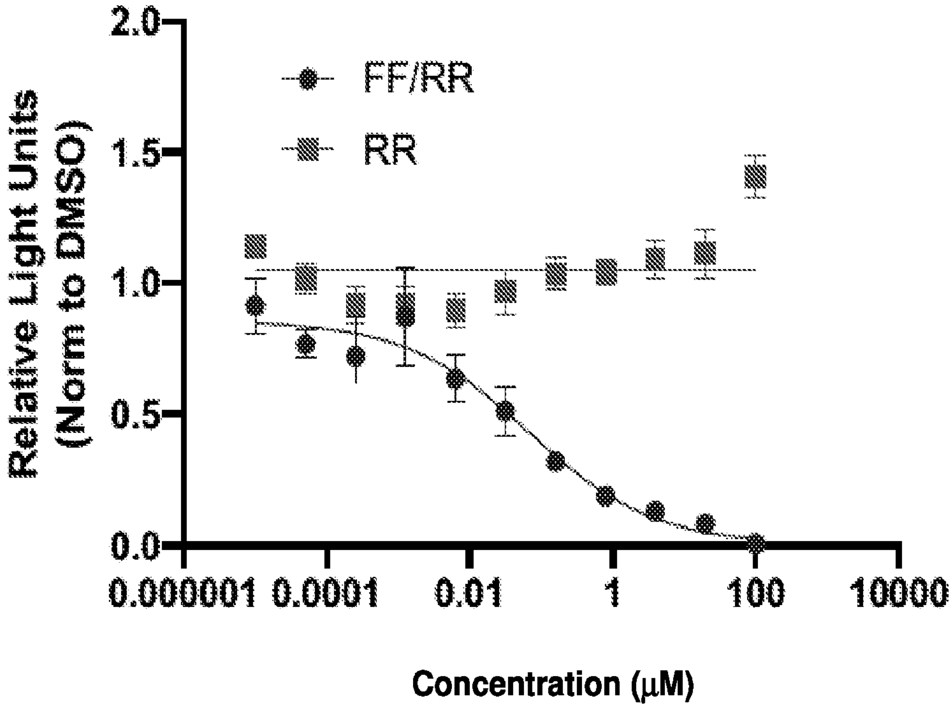
(57) **ABSTRACT**

The invention provides substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds, pharmaceutical compositions, their use for inhibiting LINE1 reverse transcriptase activity, their use for inhibiting HERV-K reverse transcriptase activity, and their use in the treatment of medical disorders, such as cancer.

Specification includes a Sequence Listing.

Figure 1.


FF/RR
RR
Relative Light Units
(Norm to DMSO)
2.0
1.5
1.0
0.5
0.0
0.000001
0.0001
0.01
1
100
10000
Concentration (μM)

DIDEHYDRO-3'-DEOXY-4'-ETHYNYLTHYMIDINES AND RELATED COMPOUNDS AND THEIR USE IN TREATING MEDICAL CONDITIONS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/333,914, filed Apr. 22, 2022, the entire disclosure of which is incorporated by reference herein in its entirety.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

[0002] This application contains a Sequence Listing which has been submitted electronically via Patent Center in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Apr. 21, 2023, is named 199646_seglist.xml and is 2,700 bytes in size.

FIELD OF THE INVENTION

[0003] The invention provides substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds, pharmaceutical compositions, their use for inhibiting LINE1 reverse transcriptase activity, their use for inhibiting HERV-K reverse transcriptase activity, and their use in the treatment of medical disorders, such as cancer.

BACKGROUND

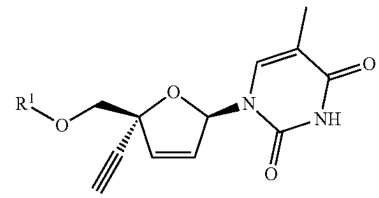
[0004] Cancer continues to be a significant health problem despite the substantial research efforts and scientific advances reported in the literature for treating this disease. Solid tumors, including prostate cancer, breast cancer, and lung cancer remain highly prevalent among the world population. Leukemias and lymphomas also account for a significant proportion of new cancer diagnoses. Current treatment options for these cancers are not effective for all patients and/or can have substantial adverse side effects. New therapies are needed to address this unmet need in cancer therapy.

[0005] Accordingly, the need exists for new therapeutic methods that provide improved efficacy and/or reduced side effects for treating medical disorders, such as cancer. The present invention addresses the foregoing needs and provides other related advantages.

SUMMARY

[0006] The invention provides substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds, pharmaceutical compositions, their use for inhibiting LINE1 reverse transcriptase activity, their use for inhibiting HERV-K reverse transcriptase activity, and their use in the treatment of medical disorders, such as cancer. In particular, one aspect of the invention provides a collection of substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds, such as a compound represented by Formula I:

(I)

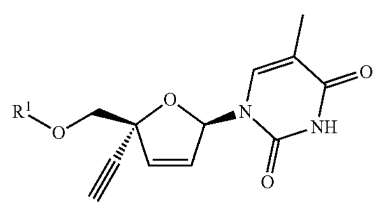


or a pharmaceutically acceptable salt thereof, where the variables are as defined in the detailed description. Further description of additional collections of substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds are described in the detailed description. The compounds may be part of a pharmaceutical composition comprising a pharmaceutically acceptable carrier.

[0007] Another aspect of the invention provides a method of treating a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder. The method comprises administering a therapeutically effective amount of a compound described herein, such as a compound of Formula I, to a subject in need thereof to treat the disorder, as further described in the detailed description.

[0008] Another aspect of the invention provides a method of treating a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises administering to a subject in need thereof a therapeutically effective amount of a compound of Formula II to treat the disorder; wherein Formula II is represented by:

(II)



or a pharmaceutically acceptable salt thereof, where the variables are as defined in the detailed description. Additional features of the method are described in the detailed description.

[0009] Another aspect of the invention provides a method of inhibiting LINE1 reverse transcriptase activity. The method comprises contacting a LINE1 reverse transcriptase with an effective amount of a compound described herein, such as a compound of Formula I, in order to inhibit the activity of said LINE1 reverse transcriptase, as further described in the detailed description.

[0010] Another aspect of the invention provides a method of inhibiting LINE1 reverse transcriptase activity in a subject suffering from a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises contacting a LINE1 reverse transcriptase with an effective amount of a compound described herein, such as a compound of Formula II, in order to inhibit the activity of said LINE1 reverse transcriptase, as further described in the detailed description.

[0011] Another aspect of the invention provides a method of inhibiting HERV-K reverse transcriptase activity. The method comprises contacting a HERV-K reverse transcriptase with an effective amount of a compound described herein, such as a compound of Formula I, in order to inhibit the activity of said HERV-K reverse transcriptase, as further described in the detailed description.

[0012] Another aspect of the invention provides a method of inhibiting HERV-K reverse transcriptase activity in a subject suffering from a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises contacting a HERV-K reverse transcriptase with an effective amount of a compound described herein, such as a compound of Formula II, in order to inhibit the activity of said HERV-K reverse transcriptase, as further described in the detailed description.

BRIEF DESCRIPTION OF FIGURES

[0013] FIG. **1** is a graph depicting inhibition of LINE1 reverse transcriptase by a test compound in an artificial-intron Cis LINE1 reporter assay, as described in Example 5.

DETAILED DESCRIPTION

[0014] The invention provides substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds, pharmaceutical compositions, their use for inhibiting LINE1 reverse transcriptase activity, their use for inhibiting HERV-K reverse transcriptase activity, and their use in the treatment of medical disorders, such as cancer. The practice of the present invention employs, unless otherwise indicated, conventional techniques of organic chemistry, pharmacology, molecular biology (including recombinant techniques), cell biology, biochemistry, and immunology. Such techniques are explained in the literature, such as in "Comprehensive Organic Synthesis" (B. M. Trost & I. Fleming, eds., 1991-1992); "Handbook of experimental immunology" (D. M. Weir & C. C. Blackwell, eds.); "Current protocols in molecular biology" (F. M. Ausubel et al., eds., 1987, and periodic updates); and "Current protocols in immunology" (J. E. Coligan et al., eds., 1991), each of which is herein incorporated by reference in its entirety.

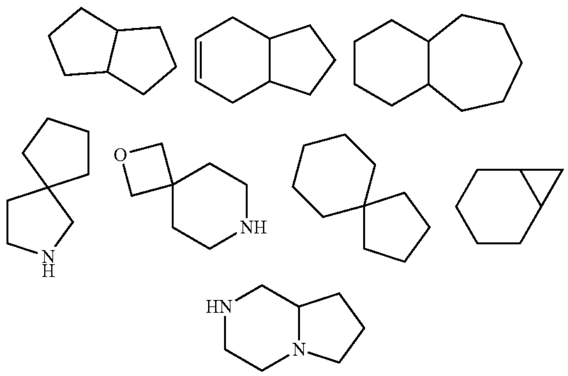
[0015] Various aspects of the invention are set forth below in sections; however, aspects of the invention described in one particular section are not to be limited to any particular section. Further, when a variable is not accompanied by a definition, the previous definition of the variable controls.

Definitions

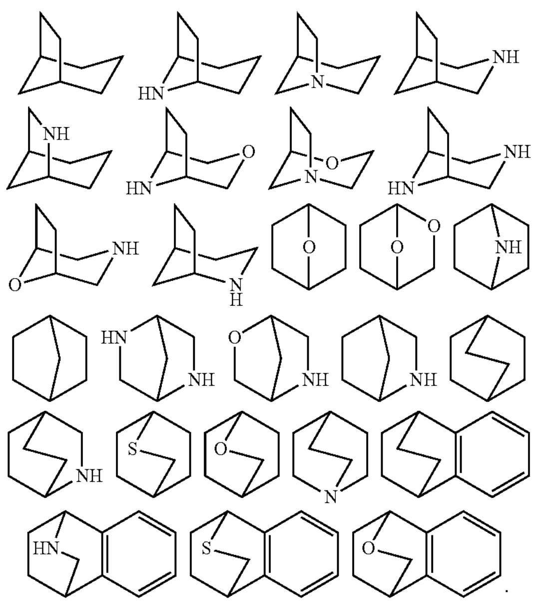
[0016] Compounds of the present invention include those described generally herein, and are further illustrated by the classes, subclasses, and species disclosed herein. As used herein, the following definitions shall apply unless otherwise indicated. These definitions apply regardless of whether a term is used by itself or in combination with other terms, unless otherwise indicated. Hence, the definition of "alkyl" applies to "alkyl" as well as the "alkyl" portions of "—O-alkyl" etc. For purposes of this invention, the chemical elements are identified in accordance with the Periodic Table of the Elements, CAS version, Handbook of Chemistry and Physics, 75^{th} Ed. Additionally, general principles of organic chemistry are described in "Organic Chemistry", Thomas Sorrell, University Science Books, Sausalito: 1999, and "March's Advanced Organic Chemistry", 5^{th} Ed., Ed.: Smith, M. B. and March, J., John Wiley & Sons, New York: 2001, the entire contents of which are hereby incorporated by reference.

[0017] The term "aliphatic" or "aliphatic group", as used herein, means a straight-chain (i.e., unbranched) or branched, substituted or unsubstituted hydrocarbon chain that is completely saturated or that contains one or more units of unsaturation, or a monocyclic hydrocarbon or bicyclic hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic (also referred to herein as "cycloaliphatic"), that has a single point of attachment to the rest of the molecule. Unless otherwise specified, aliphatic groups contain 1-6 aliphatic carbon atoms. In some embodiments, aliphatic groups contain 1-5 aliphatic carbon atoms. In other embodiments, aliphatic groups contain 1-4 aliphatic carbon atoms. In still other embodiments, aliphatic groups contain 1-3 aliphatic carbon atoms, and in yet other embodiments, aliphatic groups contain 1-2 aliphatic carbon atoms. In some embodiments, "cycloaliphatic" refers to a monocyclic C_3-C_6 hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic, that has a single point of attachment to the rest of the molecule. Suitable aliphatic groups include, but are not limited to, linear or branched, substituted or unsubstituted alkyl, alkenyl, alkynyl groups and hybrids thereof such as (cycloalkyl)alkyl, (cycloalkenyl)alkyl or (cycloalkyl)alkenyl.

[0018] As used herein, the term "bicyclic ring" or "bicyclic ring system" refers to any bicyclic ring system, i.e. carbocyclic or heterocyclic, saturated or having one or more units of unsaturation, having one or more atoms in common between the two rings of the ring system. Thus, the term includes any permissible ring fusion, such as ortho-fused or spirocyclic. As used herein, the term "heterobicyclic" is a subset of "bicyclic" that requires that one or more heteroatoms are present in one or both rings of the bicycle. Such heteroatoms may be present at ring junctions and are optionally substituted, and may be selected from nitrogen (including N-oxides), oxygen, sulfur (including oxidized forms such as sulfones and sulfonates), phosphorus (including oxidized forms such as phosphates), boron, etc. In some embodiments, a bicyclic group has 7-12 ring members and 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. As used herein, the term "bridged bicyclic" refers to any bicyclic ring system, i.e. carbocyclic or heterocyclic, saturated or partially unsaturated, having at least one bridge. As defined by IUPAC, a "bridge" is an unbranched chain of atoms or an atom or a valence bond connecting two bridgeheads, where a "bridgehead" is any skeletal atom of the ring system which is bonded to three or more skeletal atoms (excluding hydrogen). In some embodiments, a bridged bicyclic group has 7-12 ring members and 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur. Such bridged bicyclic groups are well known in the art and include those groups set forth below where each group is attached to the rest of the molecule at any substitutable carbon or nitrogen atom. Unless otherwise specified, a bridged bicyclic group is optionally substituted with one or more substituents as set forth for aliphatic groups. Additionally or alternatively, any substitutable nitrogen of a bridged bicyclic group is optionally substituted. Exemplary bicyclic rings include:



[0019] Exemplary bridged bicyclics include:



[0020] The term "lower alkyl" refers to a C_{1-4} straight or branched alkyl group. Exemplary lower alkyl groups are methyl, ethyl, propyl, isopropyl, butyl, isobutyl, and tert-butyl.

[0021] The term "lower haloalkyl" refers to a C_{1-4} straight or branched alkyl group that is substituted with one or more halogen atoms.

[0022] The term "heteroatom" means one or more of oxygen, sulfur, nitrogen, phosphorus, or silicon (including, any oxidized form of nitrogen, sulfur, phosphorus, or silicon; the quaternized form of any basic nitrogen or; a substitutable nitrogen of a heterocyclic ring, for example N (as in 3,4-dihydro-2H-pyrrolyl), NH (as in pyrrolidinyl) or NR^+ (as in N-substituted pyrrolidinyl)).

[0023] The term "unsaturated," as used herein, means that a moiety has one or more units of unsaturation.

[0024] As used herein, the term "bivalent C_{1-8} (or C_{1-6}) saturated or unsaturated, straight or branched, hydrocarbon chain", refers to bivalent alkylene, alkenylene, and alkynylene chains that are straight or branched as defined herein.

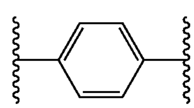
[0025] The term "alkylene" refers to a bivalent alkyl group. An "alkylene chain" is a polymethylene group, i.e., —$(CH_2)_n$—, wherein n is a positive integer, preferably from 1 to 6, from 1 to 4, from 1 to 3, from 1 to 2, or from 2 to 3. A substituted alkylene chain is a polymethylene group in which one or more methylene hydrogen atoms are replaced with a substituent. Suitable substituents include those described below for a substituted aliphatic group.

[0026] The term "—(C_0 alkylene)-" refers to a bond. Accordingly, the term "—(C_{0-3} alkylene)-" encompasses a bond (i.e., C_0) and a —(C_{1-3} alkylene)- group.

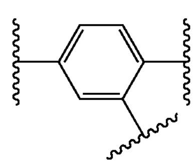
[0027] The term "alkenylene" refers to a bivalent alkenyl group. A substituted alkenylene chain is a polymethylene group containing at least one double bond in which one or more hydrogen atoms are replaced with a substituent. Suitable substituents include those described below for a substituted aliphatic group.

[0028] The term "halogen" means F, Cl, Br, or I.

[0029] The term "aryl" used alone or as part of a larger moiety as in "aralkyl," "aralkoxy," or "aryloxyalkyl," refers to monocyclic or bicyclic ring systems having a total of five to fourteen ring members, wherein at least one ring in the system is aromatic and wherein each ring in the system contains 3 to 7 ring members. The term "aryl" may be used interchangeably with the term "aryl ring." In certain embodiments of the present invention, "aryl" refers to an aromatic ring system which includes, but not limited to, phenyl, biphenyl, naphthyl, anthracyl and the like, which may bear one or more substituents. Also included within the scope of the term "aryl," as it is used herein, is a group in which an aromatic ring is fused to one or more non-aromatic rings, such as indanyl, phthalimidyl, naphthimidyl, phenanthridinyl, or tetrahydronaphthyl, and the like. The term "phenylene" refers to a multivalent phenyl group having the appropriate number of open valences to account for groups attached to it. For example, "phenylene" is a bivalent phenyl group when it has two groups attached to it

(e.g., );

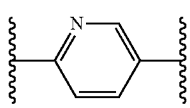
"phenylene" is a trivalent phenyl group when it has three groups attached to it

(e.g., ).

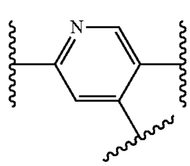
The term "arylene" refers to a bivalent aryl group.

[0030] The terms "heteroaryl" and "heteroar-," used alone or as part of a larger moiety, e.g., "heteroaralkyl," or "heteroaralkoxy," refer to groups having 5 to 10 ring atoms, preferably 5, 6, or 9 ring atoms; having 6, 10, or 14 π electrons shared in a cyclic array; and having, in addition to carbon atoms, from one to five heteroatoms. The term "heteroatom" refers to nitrogen, oxygen, or sulfur, and includes any oxidized form of nitrogen or sulfur, and any quaternized form of a basic nitrogen. Heteroaryl groups include, without limitation, thienyl, furanyl, pyrrolyl, imidazolyl, pyrazolyl, triazolyl, tetrazolyl, oxazolyl, isoxazolyl, oxadiazolyl, thiazolyl, isothiazolyl, thiadiazolyl, pyridyl, pyridazinyl, pyrimidinyl, pyrazinyl, indolizinyl, purinyl, naphthyridinyl, and pteridinyl. The terms "heteroaryl" and "heteroar-", as used herein, also include groups in which a heteroaromatic ring is fused to one or more aryl, cycloaliphatic, or heterocyclyl rings, where unless otherwise specified, the radical or point of attachment is on the heteroaromatic ring or on one of the rings to which the heteroaromatic ring is fused. Nonlimiting examples include indolyl, isoindolyl, benzothienyl, benzofuranyl, dibenzofuranyl, indazolyl, benzimidazolyl, benzthiazolyl, quinolyl, isoquinolyl, cinnolinyl, phthalazinyl, quinazolinyl, quinoxalinyl, 4H-quinolizinyl, carbazolyl, acridinyl, phenazinyl, phenothiazinyl, phenoxazinyl, tetrahydroquinolinyl, and tetrahydroisoquinolinyl. A heteroaryl group may be mono- or bicyclic. The term "heteroaryl" may be used interchangeably with the terms "heteroaryl ring," "heteroaryl group," or "heteroaromatic," any of which terms include rings that are optionally substituted. The term "heteroaralkyl" refers to an alkyl group substituted by a heteroaryl, wherein the alkyl and heteroaryl portions independently are optionally substituted.

[0031] The term "heteroarylene" refers to a multivalent heteroaryl group having the appropriate number of open valences to account for groups attached to it. For example, "heteroarylene" is a bivalent heteroaryl group when it has two groups attached to it; "heteroarylene" is a trivalent heteroaryl group when it has three groups attached to it. The term "pyridinylene" refers to a multivalent pyridine radical having the appropriate number of open valences to account for groups attached to it. For example, "pyridinylene" is a bivalent pyridine radical when it has two groups attached to it

(e.g., );

"pyridinylene" is a trivalent pyridine radical when it has three groups attached to it

(e.g., ).

[0032] As used herein, the terms "heterocycle," "heterocyclyl," "heterocyclic radical," and "heterocyclic ring" are used interchangeably and refer to a stable 5- to 7-membered monocyclic or 7-10-membered bicyclic heterocyclic moiety that is either saturated or partially unsaturated, and having, in addition to carbon atoms, one or more, preferably one to four, heteroatoms, as defined above. When used in reference to a ring atom of a heterocycle, the term "nitrogen" includes a substituted nitrogen. As an example, in a saturated or partially unsaturated ring having 0-3 heteroatoms selected from oxygen, sulfur or nitrogen, the nitrogen may be N (as in 3,4-dihydro-2H-pyrrolyl), NH (as in pyrrolidinyl), or $^{+}$NR (as in N-substituted pyrrolidinyl).

[0033] A heterocyclic ring can be attached to its pendant group at any heteroatom or carbon atom that results in a stable structure and any of the ring atoms can be optionally substituted. Examples of such saturated or partially unsaturated heterocyclic radicals include, without limitation, tetrahydrofuranyl, tetrahydrothiophenyl pyrrolidinyl, piperidinyl, pyrrolinyl, tetrahydroquinolinyl, tetrahydroisoquinolinyl, decahydroquinolinyl, oxazolidinyl, piperazinyl, dioxanyl, dioxolanyl, diazepinyl, oxazepinyl, thiazepinyl, morpholinyl, 2-oxa-6-azaspiro[3.3]heptane, and quinuclidinyl. The terms "heterocycle," "heterocyclyl," "heterocyclyl ring," "heterocyclic group," "heterocyclic moiety," and "heterocyclic radical," are used interchangeably herein, and also include groups in which a heterocyclyl ring is fused to one or more aryl, heteroaryl, or cycloaliphatic rings, such as indolinyl, 3H-indolyl, chromanyl, phenanthridinyl, or tetrahydroquinolinyl. A heterocyclyl group may be mono- or bicyclic. The term "heterocyclylalkyl" refers to an alkyl group substituted by a heterocyclyl, wherein the alkyl and heterocyclyl portions independently are optionally substituted. The term "oxo-heterocyclyl" refers to a heterocyclyl substituted by an oxo group. The term "heterocyclylene" refers to a multivalent heterocyclyl group having the appropriate number of open valences to account for groups attached to it. For example, "heterocyclylene" is a bivalent heterocyclyl group when it has two groups attached to it; "heterocyclylene" is a trivalent heterocyclyl group when it has three groups attached to it.

[0034] As used herein, the term "partially unsaturated" refers to a ring moiety that includes at least one double or triple bond. The term "partially unsaturated" is intended to encompass rings having multiple sites of unsaturation, but is not intended to include aryl or heteroaryl moieties, as herein defined.

[0035] As described herein, compounds of the invention may contain "optionally substituted" moieties. In general, the term "substituted," whether preceded by the term "optionally" or not, means that one or more hydrogens of the designated moiety are replaced with a suitable substituent. Unless otherwise indicated, an "optionally substituted" group may have a suitable substituent at each substitutable position of the group, and when more than one position in any given structure may be substituted with more than one substituent selected from a specified group, the substituent may be either the same or different at every position. Combinations of substituents envisioned by this invention are preferably those that result in the formation of stable or chemically feasible compounds. The term "stable," as used herein, refers to compounds that are not substantially altered when subjected to conditions to allow for their production, detection, and, in certain embodiments, their recovery, purification, and use for one or more of the purposes disclosed herein.

[0036] Each optional substituent on a substitutable carbon is a monovalent substituent independently selected from halogen; $—(CH_2)_{0-4}R^{\circ}$; $—(CH_2)_{0-4}OR^{\circ}$; $—O(CH_2)_{0-4}R^{\circ}$, $—O—(CH_2)_{0-4}C(O)OR^{\circ}$; $—(CH_2)_{0-4}CH(OR^{\circ})_2$; $—(CH_2)_{0-4}SR^{\circ}$; $—(CH_2)_{0-4}Ph$, which may be substituted with R°; $—(CH_2)_{0-4}O(CH_2)_{0-1}Ph$ which may be substituted with R°;

—CH═CHPh, which may be substituted with R°; —$(CH_2)_{0-4}O(CH_2)_{0-1}$-pyridyl which may be substituted with R°; —NO_2; —CN; —N_3; —$(CH_2)_{0-4}N(R°)_2$; —$(CH_2)_{0-4}N(R°)C(O)R°$; —N(R°)C(S)R°; —$(CH_2)_{0-4}N(R°)C(O)NR°_2$; —$N(R°)C(S)NR°_2$; —$(CH_2)_{0-4}N(R°)C(O)OR°$; —N(R°)N(R°)C(O)R°; —$N(R°)N(R°)C(O)NR°_2$; —N(R°)N(R°)C(O)OR°; —$(CH_2)_{0-4}C(O)R°$; —C(S)R°; —$(CH_2)_{0-4}C(O)OR°$; —$(CH_2)_{0-4}C(O)SR°$; —$(CH_2)_{0-4}C(O)OSiR°_3$; —$(CH_2)_{0-4}OC(O)R°$; —$OC(O)(CH_2)_{0-4}SR$—, SC(S)SR°; —$(CH_2)_{0-4}SC(O)R°$; —$(CH_2)_{0-4}C(O)NR°_2$; —$C(S)NR°_2$; —C(S)SR°; —SC(S)SR°, —$(CH_2)_{0-4}OC(O)NR°_2$; —C(O)N(OR°)R°; —C(O)C(O)R°; —$C(O)CH_2C(O)R°$; —C(NOR°)R°; —$(CH_2)_{0-4}SSR°$; —$(CH_2)_{0-4}S(O)_2R°$; —$(CH_2)_{0-4}S(O)_2OR°$; —$(CH_2)_{0-4}OS(O)_2R°$; —$S(O)_2NR°_2$; —$S(O)(NR°)R°$; —$S(O)_2N{=}C(NR°_2)_2$; —$(CH_2)_{0-4}S(O)R°$; —$N(R°)S(O)_2NR°_2$; —$N(R°)S(O)_2R°$; —N(OR°)R°; —$C(NH)NR°_2$; —$P(O)_2R°$; —$P(O)R°_2$; —$OP(O)R°_2$; —$OP(O)(OR°)_2$; $SiR°_3$; —(C_{1-4} straight or branched alkylene)O—$N(R°)_2$; or —(C_{1-4} straight or branched alkylene)C(O)O—$N(R°)_2$.

[0037] Each R° is independently hydrogen, C_{1-6} aliphatic, —CH_2Ph, —$O(CH_2)_{0-1}Ph$, —CH_2-(5-6 membered heteroaryl ring), or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, notwithstanding the definition above, two independent occurrences of R°, taken together with their intervening atom(s), form a 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, which may be substituted by a divalent substituent on a saturated carbon atom of R° selected from ═O and ═S; or each R° is optionally substituted with a monovalent substituent independently selected from halogen, —$(CH_2)_{0-2}R^●$, -(halo$R^●$), —$(CH_2)_{0-2}OH$, —$(CH_2)_{0-2}OR^●$, —$(CH_2)_{0-2}CH(OR^●)_2$; —O(halo$R^●$), —CN, —N_3, —$(CH_2)_{0-2}C(O)R^●$, —$(CH_2)_{0-2}C(O)OH$, —$(CH_2)_{0-2}C(O)OR^●$, —$(CH_2)_{0-2}SR^●$, —$(CH_2)_{0-2}SH$, —$(CH_2)_{0-2}NH_2$, —$(CH_2)_{0-2}NHR^●$, —$(CH_2)_{0-2}NR^●_2$, —NO_2, —$SiR^●_3$, —$OSiR^●_3$, —$C(O)SR^●$, —(C_{1-4} straight or branched alkylene)$C(O)OR^●$, or —$SSR^●$.

[0038] Each $R^●$ is independently selected from C_{1-4} aliphatic, —CH_2Ph, —$O(CH_2)_{0-1}Ph$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, and wherein each $R^●$ is unsubstituted or where preceded by halo is substituted only with one or more halogens; or wherein an optional substituent on a saturated carbon is a divalent substituent independently selected from ═O, ═S, ═NNR^*_2, ═NNHC(O)R*, ═NNHC(O)OR*, ═$NNHS(O)_2R^*$, ═NR*, ═NOR*, —$O(C(R^*_2))_{2-3}O$—, or —$S(C(R^*_2))_{2-3}S$—, or a divalent substituent bound to vicinal substitutable carbons of an "optionally substituted" group is —$O(CR^*_2)_{2-3}O$—, wherein each independent occurrence of R* is selected from hydrogen, C_{1-6} aliphatic or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur.

[0039] When R* is C_{1-6} aliphatic, R* is optionally substituted with halogen, —$R^●$, —(halo$R^●$), —OH, —$OR^●$, —O(halo$R^●$), —CN, —C(O)OH, —$C(O)OR^●$, —NH_2, —$NHR^●$, —$NR^●_2$, or —NO_2, wherein each $R^●$ is independently selected from C_{1-4} aliphatic, —CH_2Ph, —$O(CH_2)_{0-1}Ph$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, and wherein each $R^●$ is unsubstituted or where preceded by halo is substituted only with one or more halogens.

[0040] An optional substituent on a substitutable nitrogen is independently —$R^†$, —$NR^†_2$, —$C(O)R^†$, —$C(O)OR^†$, —$C(O)C(O)R^†$, —$C(O)CH_2C(O)R^†$, —$S(O)_2R^†$, —$S(O)_2NR^†_2$, —$C(S)NR^†_2$, —$C(NH)NR^†_2$, or —$N(R^†)S(O)_2R^†$; wherein each $R^†$ is independently hydrogen, C_{1-6} aliphatic, unsubstituted —OPh, or an unsubstituted 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, or, two independent occurrences of $R^†$, taken together with their intervening atom(s) form an unsubstituted 3-12-membered saturated, partially unsaturated, or aryl mono- or bicyclic ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur; wherein when $R^†$ is C_{1-6} aliphatic, $R^†$ is optionally substituted with halogen, —$R^●$, —(halo$R^●$), —OH, —$OR^●$, —O(halo$R^●$), —CN, —C(O)OH, —$C(O)OR^●$, —NH_2, —$NHR^●$, —$NR^●_2$, or —NO_2, wherein each $R^●$ is independently selected from C_{1-4} aliphatic, —CH_2Ph, —$O(CH_2)_{0-1}Ph$, or a 5-6-membered saturated, partially unsaturated, or aryl ring having 0-4 heteroatoms independently selected from nitrogen, oxygen, or sulfur, and wherein each $R^●$ is unsubstituted or where preceded by halo is substituted only with one or more halogens.

[0041] As used herein, the term "pharmaceutically acceptable salt" refers to those salts which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response and the like, and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well known in the art. For example, S. M. Berge et al., describe pharmaceutically acceptable salts in detail in J. Pharmaceutical Sciences, 1977, 66, 1-19, incorporated herein by reference. Pharmaceutically acceptable salts of the compounds of this invention include those derived from suitable inorganic and organic acids and bases. Examples of pharmaceutically acceptable, nontoxic acid addition salts are salts of an amino group formed with inorganic acids such as hydrochloric acid, hydrobromic acid, phosphoric acid, sulfuric acid and perchloric acid or with organic acids such as acetic acid, oxalic acid, maleic acid, tartaric acid, citric acid, succinic acid or malonic acid or by using other methods used in the art such as ion exchange. Other pharmaceutically acceptable salts include adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, formate, fumarate, glucoheptonate, glycerophosphate, gluconate, hemisulfate, heptanoate, hexanoate, hydroiodide, 2-hydroxy-ethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, p-toluenesulfonate, undecanoate, valerate salts, and the like.

[0042] Further, acids which are generally considered suitable for the formation of pharmaceutically useful salts from basic pharmaceutical compounds are discussed, for example, by P. Stahl et al., Camille G. (eds.) *Handbook of Pharmaceutical Salts. Properties, Selection and Use*. (2002) Zurich: Wiley-VCH; S. Berge et al., *Journal of Pharmaceu-*

tical Sciences (1977) 66(1) 1-19; P. Gould, *International J. of Pharmaceutics* (1986) 33 201-217; Anderson et al., *The Practice of Medicinal Chemistry* (1996), Academic Press, New York; and in *The Orange Book* (Food & Drug Administration, Washington, D.C. on their website). These disclosures are incorporated herein by reference.

[0043] Salts derived from appropriate bases include alkali metal, alkaline earth metal, ammonium and $N^{+}(C_{1-4}alkyl)_4$ salts. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, magnesium, and the like. Further pharmaceutically acceptable salts include, when appropriate, nontoxic ammonium, quaternary ammonium, and amine cations formed using counterions such as halide, hydroxide, carboxylate, sulfate, phosphate, nitrate, loweralkyl sulfonate and aryl sulfonate.

[0044] Unless otherwise stated, structures depicted herein are also meant to include all isomeric (e.g., enantiomeric, diastereomeric, and geometric (or conformational)) forms of the structure; for example, the R and S configurations for each asymmetric center, Z and E double bond isomers, and Z and E conformational isomers. Therefore, single stereochemical isomers as well as enantiomeric, diastereomeric, and geometric (or conformational) mixtures of the present compounds are within the scope of the invention. Unless otherwise stated, all tautomeric forms of the compounds of the invention are within the scope of the invention. Additionally, unless otherwise stated, structures depicted herein are also meant to include compounds that differ only in the presence of one or more isotopically enriched atoms. For example, compounds having the present structures including the replacement of hydrogen by deuterium or tritium, or the replacement of a carbon by a ^{13}C- or ^{14}C-enriched carbon are within the scope of this invention. Such compounds are useful, for example, as analytical tools, as probes in biological assays, or as therapeutic agents in accordance with the present invention.

[0045] Diastereomeric mixtures can be separated into their individual diastereomers on the basis of their physical chemical differences by methods known to those skilled in the art, such as, for example, by chromatography and/or fractional crystallization. Enantiomers can be separated by converting the enantiomeric mixture into a diastereomeric mixture by reaction with an appropriate optically active compound (e.g., chiral auxiliary such as a chiral alcohol or Mosher's acid chloride), separating the diastereomers and converting (e.g., hydrolyzing) the individual diastereomers to the corresponding pure enantiomers. Alternatively, a particular enantiomer of a compound of the present invention may be prepared by asymmetric synthesis. Still further, where the molecule contains a basic functional group (such as amino) or an acidic functional group (such as carboxylic acid) diastereomeric salts are formed with an appropriate optically-active acid or base, followed by resolution of the diastereomers thus formed by fractional crystallization or chromatographic means known in the art, and subsequent recovery of the pure enantiomers.

[0046] Individual stereoisomers of the compounds of the invention may, for example, be substantially free of other isomers, or may be admixed, for example, as racemates or with all other, or other selected, stereoisomers. Chiral center (s) in a compound of the present invention can have the S or R configuration as defined by the *IUPAC* 1974 Recommendations. Further, to the extent a compound described herein may exist as a atropisomer (e.g., substituted biaryls), all forms of such atropisomer are considered part of this invention.

[0047] Chemical names, common names, and chemical structures may be used interchangeably to describe the same structure. If a chemical compound is referred to using both a chemical structure and a chemical name, and an ambiguity exists between the structure and the name, the structure predominates. It should also be noted that any carbon as well as heteroatom with unsatisfied valences in the text, schemes, examples and tables herein is assumed to have the sufficient number of hydrogen atom(s) to satisfy the valences.

[0048] The terms "a" and "an" as used herein mean "one or more" and include the plural unless the context is inappropriate.

[0049] The term "alkyl" refers to a saturated straight or branched hydrocarbon, such as a straight or branched group of 1-12, 1-10, or 1-6 carbon atoms, referred to herein as C_1-C_{12} alkyl, C_1-C_{10} alkyl, and C_1-C_6 alkyl, respectively. Exemplary alkyl groups include, but are not limited to, methyl, ethyl, propyl, isopropyl, 2-methyl-1-propyl, 2-methyl-2-propyl, 2-methyl-1-butyl, 3-methyl-1-butyl, 2-methyl-3-butyl, 2,2-dimethyl-1-propyl, 2-methyl-1-pentyl, 3-methyl-1-pentyl, 4-methyl-1-pentyl, 2-methyl-2-pentyl, 3-methyl-2-pentyl, 4-methyl-2-pentyl, 2,2-dimethyl-1-butyl, 3,3-dimethyl-1-butyl, 2-ethyl-1-butyl, butyl, isobutyl, t-butyl, pentyl, isopentyl, neopentyl, hexyl, heptyl, octyl, etc.

[0050] The term "cycloalkyl" refers to a monovalent saturated cyclic, bicyclic, or bridged cyclic (e.g., adamantyl) hydrocarbon group of 3-12, 3-8, 4-8, or 4-6 carbons, referred to herein, e.g., as "C_3-C_6 cycloalkyl," derived from a cycloalkane. Exemplary cycloalkyl groups include cyclohexyl, cyclopentyl, cyclobutyl, and cyclopropyl. The term "cycloalkylene" refers to a bivalent cycloalkyl group.

[0051] The term "haloalkyl" refers to an alkyl group that is substituted with at least one halogen. Exemplary haloalkyl groups include $—CH_2F$, $—CHF_2$, $—CF_3$, $—CH_2CF_3$, $—CF_2CF_3$, and the like. The term "haloalkylene" refers to a bivalent haloalkyl group.

[0052] The term "hydroxyalkyl" refers to an alkyl group that is substituted with at least one hydroxyl. Exemplary hydroxyalkyl groups include $—CH_2CH_2OH$, $—C(H)(OH)CH_3$, $—CH_2C(H)(OH)CH_2CH_2OH$, and the like.

[0053] The terms "alkenyl" and "alkynyl" are art-recognized and refer to unsaturated aliphatic groups analogous in length and possible substitution to the alkyls described above, but that contain at least one double or triple bond respectively.

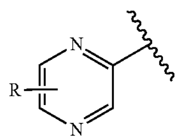
[0054] The term "carbocyclylene" refers to a multivalent carbocyclyl group having the appropriate number of open valences to account for groups attached to it. For example, "carbocyclylene" is a bivalent carbocyclyl group when it has two groups attached to it; "carbocyclylene" is a trivalent carbocyclyl group when it has three groups attached to it.

[0055] The terms "alkoxyl" or "alkoxy" are art-recognized and refer to an alkyl group, as defined above, having an oxygen radical attached thereto. Representative alkoxyl groups include methoxy, ethoxy, propyloxy, tert-butoxy and the like. The term "haloalkoxyl" refers to an alkoxyl group that is substituted with at least one halogen. Exemplary haloalkoxyl groups include $—OCH_2F$, $—OCHF_2$, $—OCF_3$, $—OCH_2CF_3$, $—OCF_2CF_3$, and the like.

[0056] The term "oxo" is art-recognized and refers to a "═O" substituent. For example, a cyclopentane substituted with an oxo group is cyclopentanone.

[0057] The symbol "ᴧᴧᴧ" indicates a point of attachment.

[0058] When a chemical structure containing a ring is depicted with a substituent having a bond that crosses a ring bond, the substituent may be attached at any available position on the ring. For example, the chemical structure



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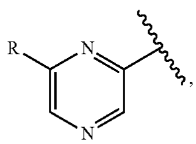

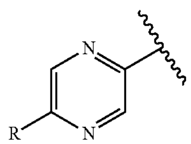

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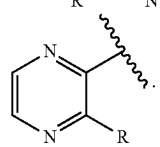
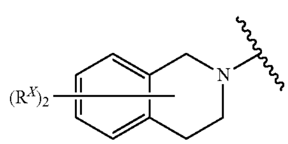
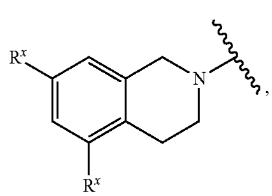
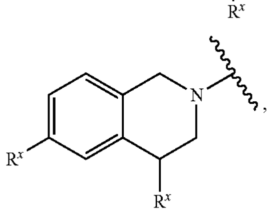


In the context of a polycyclic fused ring, when a chemical structure containing a polycyclic fused ring is depicted with one or more substituent(s) having a bond that crosses multiple rings, the one or more substituent(s) may be independently attached to any of the rings crossed by the bond. To illustrate, the chemical structure



encompasses, for example,





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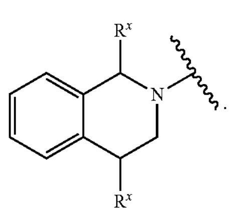


[0059] When any substituent or variable occurs more than one time in any constituent or the compound of the invention, its definition on each occurrence is independent of its definition at every other occurrence, unless otherwise indicated.

[0060] One or more compounds of the invention may exist in unsolvated as well as solvated forms with pharmaceutically acceptable solvents such as water, ethanol, and the like, and it is intended that the invention embrace both solvated and unsolvated forms. "Solvate" means a physical association of a compound of this invention with one or more solvent molecules. This physical association involves varying degrees of ionic and covalent bonding, including hydrogen bonding. In certain instances the solvate will be capable of isolation, for example when one or more solvent molecules are incorporated in the crystal lattice of the crystalline solid. "Solvate" encompasses both solution-phase and isolatable solvates. Non-limiting examples of suitable solvates include ethanolates, methanolates, and the like. "Hydrate" is a solvate wherein the solvent molecule is H_2O.

[0061] As used herein, the terms "subject" and "patient" are used interchangeable and refer to organisms to be treated by the methods of the present invention. Such organisms preferably include, but are not limited to, mammals (e.g., murines, simians, equines, bovines, porcines, canines, felines, and the like), and most preferably includes humans.

[0062] The term "IC_{50}" is art-recognized and refers to the concentration of a compound that is required to achieve 50% inhibition of the target.

[0063] As used herein, the term "effective amount" refers to the amount of a compound sufficient to effect beneficial or desired results (e.g., a therapeutic, ameliorative, inhibitory or preventative result). An effective amount can be administered in one or more administrations, applications or dosages and is not intended to be limited to a particular formulation or administration route. As used herein, the term "treating" includes any effect, e.g., lessening, reducing, modulating, ameliorating or eliminating, that results in the improvement of the condition, disease, disorder, and the like, or ameliorating a symptom thereof.

[0064] As used herein, the term "pharmaceutical composition" refers to the combination of an active agent with a carrier, inert or active, making the composition especially suitable for diagnostic or therapeutic use in vivo or ex vivo.

[0065] As used herein, the term "pharmaceutically acceptable carrier" refers to any of the standard pharmaceutical carriers, such as a phosphate buffered saline solution, water, emulsions (e.g., such as an oil/water or water/oil emulsions), and various types of wetting agents. The compositions also can include stabilizers and preservatives. For examples of carriers, stabilizers and adjuvants, see e.g., Martin, Remington's Pharmaceutical Sciences, 15th Ed., Mack Publ. Co., Easton, PA [1975].

[0066] For therapeutic use, salts of the compounds of the present invention are contemplated as being pharmaceutically acceptable. However, salts of acids and bases that are non-pharmaceutically acceptable may also find use, for example, in the preparation or purification of a pharmaceutically acceptable compound.

[0067] In addition, when a compound of the invention contains both a basic moiety (such as, but not limited to, a pyridine or imidazole) and an acidic moiety (such as, but not limited to, a carboxylic acid) zwitterions ("inner salts") may be formed. Such acidic and basic salts used within the scope of the invention are pharmaceutically acceptable (i.e., non-toxic, physiologically acceptable) salts. Such salts of the compounds of the invention may be formed, for example, by reacting a compound of the invention with an amount of acid or base, such as an equivalent amount, in a medium such as one in which the salt precipitates or in an aqueous medium followed by lyophilization.

[0068] Throughout the description, where compositions are described as having, including, or comprising specific components, or where processes and methods are described as having, including, or comprising specific steps, it is contemplated that, additionally, there are compositions of the present invention that consist essentially of, or consist of, the recited components, and that there are processes and methods according to the present invention that consist essentially of, or consist of, the recited processing steps.

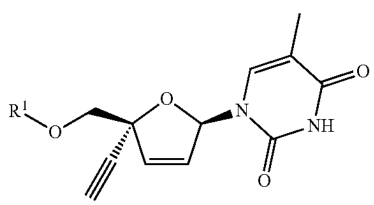
[0069] As a general matter, compositions specifying a percentage are by weight unless otherwise specified.

I. Substituted 2',3'-Didehydro-3'-deoxy-4'-ethynylthymidines and Related Compounds

[0070] The invention provides substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds. The compounds may be used in the pharmaceutical compositions and therapeutic methods described herein. Exemplary compounds are described in the following sections, along with exemplary procedures for making the compounds.

[0071] One aspect of the invention provides a compound represented by Formula I:

(I)



[0072] or a pharmaceutically acceptable salt thereof; wherein:

[0073] R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —P(O)(N(R^3)(R^4))$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^8;

[0074] R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen; or

[0075] R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

[0076] R^3 and R^9 each represent independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0077] R^4 represents independently for each occurrence —C(R^5)$_2$—CO$_2$R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0078] R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

[0079] R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0080] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —N(R^9)$_2$;

[0081] R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—

(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0082] m and p are independently for each occurrence 0, 1, 2, or 3; and

[0083] n is 1, 2, or 3.

[0084] The definitions of variables in Formula I above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

[0085] In certain embodiments, the compound is a compound of Formula I.

[0086] As defined generally above, R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —P(O)(N(R^3)(R^4))$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^8.

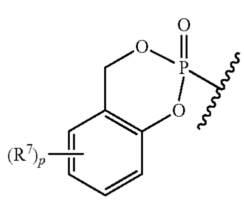
[0087] In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, or —P(O)(N(R^3)(R^4))$_2$. In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —P(O)(N(R^3)(R^4))$_2$.

[0088] In certain embodiments, R^1 is —P(O)(OR^2)$_2$ or —P(O)(N(R^3)(R^4))$_2$.

[0089] In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^8. In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N(R^3)$_2$. In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —P(O)(OR^2)$_2$. In certain embodiments, R^1 is —C(O)—C(H)(R^5)—N(R^3)$_2$ or —C(O)R^8.

[0090] In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)). In certain embodiments, R^1 is —P(O)(OR^2)$_2$. In certain embodiments, R^1 is —P(O)(N(R^3)(R^4))$_2$. In certain embodiments, R^1 is —C(O)—C(H)(R^5)—N(R^3)$_2$. In certain embodiments, R^1 is —C(O)R^8.

[0091] In certain embodiments, R^1 is or



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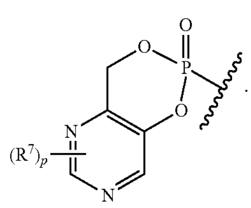
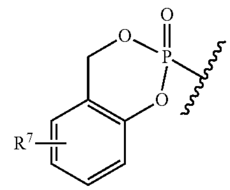


In certain embodiments, R^1 is



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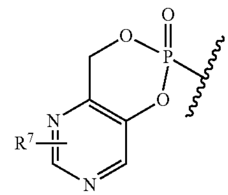
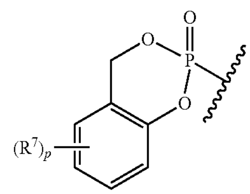


In certain embodiments, R^1 is



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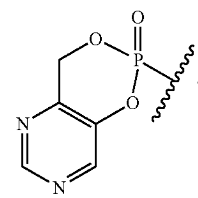
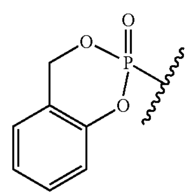


In certain embodiments, R^1 is



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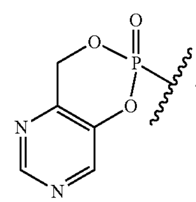
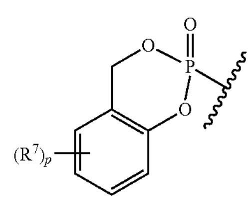
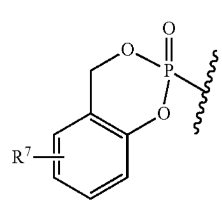


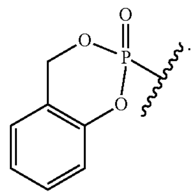
[0092] In certain embodiments, R^1 is



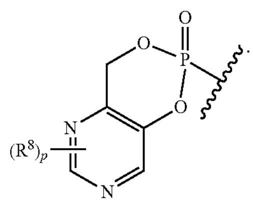
In certain embodiments, R^1 is



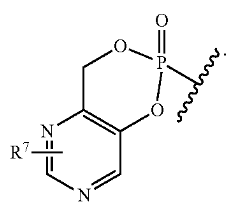
In certain embodiments, R^1 is



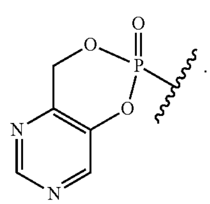
[0093] In certain embodiments, R^1 is



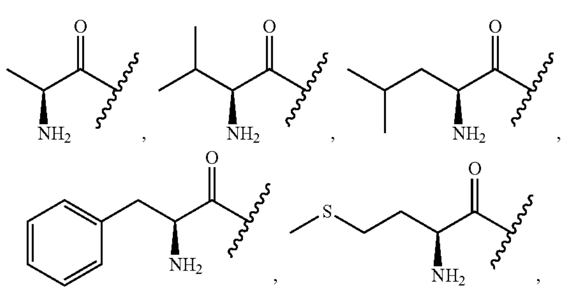
In certain embodiments, R^1 is



In certain embodiments, R^1 is



[0094] In certain embodiments, R^1 is



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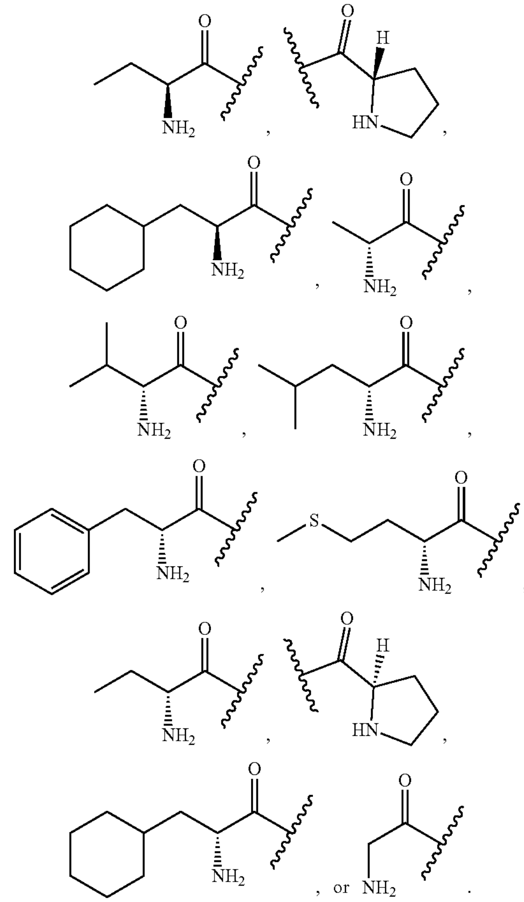
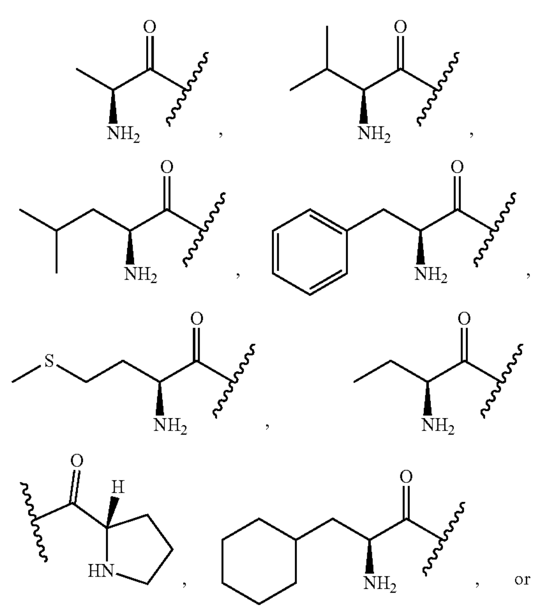


[0095] In certain embodiments, R^1 is



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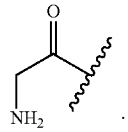
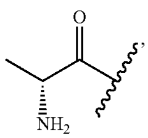
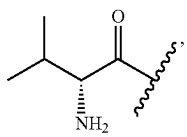
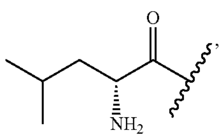
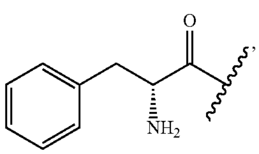


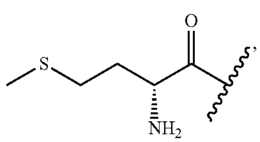
In certain embodiments, R^1 is

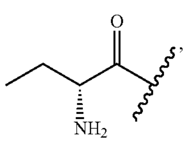


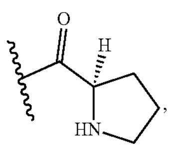


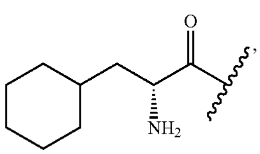












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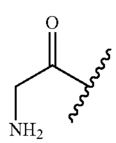
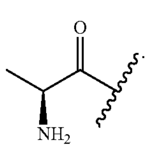
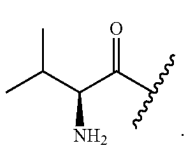


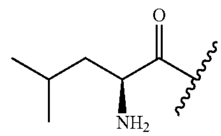
[0096] In certain embodiments, R^1 is



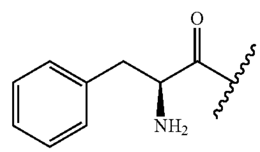
In certain embodiments, R^1 is



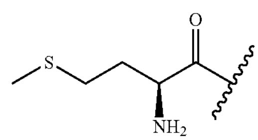
In certain embodiments, R^1 is



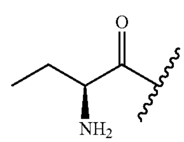
In certain embodiments, R^1 is



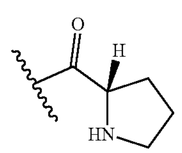
In certain embodiments, R^1 is



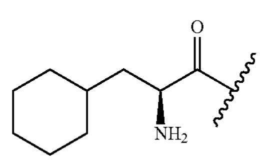
In certain embodiments, R^1 is



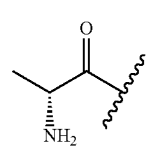
In certain embodiments, R^1 is



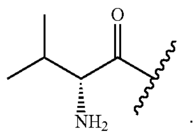
In certain embodiments, R^1 is



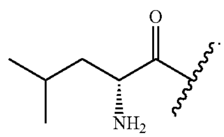
[0097] In certain embodiments, R^1 is



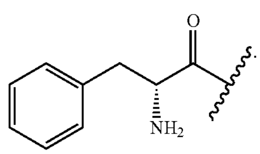
In certain embodiments, R^1 is



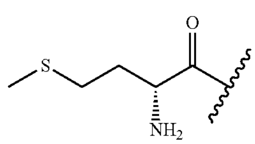
In certain embodiments, R^1 is



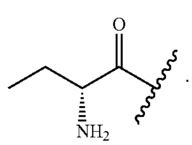
In certain embodiments, R^1 is



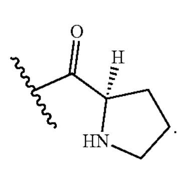
In certain embodiments, R^1 is



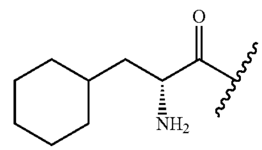
In certain embodiments, R^1 is



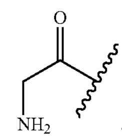
In certain embodiments, R^1 is



In certain embodiments, R^1 is



[0098] In certain embodiments, R^1 is



In certain embodiments, R^1 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

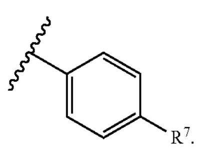
[0099] As defined generally above, R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen; or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7.

[0100] In certain embodiments, R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0101] In certain embodiments, R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence phenyl or naphthyl; each of which is substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7.

[0102] In certain embodiments, R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^2 represents independently for each occurrence phenyl or naphthyl. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0103] In certain embodiments, R^2 is phenyl substituted with m instances of R^7. In certain embodiments, R^2 is



In certain embodiments, R^2 is naphthyl substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7.

[0104] In certain embodiments, R^2 is phenyl. In certain embodiments, R^2 is naphthyl. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^2 represents independently for each occurrence an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0105] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen; or two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7.

[0106] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0107] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0108] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen.

[0109] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl); wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—N(H)—(C_{1-10} alkyl); wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-10} alkyl) or —CH_2—OC(O)—N(H)—(C_{1-10} alkyl); wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-6} alkyl) or —CH_2—OC(O)—N(H)—(C_{1-6} alkyl). In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{3-5} cycloalkyl) or —CH_2—OC(O)—N(H)—(C_{3-5} cycloalkyl).

[0110] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-6} alkyl). In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{3-5} cycloalkyl).

[0111] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—N(H)—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)—N(H)—(C_{1-10} alkyl); wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)—N(R^9)—(C_{1-6} alkyl). In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)—N(R^9)—(C_{3-5} cycloalkyl).

[0112] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl).

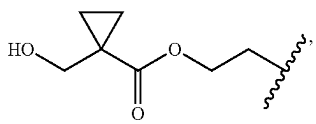
[0113] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl.

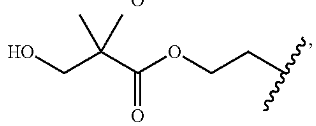
[0114] In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl.

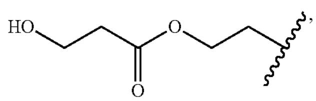
[0115] In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is substituted with one hydroxyl.

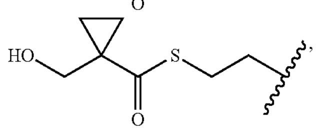
[0116] In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is substituted with one hydroxyl.

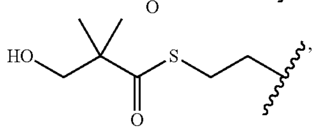
[0117] In certain embodiments, R^2 represents independently for each occurrence











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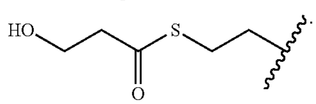
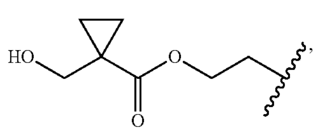
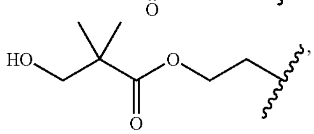


In certain embodiments, R^2 represents independently for each occurrence





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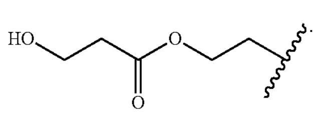
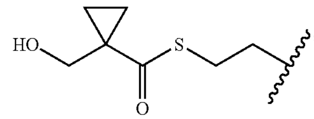
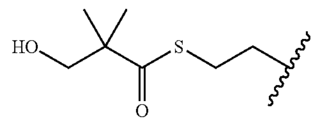


In certain embodiments, R^2 represents independently for each occurrence





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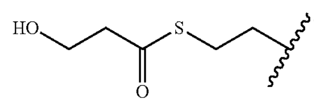
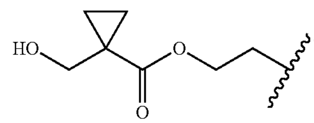
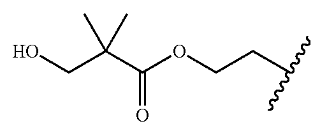


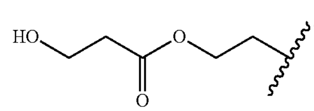
[0118] In certain embodiments, R^2 is



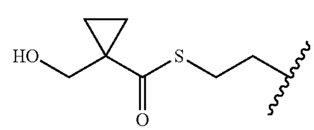
In certain embodiments, R^2 is



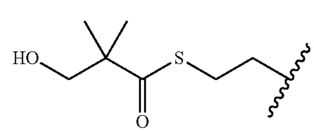
In certain embodiments, R^2 is



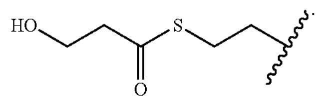
In certain embodiments, R^2 is



In certain embodiments, R^2 is



In certain embodiments, R^2 is



[0119] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-20} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen. In certain embodiments, one instance of R^2 is —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-10} alkylene is substituted with one —O—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen.

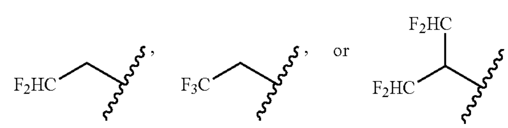
[0120] In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—O—(C_{1-20} alkyl), —$(CH_2)_3$—O—(C_{1-20} alkyl), —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—S—(C_{1-20} alkyl), or —$(CH_2)_3$—S—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—O—(C_{1-20} alkyl) or —$(CH_2)_3$—O—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—O—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —$(CH_2)_3$—O—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—S—(C_{1-20} alkyl) or —$(CH_2)_3$—S—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—S—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —$(CH_2)_3$—S—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen.

[0121] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-7} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-10} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} haloalkyl optionally substituted with one hydroxyl.

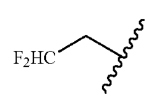
[0122] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0123] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-7} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-10} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

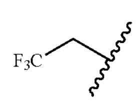
[0124] In certain embodiments, R^2 represents independently for each occurrence



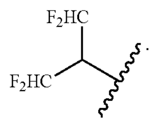
In certain embodiments, R^2 is



In certain embodiments, R^2 is



In certain embodiments, R^2 is



[0125] In certain embodiments, R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0126] In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring; and wherein said benzene ring is substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring.

[0127] In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 5-6 membered heteroaromatic ring; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms.

[0128] In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0129] In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring; and wherein said benzene ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring.

[0130] In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 5-6 membered heteroaromatic ring; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms.

[0131] In certain embodiments, R^2 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0132] As defined generally above, R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and

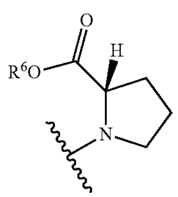
R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl. In certain embodiments, R^3 represents independently for each occurrence C_{1-4} alkyl.

[0133] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl; or R^3 and R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

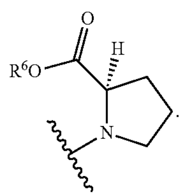
[0134] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl. In certain embodiments, R^3 is hydrogen. In certain embodiments, R^3 is methyl. In certain embodiments, R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, two instances of R^3 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0135] In certain embodiments, R^3 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

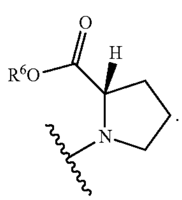
[0136] In certain embodiments, —N(R^3)(R^4) is



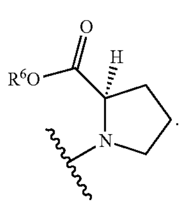
or



In certain embodiments, —N(R^3)(R^4) is



In certain embodiments, —N(R^3)(R^4) is



[0137] As defined generally above, R^4 represents independently for each occurrence —$C(R^5)_2$—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7.

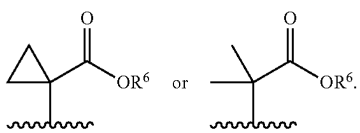
[0138] In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0139] In certain embodiments, R^4 represents independently for each occurrence —$C(R^5)_2$—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl.

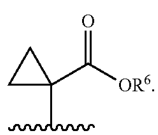
[0140] In certain embodiments, R^4 represents independently for each occurrence —$C(R^5)_2$—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—

(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

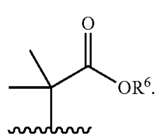
[0141] In certain embodiments, R^4 represents independently for each occurrence —C(R^5)$_2$—CO_2R^6. In certain embodiments, R^4 represents independently for each occurrence



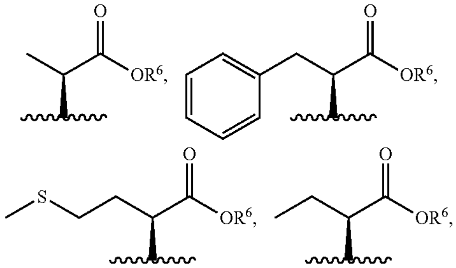
In certain embodiments, R^4 represents independently for each occurrence



In certain embodiments, R^4 represents independently for each occurrence



[0142] In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6. In certain embodiments, R^4 is



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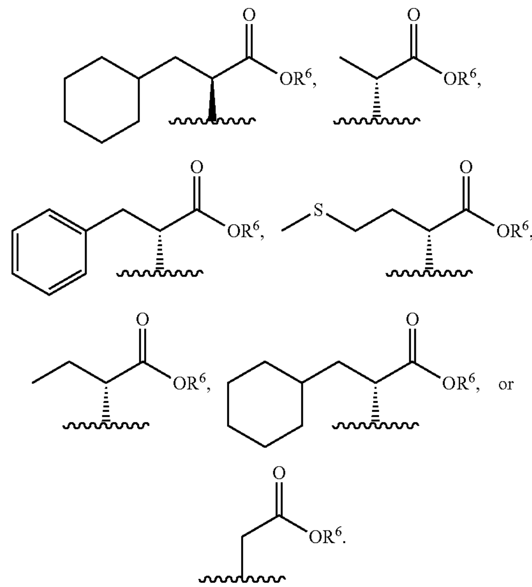
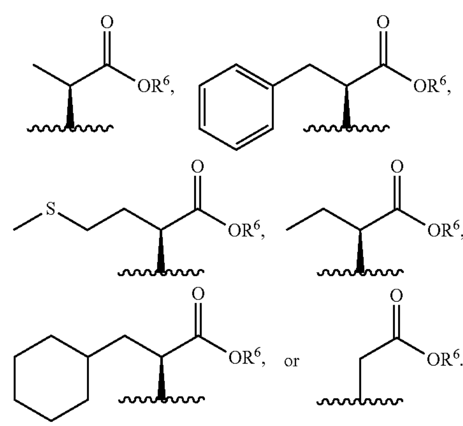
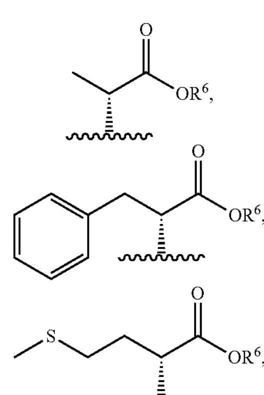


In certain embodiments, R^4 is



In certain embodiments, R^4 is



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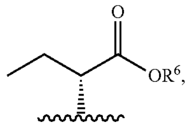

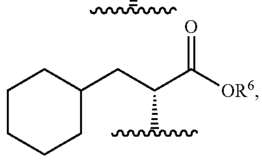

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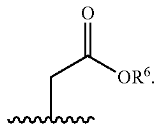
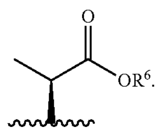
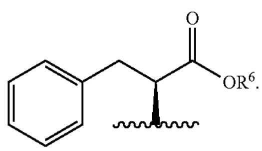


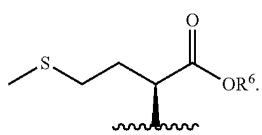
[0143] In certain embodiments, R^4 is



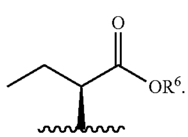
In certain embodiments, R^4 is



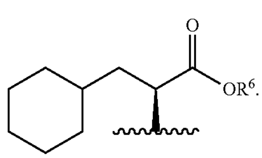
In certain embodiments, R^4 is



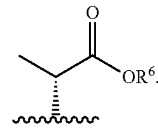
In certain embodiments, R^4 is



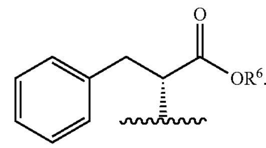
In certain embodiments, R^4 is



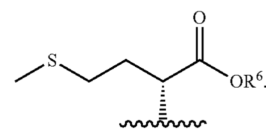
In certain embodiments, R^4 is



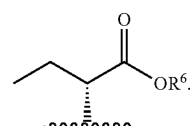
In certain embodiments, R^4 is



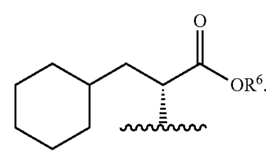
In certain embodiments, R^4 is



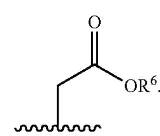
In certain embodiments, R^4 is



In certain embodiments, R^4 is



In certain embodiments, R^4 is



[0144] In certain embodiments, R^4 is C_{1-20} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-7} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-4} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-20} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0145] In certain embodiments, R^4 is C_{1-20} alkyl. In certain embodiments, R^4 is C_{1-7} alkyl. In certain embodiments, R^4 is C_{1-4} alkyl. In certain embodiments, R^4 is C_{1-20} haloalkyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0146] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7. In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, or naphthyl; wherein each of said phenyl and naphthyl is substituted with m instances of R^7. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7.

[0147] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, or naphthyl. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0148] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, wherein said phenyl is substituted with m instances of R^7. In certain embodiments, R^4 is phenyl substituted with m instances of R^7. In certain embodiments, R^4 is naphthyl substituted with m instances of R^7. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7. In certain embodiments, R^4 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7.

[0149] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl. In certain embodiments, R^4 is phenyl. In certain embodiments, R^4 is naphthyl. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^4 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0150] In certain embodiments, R^4 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0151] As defined generally above, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring; or R^3 and R^5 are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0152] In certain embodiments, R^5 is C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or R^3 and R^5 are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0153] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, or C_{3-5} cycloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} haloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{3-5} cycloalkyl.

[0154] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl) or —SH. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl or hydroxyl.

[0155] In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl) or —SH. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl or hydroxyl.

[0156] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with phenyl or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0157] In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0158] In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl) or —SH. In certain embodiments, R^5 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl or hydroxyl.

[0159] In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0160] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring.

[0161] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl.

[0162] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0163] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl). In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl.

[0164] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl or hydrogen. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-4} alkyl.

[0165] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen. In certain embodiments, R^5 is C_{1-6} alkyl. In certain embodiments, R^5 is C_{1-4} alkyl. In certain embodiments, R^5 is methyl. In certain embodiments, R^5 is hydrogen.

[0166] In certain embodiments, two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring. In certain embodiments, two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-membered saturated carbocyclic ring.

[0167] In certain embodiments, R^5 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0168] As defined generally above, R^6 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0169] In certain embodiments, R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

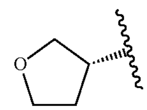
[0170] In certain embodiments, R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^6 is C_{1-6} haloalkyl. In certain embodiments, R^6 is a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0171] In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0172] In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^6 is C_{1-6} alkyl substituted with a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0173] In certain embodiments, R^6 represents independently for each occurrence C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0174] In certain embodiments, R^6 represents independently for each occurrence C_{1-6} alkyl, allyl, C_{3-5} cycloalkyl,



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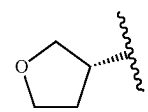
—CH_2-phenyl, or —CH_2—(C_{3-5} cycloalkyl). In certain embodiments, R^6 is C_{1-6} alkyl, allyl, C_{3-5} cycloalkyl,



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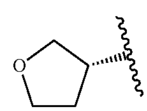
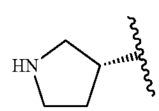
—CH_2-phenyl, or —CH_2—(C_{3-5} cycloalkyl).

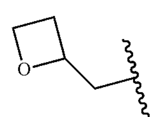
[0175] In certain embodiments, R^6 is C_{1-4} alkyl represents independently for each occurrence C_{3-5} cycloalkyl. In certain embodiments, R^6 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^6 represents independently for each occurrence methyl or ethyl. In certain embodiments, R^6 represents independently for each occurrence C_{3-5} cycloalkyl.

[0176] In certain embodiments, R^6 is C_{1-4} alkyl or C_{3-5} cycloalkyl. In certain embodiments, R^6 is C_{1-4} alkyl. In certain embodiments, R^6 is methyl or ethyl. In certain embodiments, R^6 is C_{3-5} cycloalkyl. In certain embodiments, R^6 is cyclobutyl.

[0177] In certain embodiments, R^6 represents independently for each occurrence C_{3-5} cycloalkyl,



, 

, 

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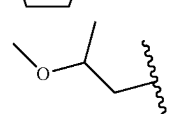

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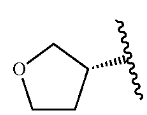
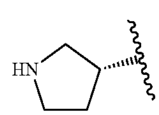
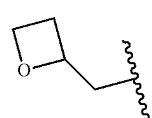
In certain embodiments, R^6 is C_{3-5} cycloalkyl,



, 

, 

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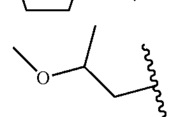

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[0178] In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0179] In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0180] In certain embodiments, R^6 is C_{2-6} alkenyl. In certain embodiments, R^6 is C_{3-7} cycloalkyl. In certain embodiments, R^6 is a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0181] In certain embodiments, R^6 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0182] As defined generally above, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —$N(R^9)_2$. In certain embodiments, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl. In certain embodiments, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, or C_{1-4} haloalkyl.

[0183] In certain embodiments, R^7 represents independently for each occurrence halo. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} haloalkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkoxyl. In certain embodiments, R^7 represents independently for each occurrence —$N(R^9)_2$. In certain embodiments, R^7 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0184] As defined generally above, R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0185] In certain embodiments, R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7. In certain embodiments, R^8 is phenyl or naphthyl; wherein said phenyl is substituted with n instances of R^7 and said naphthyl is substituted with m instances of R^7. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7.

[0186] In certain embodiments, R^8 is phenyl substituted with n instances of R^7, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^8 is phenyl substituted with n instances of R^7 or naphthyl. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0187] In certain embodiments, R^8 is phenyl substituted with n instances of R^7. In certain embodiments, R^8 is naphthyl substituted with m instances of R^7. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7. In certain embodiments, R^8 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7.

[0188] In certain embodiments, R^8 is naphthyl. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^8 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0189] In certain embodiments, R^8 is C_{1-4} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{10} alkylene)-OC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0190] In certain embodiments, R^8 is C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0191] In certain embodiments, R^8 is C_{1-20} alkyl substituted with one hydroxyl. In certain embodiments, R^8 is C_{1-7} alkyl substituted with one hydroxyl. In certain embodiments, R^8 is $C_1$0.4 alkyl substituted with one hydroxyl. In certain embodiments, R^8 is C_{1-10} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0192] In certain embodiments, R^8 is C_{1-7} alkyl. In certain embodiments, R^8 is C_{1-4} alkyl. In certain embodiments, R^8 is C_{1-10} haloalkyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0193] In certain embodiments, R^8 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0194] As defined generally above, R^9 represents independently for each occurrence hydrogen or C_{1-4} alkyl, or two instances of R^9 are taken together with the atom to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^9 represents independently for each occurrence hydrogen or C_{1-4} alkyl. In certain embodiments, R^9 represents independently for each occurrence hydrogen or methyl. In certain embodiments, R^9 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^9 is methyl. In certain embodiments, R^9 is hydrogen.

[0195] In certain embodiments, two instances of R^9 are taken together with the atom to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^9 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0196] As defined generally above, m is independently for each occurrence 0, 1, 2, or 3. In certain embodiments, m is 0. In certain embodiments, m is 1. In certain embodiments, m is 2. In certain embodiments, m is 3. In certain embodiments, m is independently for each occurrence 0 or 1. In certain embodiments, m is independently for each occurrence 1 or 2. In certain embodiments, m is independently for each occurrence 2 or 3. In certain embodiments, m is independently for each occurrence 0, 1, or 2. In certain embodiments m is independently for each occurrence 1, 2, or 3. In certain embodiments, m is selected from the values represented in the compounds in Tables 1, 2, 3, 4, and 5, below.

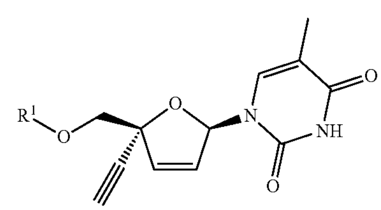
[0197] As defined generally above, n is 1, 2, or 3. In certain embodiments, n is 1. In certain embodiments, n is 2. In certain embodiments, n is 3. In certain embodiments, n is 1 or 2. In certain embodiments, n is 2 or 3. In certain embodiments, n is selected from the values represented in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0198] As defined generally above, p is 0, 1, 2, or 3. In certain embodiments, p is 0. In certain embodiments, p is 1. In certain embodiments, p is 2. In certain embodiments, p is 3. In certain embodiments, p is 0 or 1. In certain embodiments, p is 1 or 2. In certain embodiments, p is 2 or 3. In certain embodiments, p is 0, 1, or 2. In certain embodiments p is 1, 2, or 3. In certain embodiments, p is selected from the values represented in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0199] The description above describes multiple embodiments relating to compounds of Formula I. The patent application specifically contemplates all combinations of the embodiments.

[0200] One aspect of the invention provides a compound represented by Formula I-1:

(I-1)



[0201] or a pharmaceutically acceptable salt thereof; wherein:

[0202] R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)$(OR^2)_2$, —C(O)—C(H)(R^5)—N$(R^3)_2$, or —C(O)R^1;

[0203] R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring;

[0204] R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0205] R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0206] R^5 is C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0207] R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl;

[0208] R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0209] m is independently for each occurrence 0, 1, 2, or 3; and

[0210] n is 1, 2, or 3.

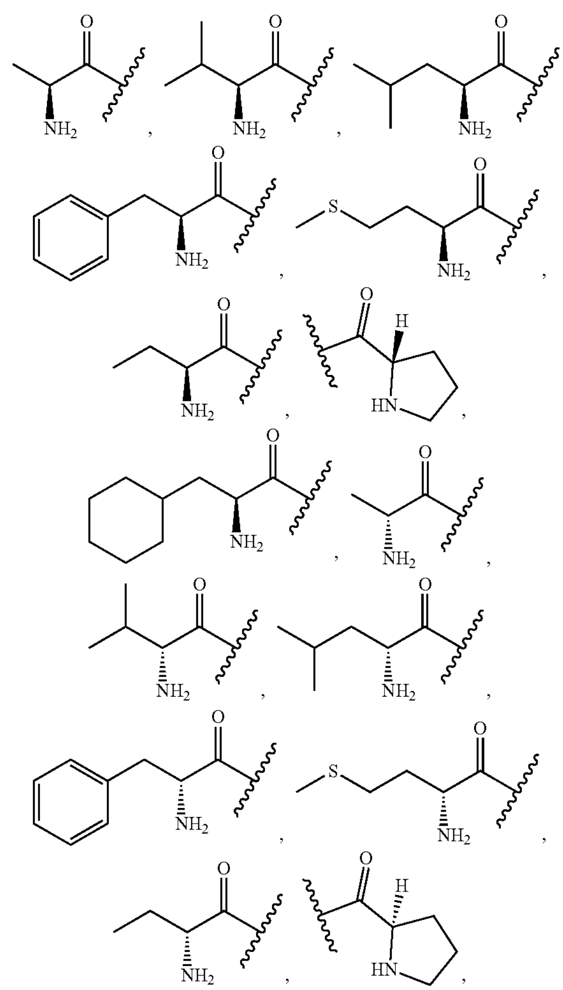
[0211] The definitions of variables in Formula I-1 above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

[0212] In certain embodiments, the compound is a compound of Formula I-1.

[0213] As defined generally above, R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)$(OR^2)_2$, —C(O)—C(H)(R^5)—N$(R^3)_2$, or —C(O)R^8. In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N$(R^3)_2$. In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —P(O)$(OR^2)_2$. In certain embodiments, R^1 is —C(O)—C(H)(R^5)—N$(R^3)_2$ or —C(O)R^8.

[0214] In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)). In certain embodiments, R^1 is —P(O)$(OR^2)_2$. In certain embodiments, R^1 is —C(O)—C(H)(R^5)—N$(R^3)_2$. In certain embodiments, R^1 is —C(O)R^8.

[0215] In certain embodiments, R^1 is



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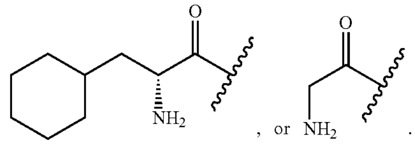
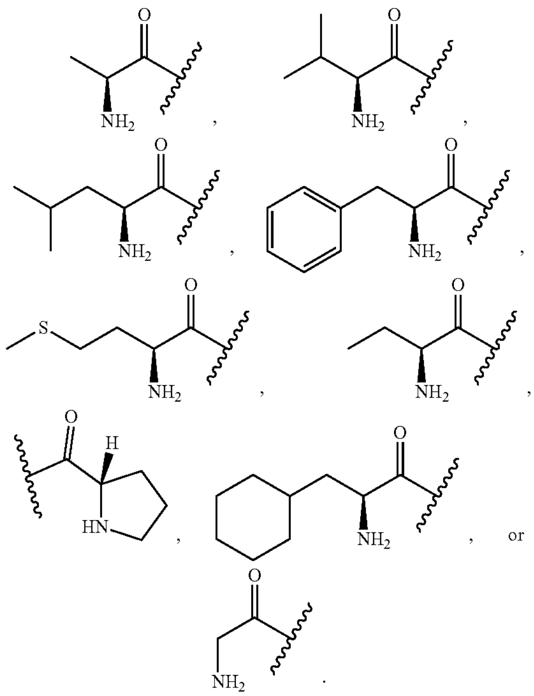
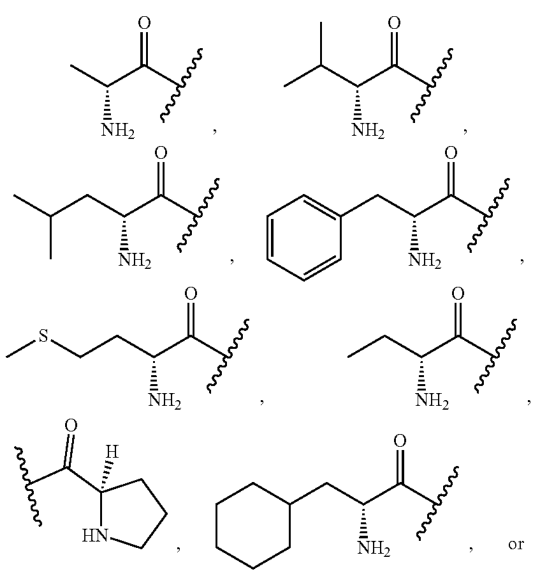


[0216] In certain embodiments, R^1 is



In certain embodiments, R^1 is



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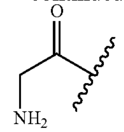
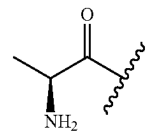
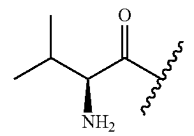


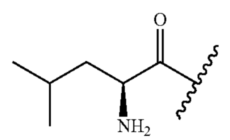
[0217] In certain embodiments, R^1 is



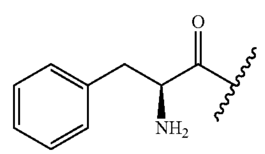
In certain embodiments, R^1 is



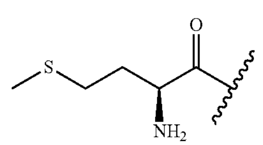
In certain embodiments, R^1 is



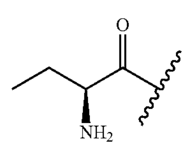
In certain embodiments, R^1 is



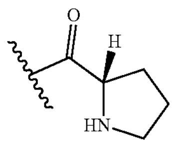
In certain embodiments, R^1 is



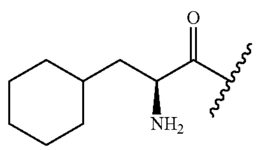
In certain embodiments, R^1 is



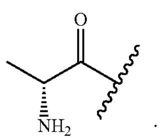
In certain embodiments, R^1 is



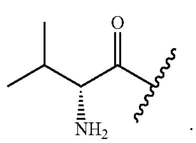
In certain embodiments, R^1 is



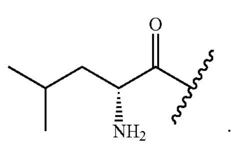
[0218] In certain embodiments, R^1 is



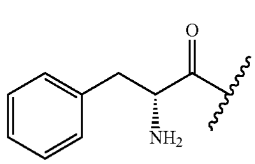
In certain embodiments, R^1 is



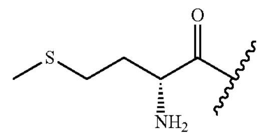
In certain embodiments, R^1 is



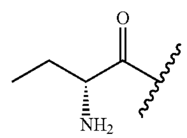
In certain embodiments, R^1 is



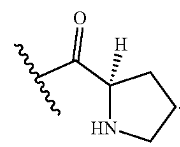
In certain embodiments, R^1 is



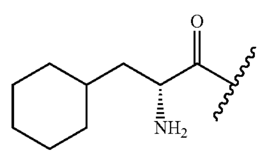
In certain embodiments, R^1 is



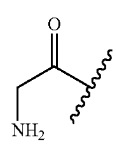
In certain embodiments, R^1 is



In certain embodiments, R^1 is



[0219] In certain embodiments, R^1 is



In certain embodiments, R^1 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^5 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

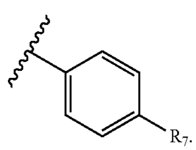
[0220] As defined generally above, R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0221] In certain embodiments, R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence phenyl or naphthyl; each of which is substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7.

[0222] In certain embodiments, R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^2 represents independently for each occurrence phenyl or naphthyl. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0223] In certain embodiments, R^2 is phenyl substituted with m instances of R^7. In certain embodiments, R^2 is



In certain embodiments, R^2 is naphthyl substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7. In certain embodiments, R^2 represents independently for each occurrence an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7.

[0224] In certain embodiments, R^2 is phenyl. In certain embodiments, R^2 is naphthyl. In certain embodiments, R^2 represents independently for each occurrence a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^2 represents independently for each occurrence an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0225] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0226] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

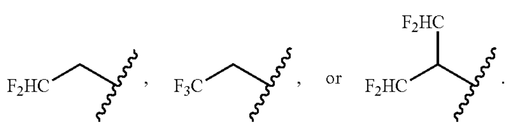
[0227] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-7} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-10} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} haloalkyl optionally substituted with one hydroxyl.

[0228] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence

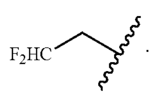
—(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0229] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-7} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-10} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

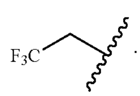
[0230] In certain embodiments, R^2 represents independently for each occurrence



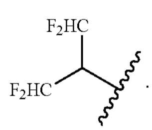
[0231] In certain embodiments, R^2 is



In certain embodiments, R^2 is



In certain embodiments, R^2 is



[0232] In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0233] In certain embodiments, R^2 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^2 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

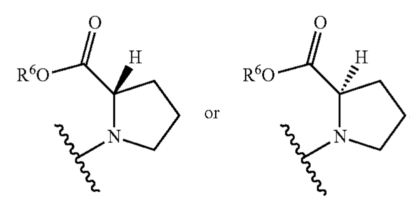
[0234] As defined generally above, R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl. In certain embodiments, R^3 represents independently for each occurrence C_{1-4} alkyl.

[0235] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl; or R^3 and R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

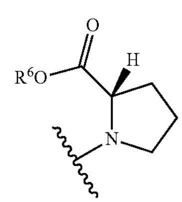
[0236] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl. In certain embodiments, R^3 is hydrogen. In certain embodiments, R^3 is methyl. In certain embodiments, R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, two instances of R^3 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0237] In certain embodiments, R^3 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^3 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

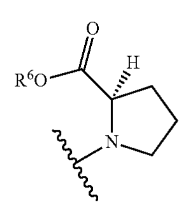
[0238] In certain embodiments, —N(R^3)(R^4) is



In certain embodiments, —N(R^3)(R^4) is



In certain embodiments, —N(R^3)(R^4) is



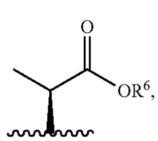
[0239] As defined generally above, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl;

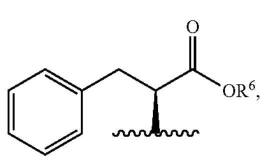
naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0240] In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0241] In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0242] In certain embodiments, R^4 is —C(H)(R^5)—CO_2R^6. In certain embodiments, R^4 is





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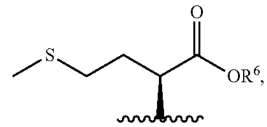

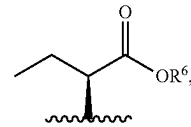

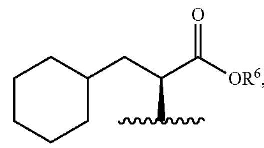

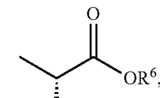

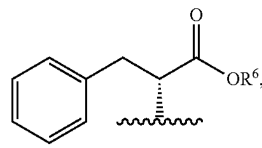

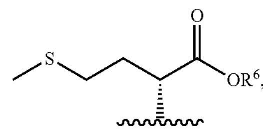

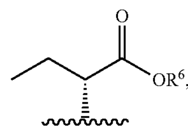

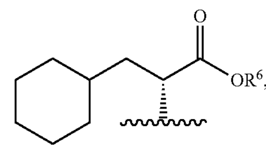

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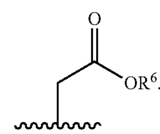
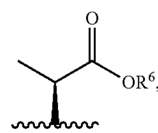
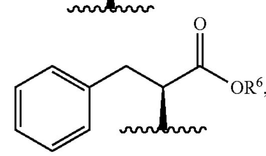
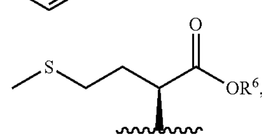
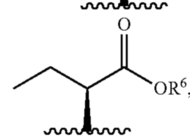


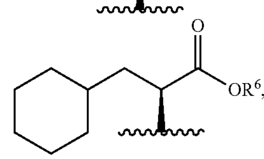
In certain embodiments, R^4 is











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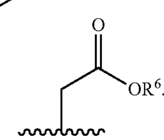
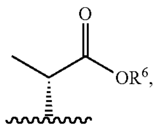
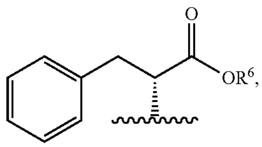
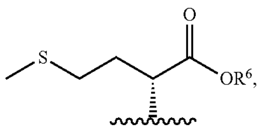
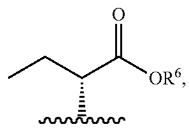
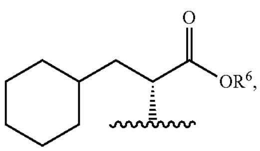


In certain embodiments, R^4 is

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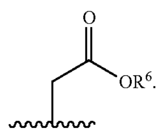
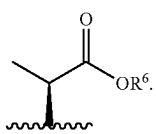
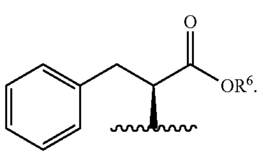


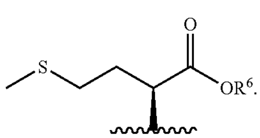
[0243] In certain embodiments, R^4 is



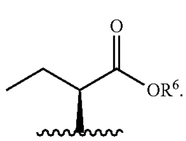
In certain embodiments, R^4 is



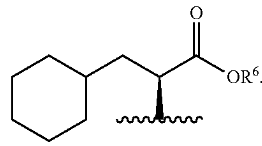
In certain embodiments, R^4 is



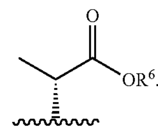
In certain embodiments, R^4 is



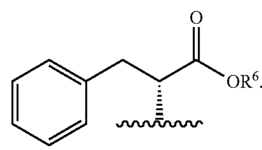
In certain embodiments, R^4 is



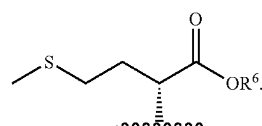
In certain embodiments, R^4 is



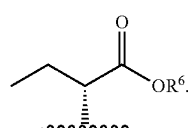
In certain embodiments, R^4 is



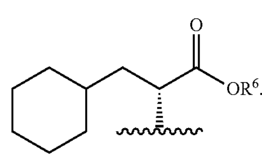
In certain embodiments, R^4 is



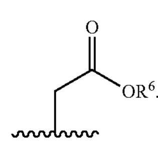
In certain embodiments, R^4 is



In certain embodiments, R^4 is



In certain embodiments, R^4 is



[0244] In certain embodiments, R^4 is C_{1-20} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-7} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-4} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is C_{1-20} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0245] In certain embodiments, R^4 is C_{1-20} alkyl. In certain embodiments, R^4 is C_{1-7} alkyl. In certain embodiments, R^4 is C_{1-4} alkyl. In certain embodiments, R^4 is C_{1-20} haloalkyl. In certain embodiments, R^4 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-S—(C_{1-10} alkyl). In certain embodiments, R^4 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0246] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7. In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, or naphthyl; wherein each of said phenyl and naphthyl is substituted with m instances of R^7. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7.

[0247] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, phenyl, or naphthyl. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0248] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl, wherein said phenyl is substituted with m instances of R^7. In certain embodiments, R^4 is phenyl substituted with m instances of R^7. In certain embodiments, R^4 is naphthyl substituted with m instances of R^7. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7. In certain embodiments, R^4 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7.

[0249] In certain embodiments, R^4 is —(C_{1-10} alkylene)-phenyl. In certain embodiments, R^4 is phenyl. In certain embodiments, R^4 is naphthyl. In certain embodiments, R^4 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^4 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0250] In certain embodiments, R^2 and R^4 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0251] In certain embodiments, R^4 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^4 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

[0252] As defined generally above, R^5 is C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or R^3 and R^5 are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0253] In certain embodiments, R^5 is C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen. In certain embodiments, R^5 is C_{1-6} haloalkyl.

[0254] In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl) or —SH. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl or hydroxyl.

[0255] In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with phenyl or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0256] In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl) or —SH. In certain embodiments, R^5 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl or hydroxyl.

[0257] In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl or a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^5 is C_{1-6} alkyl substituted with an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0258] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0259] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl). In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl.

[0260] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen. In certain embodiments, R^S is C_{1-6} alkyl. In certain embodiments, R^5 is C_{1-4} alkyl. In certain embodiments, R^5 is methyl. In certain embodiments, R^5 is hydrogen.

[0261] In certain embodiments, R^5 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^5 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

[0262] As defined generally above, R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

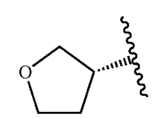
[0263] In certain embodiments, R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^6 is C_{1-6} haloalkyl. In certain embodiments, R^6 is a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0264] In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0265] In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^6 is C_{1-6} alkyl substituted with a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0266] In certain embodiments, R^6 is C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

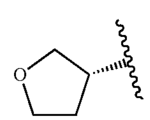
[0267] In certain embodiments, R^6 is C_{1-6} alkyl, allyl, C_{3-5} cycloalkyl,

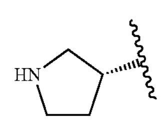


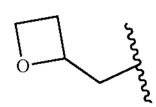
—CH_2-phenyl, or —CH_2—(C_{3-5} cycloalkyl).

[0268] In certain embodiments, R^6 is C_{1-4} alkyl or C_{3-5} cycloalkyl. In certain embodiments, R^6 is C_{1-4} alkyl. In certain embodiments, R^6 is methyl or ethyl. In certain embodiments, R^6 is C_{3-5} cycloalkyl. In certain embodiments, R^6 is cyclobutyl.

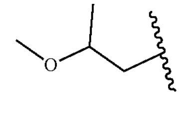
[0269] In certain embodiments, R^6 is C_{3-5} cycloalkyl,







or



[0270] In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0271] In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0272] In certain embodiments, R^6 is C_{2-6} alkenyl. In certain embodiments, R^6 is C_{3-7} cycloalkyl. In certain embodiments, R^6 is a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0273] In certain embodiments, R^6 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^6 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

[0274] As defined generally above, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl. In certain embodiments, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, or C_{1-4} haloalkyl. In certain embodiments, R^7 represents independently for each occurrence halo. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} haloalkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkoxyl. In certain embodiments, R^7 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^7 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

[0275] As defined generally above, R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0276] In certain embodiments, R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7. In certain embodiments, R^8 is phenyl or naphthyl; wherein said phenyl is substituted with n instances of R^7 and said naphthyl is substituted with m instances of R^7. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7.

[0277] In certain embodiments, R^8 is phenyl substituted with n instances of R^7, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^8 is phenyl substituted with n instances of R^7 or naphthyl. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0278] In certain embodiments, R^8 is phenyl substituted with n instances of R^7. In certain embodiments, R^8 is naphthyl substituted with m instances of R^7. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7. In certain embodiments, R^8 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said heteroaryl is substituted with m instances of R^7.

[0279] In certain embodiments, R^8 is naphthyl. In certain embodiments, R^8 is a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur. In certain embodiments, R^8 is an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur.

[0280] In certain embodiments, R^8 is C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0281] In certain embodiments, R^8 is C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0282] In certain embodiments, R^8 is C_{1-20} alkyl substituted with one hydroxyl. In certain embodiments, R^8 is C_{1-7} alkyl substituted with one hydroxyl. In certain embodiments, R^8 is C_{1-4} alkyl substituted with one hydroxyl. In certain embodiments, R^8 is C_{1-10} haloalkyl optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0283] In certain embodiments, R^8 is C_{1-7} alkyl. In certain embodiments, R^8 is C_{1-4} alkyl. In certain embodiments, R^8 is C_{1-10} haloalkyl. In certain embodiments, R^8 is —(C_{1-10} alkylene)-O—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-8} alkylene)-S—(C_{1-10} alkyl). In certain embodiments, R^8 is —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0284] In certain embodiments, R^8 is selected from the groups depicted in the compounds in Tables 1 and 2, below. In certain embodiments, R^8 is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

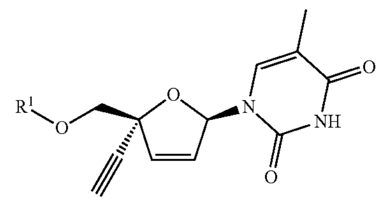
[0285] As defined generally above, m is 0, 1, 2, or 3. In certain embodiments, m is 0. In certain embodiments, m is 1. In certain embodiments, m is 2. In certain embodiments, m is 3. In certain embodiments, m is 0 or 1. In certain embodiments, m is 1 or 2. In certain embodiments, m is 2 or 3. In certain embodiments, m is 0, 1, or 2. In certain embodiments m is 1, 2, or 3. In certain embodiments, m is selected from the values represented in the compounds in Tables 1 and 2, below. In certain embodiments, m is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

[0286] As defined generally above, n is 1, 2, or 3. In certain embodiments, n is 1. In certain embodiments, n is 2. In certain embodiments, n is 3. In certain embodiments, n is 1 or 2. In certain embodiments, n is 2 or 3. In certain embodiments, n is selected from the values represented in the compounds in Tables 1 and 2, below. In certain embodiments, n is selected from the values represented in the compounds in Tables 1, 2, and 3, below.

[0287] The description above describes multiple embodiments relating to compounds of Formula I-1. The patent application specifically contemplates all combinations of the embodiments.

[0288] Another aspect of the invention provides a compound represented by Formula I-A:

(I-A)



[0289] or a pharmaceutically acceptable salt thereof; wherein:

[0290] R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N(R^3)$_2$;

[0291] R^2 is phenyl or naphthyl, each of which is substituted with m instances of R^7;

[0292] R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0293] R^4 is —C(H)(R^5)—CO_2R^6;

[0294] R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl,

[0295] R^6 is C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom;

[0296] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl; and

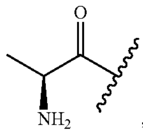
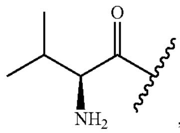
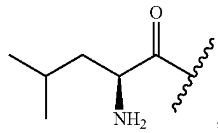
[0297] m is 0, 1, 2, or 3.

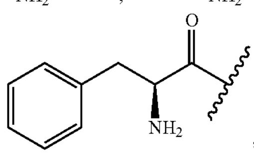
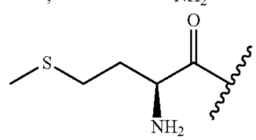
[0298] The definitions of variables in Formula I-A above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

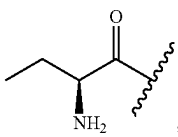
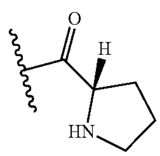
[0299] In certain embodiments, the compound is a compound of Formula I-A.

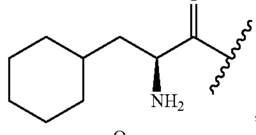
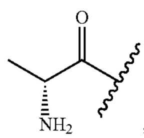
[0300] As defined generally above, R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N(R^3)$_2$. In certain embodiments, R^1 is —P(O)(OR^2)(N(R^3)(R^4)).

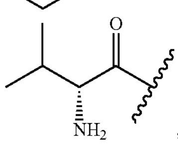
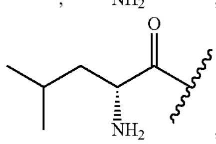
[0301] In certain embodiments, R^1 is —C(O)—C(H)(R^5)—N(R^3)$_2$. In certain embodiments, R^1 is

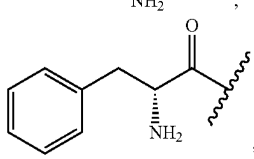
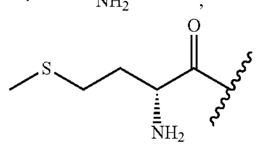
, , ,

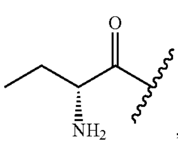
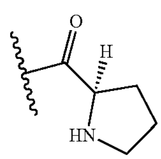
, ,

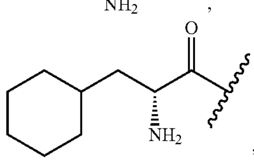
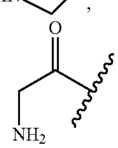
, ,

, ,

, ,

, ,

, ,

, or .

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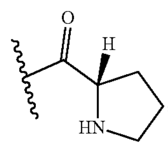
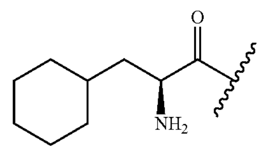
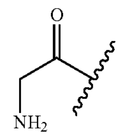
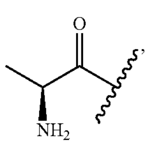
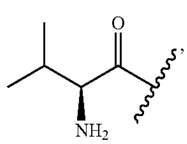
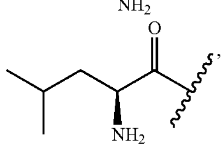
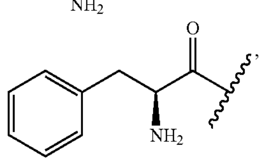
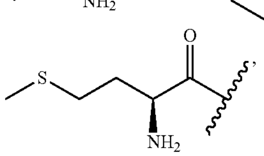
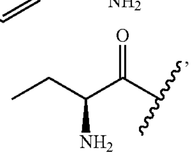
, , or

.

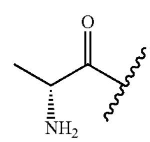
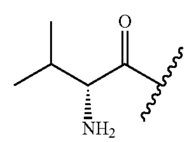
[0302] In certain embodiments, R^1 is

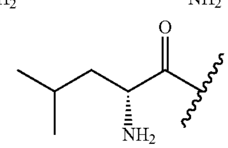
 

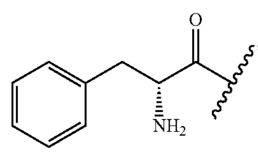
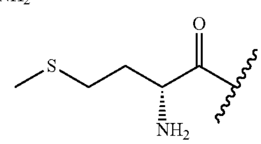
 

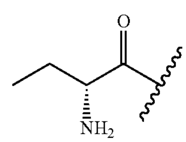
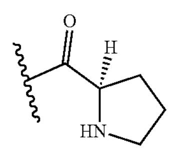
 

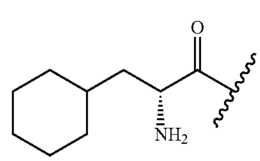
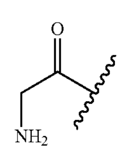
In certain embodiments, R^1 is

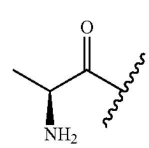


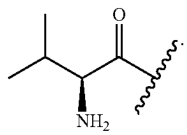
 

 or 

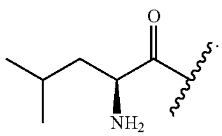
[0303] In certain embodiments, R^1 is



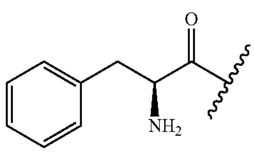
In certain embodiments, R^1 is



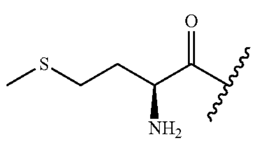
In certain embodiments, R^1 is



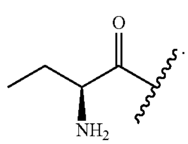
In certain embodiments, R^1 is



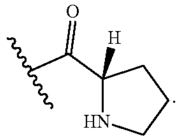
In certain embodiments, R^1 is



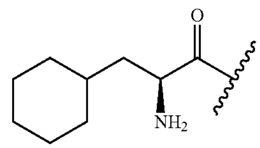
In certain embodiments, R^1 is



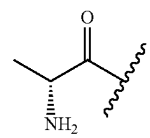
In certain embodiments, R^1 is



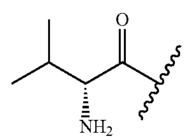
In certain embodiments, R^1 is



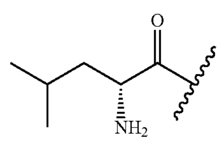
[0304] In certain embodiments, R^1 is



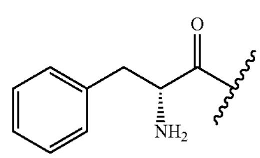
In certain embodiments, R^1 is



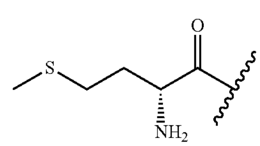
In certain embodiments, R^1 is



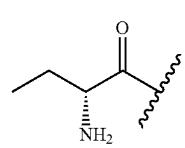
In certain embodiments, R^1 is



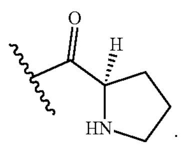
In certain embodiments, R^1 is



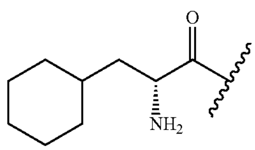
In certain embodiments, R^1 is



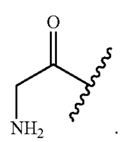
In certain embodiments, R^1 is



In certain embodiments, R^1 is



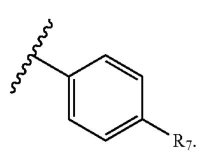
[0305] In certain embodiments, R^1 is



In certain embodiments, R^1 is selected from the groups depicted in the compounds in Tables 1 and 2, below.

[0306] In certain embodiments, R^1 is selected from the groups depicted in the compounds in Table 1, below. In certain embodiments, R^1 is selected from the groups depicted in the compounds in Tables 1 and 3, below.

[0307] As defined generally above, R^2 is phenyl or naphthyl; each of which is substituted with m instances of R^7. In certain embodiments, R^2 is phenyl substituted with m instances of R^7. In certain embodiments, R^2 is



In certain embodiments, R^2 is naphthyl substituted with m instances of R^7.

[0308] In certain embodiments, R^2 is phenyl or naphthyl. In certain embodiments, R^2 is phenyl. In certain embodiments, R^2 is naphthyl. In certain embodiments, R^2 is selected from the groups depicted in the compounds in Table 1, below. In certain embodiments, R^2 is selected from the groups depicted in the compounds in Tables 1 and 3, below.

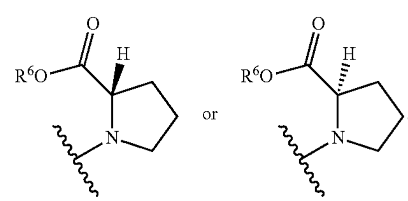
[0309] As defined generally above, R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl. In certain embodiments, R^3 represents independently for each occurrence C_{1-4} alkyl.

[0310] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl; or R^3 and R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

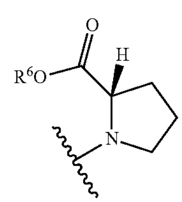
[0311] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl. In certain embodiments, R^3 is hydrogen. In certain embodiments, R^3 is methyl. In certain embodiments, R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, two instances of R^3 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0312] In certain embodiments, R^3 is selected from the groups depicted in the compounds in Table 1, below. In certain embodiments, R^3 is selected from the groups depicted in the compounds in Tables 1 and 3, below.

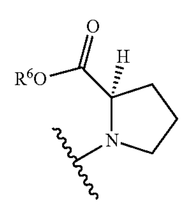
[0313] In certain embodiments, —$N(R^3)(R^4)$ is N



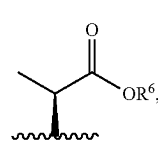
In certain embodiments, —$N(R^3)(R^4)$ is

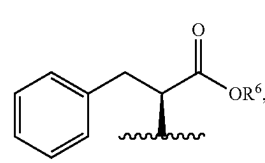


In certain embodiments, —$N(R^3)(R^4)$ is



[0314] As defined generally above, R^4 is —$C(H)(R^5)$—CO_2R^6. In certain embodiments, R^4 is





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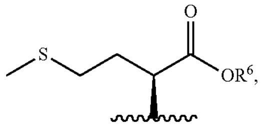
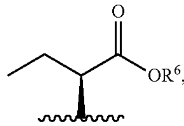
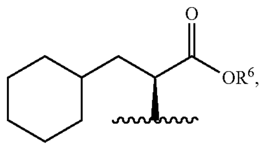
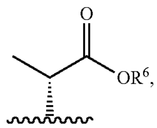
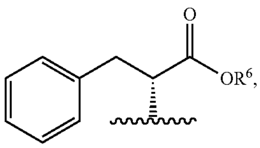
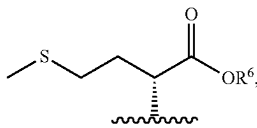
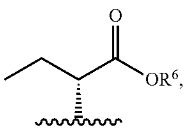
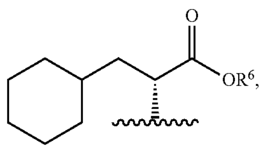

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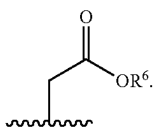
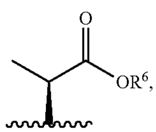
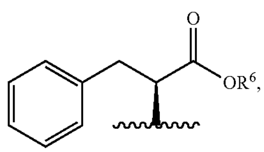
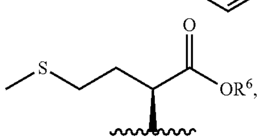
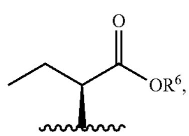
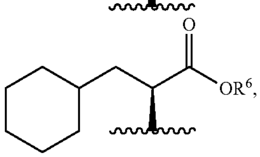


In certain embodiments, R^4 is

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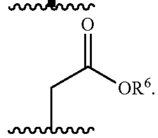
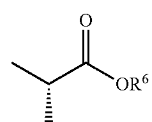


In certain embodiments, R^4 is



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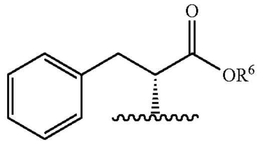

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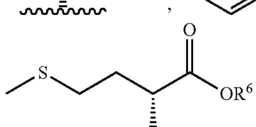

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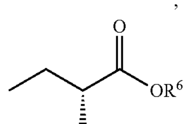

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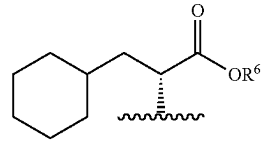

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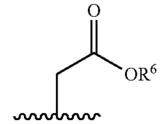

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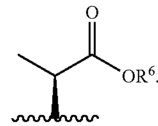
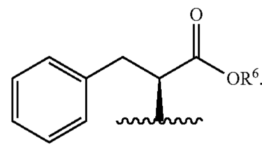
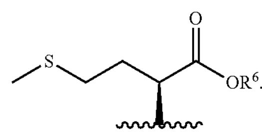
[0315] In certain embodiments, R^4 is



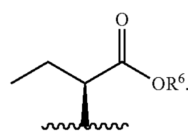
In certain embodiments, R^4 is



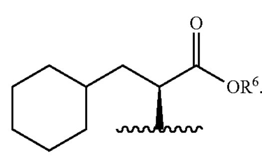
In certain embodiments, R^4 is



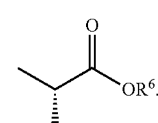
In certain embodiments, R^4 is



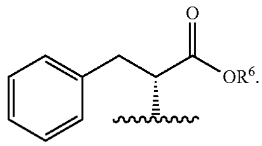
In certain embodiments, R^4 is



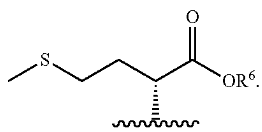
In certain embodiments, R^4 is



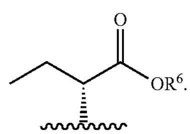
In certain embodiments, R^4 is



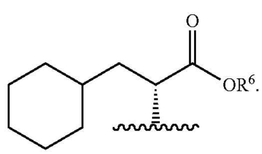
In certain embodiments, R^4 is



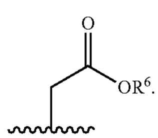
In certain embodiments, R^4 is



In certain embodiments, R^4 is



In certain embodiments, R^4 is



[0316] In certain embodiments, R^4 is selected from the groups depicted in the compounds in Table 1, below. In certain embodiments, R^4 is selected from the groups depicted in the compounds in Tables 1 and 3, below.

[0317] As defined generally above, R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

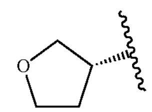
[0318] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl). In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl.

[0319] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen. In certain embodiments, R^5 is C_{1-6} alkyl. In certain embodiments, R^5 is C_{1-4} alkyl. In certain embodiments, R^5 is methyl. In certain embodiments, R^5 is hydrogen.

[0320] In certain embodiments, R^5 is selected from the groups depicted in the compounds in Table 1, below. In certain embodiments, R^5 is selected from the groups depicted in the compounds in Tables 1 and 3, below.

[0321] As defined generally above, R^6 is C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

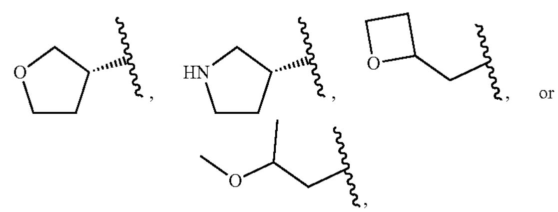
[0322] In certain embodiments, R^6 is C_{1-6} alkyl, allyl, C_{3-5} cycloalkyl,



—CH_2—, phenyl, or —CH_2—(C_{3-5} cycloalkyl).

[0323] In certain embodiments, R^6 is C_{1-4} alkyl or C_{3-5} cycloalkyl. In certain embodiments, R^6 is C_{1-4} alkyl. In certain embodiments, R^6 is methyl or ethyl. In certain embodiments, R^6 is C_{3-5} cycloalkyl. In certain embodiments, R^6 is cyclobutyl.

[0324] In certain embodiments, R^6 is C_{3-5} cycloalkyl,



[0325] In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0326] In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0327] In certain embodiments, R^6 is C_{2-6} alkenyl. In certain embodiments, R^6 is C_{3-7} cycloalkyl. In certain embodiments, R^6 is a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0328] In certain embodiments, R^6 is selected from the groups depicted in the compounds in Table 1, below. In certain embodiments, R^6 is selected from the groups depicted in the compounds in Tables 1 and 3, below.

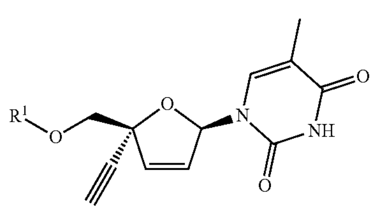
[0329] As defined generally above, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl. In certain embodiments, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, or C_{1-4} haloalkyl. In certain embodiments, R^7 represents independently for each occurrence halo. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} haloalkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkoxyl. In certain embodiments, R^7 is selected from the groups depicted in the compounds in Table 1, below. In certain embodiments, R^7 is selected from the groups depicted in the compounds in Tables 1 and 3, below.

[0330] As defined generally above, m is 0, 1, 2, or 3. In certain embodiments, m is 0. In certain embodiments, m is 1. In certain embodiments, m is 2. In certain embodiments, m is 3. In certain embodiments, m is 0 or 1. In certain embodiments, m is 1 or 2. In certain embodiments, m is 2 or 3. In certain embodiments, m is 0, 1, or 2. In certain embodiments m is 1, 2, or 3. In certain embodiments, m is selected from the values represented in the compounds in Table 1, below. In certain embodiments, m is selected from the groups depicted in the compounds in Tables 1 and 3, below.

[0331] The description above describes multiple embodiments relating to compounds of Formula I-A. The patent application specifically contemplates all combinations of the embodiments.

[0332] Another aspect of the invention provides a compound represented by Formula I-B:

(I-B)



[0333] or a pharmaceutically acceptable salt thereof; wherein:

[0334] R^1 is —P(O)(OR^2)$_2$ or —P(O)(N(R^3)(R^4))$_2$;

[0335] R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen; or two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

[0336] R^3 and R^9 each represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0337] R^4 represents independently for each occurrence —C(R^5)$_2$—CO$_2$R^6;

[0338] R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

[0339] R^6 is C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom;

[0340] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —N(R^9)$_2$; and

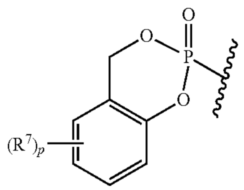
[0341] p is 0, 1, 2, or 3.

[0342] The definitions of variables in Formula I-B above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

[0343] In certain embodiments, the compound is a compound of Formula I-B.

[0344] As defined generally above, R^1 is —P(O)(OR^2)$_2$ or —P(O)(N(R^3)(R^4))$_2$. In certain embodiments, R^1 is —P(O)(OR^2)$_2$. In certain embodiments, R^1 is —P(O)(N(R^3)(R^4))$_2$.

[0345] In certain embodiments, R^1 is



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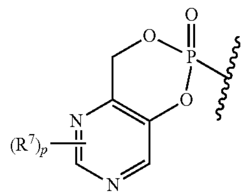
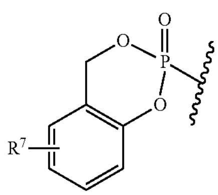


In certain embodiments, R^1 is



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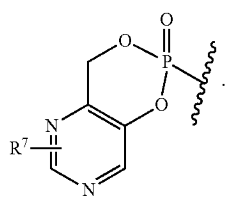
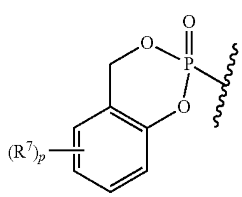


In certain embodiments, R^1 is



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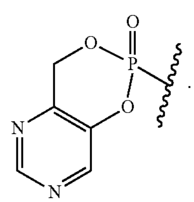
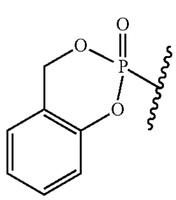


In certain embodiments, R^1 is



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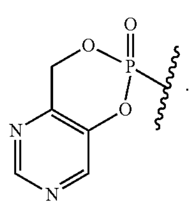
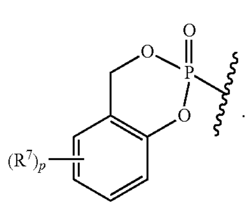
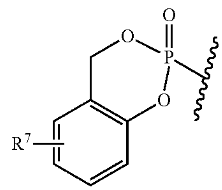


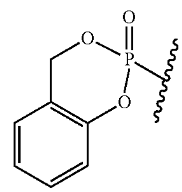
[0346] In certain embodiments, R^1 is



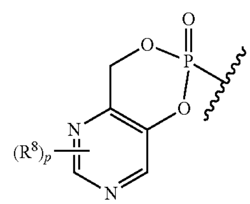
In certain embodiments, R^1 is



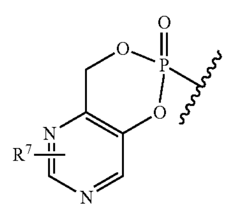
In certain embodiments, R^1 is



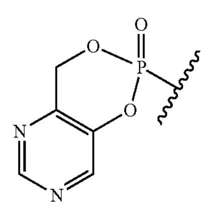
[0347] In certain embodiments, R^1 is



In certain embodiments, R^1 is



In certain embodiments, R^1 is



In certain embodiments, R^1 is selected from the groups depicted in the compounds in Tables 4 and 5, below.

[0348] As defined generally above, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10}

alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen; or two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7.

[0349] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0350] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0351] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen.

[0352] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl); wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—N(H)—(C_{1-10} alkyl); wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-10} alkyl) or —CH_2—OC(O)—N(H)—(C_{1-10} alkyl); wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-6} alkyl) or —CH_2—OC(O)—N(H)—(C_{1-6} alkyl). In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{3-5} cycloalkyl) or —CH_2—OC(O)—N(H)—(C_{3-5} cycloalkyl).

[0353] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{1-6} alkyl). In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)O—(C_{3-5} cycloalkyl).

[0354] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—N(H)—(C_{1-10} alkyl), wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)—N(H)—(C_{1-10} alkyl); wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)—N(R^9)—(C_{1-6} alkyl). In certain embodiments, R^2 represents independently for each occurrence —CH_2—OC(O)—N(R^9)—(C_{3-5} cycloalkyl).

[0355] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl).

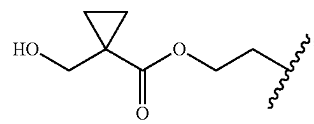
[0356] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is replaced with a C_{3-5} cycloalkylene. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl.

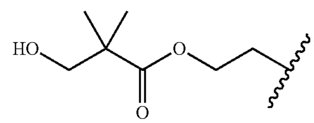
[0357] In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl) or —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is substituted with one hydroxyl.

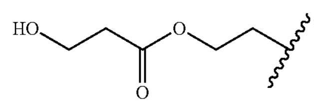
[0358] In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is substituted with one hydroxyl.

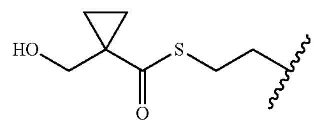
[0359] In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is optionally replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl); wherein said C_{1-10} alkyl is substituted with one hydroxyl; and wherein one methylene unit in said C_{1-10} alkyl is replaced with a cyclopropylene. In certain embodiments, R^2 represents independently for each occurrence —$(CH_2)_{1-2}$—SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is substituted with one hydroxyl.

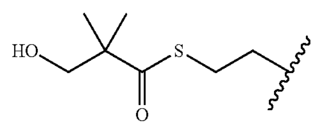
[0360] In certain embodiments, R^2 represents independently for each occurrence







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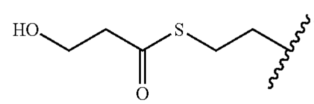
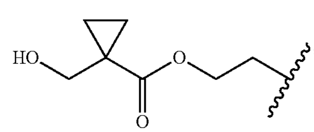
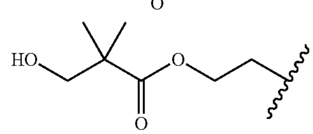


In certain embodiments, R^2 represents independently for each occurrence

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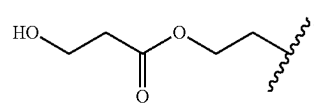
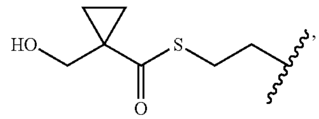
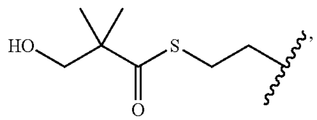


In certain embodiments, R^2 represents independently for each occurrence





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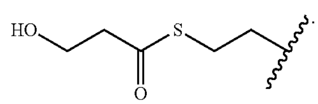
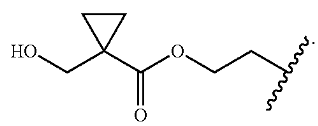
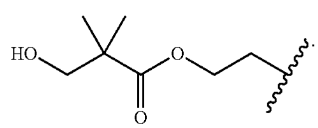


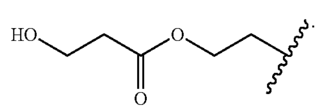
[0361] In certain embodiments, R^2 is



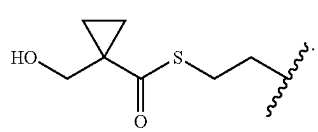
In certain embodiments, R^2 is



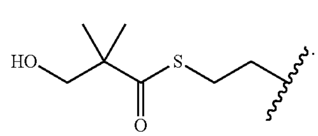
In certain embodiments, R^2 is



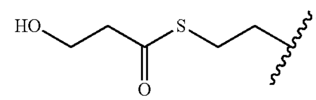
In certain embodiments, R^2 is



In certain embodiments, R^2 is



In certain embodiments, R^2 is



[0362] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-20} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen. In certain embodiments, one instance of R^2 is —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-10} alkylene is substituted with one —O—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen.

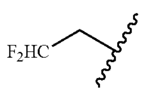
[0363] In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—O—(C_{1-20} alkyl), —$(CH_2)_3$—O—(C_{1-20} alkyl), —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—S—(C_{1-20} alkyl), or —$(CH_2)_3$—S—(C_{1-20} alkyl); and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—O—(C_{1-20} alkyl) or —$(CH_2)_3$—O—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—O—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —$(CH_2)_3$—O—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—S—(C_{1-20} alkyl) or —$(CH_2)_3$—S—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —CH_2—C(H)(—O—(C_{1-20} alkyl))-CH_2—S—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen. In certain embodiments, one instance of R^2 is —$(CH_2)_3$—S—(C_{1-20} alkyl), and any second instance of R^2 is hydrogen.

[0364] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-7} alkyl optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} alkyl optionally substituted with one hydroxyl.

[0365] In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), wherein said C_{1-10} alkyl is optionally substituted with one hydroxyl.

[0366] In certain embodiments, R^2 represents independently for each occurrence C_{1-20} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-7} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-20} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-10} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence C_{1-4} haloalkyl. In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-S—(C_{1-10} alkyl). In certain embodiments, R^2 represents independently for each occurrence —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl).

[0367] In certain embodiments, R^2 represents independently for each occurrence



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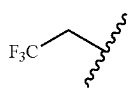

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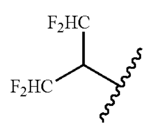

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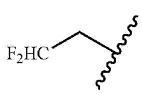
In certain embodiments, R^2 is



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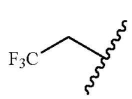
In certain embodiments, R^2 is



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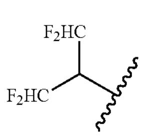
In certain embodiments, R^2 is



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[0368] In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring.

[0369] In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring; and wherein said benzene ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a benzene ring.

[0370] In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 5-6 membered heteroaromatic ring; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 7-12 membered bicyclic heterocyclic ring; wherein said ring is a 5-7 membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms; and wherein said bicyclic heterocyclic ring is substituted with p instances of R^7. In certain embodiments, two instances of R^2 are taken together with their intervening atoms to form a 10-membered bicyclic heterocyclic ring; wherein said ring is a 6-membered monocyclic partially unsaturated heterocyclic ring ortho-fused to a 6-membered heteroaromatic ring containing one or two nitrogen atoms.

[0371] In certain embodiments, R^2 is selected from the groups depicted in the compounds in Tables 4 and 5, below.

[0372] As defined generally above, R^3 represents independently for each occurrence hydrogen or C_4 alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl. In certain embodiments, R^3 represents independently for each occurrence C_{1-4} alkyl.

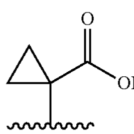
[0373] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl; or R^3 and

R^5, or two instances of R^3, are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

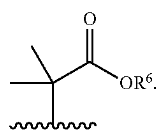
[0374] In certain embodiments, R^3 represents independently for each occurrence hydrogen or methyl. In certain embodiments, R^3 is hydrogen. In certain embodiments, R^3 is methyl. In certain embodiments, R^3 and R^5 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, two instances of R^3 are taken together with the atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom.

[0375] In certain embodiments, R^3 is selected from the groups depicted in the compounds in Tables 4 and 5, below.

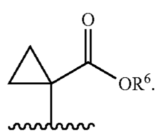
[0376] As defined generally above, R^4 represents independently for each occurrence —$C(R^5)_2$—CO_2R^6. In certain embodiments, R^4 represents independently for each occurrence



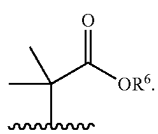
or



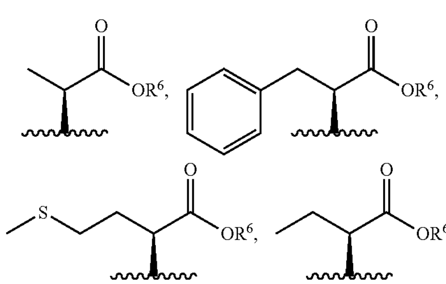
In certain embodiments, R^4 represents independently for each occurrence



In certain embodiments, R^4 represents independently for each occurrence



[0377] In certain embodiments, R^4 represents independently for each occurrence —C(H)(R^5)—CO_2R^6. In certain embodiments, R^4 represents independently for each occurrence



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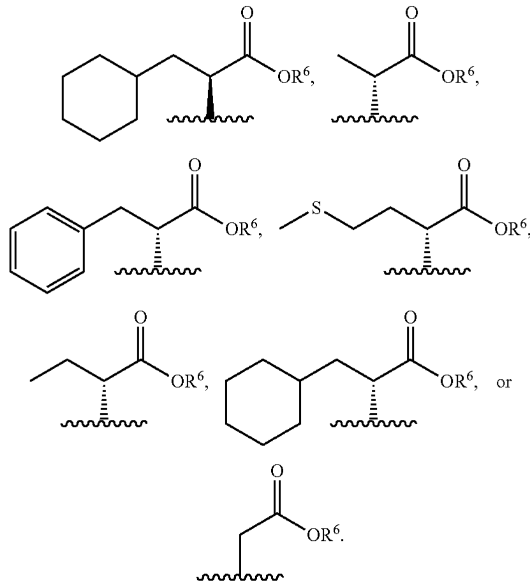
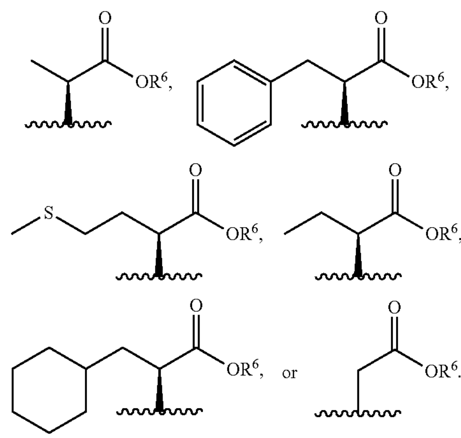
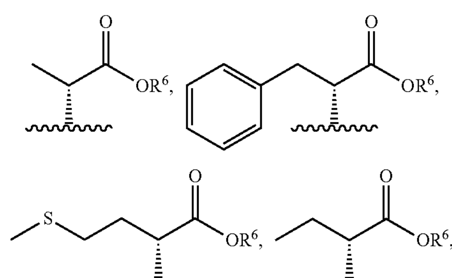


In certain embodiments, R^4 represents independently for each occurrence



In certain embodiments, R^4 represents independently for each occurrence



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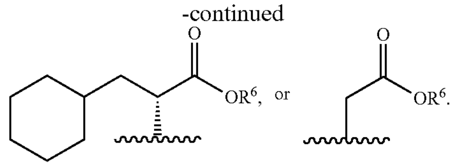
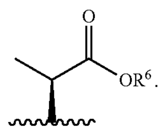
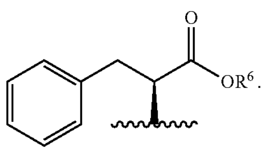


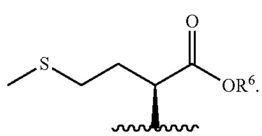
[0378] In certain embodiments, R^4 is



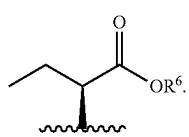
In certain embodiments, R^4 is



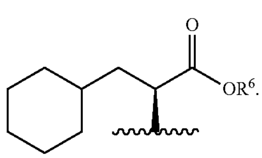
In certain embodiments, R^4 is



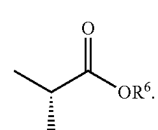
In certain embodiments, R^4 is



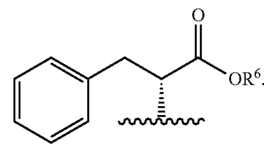
In certain embodiments, R^4 is



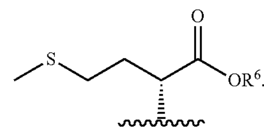
In certain embodiments, R^4 is



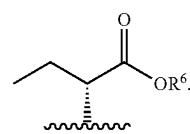
In certain embodiments, R^4 is



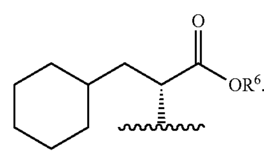
In certain embodiments, R^4 is



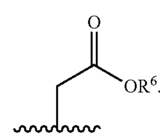
In certain embodiments, R^4 is



In certain embodiments, R^4 is



In certain embodiments, R^4 is



[0379] In certain embodiments, R^4 is selected from the groups depicted in the compounds in Tables 4 and 5, below.

[0380] As defined generally above, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring.

[0381] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, or C_{3-5} cycloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen. In certain embodiments, R^5

represents independently for each occurrence C_{1-6} haloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{3-5} cycloalkyl.

[0382] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring.

[0383] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl.

[0384] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with —S—(C_{1-4} alkyl). In certain embodiments, R^5 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^5 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl.

[0385] In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl or hydrogen. In certain embodiments, R^5 represents independently for each occurrence C_{1-6} alkyl. In certain embodiments, R^5 represents independently for each occurrence C_{1-4} alkyl.

[0386] In certain embodiments, R^5 is C_{1-6} alkyl or hydrogen. In certain embodiments, R^5 is C_{1-6} alkyl. In certain embodiments, R^5 is C_{1-4} alkyl. In certain embodiments, R^5 is methyl. In certain embodiments, R^5 is hydrogen.

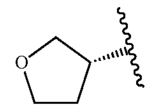
[0387] In certain embodiments, two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring. In certain embodiments, two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-membered saturated carbocyclic ring.

[0388] In certain embodiments, R^5 is selected from the groups depicted in the compounds in Tables 4 and 5, below.

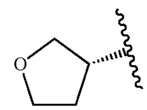
[0389] As defined generally above, R^6 represents independently for each occurrence C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0390] In certain embodiments, R^6 is C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0391] In certain embodiments, R^6 represents independently for each occurrence C_{1-6} alkyl, allyl, C_{3-5} cycloalkyl,



—CH_2-phenyl, or —CH_2—(C_{3-5} cycloalkyl). In certain embodiments, R^6 is C_{1-6} alkyl, allyl, C_{3-5} cycloalkyl,

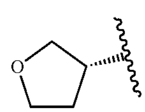
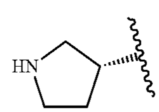
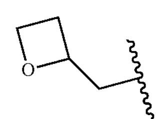


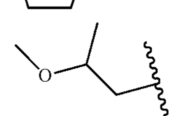
—CH_2-phenyl, or —CH_2—(C_{3-5} cycloalkyl).

[0392] In certain embodiments, R^6 is C_{1-4} alkyl represents independently for each occurrence C_{3-5} cycloalkyl. In certain embodiments, R^6 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^6 represents independently for each occurrence methyl or ethyl. In certain embodiments, R^6 represents independently for each occurrence C_{3-5} cycloalkyl.

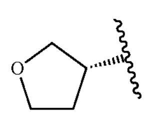
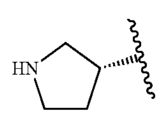
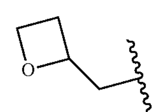
[0393] In certain embodiments, R^6 is C_{1-4} alkyl or C_{3-5} cycloalkyl. In certain embodiments, R^6 is C_{1-4} alkyl. In certain embodiments, R^6 is methyl or ethyl. In certain embodiments, R^6 is C_{3-5} cycloalkyl. In certain embodiments, R^6 is cyclobutyl.

[0394] In certain embodiments, R^6 represents independently for each occurrence C_{3-5} cycloalkyl,

 ,  ,  , or

 .

In certain embodiments, R^6 is C_{3-5} cycloalkyl,

 ,  ,  , or

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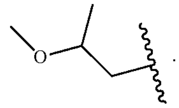


[0395] In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl optionally substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0396] In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{1-4} alkoxyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl or C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with phenyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with C_{3-7} cycloalkyl. In certain embodiments, R^6 is C_{1-6} alkyl substituted with a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom.

[0397] In certain embodiments, R^6 is C_{2-6} alkenyl. In certain embodiments, R^6 is C_{3-7} cycloalkyl. In certain embodiments, R^6 is a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom. In certain embodiments, R^6 is selected from the groups depicted in the compounds in Tables 4 and 5, below.

[0398] As defined generally above, R^7 represents independently for each occurrence halo, C_1-4 alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —$N(R^9)_2$. In certain embodiments, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl. In certain embodiments, R^7 represents independently for each occurrence halo, C_{1-4} alkyl, or C_{1-4} haloalkyl.

[0399] In certain embodiments, R^7 represents independently for each occurrence halo. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} haloalkyl. In certain embodiments, R^7 represents independently for each occurrence C_{1-4} alkoxyl. In certain embodiments, R^7 represents independently for each occurrence —$N(R^9)_2$. In certain embodiments, R^7 is selected from the groups depicted in the compounds in Tables 4 and 5, below.

[0400] As defined generally above, R^9 represents independently for each occurrence hydrogen or C_{1-4} alkyl, or two instances of R^9 are taken together with the atom to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^9 represents independently for each occurrence hydrogen or C_{1-4} alkyl. In certain embodiments, R^9 represents independently for each occurrence hydrogen or methyl. In certain embodiments, R^9 represents independently for each occurrence C_{1-4} alkyl. In certain embodiments, R^9 is methyl. In certain embodiments, R^9 is hydrogen.

[0401] In certain embodiments, two instances of R^9 are taken together with the atom to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom. In certain embodiments, R^9 is selected from the groups depicted in the compounds in Tables 1, 2, 3, 4, and 5, below.

[0402] As defined generally above, p is 0, 1, 2, or 3. In certain embodiments, p is 0. In certain embodiments, p is 1. In certain embodiments, p is 2. In certain embodiments, p is 3. In certain embodiments, p is 0 or 1. In certain embodiments, p is 1 or 2. In certain embodiments, p is 2 or 3. In certain embodiments, p is 0, 1, or 2. In certain embodiments p is 1, 2, or 3. In certain embodiments, p is selected from the values represented in the compounds in Tables 4 and 5, below.

[0403] The description above describes multiple embodiments relating to compounds of Formula I-B. The patent application specifically contemplates all combinations of the embodiments.

[0404] In certain other embodiments, the compound is a compound in Table 1, 2, 3, 4, or 5, below, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound in Table 1, 2, 3, 4, or 5, below. In certain other embodiments, the compound is a compound in Table 1, 2, or 3, below, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound in Table 1, 2, or 3, below. In certain other embodiments, the compound is a compound in Table 1 or 2 below, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound in Table 1 or 2 below. In certain embodiments, the compound is a compound in Table 1 below, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound in Table 1 below. In certain other embodiments, the compound is a compound in Table 4 or 5, below, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound in Table 4 or 5, below.

TABLE 1

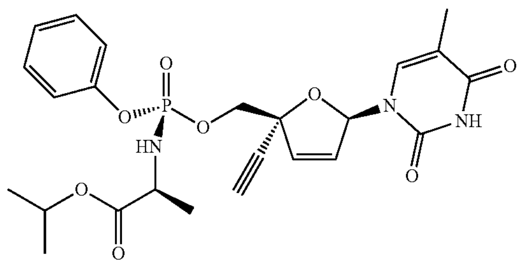
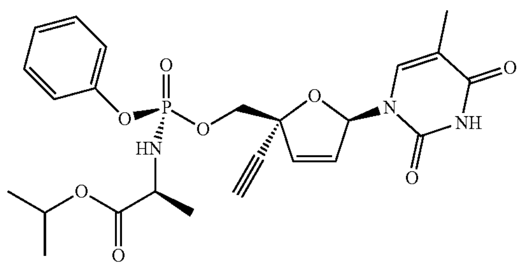
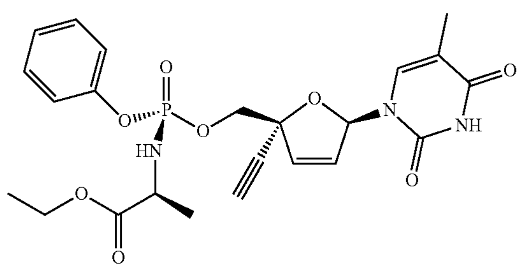
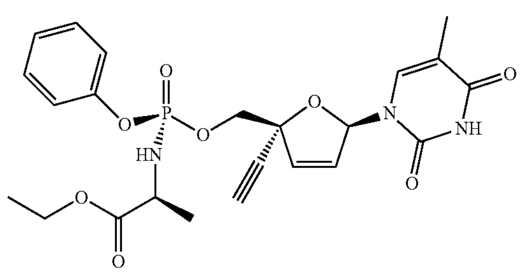
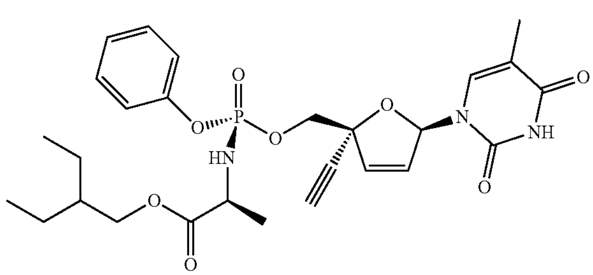
Compound No.	Chemical Structure
I-1	
I-2	
I-3	
I-4	
I-5	

TABLE 1-continued

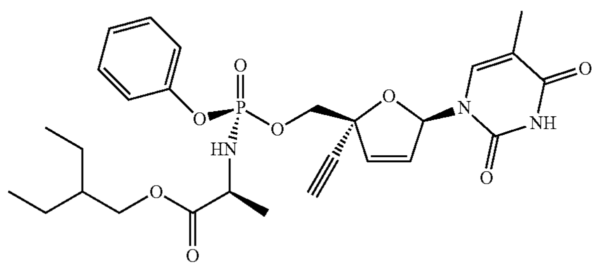
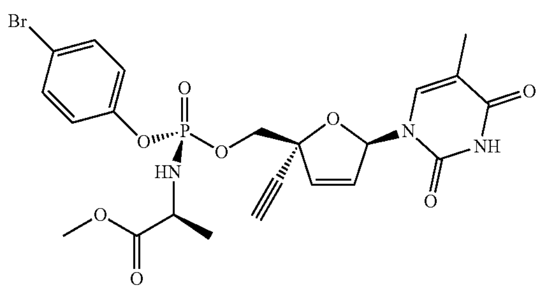
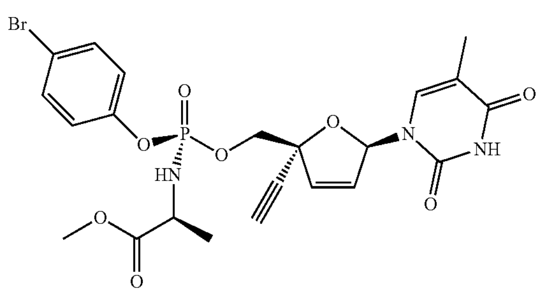
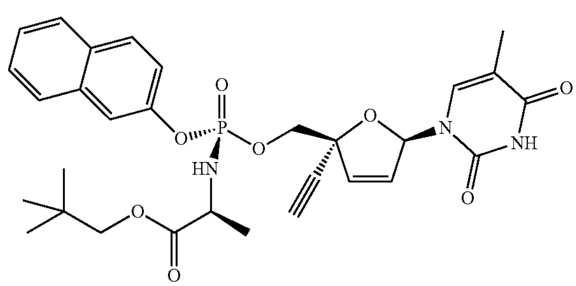
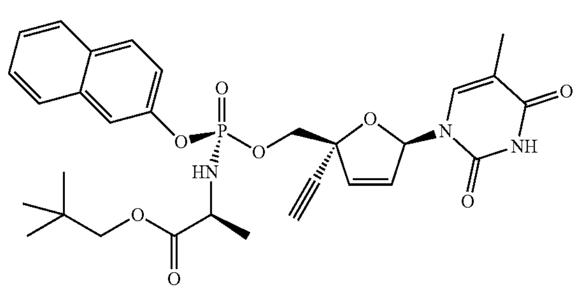
Compound No.	Chemical Structure
I-6	
I-7	
I-8	
I-9	
I-10	

TABLE 1-continued

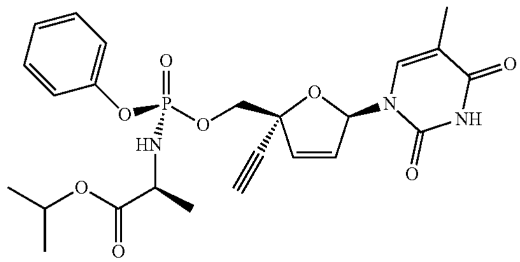
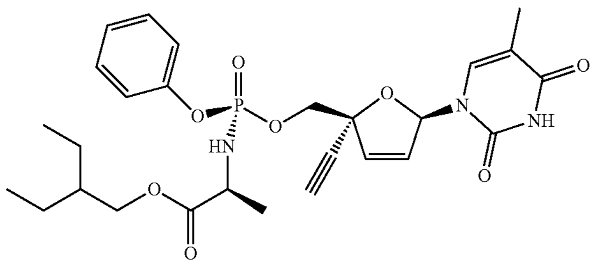
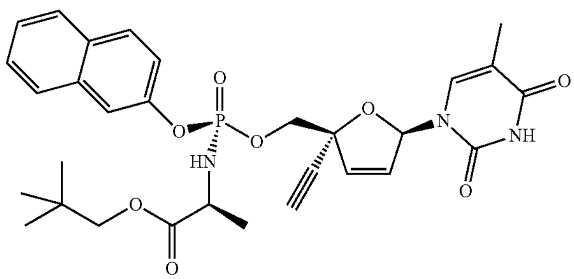
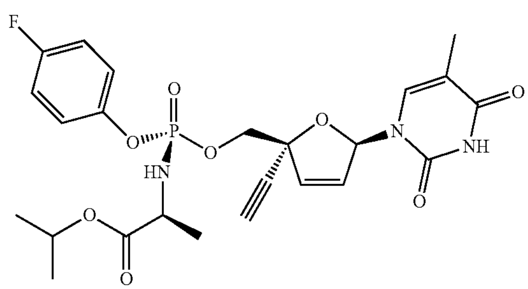
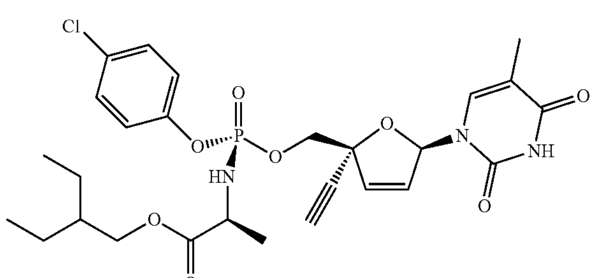
Compound No.	Chemical Structure
I-11	
I-12	
I-13	
I-14	
I-15	

TABLE 1-continued

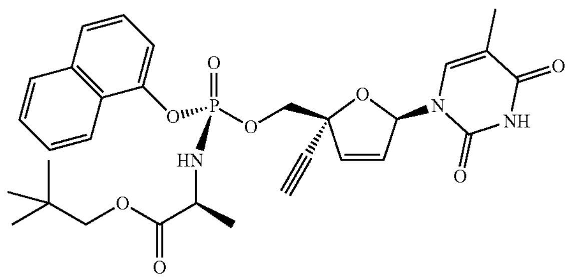
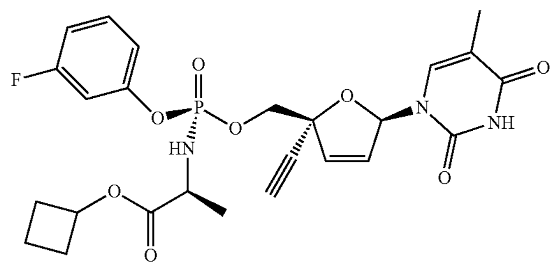
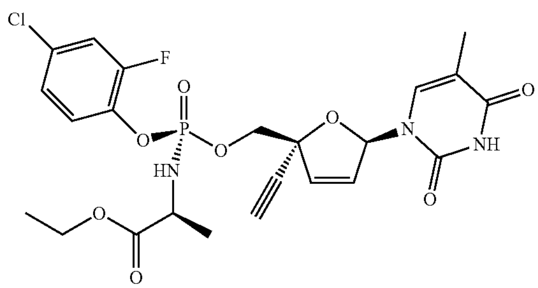
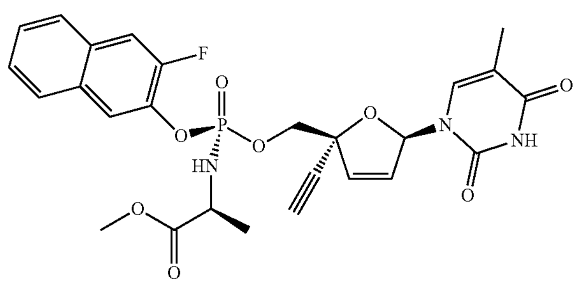
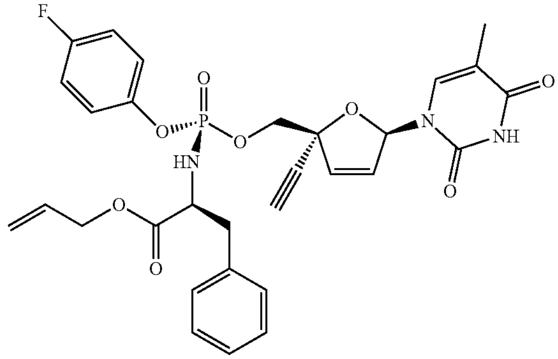
Compound No.	Chemical Structure
I-16	
I-17	
I-18	
I-19	
I-20	

TABLE 1-continued

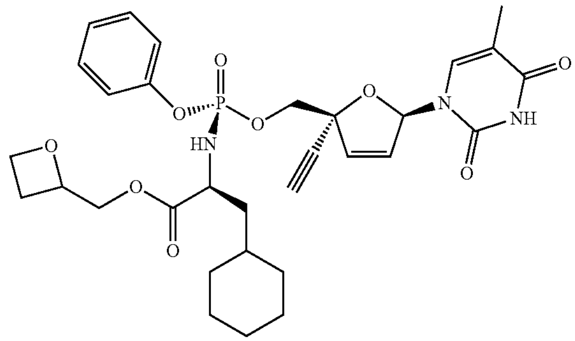
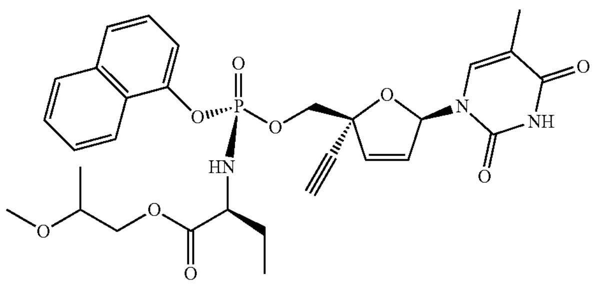
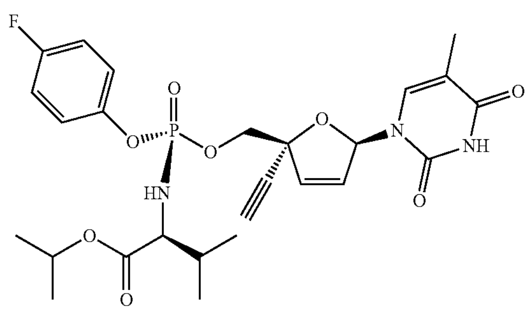
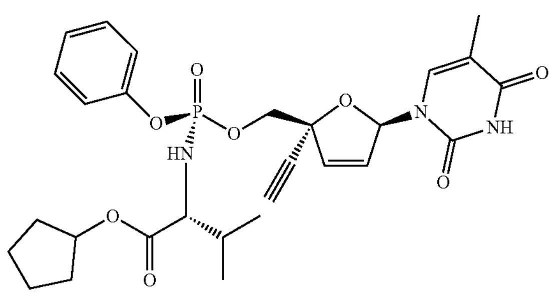
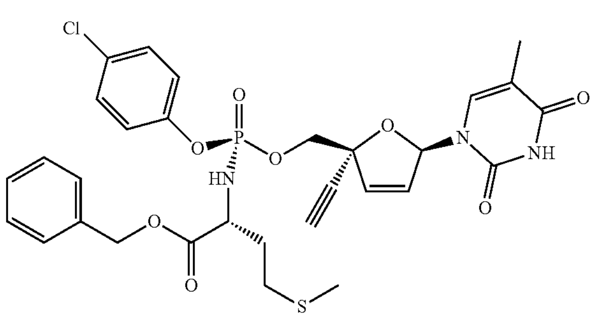
Compound No.	Chemical Structure
I-21	
I-22	
I-23	
I-24	
I-25	

TABLE 1-continued

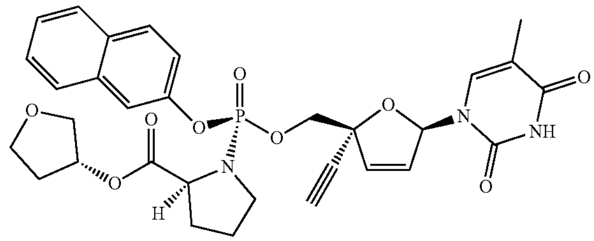
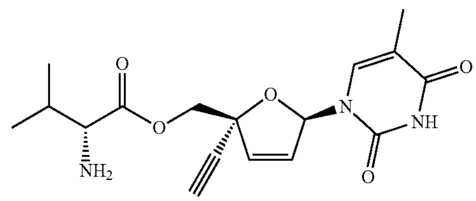
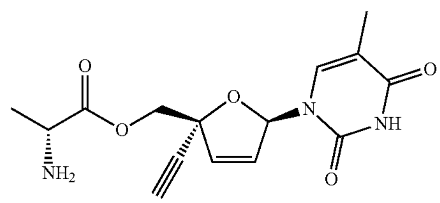
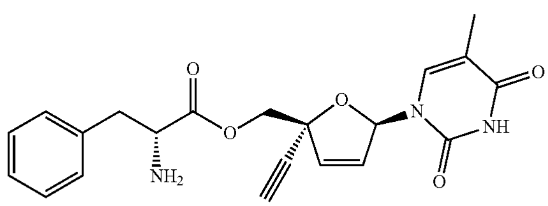
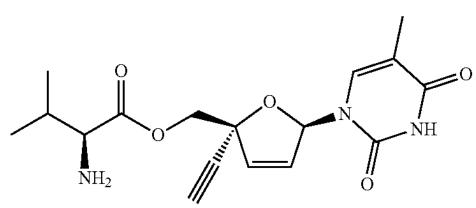
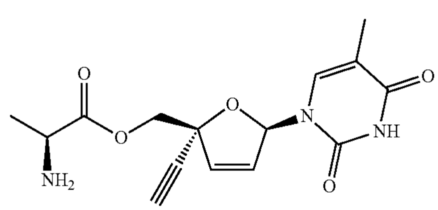
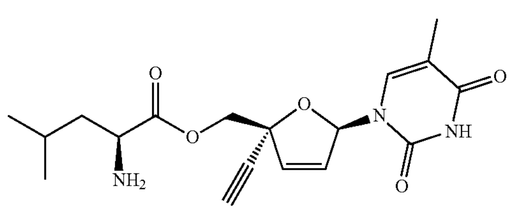
Compound No.	Chemical Structure
I-26	
I-27	
I-28	
I-29	
I-30	
I-31	
I-32	

TABLE 1-continued

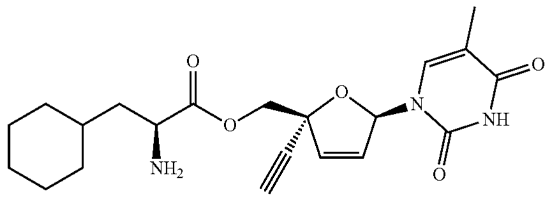
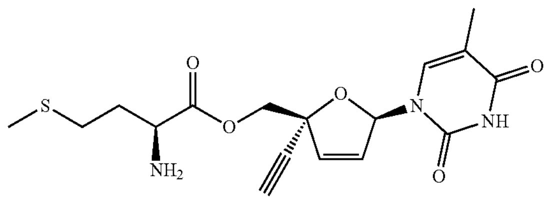
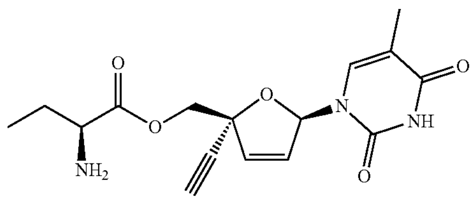
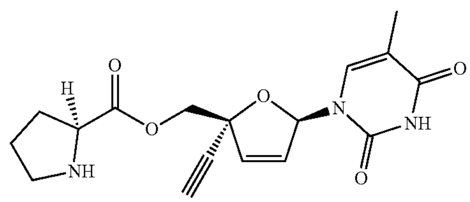
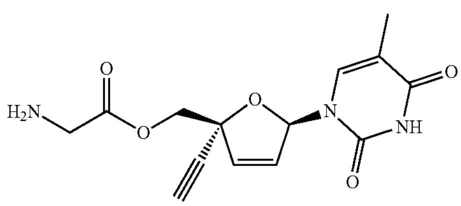
Compound No.	Chemical Structure
I-33	
I-34	
I-35	
I-36	
I-37	

TABLE 2

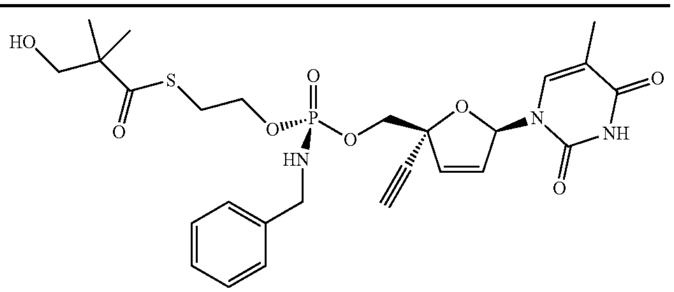
Compound No.	Chemical Structure
II-1	

TABLE 2-continued

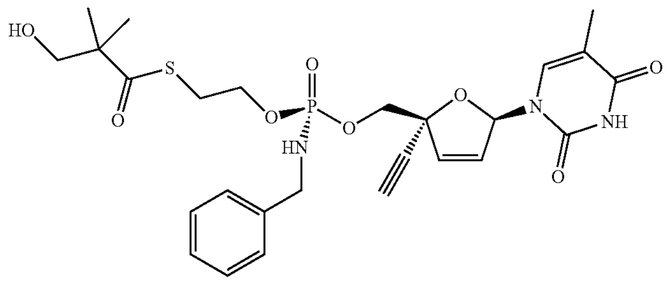
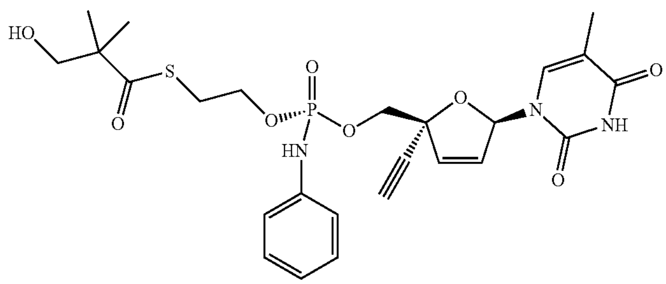
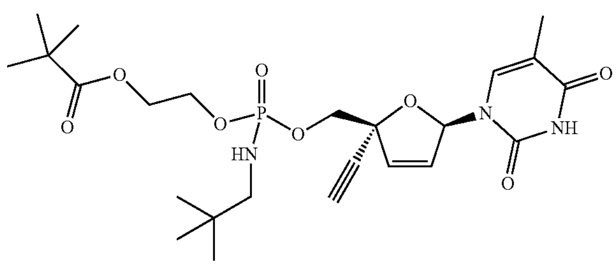
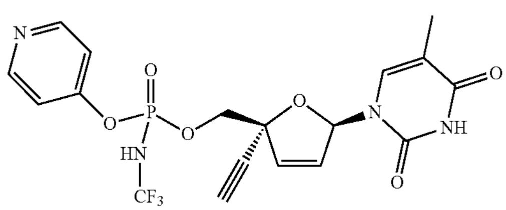
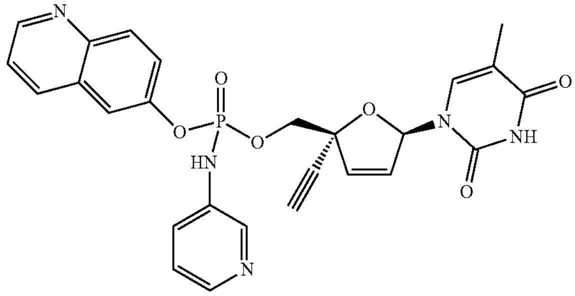
Compound No.	Chemical Structure
II-2	
II-3	
II-4	
II-5	
II-6	

TABLE 2-continued

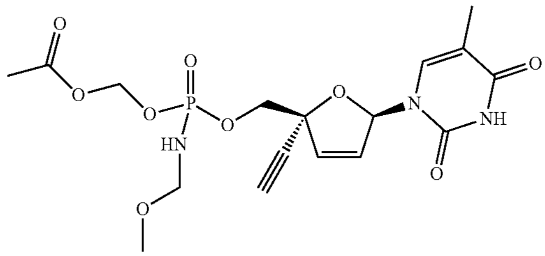
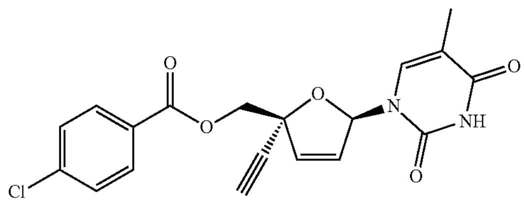
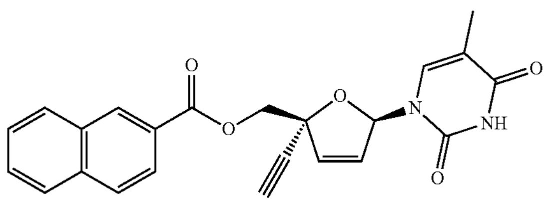
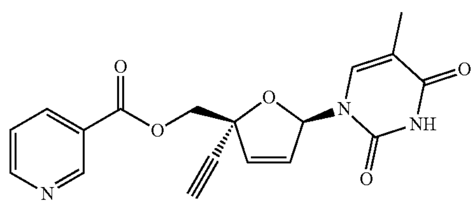
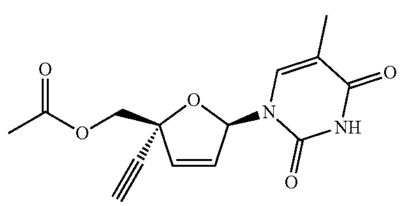
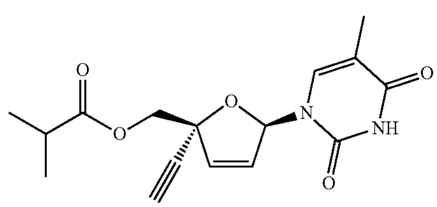
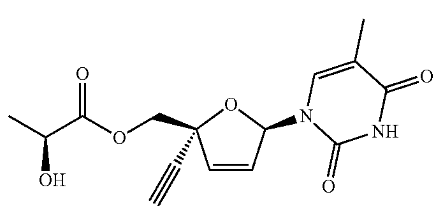
Compound No.	Chemical Structure
II-7	
II-8	
II-9	
II-10	
II-11	
II-12	
II-13	

TABLE 2-continued

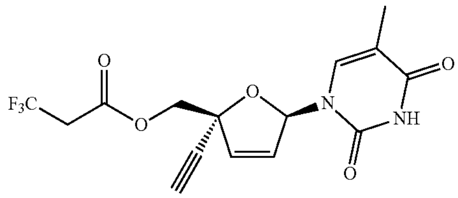
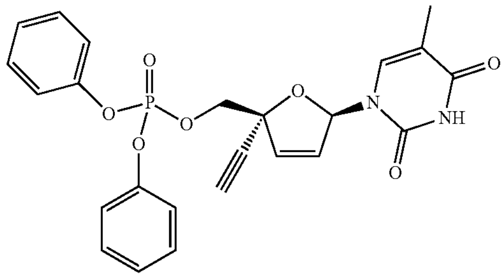
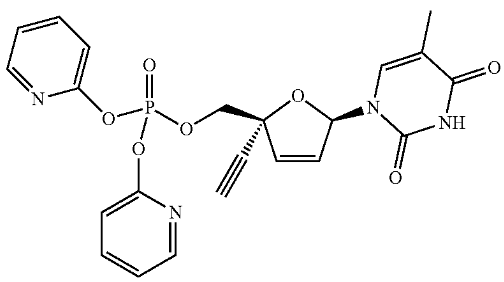
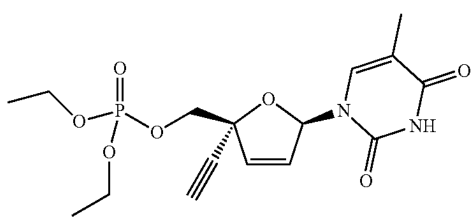
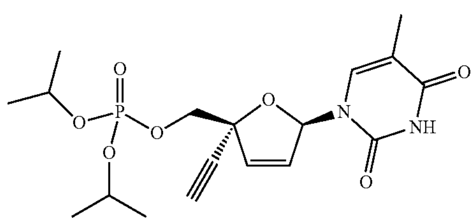
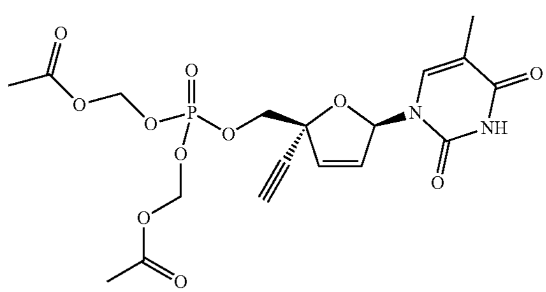
Compound No.	Chemical Structure
II-14	
II-15	
II-16	
II-17	
II-18	
II-19	

TABLE 2-continued

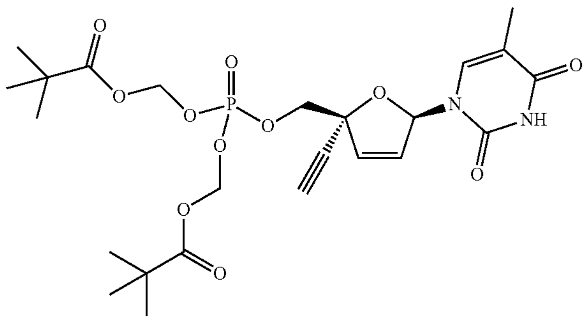
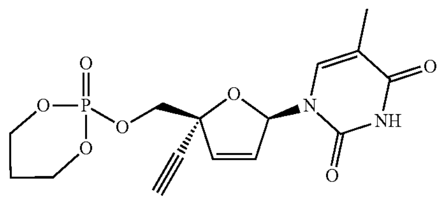
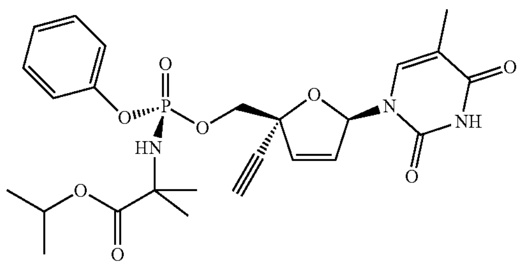
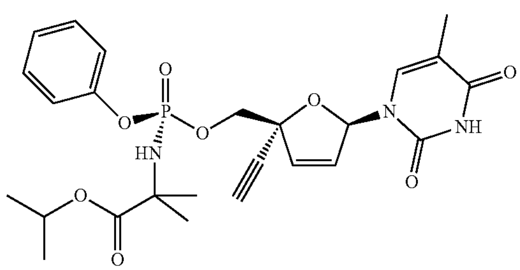
Compound No.	Chemical Structure
II-20	
II-21	
II-22	
II-23	

TABLE 3

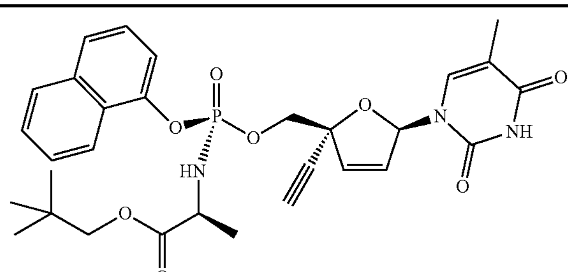
Compound No.	Chemical Structure
III-1	

TABLE 3-continued

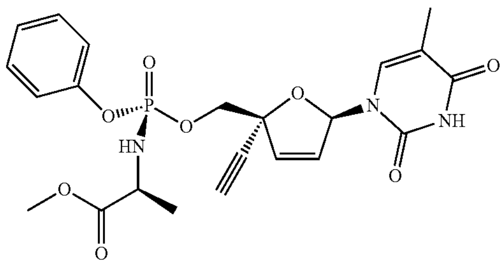
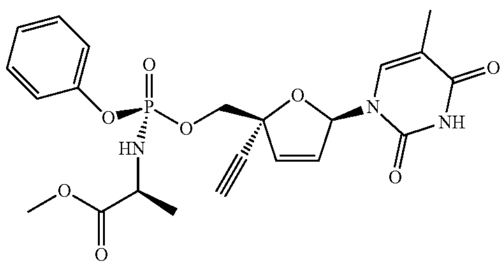
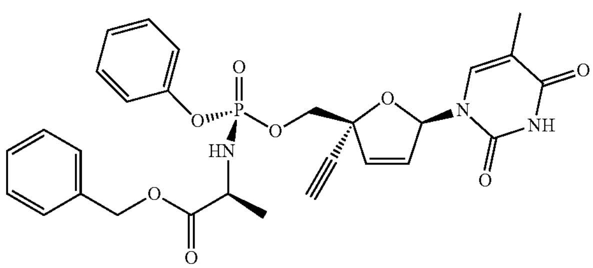
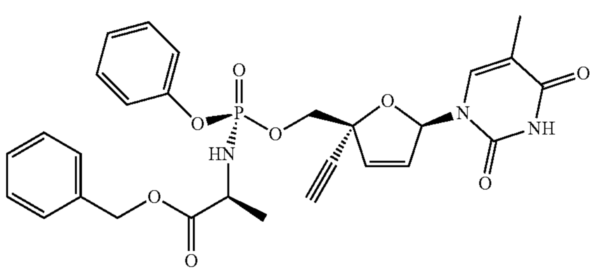
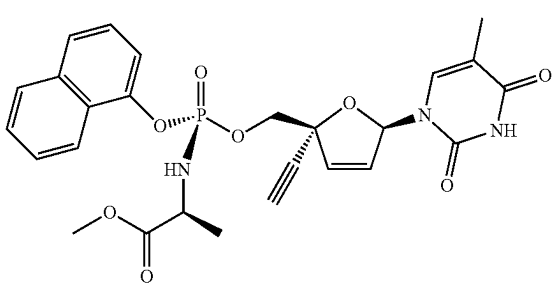
Compound No.	Chemical Structure
III-2	
III-3	
III-4	
III-5	
III-6	

TABLE 3-continued

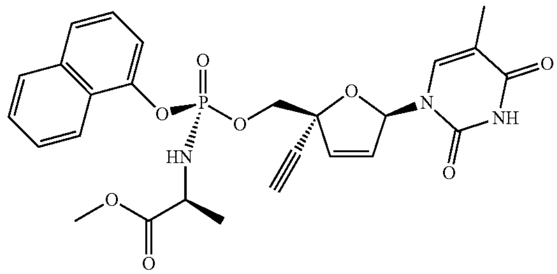
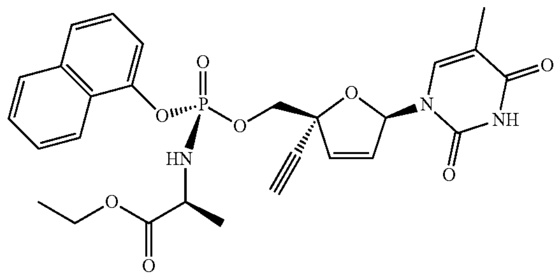
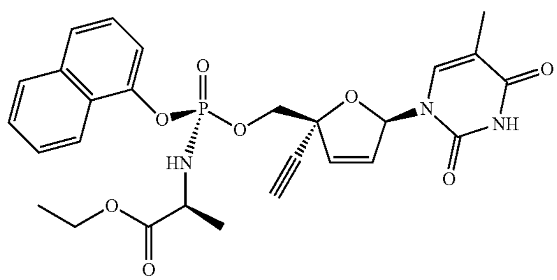
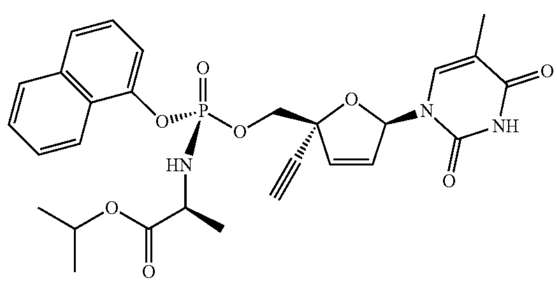
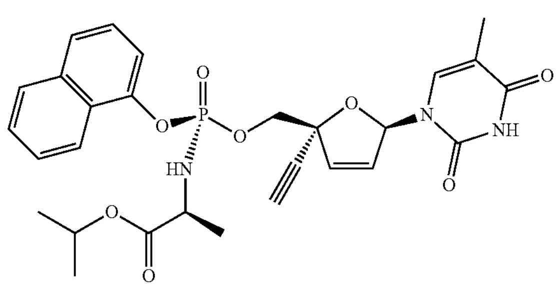
Compound No.	Chemical Structure
III-7	
III-8	
III-9	
III-10	
III-11	

TABLE 3-continued

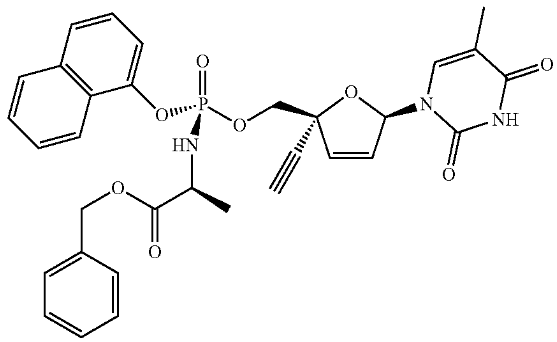
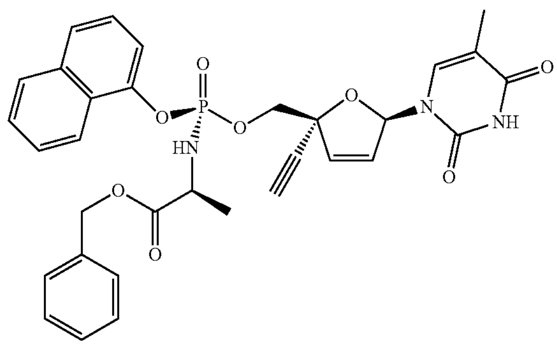
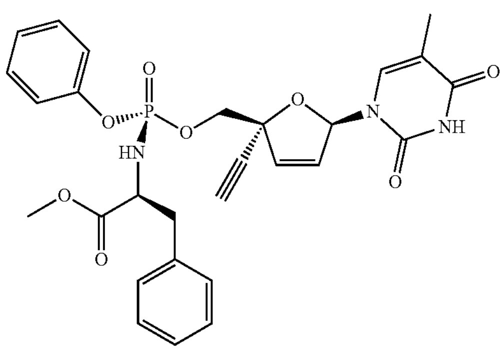
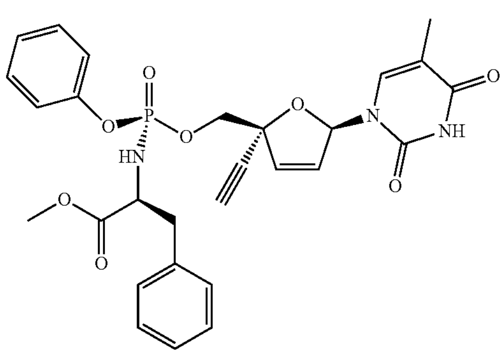
Compound No.	Chemical Structure
III-12	
III-13	
III-14	
III-15	

TABLE 3-continued

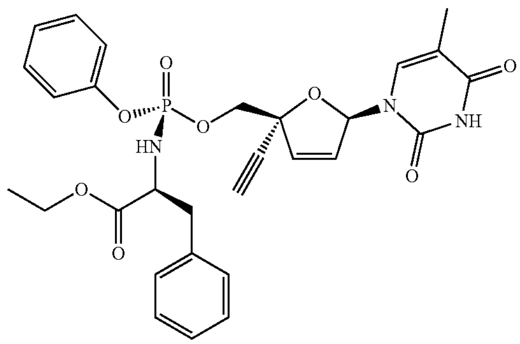
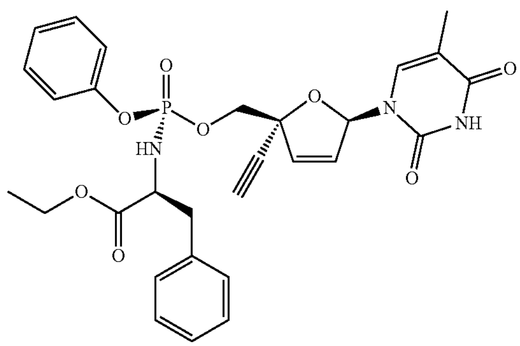
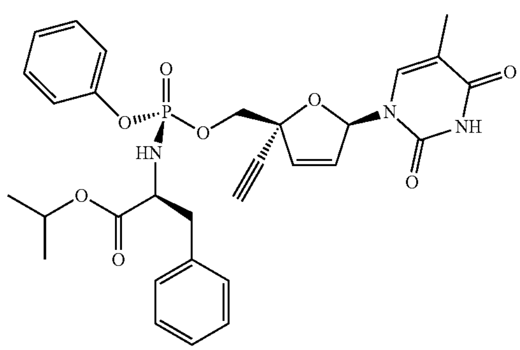
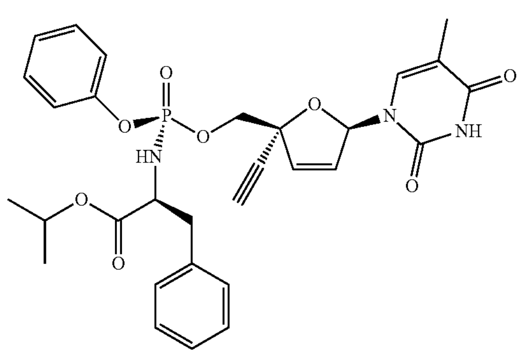
Compound No.	Chemical Structure
III-16	
III-17	
III-18	
III-19	

TABLE 3-continued

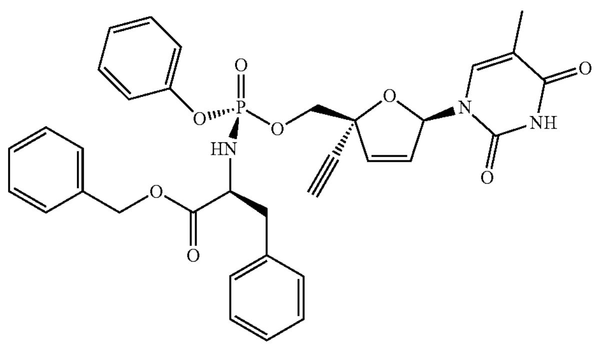
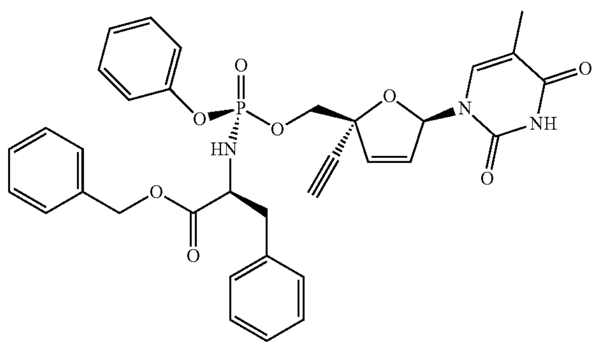
Compound No.	Chemical Structure
III-20	
III-21	

TABLE 4

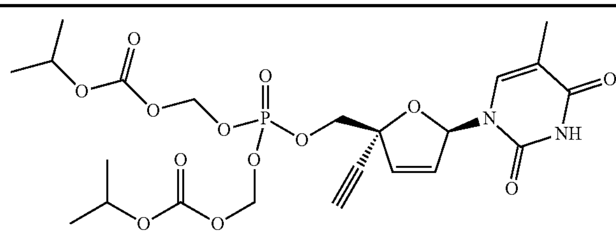
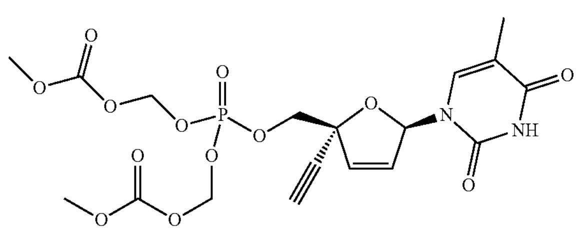
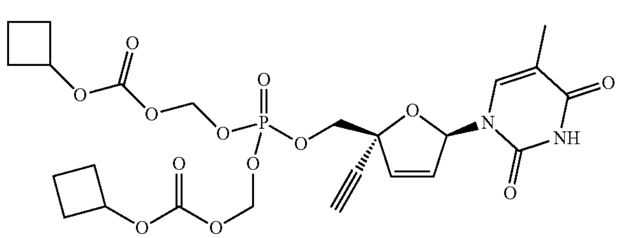
Compound No.	Chemical Structure
IV-1	
IV-2	
IV-3	

TABLE 4-continued

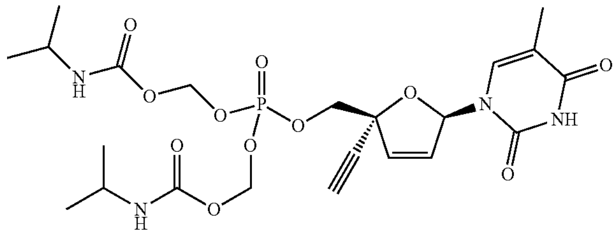
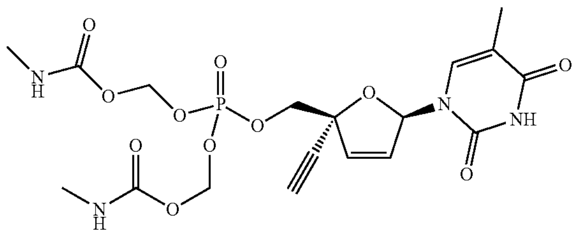
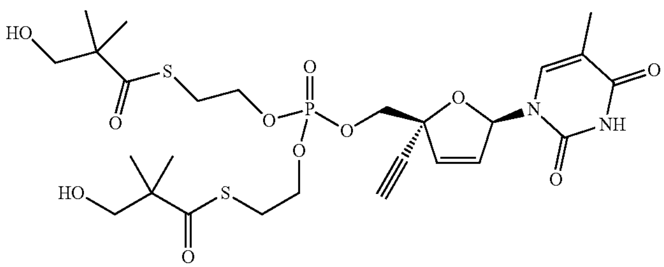
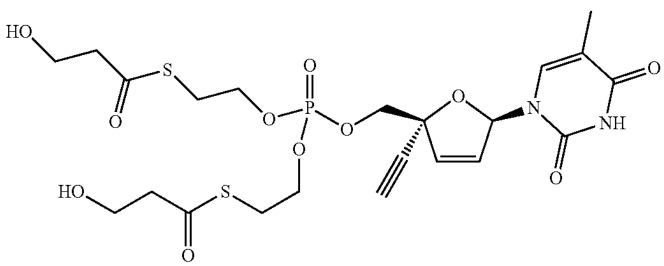
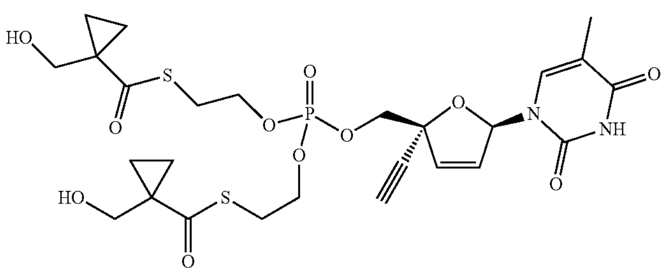
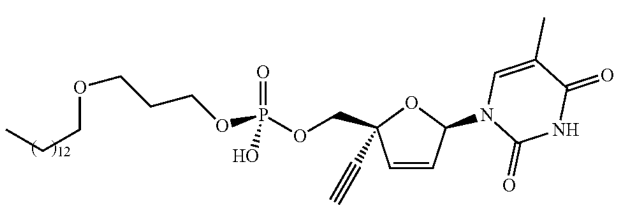
Compound No.	Chemical Structure
IV-4	
IV-5	
IV-6	
IV-7	
IV-8	
IV-9	

TABLE 4-continued

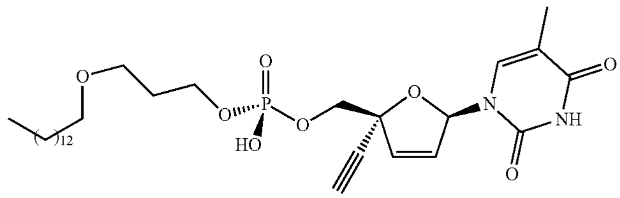
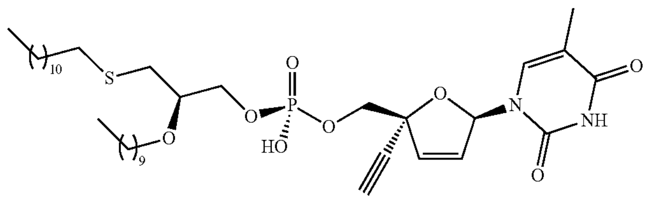
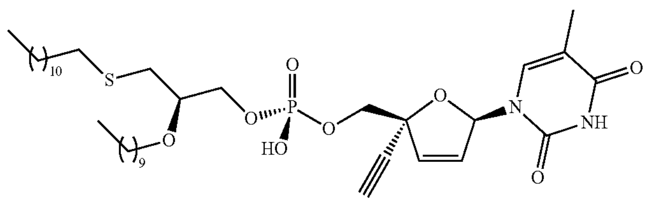
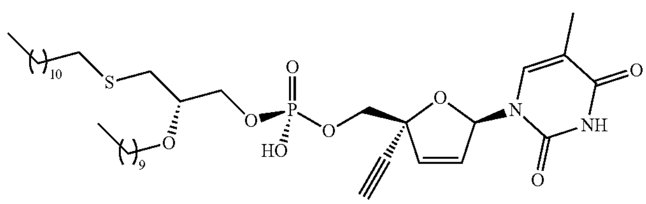
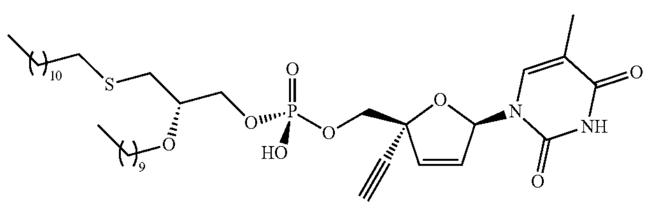
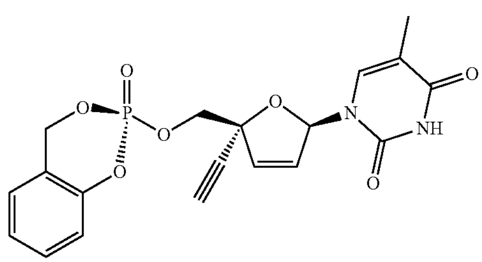
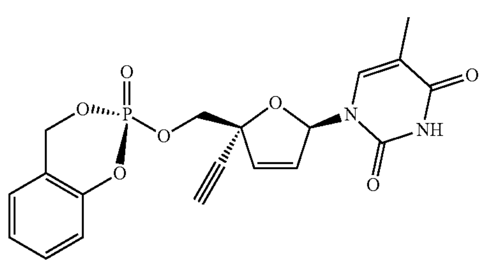
Compound No.	Chemical Structure
IV-10	
IV-11	
IV-12	
IV-13	
IV-14	
IV-15	
IV-16	

TABLE 4-continued

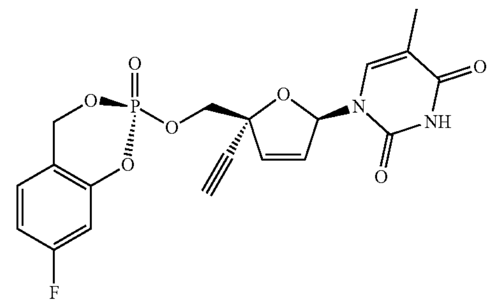
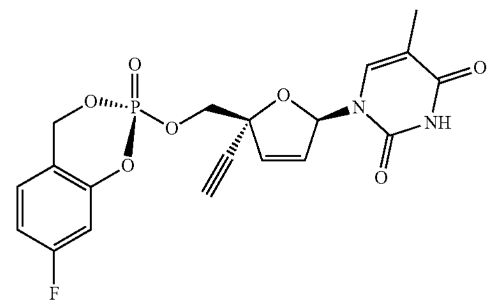
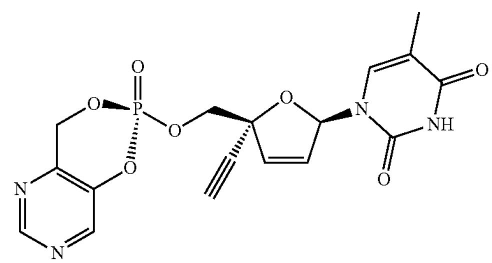
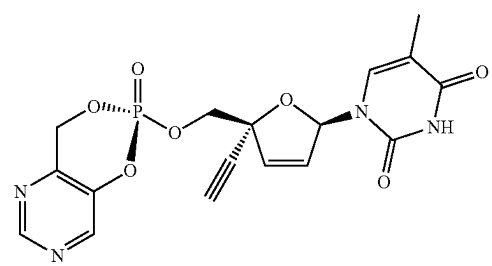
Compound No.	Chemical Structure
IV-17	
IV-18	
IV-19	
IV-20	

TABLE 5

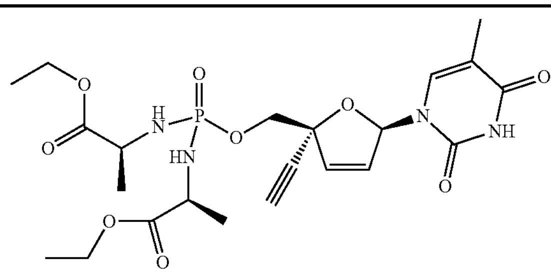
Compound No.	Chemical Structure
V-1	

TABLE 5-continued

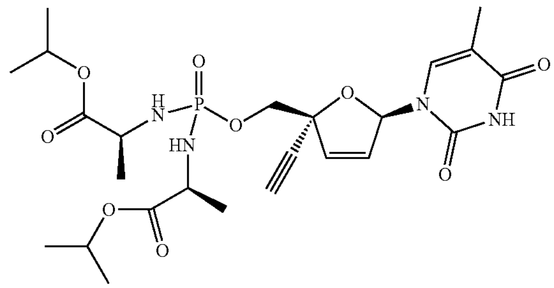
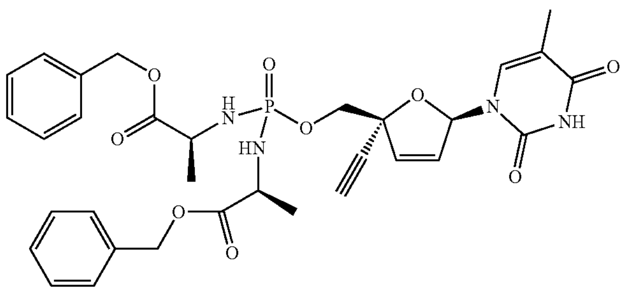
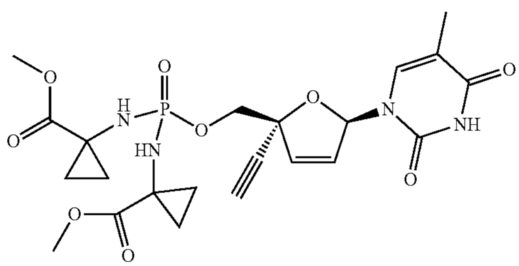
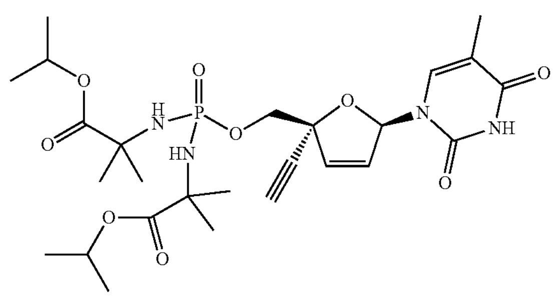
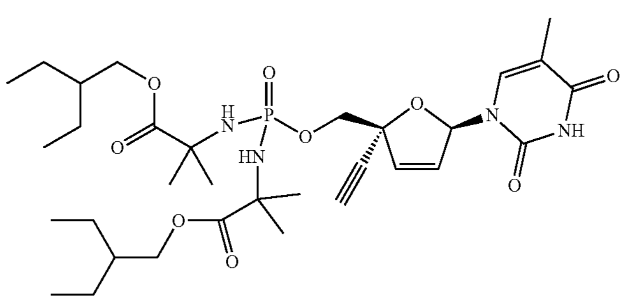
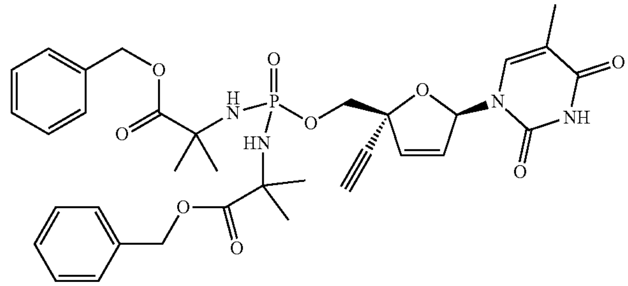
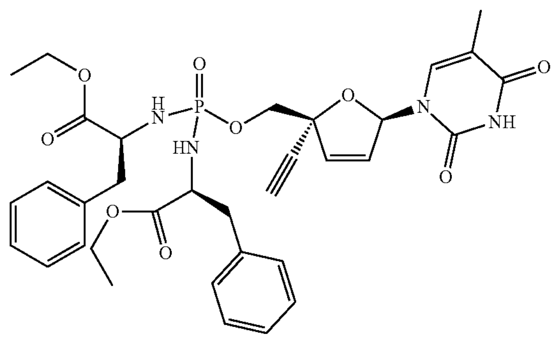
Compound No.	Chemical Structure
V-2	
V-3	
V-4	
V-5	
V-6	

TABLE 5-continued

Compound No.	Chemical Structure
V-7	
V-8	

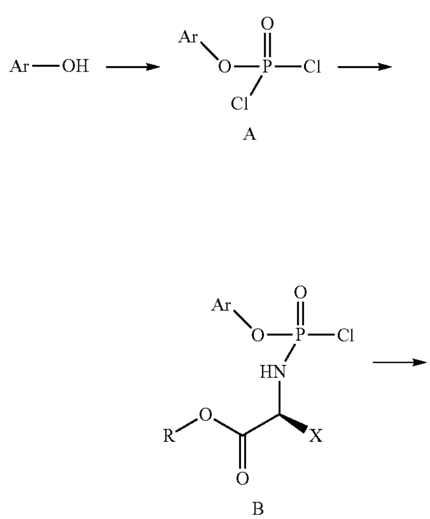
[0405] Methods for preparing 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine are provided in, for example, U.S. Pat. Nos. 7,589,078; 9,051,288; U.S. 2016/0060252; and references therein. Methods for preparing phosphoramidate and phosphonate compounds described herein, starting for example from 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine, are provided in, for example, U.S. Pat. No. 8,022,083; WO 2006/110157; WO 2006/015261; and related patents and applications, such as U.S. Pat. Nos. 7,871,991; 8,318,701, and 8,329,926. Additional strategies for preparing phosphoramidate compounds described herein are described in, for example, Slusarczyk, M. et al. "Phosphoramidates and phosphonamidates (ProTides) with antiviral activity," *Antiviral Chemistry and Chemotherapy* (2018), Vol. 26, p. 1-31, and references therein. Each of the foregoing references is hereby incorporated by reference in its entirety.

[0406] Methods for preparing compounds described herein are also illustrated in the following synthetic Schemes, and in the Examples below. The Schemes are given for the purpose of illustrating the invention, and are not intended to limit the scope or spirit of the invention. Starting materials shown in the Schemes can be obtained from commercial sources or can be prepared based on procedures described in the literature.

[0407] In the Schemes, it is understood by one skilled in the art of organic synthesis that the functionality present on various portions of the molecule should be compatible with the reagents and reactions proposed. Substituents not compatible with the reaction conditions will be apparent to one skilled in the art, and alternate methods are therefore indicated (for example, use of protecting groups or alternative reactions). Protecting group chemistry and strategy is well known in the art, for example, as described in detail in "Protecting Groups in Organic Synthesis", T. W. Greene and P. G. M. Wuts, 3rd edition, John Wiley & Sons, 1999, the entire contents of which are hereby incorporated by reference.

[0408] The synthetic route illustrated in Scheme 1 is a general method for preparing phosphoramidate compounds C. Reaction of an aryl alcohol with phosphorus oxychloride affords intermediate A. Reaction of intermediate A with an amino acid ester affords phosphoryl chloride intermediate B, which is coupled with 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine to afford phosphoramidate compounds C.

SCHEME 1.



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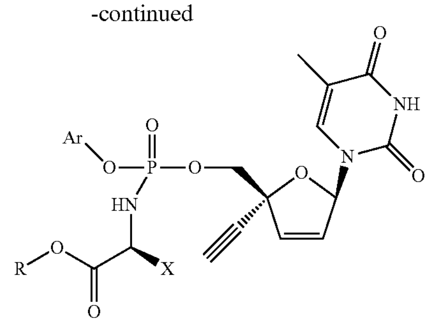
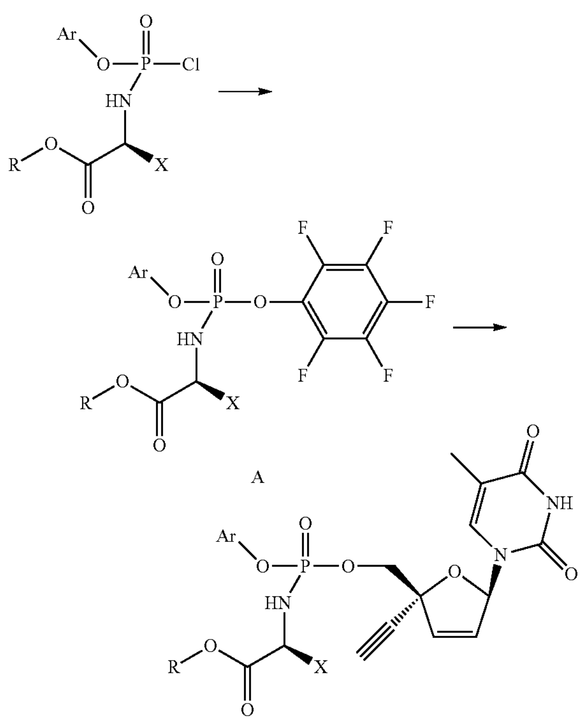


C

Ar may be aryl or heteroaryl, optionally substituted
R may be alkyl
X may be an amino acid substituent

[0409] As depicted in Scheme 2, the phosphoryl chloride intermediate of Scheme 1 can be coupled with pentafluorophenol to afford intermediate A, which is further condensed with 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine to afford phosphoramidate compounds B. Intermediate A, which is initially a mixture of diastereomers, can be separated into each of the pure diastereomers either by chromatography or recrystallization. Each pure diastereomer of intermediate A can then be separately condensed with 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine to afford phosphoramidate compounds B, which are stereochemically pure at the phosphorus stereocenter.

SCHEME 2.

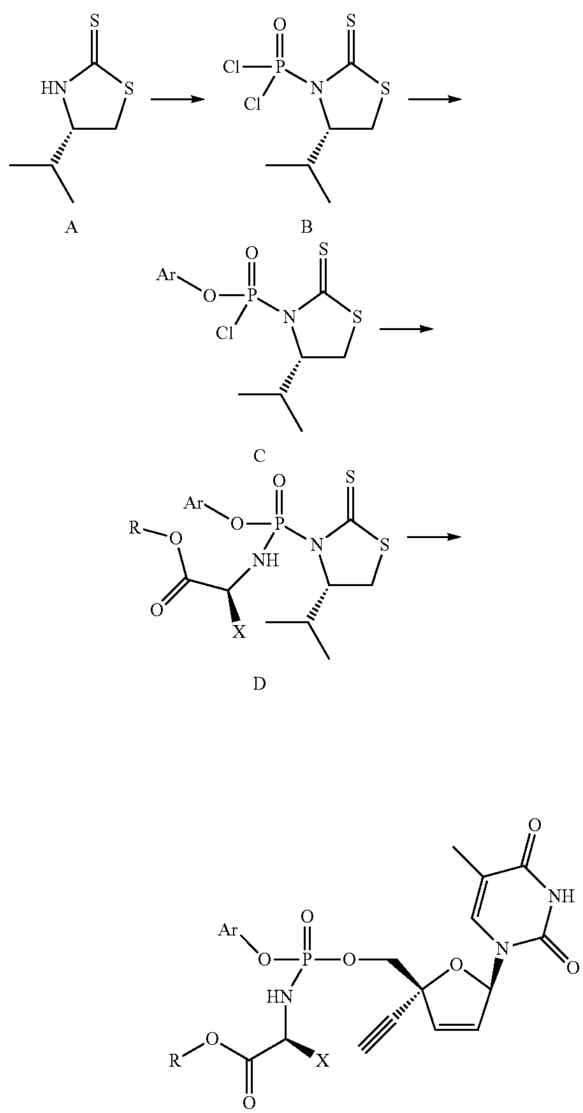


B

Ar may be aryl or heteroaryl, optionally substituted
R may be alkyl
X may be an amino acid substituent

[0410] The synthetic route illustrated in Scheme 3 is another general method for preparing phosphoramidate compounds E. Condensation of chiral auxiliary A with phosphorus oxychloride affords dichloride intermediate B, which is further condensed with an aryl alcohol to provide intermediate C. Intermediate C is condensed with an amino acid ester to provide intermediate D, which can be used as the stereochemical mixture at phosphorus or can be separated by chromatography into individual phosphorus diastereomers. Intermediate D is reacted with 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine to afford phosphoramidate compounds E, either stereochemically pure or a mixture of diastereomers at phosphorus.

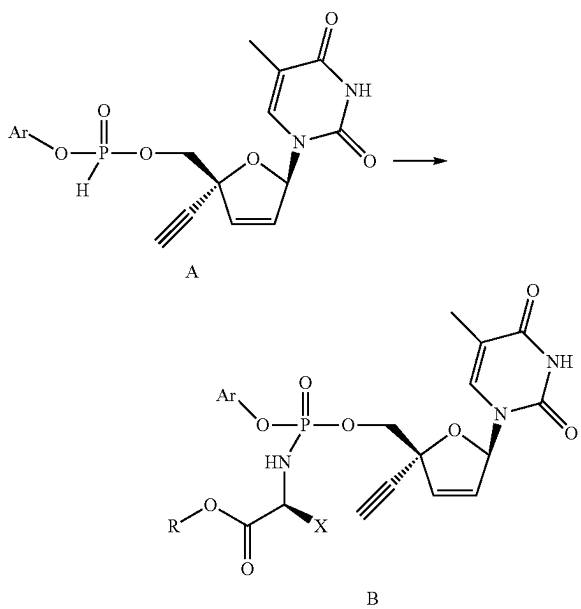
SCHEME 3.



E

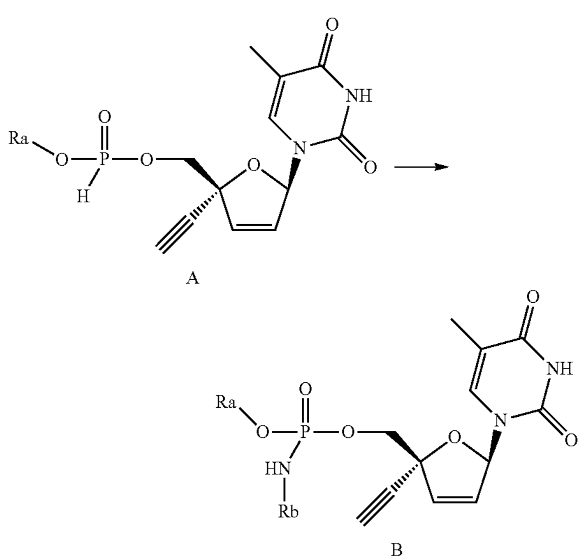
Ar may be aryl or heteroaryl, optionally substituted
R may be alkyl
X may be an amino acid substituent

[0411] Schemes 4 and 5 illustrate an additional general method for preparing phosphoramidate compounds B. 2',3'-Didehydro-3'-deoxy-4'-ethynylthymidine is reacted via a phosphite transesterification to afford intermediate A, which is further reacted via oxidative amination with an amino acid ester to afford desired product B.



Ar may be aryl or heteroaryl, optionally substituted
R may be alkyl
X may be an amino acid substituent

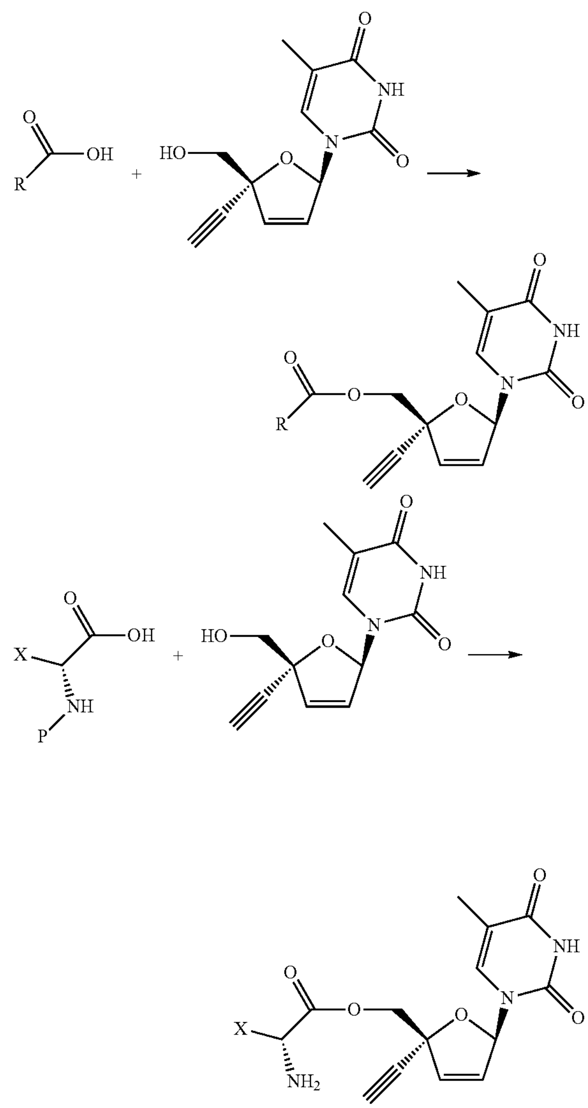
SCHEME 5.



Ra and Rb may be alkyl, optionally substituted.

[0412] As illustrated in Scheme 6, ester compounds described herein may be prepared by coupling 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine with an appropriate carboxylic acid, using, for example, an amide coupling reagent, such as DCC or HATU. To prepare amino acid esters, an N-protected amino acid (for example, NBoc or NFmoc) may be used in the coupling reaction, followed by deprotection under appropriate conditions (for example, acid deprotection for NBoc, or base deprotection for NFmoc). Protecting group chemistry and strategy is well known in the art, for example, as described in detail in "Protecting Groups in Organic Synthesis", T. W. Greene and P. G. M. Wuts, 3rd edition, John Wiley & Sons, 1999, the entire contents of which are hereby incorporated by reference.

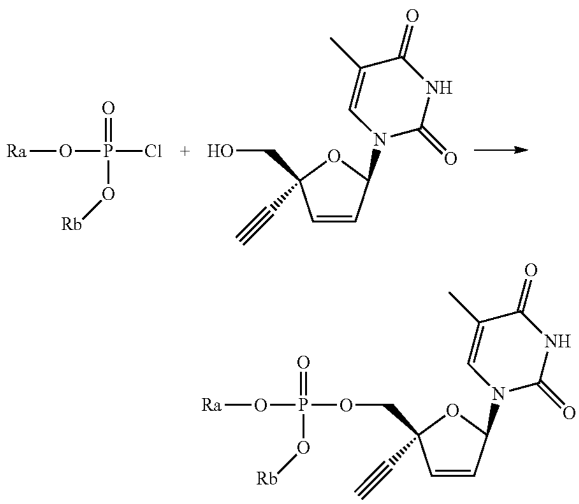
SCHEME 6.



R may be alkyl or aryl
X may be an amino acid substituent
P is an amino protecting group

[0413] The synthetic route illustrated in Scheme 7 is a general method for preparing phosphate ester compounds described herein, by condensing 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine with a dialkyl chlorophosphite.

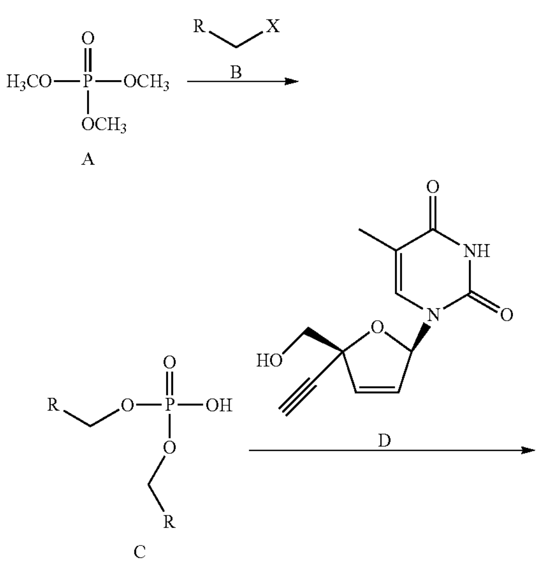
SCHEME 7.



Ra and Rb may be alkyl or aryl, or together form an alkyl ring

[0414] The synthetic scheme illustrated in Scheme 8 is a general method for preparing dialkyl phosphonates and related compounds E. Alkylating trimethyl phosphate A with electrophile B (wherein X is a leaving group, such as chloride or a sulfonate) under nucleophilic substitution conditions (using, for example, a polar aprotic solvent, such as acetone, and optionally in the presence of NaI) produces dialkyl phosphate C. Coupling phosphate C with 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine D (using, for example, amide coupling conditions, such as BOPCl and NMI in a polar aprotic solvent, such as acetonitrile) produces dialkyl phosphonates E.

SCHEME 8.



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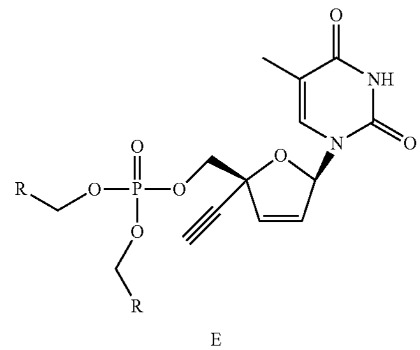
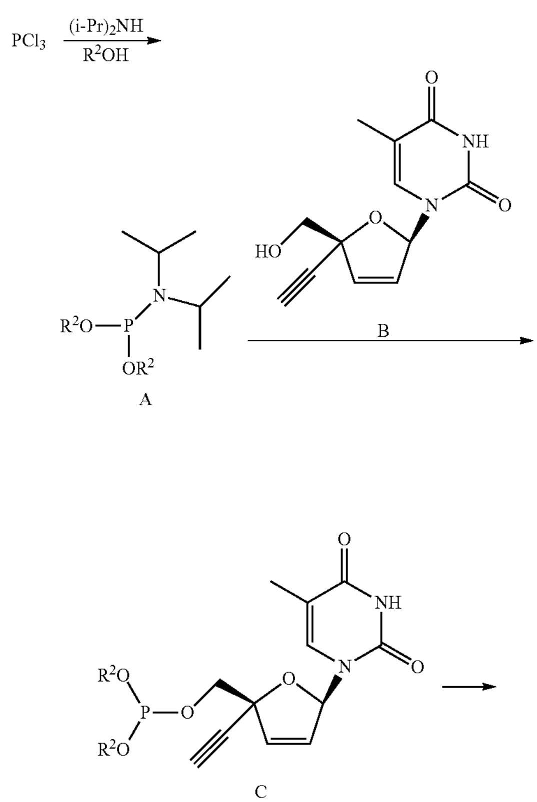


[0415] The synthetic scheme illustrated in Scheme 9 is a general method for preparing phosphonates D. Coupling PCl_3, diisopropyl amine, and an alcohol R^2OH (or diol, when two instances of R^2 are taken together to form a ring), for example, in a polar aprotic solvent (such as THF) and in the presence of a base (such as triethylamine) affords aminophosphine A. Coupling aminophosphine A with 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine B (using, for example, tetrazole or pyridinium chloride in a polar aprotic solvent, such as acetonitrile) provides phosphite C. Oxidation of phosphite C (using, for example, mCPBA in dichloromethane or t-butyl hydroperoxide) affords phosphonates D.

SCHEME 9.



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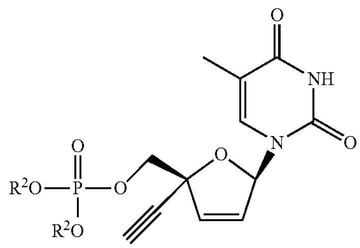
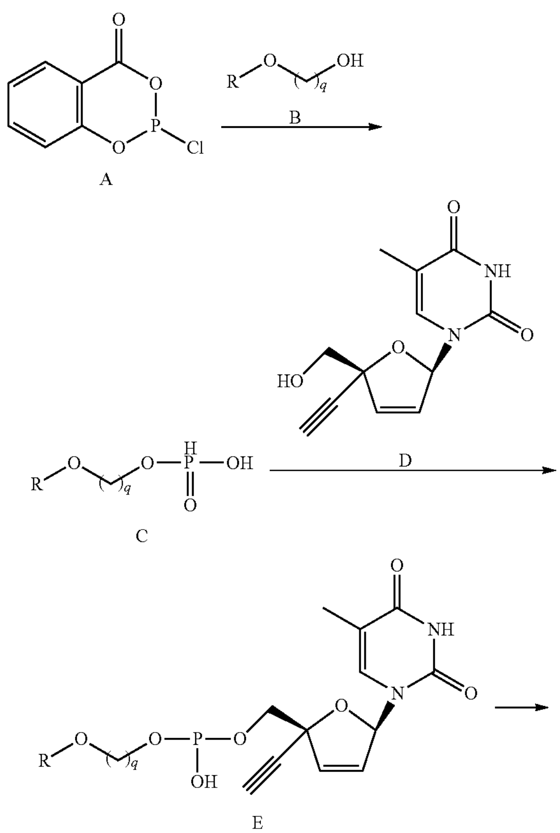


D

[0416] The synthetic scheme illustrated in Scheme 10 is a general method for preparing dialkyl hydrogen phosphates F. Coupling chloro-phosphorous reagent A with alcohol B (using, for example, a base, such as pyridine, in a polar aprotic solvent, such as 1,4-dioxane), followed by hydrolysis (using, for example, a base, such as sodium bicarbonate, with water and an organic solvent, such as chloroform) affords alkyl hydrogen phosphonate C. Coupling hydrogen phosphonate C with 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine D (using, for example, an activating agent, such as pivaloyl chloride, and a base, such as pyridine) affords dialkyl hydrogen phosphite E. Oxidation of E (using, for example, iodine in THF and water) affords dialkyl hydrogen phosphates F.

SCHEME 10.



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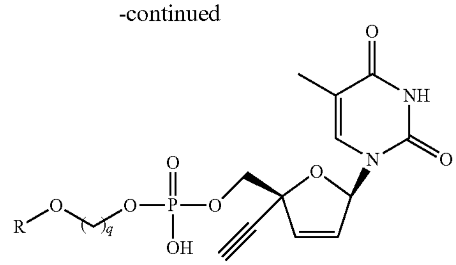
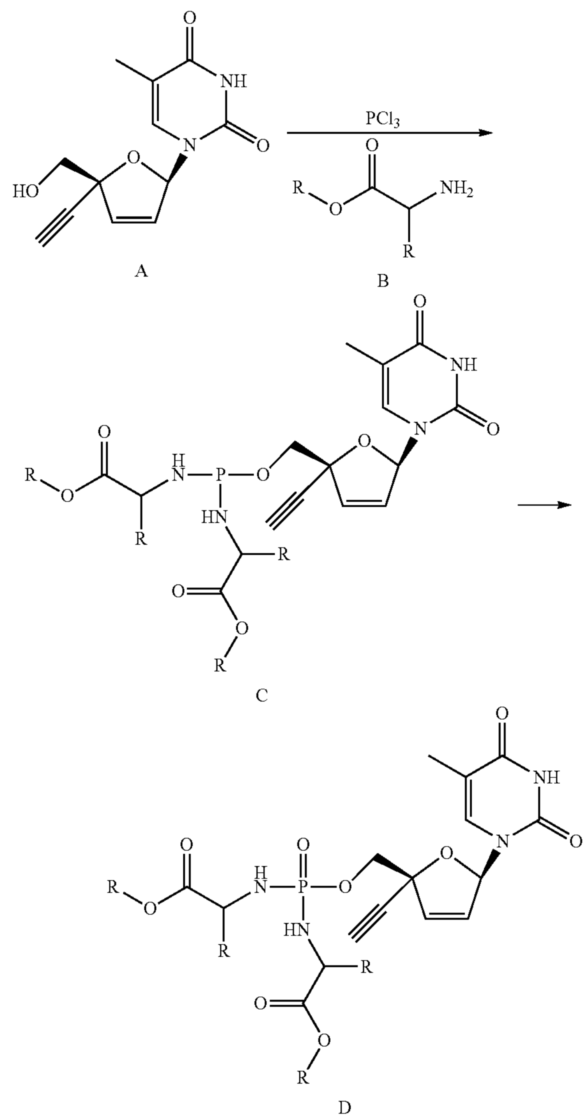


F

[0417] The synthetic scheme illustrated in Scheme 11 is a general method for preparing diamino phosphonate D. Coupling PCl_3, an amino ester B, and 2',3'-didehydro-3'-deoxy-4'-ethynylthymidine A affords diamino phosphite C. Oxidation of C (using, for example, mCPBA in dichloromethane) affords diamino phosphonate D.

SCHEME 11.



[0418] The modular synthetic routes described herein and in the foregoing references can also be readily modified by one of skill in the art of organic synthesis to provide additional substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds using strategies and reactions well known in the art, as described in, for example, "Comprehensive Organic Synthesis" (B. M. Trost & I. Fleming, eds., 1991-1992).

II. Methods of Treating Medical Disorders

[0419] Another aspect of the invention provides methods for treating medical disorders. This is described in more detail below.

A. First Therapeutic Method

[0420] It is contemplated that the substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds described herein, such as a compound of Formula I, I-1, I-A, or I-B, or other compounds in Section I, above, provide therapeutic benefits to subjects suffering from cancer and other disorders. Accordingly, one aspect of the invention provides a method of treating a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder. The method comprises administering a therapeutically effective amount of a compound described in Section I above, such as a compound of Formula I, I-1, I-A, or I-B, to a subject in need thereof to treat the disorder. In certain embodiments, the particular compound of Formula I, I-1, I-A, or I-B, is a compound defined by one of the embodiments described in Section I, above.

Viral Infection

[0421] In certain embodiments, the disorder is an immune disorder that is a viral infection. In certain embodiments, the viral infection is an infection by human immunodeficiency viruses 1 or 2 (HIV-1 or HIV-2), human T-cell leukemia viruses 1 or 2 (HTLV-1 or HTLV-2), respiratory syncytial virus (RSV), human papilloma virus (HPV), adenovirus, hepatitis B virus (HBV), hepatitis C virus (HCV), Epstein-Barr virus (EBV), varicella zoster virus (VZV), cytomegalovirus (CMV), herpes simplex viruses 1 or 2 (HSV-1 or HSV-2), human herpes virus 8 (HHV-8, also known as Kaposi's sarcoma-associated virus), or a flavivirus selected from Yellow Fever virus, Dengue virus, Japanese Encephalitis, and West Nile virus.

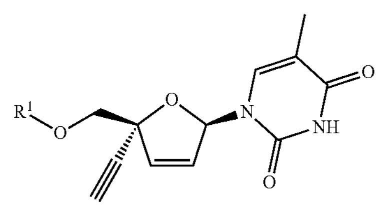
[0422] In certain embodiments, the viral infection is an infection by human immunodeficiency viruses 1 or 2 (HIV-1 or HIV-2). In certain embodiments, the viral infection is an infection by human immunodeficiency virus 1 (HIV-1). In certain embodiments, the viral infection is an infection by human immunodeficiency virus 2 (HIV-2). In certain embodiments, the viral infection is an infection by human T-cell leukemia viruses 1 or 2 (HTLV-1 or HTLV-2). In certain embodiments, the viral infection is an infection by respiratory syncytial virus (RSV). In certain embodiments, the viral infection is an infection by human papilloma virus (HPV). In certain embodiments, the viral infection is an infection by adenovirus. In certain embodiments, the viral infection is an infection by hepatitis B virus (HBV). In certain embodiments, the viral infection is an infection by hepatitis C virus (HCV). In certain embodiments, the viral infection is an infection by Epstein-Barr virus (EBV). In certain embodiments, the viral infection is an infection by varicella zoster virus (VZV). In certain embodiments, the viral infection is an infection by cytomegalovirus (CMV). In certain embodiments, the viral infection is an infection by herpes simplex viruses 1 or 2 (HSV-1 or HSV-2). In certain embodiments, the viral infection is an infection by human herpes virus 8 (HHV-8, also known as Kaposi's sarcoma-associated virus). In certain embodiments, the viral infection is an infection by a flavivirus selected from Yellow Fever virus, Dengue virus, Japanese Encephalitis, and West Nile virus.

[0423] Additional exemplary features that may characterize the First Therapeutic Method described herein are provided below and include, for example, disorders and patients to be treated.

B. Second Therapeutic Method

[0424] Another aspect of the invention provides a method of treating a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises administering to a subject in need thereof a therapeutically effective amount of a compound of Formula II to treat the disorder; wherein Formula II is represented by:

(II)



[0425] or a stereoisomer thereof; or a pharmaceutically acceptable salt of either of the foregoing; wherein:

[0426] R^1 is hydrogen, —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —P(O)(N(R^3)(R^4))$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^2;

[0427] R^2 represents independently for each occurrence hydrogen, —P(O)(OH)$_2$, —P(O)(OH)—O—P(O)(OH)$_2$, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

[0428] R^3 and R^9 each represent independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, or two instances of R^9, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0429] R^4 represents independently for each occurrence —$C(R^5)_2$—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0430] R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

[0431] R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0432] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —$N(R^9)_2$; and

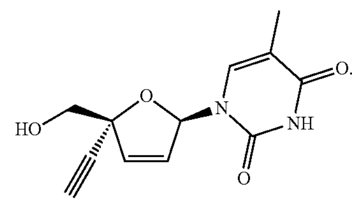
[0433] m and p are independently for each occurrence 0, 1, 2, or 3.

[0434] The definitions of variables in Formula II above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

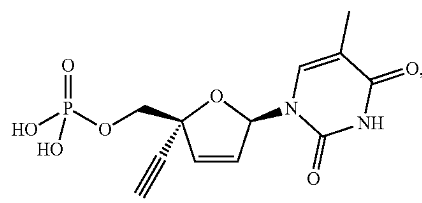
[0435] In certain embodiments, the compound is a compound of Formula II, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound of Formula II.

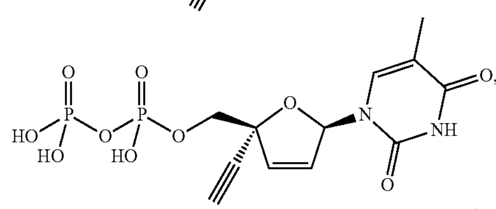
[0436] In certain embodiments, the particular compound of Formula II is a compound defined by one of the embodiments described in Section I, above, such as a compound of Formula I, I-1, I-A, or I-B. Additionally, embodiments described in Section I above for variables R^1, R^2, R^3, R^4, R^5, R^6, R^7, R^9, m, and p also apply to compounds of Formula II. For example, in certain embodiments, the compound is a compound of Formula II wherein R^1 is —$P(O)(OR^2)(N(R^3)(R^4))$ or —C(O)—$C(H)(R^5)$—$N(R^3)_2$.

[0437] In certain embodiments, the compound of Formula II is

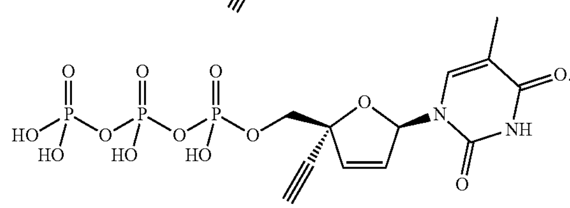


[0438] In certain embodiments, the compound of Formula II is

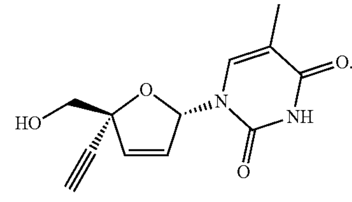




or

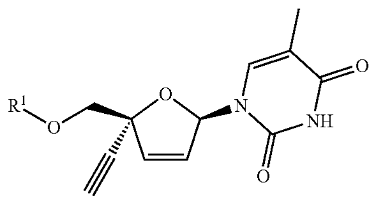


In certain embodiments, the compound of Formula II is



[0439] Another aspect of the invention provides a method of treating a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises administering to a subject in need thereof a therapeutically effective amount of a compound of Formula II-1 to treat the disorder; wherein Formula II-1 is represented by:

(II-1)



[0440] or a pharmaceutically acceptable salt thereof; wherein:

[0441] R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, —C(O)R^2, or hydrogen;

[0442] R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, hydrogen, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring;

[0443] R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0444] R^4 is —C(H)(R^5)—CO$_2$R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0445] R^5 is C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0446] R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0447] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl; and

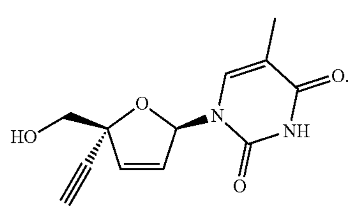
[0448] m is independently for each occurrence 0, 1, 2, or 3.

[0449] The definitions of variables in Formula 1-1 above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

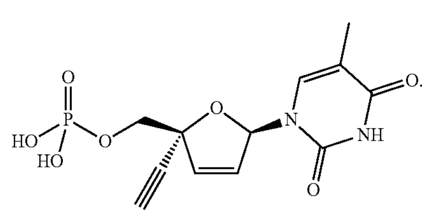
[0450] In certain embodiments, the compound is a compound of Formula II-1.

[0451] In certain embodiments, the particular compound of Formula II-1 is a compound defined by one of the embodiments described in Section I, above, such as a compound of Formula I-1 or I-A. Additionally, embodiments described in Section I above for variables R^1, R^2, R^3, R^4, R^5, R^6, R^7, and m of Formula I-1 also apply to compounds of Formula II-1. For example, in certain embodiments, the compound is a compound of Formula II-1 wherein R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N(R^3)$_2$.

[0452] In certain embodiments, the compound of Formula II-1 is



In certain embodiments, the compound of Formula II-1 is



[0453] Additional exemplary features that may characterize the Second Therapeutic Method described herein are provided below and include, for example, disorders and patients to be treated.

C. Additional Exemplary Features of the First and Second Therapeutic Methods

[0454] Additional exemplary features that may characterize the First and Second Therapeutic Methods described herein are provided below and include, for example, disorders and patients to be treated. A more thorough description of such features is provided below. The invention embraces all permutations and combinations of these features.

Pharmaceutical Compositions and Additional Therapeutic Agents

[0455] In certain embodiments, the compound of Formula II, or compound defined by one of the embodiments described in Section I, above, such as a compound of Formula I, I-1, I-A, or I-B, is administered in a pharmaceutical composition comprising the compound and a pharmaceutically acceptable carrier, as further described in Section V, below.

[0456] In certain embodiments, the method further comprises administering an effective amount of an additional therapeutic agent, as further described in Section IV, below.

Cancer

[0457] In certain embodiments, the disorder is cancer. In certain embodiments, the cancer is a solid tumor or leukemia. In certain embodiments, the cancer is a solid tumor. In certain embodiments, the cancer is a carcinoma or melanoma. In certain embodiments, the cancer is a carcinoma. In certain embodiments, the cancer is a sarcoma. In certain embodiments, the cancer is a melanoma. In certain embodiments, the cancer is a lymphoma. In certain embodiments, the cancer is a leukemia.

[0458] In certain embodiments, the cancer is breast cancer, ovarian cancer, uterine cancer, cervical cancer, prostate cancer, testicular cancer, lung cancer, leukemia, head and neck cancer, oral cancer, esophageal cancer, stomach cancer, bile duct and gallbladder cancers, bladder cancer, urinary tract cancer, colon cancer, rectal cancer, thyroid cancer, pancreatic cancer, kidney cancer, liver cancer, brain cancer, skin cancer, or eye cancer.

[0459] In certain embodiments, the cancer has (i) expression of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; (ii) activity of LINE1 reverse transcriptase; (iii) expression of HERV-K RNA, and/or (iv) activity of HERV-K reverse transcriptase.

[0460] In certain embodiments, the cancer has (i) expression of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; and/or (ii) activity of LINE1 reverse transcriptase. In certain embodiments, the cancer has expression of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide. In certain embodiments, the cancer has expression of LINE1 RNA. In certain embodiments, the cancer has expression of LINE1 ORF1 polypeptide. In certain embodiments, the cancer has expression of LINE1 ORF2 polypeptide. In certain embodiments, the cancer has activity of LINE1 reverse transcriptase.

[0461] In certain embodiments, the cancer has (i) expression of HERV-K RNA, and/or (ii) activity of HERV-K reverse transcriptase. In certain embodiments, the cancer has expression of HERV-K RNA. In certain embodiments, the cancer has activity of HERV-K reverse transcriptase.

[0462] In certain embodiments, the cancer has elevated (i) levels of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; (ii) activity of LINE1 reverse transcriptase; (iii) levels of HERV-K RNA, and/or (iv) activity of HERV-K reverse transcriptase.

[0463] In certain embodiments, the cancer has elevated (i) levels of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; and/or (ii) activity of LINE1 reverse transcriptase. In certain embodiments, the cancer has elevated levels of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide. In certain embodiments, the cancer has elevated levels of LINE1 RNA. In certain embodiments, the cancer has elevated levels of LINE1 ORF1 polypeptide. In certain embodiments, the cancer has elevated levels of LINE1 ORF2 polypeptide. In certain embodiments, the cancer has elevated activity of LINE1 reverse transcriptase.

[0464] In certain embodiments, the cancer has elevated (i) levels of HERV-K RNA, and/or (ii) activity of HERV-K reverse transcriptase. In certain embodiments, the cancer has elevated levels of HERV-K RNA. In certain embodiments, the cancer has elevated activity of HERV-K reverse transcriptase.

[0465] In certain embodiments, the cancer is pancreatic cancer, colorectal cancer, breast cancer, prostate cancer, esophageal cancer, head and neck cancer, renal cancer, ovarian cancer, or lung cancer. In certain embodiments, the cancer is pancreatic cancer, colorectal cancer, breast cancer, prostate cancer, renal cancer, ovarian cancer, or lung cancer. In certain embodiments, the cancer is pancreatic cancer. In certain embodiments, the cancer is pancreatic adenocarcinoma. In certain embodiments, the cancer is colorectal cancer. In certain embodiments, the cancer comprises microsatellite instable (MSI) colorectal cancer or microsatellite stable (MSS) colorectal cancer. In certain embodiments, the cancer is breast cancer. In certain embodiments, the cancer is prostate cancer. In certain embodiments, the cancer is esophageal cancer. In certain embodiments, the cancer is head and neck cancer. In certain embodiments, the cancer is renal cancer. In certain embodiments, the cancer is ovarian cancer. In certain embodiments, the cancer is lung cancer. In certain embodiments, the cancer is non-small cell lung carcinoma or small cell lung carcinoma. In certain embodiments, the cancer is non-small cell lung carcinoma. In certain embodiments, the cancer is small cell lung carcinoma.

[0466] In certain embodiments, the cancer is an epithelial cancer. In certain embodiments, the epithelial cancer is pancreatic cancer, colorectal cancer, breast cancer, prostate cancer, esophageal cancer, head and neck cancer, renal cancer, ovarian cancer, or lung cancer. In certain embodiments, the epithelial cancer is pancreatic cancer, colorectal cancer, breast cancer, prostate cancer, renal cancer, ovarian cancer, or lung cancer. In certain embodiments, the colorectal cancer comprises microsatellite instable (MSI) colorectal cancer or microsatellite stable (MSS) colorectal cancer.

[0467] In certain embodiments, the cancer is a preneoplastic or early cancer lesion. In certain embodiments, the cancer is intraductal papillary mucinous neoplasm (IPMN), pancreatic intraepithelial neoplasia (PanIN), ductal carcinoma in situ (DCIS), or Barrett's Esophagus. In certain embodiments, the cancer intraductal papillary mucinous neoplasm (IPMN). In certain embodiments, the cancer is pancreatic intraepithelial neoplasia (PanIN). In certain embodiments, the cancer is ductal carcinoma in situ (DCIS). In certain embodiments, the cancer is Barrett's Esophagus.

[0468] In certain embodiments, the cancer has elevated levels of pericentrometric human satellite II (HSATII) RNA. In some embodiments, the cancer is a microsatellite instable (MSI) cancer. In some embodiments, the cancer is a microsatellite stable (MSS) cancer.

[0469] In certain embodiments, the cancer is selected from B cell lymphomas (e.g., B cell chronic lymphocytic leukemia, B cell non-Hodgkin lymphoma, cutaneous B cell lymphoma, diffuse large B cell lymphoma), basal cell carcinoma, bladder cancer, blastoma, brain metastasis, breast cancer, Burkitt lymphoma, carcinoma (e.g., adenocarcinoma (e.g., of the gastroesophageal junction)), cervical cancer, colon cancer, colorectal cancer (colon cancer and rectal cancer), endometrial carcinoma, esophageal cancer, Ewing sarcoma, follicular lymphoma, gastric cancer, gastroesophageal junction carcinoma, gastrointestinal cancer, glioblastoma (e.g., glioblastoma multiforme, e.g., newly diagnosed or recurrent), glioma, head and neck cancer (e.g., head and neck squamous cell carcinoma), hepatic metastasis, Hodgkin's and non-Hodgkin's lymphoma, kidney cancer (e.g., renal cell carcinoma and Wilms' tumors), laryngeal cancer, leukemia (e.g., chronic myelocytic leukemia, hairy cell leukemia), liver cancer (e.g., hepatic carcinoma and hepatoma), lung cancer (e.g., non-small cell lung cancer and small-cell lung cancer), lymphoblastic lymphoma, lymphoma, mantle cell lymphoma, metastatic brain tumor, metastatic cancer, myeloma (e.g., multiple myeloma), neuroblastoma, ocular melanoma, oropharyngeal cancer, osteosarcoma, ovarian cancer, pancreatic cancer (e.g., pancreatis ductal adenocarcinoma), prostate cancer (e.g., hormone refractory (e.g., castration resistant), metastatic, metastatic hormone refractory (e.g., castration resistant, androgen independent)), renal cell carcinoma (e.g., metastatic), salivary gland carcinoma, sarcoma (e.g., rhabdomyosarcoma), skin cancer (e.g., melanoma (e.g., metastatic melanoma)), soft tissue sarcoma, solid tumor, squamous cell carcinoma, synovia sarcoma, testicular cancer, thyroid cancer, transitional cell cancer (urothelial cell cancer), uveal melanoma (e.g., metastatic), verrucous carcinoma, vulval cancer, and Waldenstrom macroglobulinemia.

[0470] In some embodiments, the cancer is a virus-associated cancer. As used herein, the term "virus-associated cancer" means any cancer in which a virus is known to play a role. For example, Epstein-Barr virus (EBV) has been reported to be associated with the endemic variant of Burkitt lymphoma and certain other lymphomas. Infection by human papilloma virus (HPV) is believed to be responsible for certain types of cervical and/or genital cancer. Human T-cell leukemia virus 1 (HTLV-1) has been reported to be linked adult T-cell leukemia/lymphoma (ATLL). Human T-cell leukemia virus 2 (HTLV-2) has been reported to be linked to cutaneous T-cell lymphoma. Human herpes virus 8 (HHV-8) is believed to cause Kaposi's sarcoma in patients with AIDS. In certain embodiments, the cancer is a cancer associated with EBV, HPV, HTLV-1, HTLV-2, or HHV-8. In certain embodiments, the cancer is Burkitt lymphoma, cervical cancer, genital cancer, adult T-cell leukemia/lymphoma, cutaneous T-cell lymphoma, or Kaposi's sarcoma.

[0471] In some embodiments, the cancer is a cancer other than a virus-associated cancer. In certain embodiments, the cancer is a cancer other than a cancer associated with EBV, HPV, HTLV-1, HTLV-2, or HHV-8. In certain embodiments, the cancer is a cancer other than Burkitt lymphoma, cervical cancer, genital cancer, adult T-cell leukemia/lymphoma, cutaneous T-cell lymphoma, or Kaposi's sarcoma.

[0472] In some embodiments, the cancer is mesothelioma, hepatobilliary (hepatic and billiary duct), bone cancer, pancreatic cancer, skin cancer, cancer of the head or neck, cutaneous or intraocular melanoma, ovarian cancer, colon cancer, rectal cancer, cancer of the anal region, stomach cancer, gastrointestinal (gastric, colorectal, and duodenal), uterine cancer, carcinoma of the fallopian tubes, carcinoma of the endometrium, carcinoma of the cervix, carcinoma of the vagina, carcinoma of the vulva, Hodgkin's Disease, cancer of the esophagus, cancer of the small intestine, cancer of the endocrine system, cancer of the thyroid gland, cancer of the parathyroid gland, cancer of the adrenal gland, sarcoma of soft tissue, cancer of the urethra, cancer of the penis, prostate cancer, testicular cancer, chronic or acute leukemia, chronic myeloid leukemia, lymphocytic lymphomas, cancer of the bladder, cancer of the kidney or ureter, renal cell carcinoma, carcinoma of the renal pelvis, non-Hodgkins's lymphoma, spinal axis tumors, brain stem glioma, pituitary adenoma, adrenocortical cancer, gall bladder cancer, multiple myeloma, cholangiocarcinoma, fibrosarcoma, neuroblastoma, retinoblastoma, or a combination of one or more of the foregoing cancers.

[0473] In some embodiments, the cancer is hepatocellular carcinoma, ovarian cancer, ovarian epithelial cancer, fallopian tube cancer, papillary serous cystadenocarcinoma, uterine papillary serous carcinoma (UPSC), prostate cancer, testicular cancer, gallbladder cancer, hepatocholangiocarcinoma, soft tissue and bone synovial sarcoma, rhabdomyosarcoma, osteosarcoma, chondrosarcoma, Ewing sarcoma, anaplastic thyroid cancer, adrenocortical adenoma, pancreatic cancer, pancreatic ductal carcinoma, pancreatic adenocarcinoma, gastrointestinal/stomach (GIST) cancer, lymphoma, squamous cell carcinoma of the head and neck (SCCHN), salivary gland cancer, glioma, or brain cancer, neurofibromatosis-1 associated malignant peripheral nerve sheath tumors (MPNST), Waldenstrom's macroglobulinemia, or medulloblastoma.

[0474] In some embodiments, the cancer is hepatocellular carcinoma (HCC), hepatoblastoma, colon cancer, rectal cancer, ovarian cancer, ovarian epithelial cancer, fallopian tube cancer, papillary serous cystadenocarcinoma, uterine papillary serous carcinoma (UPSC), hepatocholangiocarcinoma, soft tissue and bone synovial sarcoma, rhabdomyosarcoma, osteosarcoma, anaplastic thyroid cancer, adrenocortical adenoma, pancreatic cancer, pancreatic ductal carcinoma, pancreatic adenocarcinoma, glioma, neurofibromatosis-1 associated malignant peripheral nerve sheath tumors (MPNST), Waldenstrom's macroglobulinemia, or medulloblastoma.

[0475] In some embodiments, the cancer is selected from renal cell carcinoma, or kidney cancer; hepatocellular carcinoma (HCC) or hepatoblastoma, or liver cancer; melanoma; breast cancer; colorectal carcinoma, or colorectal cancer; colon cancer; rectal cancer; anal cancer; lung cancer, such as non-small cell lung cancer (NSCLC) or small cell lung cancer (SCLC); ovarian cancer, ovarian epithelial cancer, ovarian carcinoma, or fallopian tube cancer; papillary serous cystadenocarcinoma or uterine papillary serous carcinoma (UPSC); prostate cancer; testicular cancer; gallbladder cancer; hepatocholangiocarcinoma; soft tissue and bone synovial sarcoma; rhabdomyosarcoma; osteosarcoma; chondrosarcoma; Ewing sarcoma; anaplastic thyroid cancer; adrenocortical carcinoma; pancreatic cancer; pancreatic ductal carcinoma or pancreatic adenocarcinoma; gastrointestinal/stomach (GIST) cancer; lymphoma; squamous cell carcinoma of the head and neck (SCCHN); salivary gland cancer; glioma, or brain cancer; neurofibromatosis-1 associated malignant peripheral nerve sheath tumors (MPNST); Waldenstrom's macroglobulinemia; and medulloblastoma.

[0476] In some embodiments, the cancer is renal cell carcinoma, hepatocellular carcinoma (HCC), hepatoblastoma, colorectal carcinoma, colorectal cancer, colon cancer, rectal cancer, anal cancer, ovarian cancer, ovarian epithelial cancer, ovarian carcinoma, fallopian tube cancer, papillary serous cystadenocarcinoma, uterine papillary serous carcinoma (UPSC), hepatocholangiocarcinoma, soft tissue and bone synovial sarcoma, rhabdomyosarcoma, osteosarcoma, chondrosarcoma, anaplastic thyroid cancer, adrenocortical carcinoma, pancreatic cancer, pancreatic ductal carcinoma, pancreatic adenocarcinoma, glioma, brain cancer, neurofibromatosis-1 associated malignant peripheral nerve sheath tumors (MPNST), Waldenstrom's macroglobulinemia, or medulloblastoma.

[0477] In some embodiments, the cancer is hepatocellular carcinoma (HCC), hepatoblastoma, colon cancer, rectal cancer, ovarian cancer, ovarian epithelial cancer, ovarian carcinoma, fallopian tube cancer, papillary serous cystadenocarcinoma, uterine papillary serous carcinoma (UPSC), hepatocholangiocarcinoma, soft tissue and bone synovial sarcoma, rhabdomyosarcoma, osteosarcoma, anaplastic thyroid cancer, adrenocortical carcinoma, pancreatic cancer, pancreatic ductal carcinoma, pancreatic adenocarcinoma, glioma, neurofibromatosis-1 associated malignant peripheral nerve sheath tumors (MPNST), Waldenstrom's macroglobulinemia, or medulloblastoma.

[0478] In some embodiments, the cancer is hepatocellular carcinoma (HCC). In some embodiments, the cancer is hepatoblastoma. In some embodiments, the cancer is colon cancer. In some embodiments, the cancer is rectal cancer. In some embodiments, the cancer is ovarian cancer, or ovarian carcinoma. In some embodiments, the cancer is ovarian epithelial cancer. In some embodiments, the cancer is fallopian tube cancer. In some embodiments, the cancer is papillary serous cystadenocarcinoma. In some embodiments, the cancer is uterine papillary serous carcinoma (UPSC). In some embodiments, the cancer is hepatocholangiocarcinoma. In some embodiments, the cancer is soft tissue and bone synovial sarcoma. In some embodiments, the cancer is rhabdomyosarcoma. In some embodiments, the cancer is osteosarcoma. In some embodiments, the cancer is anaplastic thyroid cancer. In some embodiments, the cancer is adrenocortical carcinoma. In some embodiments, the cancer is pancreatic cancer, or pancreatic ductal carcinoma. In some embodiments, the cancer is pancreatic adenocarcinoma. In some embodiments, the cancer is glioma. In some embodiments, the cancer is malignant peripheral nerve sheath tumors (MPNST). In some embodiments, the cancer is neurofibromatosis-1 associated MPNST. In some embodiments, the cancer is Waldenstrom's macroglobulinemia. In some embodiments, the cancer is medulloblastoma.

[0479] In certain embodiments, the cancer is a leukemia (e.g., acute leukemia, acute lymphocytic leukemia, acute myelocytic leukemia, acute myeloblastic leukemia, acute promyelocytic leukemia, acute myelomonocytic leukemia, acute monocytic leukemia, acute erythroleukemia, chronic leukemia, chronic myelocytic leukemia, chronic lymphocytic leukemia), polycythemia vera, lymphoma (e.g., Hodgkin's disease or non-Hodgkin's disease), Waldenstrom's macroglobulinemia, multiple myeloma, heavy chain disease, or a solid tumor such as a sarcoma or carcinoma (e.g., fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, osteogenic sarcoma, chordoma, angiosarcoma, endotheliosarcoma, lymphangiosarcoma, lymphangioendotheliosarcoma, synovioma, mesothelioma, Ewing's tumor, leiomyosarcoma, rhabdomyosarcoma, colon carcinoma, pancreatic cancer, breast cancer, ovarian cancer, prostate cancer, squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, sweat gland carcinoma, sebaceous gland carcinoma, papillary carcinoma, papillary adenocarcinomas, cystadenocarcinoma, medullary carcinoma, bronchogenic carcinoma, renal cell carcinoma, hepatoma, bile duct carcinoma, choriocarcinoma, seminoma, embryonal carcinoma, Wilm's tumor, cervical cancer, uterine cancer, testicular cancer, lung carcinoma, small cell lung carcinoma, bladder carcinoma, epithelial carcinoma, glioma, astrocytoma, glioblastoma multiforme (GBM, also known as glioblastoma), medulloblastoma, craniopharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuroma, oligodendroglioma, schwannoma, neurofibrosarcoma, meningioma, melanoma, neuroblastoma, and retinoblastoma).

[0480] In some embodiments, the cancer is glioma, astrocytoma, glioblastoma multiforme (GBM, also known as glioblastoma), medulloblastoma, craniopharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuroma, oligodendroglioma, schwannoma, neurofibrosarcoma, meningioma, melanoma, neuroblastoma, or retinoblastoma.

[0481] In some embodiments, the cancer is acoustic neuroma, astrocytoma (e.g. Grade I—Pilocytic Astrocytoma, Grade II—Low-grade Astrocytoma, Grade III—Anaplastic Astrocytoma, or Grade IV—Glioblastoma (GBM)), chordoma, CNS lymphoma, craniopharyngioma, brain stem glioma, ependymoma, mixed glioma, optic nerve glioma, subependymoma, medulloblastoma, meningioma, metastatic brain tumor, oligodendroglioma, pituitary tumors, primitive neuroectodermal (PNET) tumor, or schwannoma. In some embodiments, the cancer is a type found more commonly in children than adults, such as brain stem glioma, craniopharyngioma, ependymoma, juvenile pilocytic astrocytoma (JPA), medulloblastoma, optic nerve glioma, pineal tumor, primitive neuroectodermal tumors (PNET), or rhabdoid tumor.

Inflammatory Disorders

[0482] In certain embodiments, the disorder is an inflammatory disorder. In certain embodiments, the inflammatory disorder is rheumatoid arthritis, osteoarthritis, ankylosing spondylitis, inflammatory bowel disease, Crohn's disease, ulcerative colitis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cholestatic liver disease, or sclerosing cholangitis, psoriasis, dermatitis, vasculitis, scleroderma, asthma, bronchitis, chronic obstructive pulmonary disease (COPD), pulmonary fibrosis, pulmonary hypertension, sarcoidosis, myocarditis, pericarditis, gout, myositis, Sjogren's syndrome, or systemic lupus erythematosus.

[0483] In certain embodiments, the inflammatory disorder is rheumatoid arthritis, osteoarthritis, or ankylosing spondylitis. In certain embodiments, the inflammatory disorder is inflammatory bowel disease, Crohn's disease, or ulcerative colitis. In certain embodiments, the inflammatory disorder is nonalcoholic steatohepatitis (NASH), non-alcoholic fatty liver disease (NAFLD), cholestatic liver disease, or sclerosing cholangitis. In certain embodiments, the inflammatory disorder is psoriasis, dermatitis, vasculitis, or scleroderma. In certain embodiments, the inflammatory disorder is asthma, bronchitis, chronic obstructive pulmonary disease (COPD), pulmonary fibrosis, pulmonary hypertension, sarcoidosis, myocarditis, or pericarditis. In certain embodiments, the inflammatory disorder is gout, myositis, Sjogren's syndrome, or systemic lupus erythematosus.

Immune Disorders

[0484] In certain embodiments, the disorder is an immune disorder other than a viral infection. In certain embodiments, the immune disorder is arthritis, psoriasis, systemic lupus erythematosus (SLE), graft versus host disease, scleroderma, polymyositis, inflammatory bowel disease, dermatomyositis, ulcerative colitis, Crohn's disease, vasculitis, psoriatic arthritis, Reiter's syndrome, exfoliative psoriatic dermatitis, pemphigus vulgaris, Sjogren's syndrome, autoimmune uveitis, glomerulonephritis, post myocardial infarction cardiotomy syndrome, pulmonary hemosiderosis, amyloidosis, sarcoidosis, aphthous stomatitis, thyroiditis, gastritis, adrenalitis (Addison's disease), ovaritis, primary biliary cirrhosis, myasthenia gravis, gonadal failure, hypoparathyroidism, alopecia, psoriasis, malabsorption syndrome, pernicious anemia, hepatitis, hypopituitarism, diabetes insipidus, or sicca syndrome.

[0485] In certain embodiments, the immune disorder is a type 1 interferonopathy, type 1 diabetes, Aicardi-Goutieres syndrome (AGS), arthritis, psoriasis, systemic lupus erythematosus (SLE), lupus nephritis, cutaneous lupus erythematosus (CLE), familial chilblain lupus, systemic sclerosis, STING-associated vasculopathy with onset in infancy (SAVI), graft versus host disease, scleroderma, polymyositis, inflammatory bowel disease, dermatomyositis, ulcerative colitis, Crohn's disease, vasculitis, psoriatic arthritis, Reiter's syndrome, exfoliative psoriatic dermatitis, pemphigus vulgaris, Sjogren's syndrome, autoimmune uveitis, glomerulonephritis, post myocardial infarction cardiotomy syndrome, pulmonary hemosiderosis, amyloidosis, sarcoidosis, aphthous stomatitis, thyroiditis, gastritis, adrenalitis (Addison's disease), ovaritis, primary biliary cirrhosis, myasthenia gravis, gonadal failure, hypoparathyroidism, alopecia, malabsorption syndrome, pernicious anemia, hepatitis, hypopituitarism, diabetes insipidus, or sicca syndrome.

[0486] In certain embodiments, the immune disorder is a type 1 interferonopathy, type 1 diabetes, Aicardi-Goutieres syndrome (AGS), arthritis, psoriasis, systemic lupus erythematosus (SLE), lupus nephritis, cutaneous lupus erythematosus (CLE), familial chilblain lupus, systemic sclerosis, STING-associated vasculopathy with onset in infancy (SAVI), graft versus host disease, scleroderma, polymyositis, inflammatory bowel disease, dermatomyositis, ulcerative colitis, Crohn's disease, vasculitis, psoriatic arthritis, Reiter's syndrome, exfoliative psoriatic dermatitis, pemphigus vulgaris, Sjogren's syndrome, autoimmune uveitis, glomerulonephritis, post myocardial infarction cardiotomy syndrome, pulmonary hemosiderosis, amyloidosis, sarcoidosis, aphthous stomatitis, thyroiditis, gastritis, adrenalitis (Addison's disease), ovaritis, primary biliary cirrhosis, myasthenia gravis, gonadal failure, hypoparathyroidism, alopecia, malabsorption syndrome, pernicious anemia, hypopituitarism, diabetes insipidus, or sicca syndrome.

[0487] In certain embodiments, the immune disorder is a type 1 interferonopathy, type 1 diabetes, Aicardi-Goutieres syndrome (AGS), systemic lupus erythematosus (SLE), lupus nephritis, cutaneous lupus erythematosus (CLE), familial chilblain lupus, systemic sclerosis, STING-associated vasculopathy with onset in infancy (SAVI), Sjogren's syndrome, dermatomyositis, inflammatory bowel disease, Crohn's disease, or ulcerative colitis.

[0488] In certain embodiments, the immune disorder is a type 1 interferonopathy. In certain embodiments, the immune disorder is type 1 diabetes, Aicardi-Goutieres syndrome (AGS), systemic lupus erythematosus (SLE), lupus nephritis, cutaneous lupus erythematosus (CLE), familial chilblain lupus, systemic sclerosis, STING-associated vasculopathy with onset in infancy (SAVI), Sjogren's syndrome, or dermatomyositis. In certain embodiments, the immune disorder is systemic lupus erythematosus (SLE), lupus nephritis, cutaneous lupus erythematosus (CLE), or familial chilblain lupus. In certain embodiments, the immune disorder is systemic lupus erythematosus (SLE), lupus nephritis, or cutaneous lupus erythematosus (CLE). In certain embodiments, the immune disorder is type 1 diabetes, Aicardi-Goutieres syndrome (AGS), systemic sclerosis, STING-associated vasculopathy with onset in infancy (SAVI), Sjogren's syndrome, or dermatomyositis. In certain embodiments, the immune disorder is Aicardi-Goutieres syndrome (AGS), familial chilblain lupus, or STING-associated vasculopathy with onset in infancy (SAVI).

[0489] In certain embodiments, the immune disorder is type 1 diabetes. In certain embodiments, the immune disorder is Aicardi-Goutieres syndrome (AGS). In certain embodiments, the immune disorder is systemic lupus erythematosus (SLE). In certain embodiments, the immune disorder is lupus nephritis. In certain embodiments, the immune disorder is cutaneous lupus erythematosus (CLE). In certain embodiments, the immune disorder is familial chilblain lupus. In certain embodiments, the immune disorder is systemic sclerosis. In certain embodiments, the immune disorder is STING-associated vasculopathy with onset in infancy (SAVI). In certain embodiments, the immune disorder is Sjogren's syndrome. In certain embodiments, the immune disorder is dermatomyositis.

[0490] In certain embodiments, the immune disorder is inflammatory bowel disease, Crohn's disease, or ulcerative colitis. In certain embodiments, the immune disorder is inflammatory bowel disease. In certain embodiments, the immune disorder is Crohn's disease. In certain embodiments, the immune disorder is ulcerative colitis.

Neurodegenerative Disorders

[0491] In certain embodiments, the disorder is a neurodegenerative disorder. In certain embodiments, the neurodegenerative disorder is amyotrophic lateral sclerosis (ALS), multiple sclerosis, Parkinson's disease, Huntington's disease, peripheral neuropathy, Creutzfeldt-Jacob disease, stroke, prion disease, frontotemporal dementia, Pick's disease, progressive supranuclear palsy, spinocerebellar ataxias, Lewy body disease, dementia, multiple system atrophy, epilepsy, bipolar disorder, schizophrenia, an anxiety disorder, or major depression. In certain embodiments, the neurodegenerative disorder is neurodegenerative disorder is amyotrophic lateral sclerosis (ALS), multiple sclerosis, Parkinson's disease, Huntington's disease, or dementia.

[0492] In certain embodiments, the neurodegenerative disorder is Alzheimer's disease, amyotrophic lateral sclerosis (ALS), multiple sclerosis, Parkinson's disease, Huntington's disease, peripheral neuropathy, age-related macular degeneration, Creutzfeldt-Jacob disease, stroke, prion disease, frontotemporal dementia, Pick's disease, progressive supranuclear palsy, spinocerebellar ataxias, Lewy body disease, dementia, multiple system atrophy, epilepsy, bipolar disorder, schizophrenia, an anxiety disorder, or major depression.

[0493] In certain embodiments, the neurodegenerative disorder is Alzheimer's disease, amyotrophic lateral sclerosis (ALS), multiple sclerosis, Parkinson's disease, Huntington's disease, dementia, or age-related macular degeneration. In certain embodiments, the neurodegenerative disorder is Alzheimer's disease, amyotrophic lateral sclerosis (ALS), Parkinson's disease, or age-related macular degeneration. In certain embodiments, the neurodegenerative disorder is age-related macular degeneration.

[0494] In certain embodiments, the neurodegenerative disorder is Alzheimer's disease, amyotrophic lateral sclerosis (ALS), multiple sclerosis, Parkinson's disease, Huntington's disease, or dementia. In certain embodiments, the neurodegenerative disorder is Alzheimer's disease, amyotrophic lateral sclerosis (ALS), or Parkinson's disease. In certain embodiments, the neurodegenerative disorder is Alzheimer's disease. In certain embodiments, the neurodegenerative disorder is amyotrophic lateral sclerosis (ALS). In certain embodiments, the neurodegenerative disorder is multiple sclerosis. In certain embodiments, the neurodegenerative disorder is Parkinson's disease. In certain embodiments, the neurodegenerative disorder is Huntington's disease. In certain embodiments, the neurodegenerative disorder is dementia.

Subjects

[0495] In certain embodiments, the subject has (i) expression of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; (ii) activity of LINE1 reverse transcriptase; (iii) expression of HERV-K RNA, and/or (iv) activity of HERV-K reverse transcriptase.

[0496] In certain embodiments, the subject has (i) expression of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; and/or (ii) activity of LINE1 reverse transcriptase. In certain embodiments, the subject has expression of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide. In certain embodiments, the subject has expression of LINE1 RNA. In certain embodiments, the subject has expression of LINE1 ORF1 polypeptide. In certain embodiments, the subject has expression of LINE1 ORF2 polypeptide. In certain embodiments, the subject has activity of LINE1 reverse transcriptase.

[0497] In certain embodiments, the subject has (i) expression of HERV-K RNA, and/or (ii) activity of HERV-K reverse transcriptase. In certain embodiments, the subject has expression of HERV-K RNA. In certain embodiments, the subject has activity of HERV-K reverse transcriptase.

[0498] In certain embodiments, the subject has elevated (i) levels of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; (ii) activity of LINE1 reverse transcriptase; (iii) levels of HERV-K RNA, and/or (iv) activity of HERV-K reverse transcriptase.

[0499] In certain embodiments, the subject has elevated (i) levels of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; and/or (ii) activity of LINE1 reverse transcriptase. In certain embodiments, the subject has elevated levels of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide. In certain embodiments, the subject has elevated levels of LINE1 RNA. In certain embodiments, the subject has elevated levels of LINE1 ORF1 polypeptide. In certain embodiments, the subject has elevated levels of LINE1 ORF2 polypeptide. In certain embodiments, the subject has elevated activity of LINE1 reverse transcriptase.

[0500] In certain embodiments, the subject has elevated (i) levels of HERV-K RNA, and/or (ii) activity of HERV-K reverse transcriptase. In certain embodiments, the subject has elevated levels of HERV-K RNA. In certain embodiments, the subject has elevated activity of HERV-K reverse transcriptase.

[0501] In certain embodiments, the subject is a human. In certain embodiments, the subject is an adult human. In certain embodiments, the subject is a pediatric human. In certain embodiments, the subject is a companion animal. In certain embodiments, the subject is a canine, feline, or equine.

Uses of Compounds

[0502] Another aspect of the invention provides for the use of a compound described herein (such as a compound of Formula I, I-1, I-A, or I-B, or other compounds in Section I, or a compound of Formula II) for treating a medical disorder, such as a medical disorder described herein (for example, cancer).

[0503] Another aspect of the invention provides for the use of a compound described herein (such as a compound of Formula I, I-1, I-A, or I-B, or other compounds in Section I, or a compound of Formula II) in the manufacture of a medicament. In certain embodiments, the medicament is for treating a disorder described herein, such as cancer.

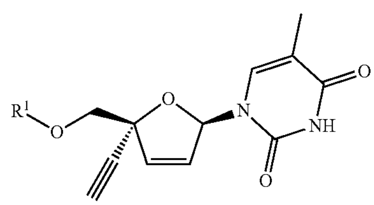
III. Methods of Inhibiting LINE1 and/or HERV-K Reverse Transcriptase Activity

[0504] It is contemplated that the substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines and related compounds described herein, such as a compound of Formula I, I-1, I-A, or I-B, or other compounds in Section I, above, can inhibit LINE1 reverse transcriptase activity. Accordingly, one aspect of the invention provides a method of inhibiting LINE1 reverse transcriptase activity. The method comprises contacting a LINE1 reverse transcriptase with an effective amount of a compound described in Section I above, such as a compound of Formula I, I-1, I-A, or I-B, in order to inhibit the activity of said LINE1 reverse transcriptase. In certain embodiments, the particular compound of Formula I, I-1, I-A, or I-B is a compound defined by one of the embodiments described in Section I, above. In certain embodiments, the method further comprises inhibiting HERV-K reverse transcriptase activity in the subject.

[0505] Another aspect of the invention provides a method of inhibiting HERV-K reverse transcriptase activity. The method comprises contacting a HERV-K reverse transcriptase with an effective amount of a compound described in Section I above, such as a compound of Formula I, I-1, I-A, or I-B, in order to inhibit the activity of said HERV-K reverse transcriptase. In certain embodiments, the particular compound of Formula I, I-1, I-A, or I-B is a compound defined by one of the embodiments described in Section I, above. In certain embodiments, the method further comprises inhibiting LINE1 reverse transcriptase activity in the subject.

[0506] Another aspect of the invention provides a method of inhibiting LINE1 reverse transcriptase activity in a subject suffering from a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises contacting a LINE1 reverse transcriptase with an effective amount of a compound of Formula II, in order to inhibit the activity of said LINE1 reverse transcriptase; wherein Formula II is represented by:

(II)



[0507] or a stereoisomer thereof; or a pharmaceutically acceptable salt of either of the foregoing; wherein:

[0508] R^1 is hydrogen, —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —P(O)(N(R^3)(R^4))$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^2;

[0509] R^2 represents independently for each occurrence hydrogen, —P(O)(OH)$_2$, —P(O)(OH)—O—P(O)(OH)$_2$, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

[0510] R^3 and R^9 each represent independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, or two instances of R^9, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0511] R^4 represents independently for each occurrence —C(R^5)$_2$—CO$_2$R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0512] R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

[0513] R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0514] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —N(R^9)$_2$; and

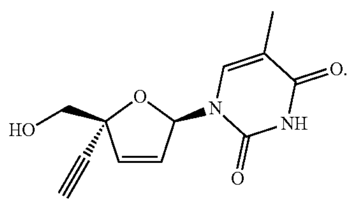
[0515] m and p are independently for each occurrence 0, 1, 2, or 3.

[0516] The definitions of variables in Formula II above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

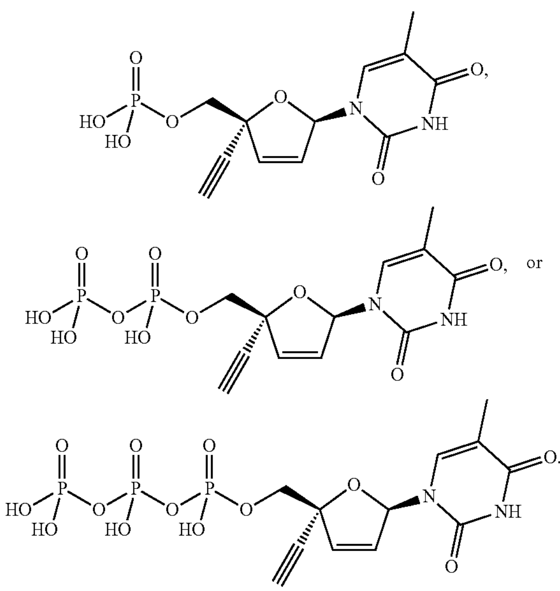
[0517] In certain embodiments, the compound is a compound of Formula II, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound of Formula II.

[0518] In certain embodiments, the particular compound of Formula II is a compound defined by one of the embodiments described in Section I, above, such as a compound of Formula I, I-1, I-A, or I-B. Additionally, embodiments described in Section I above for variables R^1, R^2, R^3, R^4, R^5, R^6, R^7, R^9, m, and p also apply to compounds of Formula II. For example, in certain embodiments, the compound is a compound of Formula II wherein R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N(R^3)$_2$.

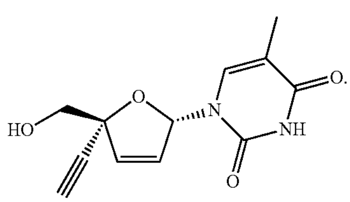
[0519] In certain embodiments, the compound of Formula II is



In certain embodiments, the compound of Formula II is



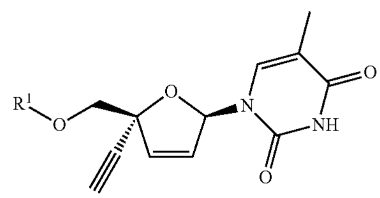
In certain embodiments, the compound of Formula II is



[0520] In certain embodiments, the method further comprises inhibiting HERV-K reverse transcriptase activity in the subject. In certain embodiments, the disorder is a disorder defined by one of the embodiments described in Section II, above, such as cancer.

[0521] Another aspect of the invention provides a method of inhibiting LINE1 reverse transcriptase activity in a subject suffering from a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises contacting a LINE1 reverse transcriptase with an effective amount of a compound of Formula II-1, in order to inhibit the activity of said LINE1 reverse transcriptase; wherein Formula II-1 is represented by:

(II-1)



[0522] or a pharmaceutically acceptable salt thereof; wherein:

[0523] R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, —C(O)R^2, or hydrogen;

[0524] R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, hydrogen, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring;

[0525] R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0526] R^4 is —C(H)(R^5)—CO$_2$R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0527] R^5 is C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0528] R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0529] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl; and

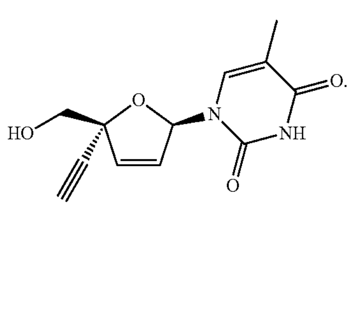
[0530] m is independently for each occurrence 0, 1, 2, or 3.

[0531] The definitions of variables in Formula 1-1 above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

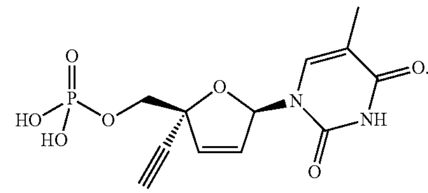
[0532] In certain embodiments, the compound is a compound of Formula II-1.

[0533] In certain embodiments, the particular compound of Formula II-1 is a compound defined by one of the embodiments described in Section I, above, such as a compound of Formula I-1 or I-A. Additionally, embodiments described in Section I above for variables R^1, R^2, R^3, R^4, R^5, R^6, R^7, and m of Formula I-1 also apply to compounds of Formula II-1. For example, in certain embodiments, the compound is a compound of Formula II-1 wherein R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N(R^3)$_2$.

[0534] In certain embodiments, the compound of Formula II-1 is



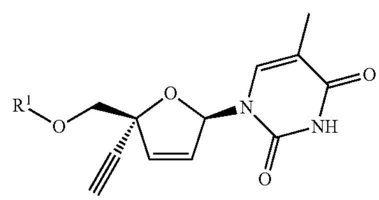
In certain embodiments, the compound of Formula II-1 is



[0535] In certain embodiments, the disorder is a disorder defined by one of the embodiments described in Section II, above, such as cancer. In certain embodiments, the method further comprises inhibiting HERV-K reverse transcriptase activity in the subject.

[0536] Another aspect of the invention provides a method of inhibiting HERV-K reverse transcriptase activity in a subject suffering from a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection. The method comprises contacting a HERV-K reverse transcriptase with an effective amount of a compound of Formula II, in order to inhibit the activity of said HERV-K reverse transcriptase; wherein Formula II is represented by:

(II)



[0537] or a stereoisomer thereof; or a pharmaceutically acceptable salt of either of the foregoing; wherein:

[0538] R^1 is hydrogen, —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —P(O)(N(R^3)(R^4))$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^2;

[0539] R^2 represents independently for each occurrence hydrogen, —P(O)(OH)$_2$, —P(O)(OH)—O—P(O)(OH)$_2$, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

[0540] R^3 and R^9 each represent independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, or two instances of R^9, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

[0541] R^4 represents independently for each occurrence —$C(R^5)_2$—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

[0542] R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

[0543] R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

[0544] R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —$N(R^9)_2$; and

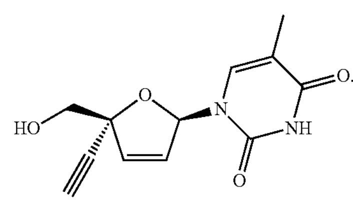
[0545] m and p are independently for each occurrence 0, 1, 2, or 3.

[0546] The definitions of variables in Formula II above encompass multiple chemical groups. The application contemplates embodiments where, for example, i) the definition of a variable is a single chemical group selected from those chemical groups set forth above, ii) the definition of a variable is a collection of two or more of the chemical groups selected from those set forth above, and iii) the compound is defined by a combination of variables in which the variables are defined by (i) or (ii).

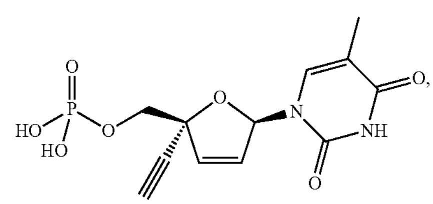
[0547] In certain embodiments, the compound is a compound of Formula II, or a pharmaceutically acceptable salt thereof. In certain embodiments, the compound is a compound of Formula II.

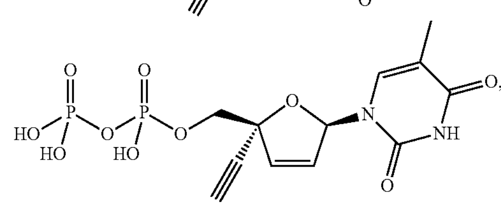
[0548] In certain embodiments, the particular compound of Formula II is a compound defined by one of the embodiments described in Section I, above, such as a compound of Formula I, I-1, I-A, or I-B. Additionally, embodiments described in Section I above for variables R^1, R^2, R^3, R^4, R^5, R^6, R^7, R^9, m, and p also apply to compounds of Formula II. For example, in certain embodiments, the compound is a compound of Formula II wherein R^1 is —$P(O)(OR^2)(N(R^3)(R^4))$ or —C(O)—C(H)(R^5)—$N(R^3)_2$.

[0549] In certain embodiments, the compound of Formula II is

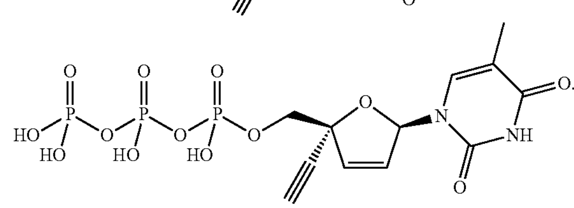


In certain embodiments, the compound of Formula II is

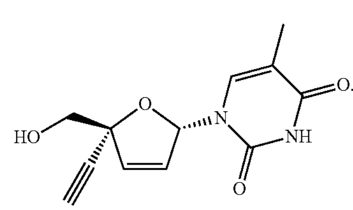




or



In certain embodiments, the compound of Formula II is



[0550] In certain embodiments, the method further comprises inhibiting LINE1 reverse transcriptase activity in the subject. In certain embodiments, the disorder is a disorder defined by one of the embodiments described in Section II, above, such as cancer.

[0551] Compounds may be tested for ability to inhibit LINE1 reverse transcriptase activity, for example, as described in the Examples. Compounds may be tested for ability to inhibit HERV-K reverse transcriptase activity, for example, as described in the Examples.

IV. Combination Therapy

[0552] Another aspect of the invention provides for combination therapy. Substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compounds described herein (e.g., a compound of Formula I, I-1, I-A, or I-B, or other compounds in Section I, or a compound of Formula II, or other compounds in Section II) or their pharmaceutically acceptable salts may be used in combination with additional therapeutic agents to treat medical disorders (e.g., according to the methods described in Section II, with disorders such as a cancer). Accordingly, in some embodiments, a method of the invention further comprises administering an effective amount of an additional therapeutic agent.

[0553] In some embodiments, the present invention provides a method of treating a disclosed disease or condition comprising administering to a patient in need thereof an effective amount of a compound disclosed herein or a pharmaceutically acceptable salt thereof and co-administering simultaneously or sequentially an effective amount of one or more additional therapeutic agents, such as those described herein. In some embodiments, the method includes co-administering one additional therapeutic agent. In some embodiments, the method includes co-administering two additional therapeutic agents. In some embodiments, the combination of the disclosed compound and the additional therapeutic agent or agents acts synergistically.

[0554] One or more other therapeutic agent may be administered separately from a compound or composition of the invention, as part of a multiple dosage regimen. Alternatively, one or more other therapeutic agents may be part of a single dosage form, mixed together with a compound of this invention in a single composition. If administered as a multiple dosage regime, one or more other therapeutic agent and a compound or composition of the invention may be administered simultaneously, sequentially or within a period of time from one another, for example within 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 18, 20, 21, 22, 23, or 24 hours from one another. In some embodiments, one or more other therapeutic agent and a compound or composition of the invention are administered as a multiple dosage regimen more than 24 hours apart.

[0555] The doses and dosage regimen of the active ingredients used in the combination therapy may be determined by an attending clinician. In certain embodiments, the substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein (e.g., a compound of Formula I, or other compounds in Section I, or a compound of Formula II, or other compounds in Section II) and the additional therapeutic agent(s) (e.g. the second, third, or fourth, or fifth anti-cancer agent, described below) are administered in doses commonly employed when such agents are used as monotherapy for treating the disorder. In other embodiments, the substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein (e.g., a compound of Formula I, or other compounds in Section I, or a compound of Formula II, or other compounds in Section II) and the additional therapeutic agent(s) (e.g. the second, third, or fourth, or fifth anti-cancer agent, described below) are administered in doses lower than the doses commonly employed when such agents are used as monotherapy for treating the disorder. In certain embodiments, the substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein (e.g., a compound of Formula I, or other compounds in Section I, or a compound of Formula II, or other compounds in Section II) and the additional therapeutic agent(s) (e.g. the second, third, or fourth, or fifth anti-cancer agent, described below) are present in the same composition, which is suitable for oral administration.

[0556] In certain embodiments, the substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein (e.g., a compound of Formula I, or other compounds in Section I, or a compound of Formula II, or other compounds in Section II) and the additional therapeutic agent(s) (e.g. the second, third, or fourth, or fifth anti-cancer agent, described below) may act additively or synergistically. A synergistic combination may allow the use of lower dosages of one or more agents and/or less frequent administration of one or more agents of a combination therapy. A lower dosage or less frequent administration of one or more agents may lower toxicity of the therapy without reducing the efficacy of the therapy.

[0557] Another aspect of this invention is a kit comprising a therapeutically effective amount of the substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein (e.g., a compound of Formula I, or other compounds in Section I, or a compound of Formula II, or other compounds in Section II), a pharmaceutically acceptable carrier, vehicle or diluent, and optionally at least one additional therapeutic agent listed above.

Cancer

[0558] Accordingly, another aspect of the invention provides a method of treating cancer in a patient. The method comprises administering to a subject in need thereof (i) a therapeutically effective amount of a substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein and (ii) a second anti-cancer agent, in order to treat the cancer.

[0559] In certain embodiments, the second anti-cancer agent is radiation therapy.

[0560] In certain embodiments, the second anti-cancer agent is a therapeutic antibody. In certain embodiments, the therapeutic antibody targets one of the following: CD20, CD30, CD33, CD52, EpCAM, CEA, gpA33, a mucin, TAG-72, CAIX, PSMA, a folate-binding protein, a ganglioside, Le, VEGF, VEGFR, VEGFR2, integrin $\alpha V\beta 3$, integrin $\alpha 5\beta 1$, EGFR, ERBB2, ERBB3, MET, IGF1R, EPHA3, TRAILR1, TRAILR2, RANKL, FAP, tenascin, CD19, KIR, NKG2A, CD47, CEACAM1, c-MET, VISTA, CD73, CD38, BAFF, interleukin-1 beta, B4GALNT1, interleukin-6, and interleukin-6 receptor.

[0561] In certain embodiments, the second anti-cancer agent is a therapeutic antibody selected from the group consisting of rituximab, ibritumomab tiuxetan, tositumomab, obinutuzumab, ofatumumab, brentuximab vedotin, gemtuzumab ozogamicin, alemtuzumab, IGN101, adecatumumab, labetuzumab, huA33, pemtumomab, oregovomab, minetumomab, cG250, J591, Mov18, farletuzumab, 3F8, ch14.18, KW-2871, hu3S193, IgN311, bevacizumab, IM-2C6, pazopanib, sorafenib, axitinib, CDP791, lenvatinib, ramucirumab, etaracizumab, volociximab, cetuximab, panitumumab, nimotuzumab, 806, afatinib, erlotinib, gefitinib, osimertinib, vandetanib, trastuzumab, pertuzumab, MM-121, AMG 102, METMAB, SCH 900105, AVE1642, IMC-A12, MK-0646, R1507, CP 751871, KB004, IIIA-4, mapatumumab, HGS-ETR2, CS-1008, denosumab, sibrotuzumab, F19, 81C6, MEDI551, lirilumab, MEDI9447, daratumumab, belimumab, canakinumab, dinutuximab, siltuximab, and tocilizumab.

[0562] In certain embodiments, the second anti-cancer agent is a cytokine. In certain embodiments, the cytokine is IL-12, IL-15, GM-CSF, or G-CSF.

[0563] In certain embodiments, the second anti-cancer agent is sipuleucel-T, aldesleukin (a human recombinant interleukin-2 product having the chemical name des-alanyl-1, serine-125 human interleukin-2), dabrafenib (a kinase inhibitor having the chemical name N-{3-[5-(2-aminopyrimidin-4-yl)-2-tert-butyl-1,3-thiazol-4-yl]-2-fluorophenyl}-2,6-difluorobenzenesulfonamide), vemurafenib (a kinase inhibitor having the chemical name propane-1-sulfonic acid {3-[5-(4-chlorophenyl)-1H-pyrrolo[2,3-b]pyridine-3-carbonyl]-2,4-difluoro-phenyl}-amide), or 2-chloro-deoxyadenosine.

[0564] In certain embodiments, the second anti-cancer agent is a placental growth factor, an antibody-drug conjugate, an oncolytic virus, or an anti-cancer vaccine. In certain embodiments, the second anti-cancer agent is a placental growth factor. In certain embodiments, the second anti-cancer agent is a placental growth factor comprising ziv-aflibercept. In certain embodiments, the second anti-cancer agent is an antibody-drug conjugate. In certain embodiments, the second anti-cancer agent is an antibody-drug conjugate selected from the group consisting of brentoxumab vedotin and trastuzumab emtransine.

[0565] In certain embodiments, the second anti-cancer agent is an oncolytic virus. In certain embodiments, the second anti-cancer agent is the oncolytic virus talimogene laherparepvec. In certain embodiments, the second anti-cancer agent is an anti-cancer vaccine. In certain embodiments, the second anti-cancer agent is an anti-cancer vaccine selected from the group consisting of a GM-CSF tumor vaccine, a STING/GM-CSF tumor vaccine, and NY-ESO-1. In certain embodiments, the second anti-cancer agent is a cytokine selected from IL-12, IL-15, GM-CSF, and G-CSF.

[0566] In certain embodiments, the second anti-cancer agent is an immune checkpoint inhibitor (also referred to as immune checkpoint blockers). Immune checkpoint inhibitors are a class of therapeutic agents that have the effect of blocking immune checkpoints. See, for example, Pardoll in Nature Reviews Cancer (2012) vol. 12, pages 252-264. In certain embodiments, the immune checkpoint inhibitor is an agent that inhibits one or more of (i) cytotoxic T-lymphocyte-associated antigen 4 (CTLA4), (ii) programmed cell death protein 1 (PD1), (iii) PDL1, (iv) LAB3, (v) B7-H3, (vi) B7-H4, and (vii) TIM3. In certain embodiments, the immune checkpoint inhibitor is ipilumumab. In certain embodiments, the immune checkpoint inhibitor is pembrolizumab.

[0567] In certain embodiments, the second anti-cancer agent is a monoclonal antibody that targets a non-checkpoint target (e.g., herceptin). In certain embodiments, the second anti-cancer agent is a non-cytoxic agent (e.g., a tyrosine-kinase inhibitor).

[0568] In certain embodiments, the second anti-cancer agent is selected from mitomycin, ribomustin, vincristine, tretinoin, etoposide, cladribine, gemcitabine, mitobronitol, methotrexate, doxorubicin, carboquone, pentostatin, nitracrine, zinostatin, cetrorelix, letrozole, raltitrexed, daunorubicin, fadrozole, fotemustine, thymalfasin, sobuzoxane, nedaplatin, aminoglutethimide, amsacrine, proglumide, elliptinium acetate, ketanserin, doxifluridine, etretinate, isotretinoin, streptozocin, nimustine, vindesine, cytarabine, bicalutamide, vinorelbine, vesnarinone, flutamide, drogenil, butocin, carmofur, razoxane, sizofilan, carboplatin, mitolactol, tegafur, ifosfamide, prednimustine, picibanil, levamisole, teniposide, improsulfan, enocitabine, lisuride, oxymetholone, tamoxifen, progesterone, mepitiostane, epitiostanol, formestane, colony stimulating factor-1, colony stimulating factor-2, denileukin diftitox, interleukin-2, leutinizing hormone releasing factor, interferon-alpha, interferon-2 alpha, interferon-beta, interferon-gamma.

[0569] In certain embodiments, the second anti-cancer agent is an ALK Inhibitor, an ATR Inhibitor, an A2A Antagonist, a Base Excision Repair Inhibitor, a Bcr-Abl Tyrosine Kinase Inhibitor, a Bruton's Tyrosine Kinase Inhibitor, a CDC7 Inhibitor, a CHK1 Inhibitor, a Cyclin-Dependent Kinase Inhibitor, a DNA-PK Inhibitor, an Inhibitor of both DNA-PK and mTOR, a DNMT1 Inhibitor, a DNMT1 Inhibitor plus 2-chloro-deoxyadenosine, an HDAC Inhibitor, a Hedgehog Signaling Pathway Inhibitor, an IDO Inhibitor, a JAK Inhibitor, a mTOR Inhibitor, a MEK Inhibitor, a MELK Inhibitor, a MTH1 Inhibitor, a PARP Inhibitor, a Phosphoinositide 3-Kinase Inhibitor, an Inhibitor of both PARP1 and DHODH, a Proteasome Inhibitor, a Topoisomerase-II Inhibitor, a Tyrosine Kinase Inhibitor, a VEGFR Inhibitor, or a WEE1 Inhibitor.

[0570] In certain embodiments, the second anti-cancer agent is an ALK Inhibitor. In certain embodiments, the second anti-cancer agent is an ALK Inhibitor comprising ceritinib or crizotinib. In certain embodiments, the second anti-cancer agent is an ATR Inhibitor. In certain embodiments, the second anti-cancer agent is an ATR Inhibitor comprising AZD6738 or VX-970. In certain embodiments, the second anti-cancer agent is an A2A Antagonist. In certain embodiments, the second anti-cancer agent is a Base Excision Repair Inhibitor comprising methoxyamine. In certain embodiments, the second anti-cancer agent is a Base Excision Repair Inhibitor, such as methoxyamine. In certain embodiments, the second anti-cancer agent is a Bcr-Abl Tyrosine Kinase Inhibitor. In certain embodiments, the second anti-cancer agent is a Bcr-Abl Tyrosine Kinase Inhibitor comprising dasatinib or nilotinib. In certain embodiments, the second anti-cancer agent is a Bruton's Tyrosine Kinase Inhibitor. In certain embodiments, the second anti-cancer agent is a Bruton's Tyrosine Kinase Inhibitor comprising ibrutinib. In certain embodiments, the second anti-cancer agent is a CDC7 Inhibitor. In certain embodiments, the second anti-cancer agent is a CDC7 Inhibitor comprising RXDX-103 or AS-141.

[0571] In certain embodiments, the second anti-cancer agent is a CHK1 Inhibitor. In certain embodiments, the second anti-cancer agent is a CHK1 Inhibitor comprising MK-8776, ARRY-575, or SAR-020106. In certain embodiments, the second anti-cancer agent is a Cyclin-Dependent

Kinase Inhibitor. In certain embodiments, the second anti-cancer agent is a Cyclin-Dependent Kinase Inhibitor comprising palbociclib. In certain embodiments, the second anti-cancer agent is a DNA-PK Inhibitor. In certain embodiments, the second anti-cancer agent is a DNA-PK Inhibitor comprising MSC2490484A. In certain embodiments, the second anti-cancer agent is Inhibitor of both DNA-PK and mTOR. In certain embodiments, the second anti-cancer agent comprises CC-115.

[0572] In certain embodiments, the second anti-cancer agent is a DNMT1 Inhibitor. In certain embodiments, the second anti-cancer agent is a DNMT1 Inhibitor comprising decitabine, RX-3117, guadecitabine, NUC-8000, or azacytidine. In certain embodiments, the second anti-cancer agent comprises a DNMT1 Inhibitor and 2-chloro-deoxyadenosine. In certain embodiments, the second anti-cancer agent comprises ASTX-727.

[0573] In certain embodiments, the second anti-cancer agent is a HDAC Inhibitor. In certain embodiments, the second anti-cancer agent is a HDAC Inhibitor comprising OBP-801, CHR-3996, etinostate, resminostate, pracinostat, CG-200745, panobinostat, romidepsin, mocetinostat, belinostat, AR-42, ricolinostat, KA-3000, or ACY-241.

[0574] In certain embodiments, the second anti-cancer agent is a Hedgehog Signaling Pathway Inhibitor. In certain embodiments, the second anti-cancer agent is a Hedgehog Signaling Pathway Inhibitor comprising sonidegib or vismodegib. In certain embodiments, the second anti-cancer agent is an IDO Inhibitor. In certain embodiments, the second anti-cancer agent is an IDO Inhibitor comprising INCB024360. In certain embodiments, the second anti-cancer agent is a JAK Inhibitor. In certain embodiments, the second anti-cancer agent is a JAK Inhibitor comprising ruxolitinib or tofacitinib. In certain embodiments, the second anti-cancer agent is a mTOR Inhibitor. In certain embodiments, the second anti-cancer agent is a mTOR Inhibitor comprising everolimus or temsirolimus. In certain embodiments, the second anti-cancer agent is a MEK Inhibitor. In certain embodiments, the second anti-cancer agent is a MEK Inhibitor comprising cobimetinib or trametinib. In certain embodiments, the second anti-cancer agent is a MELK Inhibitor. In certain embodiments, the second anti-cancer agent is a MELK Inhibitor comprising ARN-7016, APTO-500, or OTS-167. In certain embodiments, the second anti-cancer agent is a MTH1 Inhibitor. In certain embodiments, the second anti-cancer agent is a MTH1 Inhibitor comprising (S)-crizotinib, TH287, or TH588.

[0575] In certain embodiments, the second anti-cancer agent is a PARP Inhibitor. In certain embodiments, the second anti-cancer agent is a PARP Inhibitor comprising MP-124, olaparib, BGB-290, talazoparib, veliparib, niraparib, E7449, rucaparb, or ABT-767. In certain embodiments, the second anti-cancer agent is a Phosphoinositide 3-Kinase Inhibitor. In certain embodiments, the second anti-cancer agent is a Phosphoinositide 3-Kinase Inhibitor comprising idelalisib. In certain embodiments, the second anti-cancer agent is an inhibitor of both PARP1 and DHODH (i.e., an agent that inhibits both poly ADP ribose polymerase 1 and dihydroorotate dehydrogenase).

[0576] In certain embodiments, the second anti-cancer agent is a Proteasome Inhibitor. In certain embodiments, the second anti-cancer agent is a Proteasome Inhibitor comprising bortezomib or carfilzomib. In certain embodiments, the second anti-cancer agent is a Topoisomerase-II Inhibitor. In certain embodiments, the second anti-cancer agent is a Topoisomerase-II Inhibitor comprising vosaroxin.

[0577] In certain embodiments, the second anti-cancer agent is a Tyrosine Kinase Inhibitor. In certain embodiments, the second anti-cancer agent is a Tyrosine Kinase Inhibitor comprising bosutinib, cabozantinib, imatinib or ponatinib. In certain embodiments, the second anti-cancer agent is a VEGFR Inhibitor. In certain embodiments, the second anti-cancer agent is a VEGFR Inhibitor comprising regorafenib. In certain embodiments, the second anti-cancer agent is a WEE1 Inhibitor. In certain embodiments, the second anti-cancer agent is a WEE1 Inhibitor comprising AZD1775.

[0578] In certain embodiments, the second anti-cancer agent is an agonist of OX40, CD137, CD40, GITR, CD27, HVEM, TNFRSF25, or ICOS. In certain embodiments, the second anti-cancer agent is an agonist of OX40, CD137, CD40, or GITR. In certain embodiments, the second anti-cancer agent is an agonist of CD27, HVEM, TNFRSF25, or ICOS.

[0579] In certain embodiments, the method further comprises administering to the subject a third anti-cancer agent. In certain embodiments, the method further comprises administering to the subject a fourth anti-cancer agent. In certain embodiments, the method further comprises administering to the subject a fifth anti-cancer agent.

[0580] In certain embodiments, the third anti-cancer agent is one of the second anti-cancer agents described above. In certain embodiments, the fourth anti-cancer agent is one of the second anti-cancer agents described above. In certain embodiments, the fifth anti-cancer agent is one of the second anti-cancer agents described above.

Inflammatory Disorders

[0581] Another aspect of the invention provides a method of treating an inflammatory disorder in a patient. The method comprises administering to a subject in need thereof (i) a therapeutically effective amount of a substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein and (ii) a second therapeutic agent, in order to treat the inflammatory disorder.

[0582] In certain embodiments, the second therapeutic agent is a small molecule or a recombinant biologic agents. In certain embodiments, the second therapeutic agent is selected from acetaminophen, non-steroidal anti-inflammatory drugs (NSAIDS) such as aspirin, ibuprofen, naproxen, etodolac (Lodine®) and celecoxib, colchicine (Colcrys®), corticosteroids such as prednisone, prednisolone, methylprednisolone, hydrocortisone, and the like, probenecid, allopurinol, febuxostat (Uloric®), sulfasalazine (Azulfidine®), antimalarials such as hydroxychloroquine (Plaquenil®) and chloroquine (Aralen®), methotrexate (Rheumatrex®), gold salts such as gold thioglucose (Solganal®), gold thiomalate (Myochrysine®) and auranofin (Ridaura®), D-penicillamine (Depen® or Cuprimine®), azathioprine (Imuran®), cyclophosphamide (Cytoxan®), chlorambucil (Leukeran®), cyclosporine (Sandimmune®, Neoral®), tacrolimus, sirolimus, mycophenolate, leflunomide (Arava®) and "anti-TNF" agents such as etanercept (Enbrel®), infliximab (Remicade®), golimumab (Simponi®), certolizumab pegol (Cimzia®) and adalimumab (Humira®), "anti-IL-1" agents such as anakinra (Kineret®) and rilonacept (Arcalyst®), anti-T cell antibodies such as Thymoglobulin, IV Immunoglobulins (IVIg), canakinumab (Ilaris®), anti-Jak inhibitors such as tofacitinib, antibodies such as rituximab (Rituxan®), "anti-T-cell" agents such as abatacept (Orencia®), "anti-IL-6" agents such as tocilizumab (Actemra®), diclofenac, cortisone, hyaluronic acid (Synvisc® or Hyalgan®), monoclonal antibodies such as tanezumab, anticoagulants such as heparin (Calcinparine® or Liquaemin®) and warfarin (Coumadin®), antidiarrheals such as diphenoxylate (Lomotil®) and loperamide (Imodium®), bile acid binding agents such as cholestyramine, alosetron (Lotronex®), lubiprostone (Amitiza®), laxatives such as Milk of Magnesia, polyethylene glycol (MiraLax®), Dulcolax®, Correctol® and Senokot®, anticholinergics or antispasmodics such as dicyclomine (Bentyl®), Singulair®, beta-2 agonists such as albuterol (Ventolin® HFA, Proventil® HFA), levalbuterol (Xopenex®), metaproterenol (Alupent®), pirbuterol acetate (Maxair®), terbutaline sulfate (Brethaire®), salmeterol xinafoate (Serevent®) and formoterol (Foradil®), anticholinergic agents such as ipratropium bromide (Atrovent®) and tiotropium (Spiriva®), inhaled corticosteroids such as beclomethasone dipropionate (Beclovent®, Qvar®, and Vanceril®), triamcinolone acetonide (Azmacort®), mometasone (Asthmanex®), budesonide (Pulmocort®), and flunisolide (Aerobid®), Afviar®, Symbicort®, Dulera®, cromolyn sodium (Intal®), methylxanthines such as theophylline (Theo-Dur®, Theolair®, Slo-bid®, Uniphyl®, Theo-24®) and aminophylline, IgE antibodies such as omalizumab (Xolair®), nucleoside reverse transcriptase inhibitors such as zidovudine (Retrovir®), abacavir (Ziagen®), abacavir/lamivudine (Epzicom®), abacavir/lamivudine/zidovudine (Trizivir®), didanosine (Videx®), emtricitabine (Emtriva®), lamivudine (Epivir®), lamivudine/zidovudine (Combivir®), stavudine (Zerit®), and zalcitabine (Hivid®), non-nucleoside reverse transcriptase inhibitors such as delavirdine (Rescriptor®), efavirenz (Sustiva®), nevirapine (Viramune®) and etravirine (Intelence®), nucleotide reverse transcriptase inhibitors such as tenofovir (Viread®), protease inhibitors such as amprenavir (Agenerase®), atazanavir (Reyataz®), darunavir (Prezista®), fosamprenavir (Lexiva®), indinavir (Crixivan®), lopinavir and ritonavir (Kaletra®), nelfinavir (Viracept®), ritonavir (Norvir®), saquinavir (Fortovase® or Invirase®), and tipranavir (Aptivus®), entry inhibitors such as enfuvirtide (Fuzeon®) and maraviroc (Selzentry®), integrase inhibitors such as raltegravir (Isentress®), doxorubicin (Hydrodaunorubicin®), vincristine (Oncovin®), bortezomib (Velcade®), and dexamethasone (Decadron®) in combination with lenalidomide (Revlimid®), anti-IL36 agents such as BI655130, Dihydroorotate dehydrogenase inhibitors such as IMU-838, anti-OX40 agents such as KHK-4083, microbiome agents such as RBX2660, SER-287, Narrow spectrum kinase inhibitors such as TOP-1288, anti-CD40 agents such as BI-655064 and FFP-104, guanylate cyclase agonists such as dolcanatide, sphingosine kinase inhibitors such as opaganib, anti-IL-12/IL-23 agents such as AK-101, Ubiquitin protein ligase complex inhibitors such as BBT-401, sphingosine receptors modulators such as BMS-986166, P38MAPK/PDE4 inhibitors such as CBS-3595, CCR9 antagonists such as CCX-507, FimH antagonists such as EB-8018, HIF-PH inhibitors such as FG-6874, HIF-1α stabilizer such as GB-004, MAP3K8 protein inhibitors such as GS-4875, LAG-3 antibodies such as GSK-2831781, RIP2 kinase inhibitors such as GSK- 2983559, Farnesoid X receptor agonist such as MET-409, CCK2 antagonists such as PNB-001, IL-23 Receptor antagonists such as PTG-200, Purinergic P2X7 receptor antagonists such as SGM-1019, PDE4 inhibitors such as Apremilast, ICAM-1 inhibitors such as alicaforsen sodium, Anti-IL23 agents such as guselkumab, brazikumab and mirkizumab, ant-IL-15 agents such as AMG-714, TYK-2 inhibitors such as BMS-986165, NK Cells activators such as CNDO-201, RIP-1 kinase inhibitors such as GSK-2982772, anti-NKGD2 agents such as JNJ-4500, CXCL-10 antibodies such as JT-02, IL-22 receptor agonists such as RG-7880, GATA-3 antagonists such as SB-012, and Colony-stimulating factor-1 receptor inhibitors such as edicotinib.

[0583] In certain embodiments, the method further comprises administering to the subject a third therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fourth therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fifth therapeutic agent.

[0584] In certain embodiments, the third therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fourth therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fifth therapeutic agent is one of the second therapeutic agents described above.

Immune Disorders Other than a Viral Infection

[0585] Another aspect of the invention provides a method of treating an immune disorder other than a viral infection in a patient. The method comprises administering to a subject in need thereof (i) a therapeutically effective amount of a substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein and (ii) a second therapeutic agent, in order to treat the immune disorder other than a viral infection.

[0586] In certain embodiments, the second therapeutic agent is pentoxifylline, propentofylline, torbafylline, cyclosporine, methotrexate, tamoxifen, forskolin and analogs thereof, tar derivatives, steroids, vitamin A and its derivatives, vitamin D and its derivatives, a cytokine, a chemokine, a stem cell growth factor, a lymphotoxin, an hematopoietic factor, a colony stimulating factor (CSF), erythropoietin, thrombopoietin, tumor necrosis factor-a (TNF), TNF-⊖, granulocyte-colony stimulating factor (G-CSF), granulocyte macrophage-colony stimulating factor (GM-CSF), interferon-α, interferon-β, interferon-γ, interferon-λ, stem cell growth factor designated "S1 factor", human growth hormone, N-methionyl human growth hormone, bovine growth hormone, parathyroid hormone, thyroxine, insulin, proinsulin, relaxin, prorelaxin, follicle stimulating hormone (FSH), thyroid stimulating hormone (TSH), luteinizing hormone (LH), hepatic growth factor, prostaglandin, fibroblast growth factor, prolactin, placental lactogen, OB protein, mullerian-inhibiting substance, mouse gonadotropin-associated peptide, inhibin, activin, vascular endothelial growth factor, integrin, NGF-β, platelet-growth factor, TGF-α, TGF-β, insulin-like growth factor-I, insulin-like growth factor-II, macrophage-CSF (M-CSF), IL-1, IL-1α, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-11, IL-12, IL-13, IL-14, IL-15, IL-16, IL-17, IL-18, IL-21, IL-25, LIF, FLT-3, angiostatin, thrombospondin, endostatin, or lymphotoxin.

[0587] In certain embodiments, the method further comprises administering to the subject a third therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fourth therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fifth therapeutic agent.

[0588] In certain embodiments, the third therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fourth therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fifth therapeutic agent is one of the second therapeutic agents described above.

Viral Infection

[0589] Another aspect of the invention provides a method of treating an immune disorder that is a viral infection in a patient. The method comprises administering to a subject in need thereof (i) a therapeutically effective amount of a substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein and (ii) a second therapeutic agent, in order to treat the immune disorder other than a viral infection.

[0590] In certain embodiments, the second therapeutic agent is an anti-HIV agent. In certain embodiments, the second therapeutic agent is a nucleoside reverse transcriptase inhibitor (NRTI), non-nucleoside reverse transcriptase inhibitor, protease inhibitor, or fusion inhibitor. In certain embodiments, the second therapeutic agent is 3TC (Lamivudine), AZT (Zidovudine), (−)-FTC, ddI (Didanosine), ddC (zalcitabine), abacavir (ABC), tenofovir (PMPA), D-D4FC (Reverset), D4T (Stavudine), Racivir, L-FddC, L-FD4C, NVP (Nevirapine), DLV (Delavirdine), EFV (Efavirenz), SQVM (Saquinavir mesylate), RTV (Ritonavir), IDV (Indinavir), SQV (Saquinavir), NFV (Nelfinavir), APV (Amprenavir), LPV (Lopinavir), or the fusion inhibitor T20.

[0591] In certain embodiments, the second therapeutic agent is ddC, abacavir, ddI, ddA, 3TC, AZT, D4T, FTC, FddC, Fd4C, Atazanavir, Adefovir dipivoxyl, Tenofovir disoproxil, Etecavir, Indinavir, KHI-227.2-[3-[3-(S)-[[(Tetrahydrofuranyloxy)carbonyl]amino]-4-phenyl-2(R)-hydroxybutyl]]-N-(1,1-dimethylethyl)decahydro-3-isoquinolinecarboxamide, VB-11,328, KNI-174, Val-Val-Sta, CPG53820, HOEt-N2 aza-peptide isostere, 2,5-Diamino-N,N'-bis(N-benzyloxycarbonyluelyl)-1,6-diphenyl-3(S),4(S)-hexanediol BzOCValPhe[diCHOH(SS]PheValBzOC, 2,5,-Diamino-N,N'-bis(N-benzyloxycarbonyluelyl)-1,6-diphenyl-3(R),4(R)-hexanediol BzOCValPhe[diCHOH(RR]PheValBzOC, [bis(SATE)ddAMP], BILA 2186 BS, Agenerase, A-98881, A-83962, A-80987, (2-Naphthalcarbonyl)Asn[decarbonylPhe-hydroxyethyl]ProOtertButyl, A-81525, XM323, Tipranavir, SDZ PRI 053, SD146, Telinavir, (R)2QuinCOAsnPhe[CHOHCH2]PipCONHtBu, Saquinavir, R-87366, DMP 460, L685,434, L685,434-OEtNMe2, L689,502, Lasinavir, Aluviran P9941, Palinavir, or Penicillin. In certain embodiments, the second therapeutic agent is ddC, abacavir, ddI, ddA, 3TC, AZT, D4T, FTC, FddC, or Fd4C.

[0592] In certain embodiments, the method further comprises administering to the subject a third therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fourth therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fifth therapeutic agent.

[0593] In certain embodiments, the third therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fourth therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fifth therapeutic agent is one of the second therapeutic agents described above.

Neurodegenerative Disorders

[0594] Another aspect of the invention provides a method of treating a neurodegenerative disorder in a patient. The method comprises administering to a subject in need thereof (i) a therapeutically effective amount of a substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein and (ii) a second therapeutic agent, in order to treat the neurodegenerative disorder.

[0595] In certain embodiments, the second therapeutic agent is a dopaminergic treatment, a cholinesterase inhibitor, an antipsychotic drug, deep brain stimulation (for example, to stop tremor and refractory movement disorders), riluzole, a caffein A2A receptor antagonist, pramipexole, or rasagilin.

[0596] In certain embodiments, the method further comprises administering to the subject a third therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fourth therapeutic agent. In certain embodiments, the method further comprises administering to the subject a fifth therapeutic agent.

[0597] In certain embodiments, the third therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fourth therapeutic agent is one of the second therapeutic agents described above. In certain embodiments, the fifth therapeutic agent is one of the second therapeutic agents described above.

V. Pharmaceutical Compositions and Dosing Considerations

[0598] As indicated above, the invention provides pharmaceutical compositions, which comprise a therapeutically-effective amount of one or more of the compounds described above, formulated together with one or more pharmaceutically acceptable carriers (additives) and/or diluents. The pharmaceutical compositions may be specially formulated for administration in solid or liquid form, including those adapted for the following: (1) oral administration, for example, drenches (aqueous or non-aqueous solutions or suspensions), tablets, e.g., those targeted for buccal, sublingual, and systemic absorption, boluses, powders, granules, pastes for application to the tongue; (2) parenteral administration, for example, by subcutaneous, intramuscular, intravenous or epidural injection as, for example, a sterile solution or suspension, or sustained-release formulation; (3) topical application, for example, as a cream, ointment, or a controlled-release patch or spray applied to the skin; (4) intravaginally or intrarectally, for example, as a pessary, cream or foam; (5) sublingually; (6) ocularly; (7) transdermally; or (8) nasally.

[0599] In certain embodiments, the invention provides a pharmaceutical composition comprising a compound described herein (e.g., a compound of Formula I, I-1, I-A, or I-B) and a pharmaceutically acceptable carrier. In certain embodiments, the invention provides a pharmaceutical composition comprising a compound described herein (e.g., a compound of Formula I, I-1, I-A, or I-B), an additional therapeutic agent (e.g., a compound described in Section IV), and a pharmaceutically acceptable carrier.

[0600] The phrase "therapeutically effective amount" as used herein means that amount of a compound, material, or composition comprising a compound of the present invention which is effective for producing some desired therapeutic effect in at least a sub-population of cells in an animal at a reasonable benefit/risk ratio applicable to any medical treatment.

[0601] The phrase "pharmaceutically acceptable" is employed herein to refer to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

[0602] Wetting agents, emulsifiers and lubricants, such as sodium lauryl sulfate and magnesium stearate, as well as coloring agents, release agents, coating agents, sweetening, flavoring and perfuming agents, preservatives and antioxidants can also be present in the compositions.

[0603] Examples of pharmaceutically-acceptable antioxidants include: (1) water soluble antioxidants, such as ascorbic acid, cysteine hydrochloride, sodium bisulfate, sodium metabisulfite, sodium sulfite and the like; (2) oil-soluble antioxidants, such as ascorbyl palmitate, butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), lecithin, propyl gallate, alpha-tocopherol, and the like; and (3) metal chelating agents, such as citric acid, ethylenediamine tetraacetic acid (EDTA), sorbitol, tartaric acid, phosphoric acid, and the like.

[0604] Formulations of the present invention include those suitable for oral, nasal, topical (including buccal and sublingual), rectal, vaginal and/or parenteral administration. The formulations may conveniently be presented in unit dosage form and may be prepared by any methods well known in the art of pharmacy. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will vary depending upon the host being treated, the particular mode of administration. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will generally be that amount of the compound which produces a therapeutic effect. Generally, out of one hundred percent, this amount will range from about 0.1 percent to about ninety-nine percent of active ingredient, preferably from about 5 percent to about 70 percent, most preferably from about 10 percent to about 30 percent.

[0605] In certain embodiments, a formulation of the present invention comprises an excipient selected from the group consisting of cyclodextrins, celluloses, liposomes, micelle forming agents, e.g., bile acids, and polymeric carriers, e.g., polyesters and polyanhydrides; and a compound of the present invention. In certain embodiments, an aforementioned formulation renders orally bioavailable a compound of the present invention.

[0606] Methods of preparing these formulations or compositions include the step of bringing into association a compound of the present invention with the carrier and, optionally, one or more accessory ingredients. In general, the formulations are prepared by uniformly and intimately bringing into association a compound of the present invention with liquid carriers, or finely divided solid carriers, or both, and then, if necessary, shaping the product.

[0607] Formulations of the invention suitable for oral administration may be in the form of capsules, cachets, pills, tablets, lozenges (using a flavored basis, usually sucrose and acacia or tragacanth), powders, granules, or as a solution or a suspension in an aqueous or non-aqueous liquid, or as an oil-in-water or water-in-oil liquid emulsion, or as an elixir or syrup, or as pastilles (using an inert base, such as gelatin and glycerin, or sucrose and acacia) and/or as mouth washes and the like, each containing a predetermined amount of a compound of the present invention as an active ingredient. A compound of the present invention may also be administered as a bolus, electuary or paste.

[0608] In solid dosage forms of the invention for oral administration (capsules, tablets, pills, dragees, powders, granules, trouches and the like), the active ingredient is mixed with one or more pharmaceutically-acceptable carriers, such as sodium citrate or dicalcium phosphate, and/or any of the following: (1) fillers or extenders, such as starches, lactose, sucrose, glucose, mannitol, and/or silicic acid; (2) binders, such as, for example, carboxymethylcellulose, alginates, gelatin, polyvinyl pyrrolidone, sucrose and/or acacia; (3) humectants, such as glycerol; (4) disintegrating agents, such as agar-agar, calcium carbonate, potato or tapioca starch, alginic acid, certain silicates, and sodium carbonate; (5) solution retarding agents, such as paraffin; (6) absorption accelerators, such as quaternary ammonium compounds and surfactants, such as poloxamer and sodium lauryl sulfate; (7) wetting agents, such as, for example, cetyl alcohol, glycerol monostearate, and non-ionic surfactants; (8) absorbents, such as kaolin and bentonite clay; (9) lubricants, such as talc, calcium stearate, magnesium stearate, solid polyethylene glycols, sodium lauryl sulfate, zinc stearate, sodium stearate, stearic acid, and mixtures thereof; (10) coloring agents; and (11) controlled release agents such as crospovidone or ethyl cellulose. In the case of capsules, tablets and pills, the pharmaceutical compositions may also comprise buffering agents. Solid compositions of a similar type may also be employed as fillers in soft and hard-shelled gelatin capsules using such excipients as lactose or milk sugars, as well as high molecular weight polyethylene glycols and the like.

[0609] A tablet may be made by compression or molding, optionally with one or more accessory ingredients. Compressed tablets may be prepared using binder (for example, gelatin or hydroxypropylmethyl cellulose), lubricant, inert diluent, preservative, disintegrant (for example, sodium starch glycolate or cross-linked sodium carboxymethyl cellulose), surface-active or dispersing agent. Molded tablets may be made by molding in a suitable machine a mixture of the powdered compound moistened with an inert liquid diluent.

[0610] The tablets, and other solid dosage forms of the pharmaceutical compositions of the present invention, such as dragees, capsules, pills and granules, may optionally be scored or prepared with coatings and shells, such as enteric coatings and other coatings well known in the pharmaceutical-formulating art. They may also be formulated so as to provide slow or controlled release of the active ingredient therein using, for example, hydroxypropylmethyl cellulose in varying proportions to provide the desired release profile, other polymer matrices, liposomes and/or microspheres. They may be formulated for rapid release, e.g., freeze-dried. They may be sterilized by, for example, filtration through a bacteria-retaining filter, or by incorporating sterilizing agents in the form of sterile solid compositions which can be dissolved in sterile water, or some other sterile injectable medium immediately before use. These compositions may also optionally contain opacifying agents and may be of a composition that they release the active ingredient(s) only, or preferentially, in a certain portion of the gastrointestinal tract, optionally, in a delayed manner. Examples of embedding compositions which can be used include polymeric substances and waxes. The active ingredient can also be in micro-encapsulated form, if appropriate, with one or more of the above-described excipients.

[0611] Liquid dosage forms for oral administration of the compounds of the invention include pharmaceutically acceptable emulsions, microemulsions, solutions, suspensions, syrups and elixirs. In addition to the active ingredient, the liquid dosage forms may contain inert diluents commonly used in the art, such as, for example, water or other solvents, solubilizing agents and emulsifiers, such as ethyl alcohol, isopropyl alcohol, ethyl carbonate, ethyl acetate, benzyl alcohol, benzyl benzoate, propylene glycol, 1,3-butylene glycol, oils (in particular, cottonseed, groundnut, corn, germ, olive, castor and sesame oils), glycerol, tetrahydrofuryl alcohol, polyethylene glycols and fatty acid esters of sorbitan, and mixtures thereof.

[0612] Besides inert diluents, the oral compositions can also include adjuvants such as wetting agents, emulsifying and suspending agents, sweetening, flavoring, coloring, perfuming and preservative agents.

[0613] Suspensions, in addition to the active compounds, may contain suspending agents as, for example, ethoxylated isostearyl alcohols, polyoxyethylene sorbitol and sorbitan esters, microcrystalline cellulose, aluminum metahydroxide, bentonite, agar-agar and tragacanth, and mixtures thereof.

[0614] Formulations of the pharmaceutical compositions of the invention for rectal or vaginal administration may be presented as a suppository, which may be prepared by mixing one or more compounds of the invention with one or more suitable nonirritating excipients or carriers comprising, for example, cocoa butter, polyethylene glycol, a suppository wax or a salicylate, and which is solid at room temperature, but liquid at body temperature and, therefore, will melt in the rectum or vaginal cavity and release the active compound.

[0615] Formulations of the present invention which are suitable for vaginal administration also include pessaries, tampons, creams, gels, pastes, foams or spray formulations containing such carriers as are known in the art to be appropriate.

[0616] Dosage forms for the topical or transdermal administration of a compound of this invention include powders, sprays, ointments, pastes, creams, lotions, gels, solutions, patches and inhalants. The active compound may be mixed under sterile conditions with a pharmaceutically-acceptable carrier, and with any preservatives, buffers, or propellants which may be required.

[0617] The ointments, pastes, creams and gels may contain, in addition to an active compound of this invention, excipients, such as animal and vegetable fats, oils, waxes, paraffins, starch, tragacanth, cellulose derivatives, polyethylene glycols, silicones, bentonites, silicic acid, talc and zinc oxide, or mixtures thereof.

[0618] Powders and sprays can contain, in addition to a compound of this invention, excipients such as lactose, talc, silicic acid, aluminum hydroxide, calcium silicates and polyamide powder, or mixtures of these substances. Sprays can additionally contain customary propellants, such as chlorofluorohydrocarbons and volatile unsubstituted hydrocarbons, such as butane and propane.

[0619] Transdermal patches have the added advantage of providing controlled delivery of a compound of the present invention to the body. Such dosage forms can be made by dissolving or dispersing the compound in the proper medium. Absorption enhancers can also be used to increase the flux of the compound across the skin. The rate of such flux can be controlled by either providing a rate controlling membrane or dispersing the compound in a polymer matrix or gel.

[0620] Ophthalmic formulations, eye ointments, powders, solutions and the like, are also contemplated as being within the scope of this invention.

[0621] Pharmaceutical compositions of this invention suitable for parenteral administration comprise one or more compounds of the invention in combination with one or more pharmaceutically-acceptable sterile isotonic aqueous or nonaqueous solutions, dispersions, suspensions or emulsions, or sterile powders which may be reconstituted into sterile injectable solutions or dispersions just prior to use, which may contain sugars, alcohols, antioxidants, buffers, bacteriostats, solutes which render the formulation isotonic with the blood of the intended recipient or suspending or thickening agents.

[0622] Examples of suitable aqueous and nonaqueous carriers which may be employed in the pharmaceutical compositions of the invention include water, ethanol, polyols (such as glycerol, propylene glycol, polyethylene glycol, and the like), and suitable mixtures thereof, vegetable oils, such as olive oil, and injectable organic esters, such as ethyl oleate. Proper fluidity can be maintained, for example, by the use of coating materials, such as lecithin, by the maintenance of the required particle size in the case of dispersions, and by the use of surfactants.

[0623] These compositions may also contain adjuvants such as preservatives, wetting agents, emulsifying agents and dispersing agents. Prevention of the action of microorganisms upon the subject compounds may be ensured by the inclusion of various antibacterial and antifungal agents, for example, paraben, chlorobutanol, phenol sorbic acid, and the like. It may also be desirable to include isotonic agents, such as sugars, sodium chloride, and the like into the compositions. In addition, prolonged absorption of the injectable pharmaceutical form may be brought about by the inclusion of agents which delay absorption such as aluminum monostearate and gelatin.

[0624] In some cases, in order to prolong the effect of a drug, it is desirable to slow the absorption of the drug from subcutaneous or intramuscular injection. This may be accomplished by the use of a liquid suspension of crystalline or amorphous material having poor water solubility. The rate of absorption of the drug then depends upon its rate of dissolution which, in turn, may depend upon crystal size and crystalline form. Alternatively, delayed absorption of a parenterally-administered drug form is accomplished by dissolving or suspending the drug in an oil vehicle.

[0625] Injectable depot forms are made by forming microencapsule matrices of the subject compounds in biodegradable polymers such as polylactide-polyglycolide. Depending on the ratio of drug to polymer, and the nature of the particular polymer employed, the rate of drug release can be controlled. Examples of other biodegradable polymers include poly(orthoesters) and poly(anhydrides). Depot injectable formulations are also prepared by entrapping the drug in liposomes or microemulsions which are compatible with body tissue.

[0626] When the compounds of the present invention are administered as pharmaceuticals, to humans and animals, they can be given per se or as a pharmaceutical composition containing, for example, 0.1 to 99% (more preferably, 10 to 30%) of active ingredient in combination with a pharmaceutically acceptable carrier.

[0627] The preparations of the present invention may be given orally, parenterally, topically, or rectally. They are of course given in forms suitable for each administration route. For example, they are administered in tablets or capsule form, by injection, inhalation, eye lotion, ointment, suppository, etc. administration by injection, infusion or inhalation; topical by lotion or ointment; and rectal by suppositories. Oral administrations are preferred.

[0628] The phrases "parenteral administration" and "administered parenterally" as used herein means modes of administration other than enteral and topical administration, usually by injection, and includes, without limitation, intravenous, intramuscular, intraarterial, intrathecal, intracapsular, intraorbital, intracardiac, intradermal, intraperitoneal, transtracheal, subcutaneous, subcuticular, intraarticular, subcapsular, subarachnoid, intraspinal and intrasternal injection and infusion.

[0629] The phrases "systemic administration," "administered systemically," "peripheral administration" and "administered peripherally" as used herein mean the administration of a compound, drug or other material other than directly into the central nervous system, such that it enters the patient's system and, thus, is subject to metabolism and other like processes, for example, subcutaneous administration.

[0630] These compounds may be administered to humans and other animals for therapy by any suitable route of administration, including orally, nasally, as by, for example, a spray, rectally, intravaginally, parenterally, intracisternally and topically, as by powders, ointments or drops, including buccally and sublingually.

[0631] Regardless of the route of administration selected, the compounds of the present invention, which may be used in a suitable hydrated form, and/or the pharmaceutical compositions of the present invention, are formulated into pharmaceutically-acceptable dosage forms by conventional methods known to those of skill in the art.

[0632] Actual dosage levels of the active ingredients in the pharmaceutical compositions of this invention may be varied so as to obtain an amount of the active ingredient which is effective to achieve the desired therapeutic response for a particular patient, composition, and mode of administration, without being toxic to the patient.

[0633] The selected dosage level will depend upon a variety of factors including the activity of the particular compound of the present invention employed, or the ester, salt or amide thereof, the route of administration, the time of administration, the rate of excretion or metabolism of the particular compound being employed, the rate and extent of absorption, the duration of the treatment, other drugs, compounds and/or materials used in combination with the particular compound employed, the age, sex, weight, condition, general health and prior medical history of the patient being treated, and like factors well known in the medical arts.

[0634] A physician or veterinarian having ordinary skill in the art can readily determine and prescribe the effective amount of the pharmaceutical composition required. For example, the physician or veterinarian could start doses of the compounds of the invention employed in the pharmaceutical composition at levels lower than that required in order to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved.

[0635] In general, a suitable daily dose of a compound of the invention will be that amount of the compound which is the lowest dose effective to produce a therapeutic effect. Such an effective dose will generally depend upon the factors described above. Preferably, the compounds are administered at about 0.01 mg/kg to about 200 mg/kg, more preferably at about 0.1 mg/kg to about 100 mg/kg, even more preferably at about 0.5 mg/kg to about 50 mg/kg. When the compounds described herein are co-administered with another agent (e.g., as sensitizing agents), the effective amount may be less than when the agent is used alone.

[0636] If desired, the effective daily dose of the active compound may be administered as two, three, four, five, six or more sub-doses administered separately at appropriate intervals throughout the day, optionally, in unit dosage forms. Preferred dosing is one administration per day.

[0637] The invention further provides a unit dosage form (such as a tablet or capsule) comprising a substituted 2',3'-didehydro-3'-deoxy-4'-ethynylthymidines or related compound described herein in a therapeutically effective amount for the treatment of a medical disorder described herein.

EXAMPLES

[0638] The invention now being generally described, will be more readily understood by reference to the following examples, which are included merely for purposes of illustration of certain aspects and embodiments of the present invention, and are not intended to limit the invention. Starting materials described herein can be obtained from commercial sources or may be readily prepared from commercially available materials using transformations known to those of skill in the art.

[0639] All chemical reactions were carried out using commercial materials and reagents without further purification unless otherwise noted. All chemical reactions were monitored by thin layer chromatography (TLC) on silica gel plates (Kieselgel 60 F254, Merck), ultra-performance liquid chromatography (UPLC) or NMR. Visualization of the spots on TLC plates was achieved by UV light and by staining the TLC plates in potassium permanganate and charring with a heat gun.

[0640] Flash column chromatography was performed on silica gel using Fluorochem silicagel LC60A 40-63 micron and reagent grade heptane, ethyl acetate, dichloromethane and methanol mixtures as eluent. Chromatography was performed on a Biotage Isolera using silica (normal phase) (SiliCycle SiliaSep Premium 25 μm or Biotage SNAP Ultra HP-Sphere 25 μm) or C18 (reverse phase) (Biotage SNAP Ultra C18 HP Sphere 25 μm) pre-packed cartridges; or by flash-column chromatography using silica gel (Fluorochem silica gel 60A 40-63 μm).

[0641] UPLC was recorded on a Waters Acquity UPLC HClass instrument with Acquity PDA detector, ELS detector and quaternary solvent system. Acidic methods were run using a gradient of 0.1% formic acid in acetonitrile and 0.1% formic acid in water on a CSH C18 column (2.1×50 mm 1.7

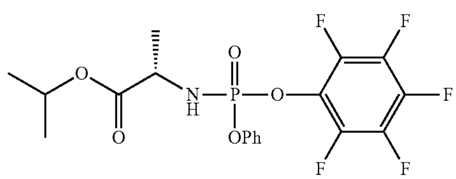
μm) at 0.8 mL/min. Basic methods were run using a gradient of 0.1% ammonia in acetonitrile and 0.1% ammonia in water on a BEH C18 column (2.1×50 mm 2.5 μm) at 0.8 mL/min.

[0642] All products were characterized by ^{1}H NMR, and where appropriate, ^{13}C, ^{31}P and ^{19}F NMR. NMR spectral data was recorded on a JEOL ECX400 MHz spectrometer. Chemical shifts are expressed in parts per million values (ppm) and are designated as s (singlet); br s (broad singlet); d (doublet); t (triplet); q (quartet); quint (quintet) or m (multiplet).

Example 1—Synthesis of Pentafluorophenyl Phosphoramidate Intermediates

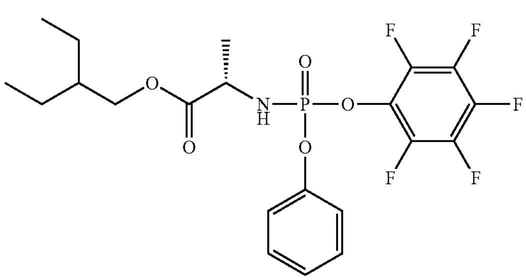
[0643] The following pentafluorophenyl phosphoramidate intermediates were prepared as described below.

Synthesis of Isopropyl ((Perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate



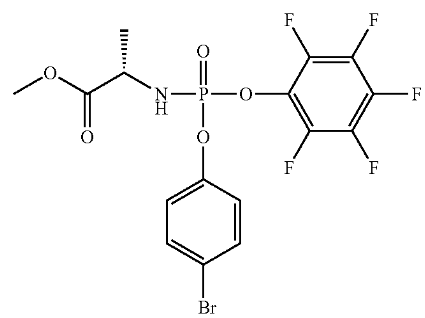
[0644] To a solution of L-alanine isopropyl ester hydrochloride (500 mg, 2.98 mmol) and phenyl dichlorophosphonate (629 mg, 2.98 mmol) in 5 mL of dichloromethane at −78° C. was added triethylamine (603 mg, 5.97 mmol) dropwise, and the solution was allowed to warm to room temperature and stirred for 30 minutes. The mixture was cooled to −78° C., and a solution of triethylamine (603 mg, 5.97 mmol) and pentafluorophenol (549 mg, 2.98 mmol) in 2 mL of dichloromethane was added. The mixture was stirred for 1 h at room temperature and then left to stand for 18 h at room temperature. Water was added, the dichloromethane layer was separated, and the water layer was washed twice with dichloromethane. The dichloromethane solutions were combined, dried over magnesium sulfate, and concentrated to provide a white solid. The solid was crystallized from 10% methyl t-butyl ether/hexanes to afford 380 mg white solid. The product was a 4:1 mixture of diastereomers at phosphorus.

Synthesis of 2-Ethylbutyl ((Perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate



[0645] 2-Ethylbutyl L-alaninate hydrochloride (558 mg, 2.66 mmol) was dissolved in dichloromethane (5 mL) and the mixture was cooled to −78° C. under Ar. Phenyl dichlorophosphate (397 μL, 2.66 mmol) and triethylamine (742 μL, 5.32 mmol) were added and the mixture was warmed to room temperature and stirred for 16 hours. Pentafluorophenol (490 mg, 2.66 mmol) and triethylamine (371 μL, 2.66 mmol) were added and the mixture was stirred for 2 hours. It was partitioned between ethyl acetate and water, and the aqueous layer was extracted with ethyl acetate. The combined organics were washed with brine, dried over magnesium sulfate, filtered and concentrated in vacuo. The resulting material was purified by normal phase purification (Biotage Isolera, 25g cartridge, eluent: 0-30% ethyl acetate in heptane) to give 2-ethylbutyl ((perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate as a mixture of two diastereomers at phosphorus (815 mg, 62%). ^{1}H-NMR (400 MHz, $CDCl_3$) δ 7.38-7.33 (m, 4H), 7.28-7.19 (m, 6H), 4.23-3.91 (m, 8H), 1.55-1.49 (m, 5H), 1.46 (dd, 6H), 1.37-1.31 (m, 5H), 0.88 (td, 12H). 23H for each diastereomer. ^{31}P-NMR (162 MHz, $CDCl_3$) δ −1.06. ^{19}F-NMR (376 MHz, $CDCl_3$) δ −153.1 (dd, J=38.5, 20.0 Hz, 2F), −159.2-−159.4 (m, 1F), −161.9-−162.1 (m, 2F).

Synthesis of Methyl ((4-Bromophenoxy)(perfluorophenoxy)phosphoryl)-L-alaninate

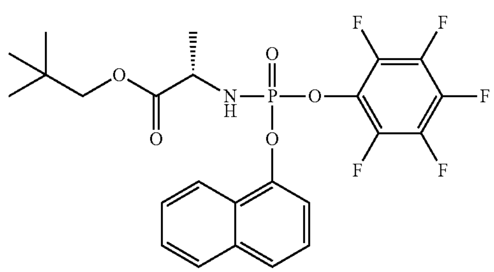


[0646] 4-Bromophenol (564 mg, 3.26 mmol) was dissolved in dichloromethane (10 mL) and cooled to −78° C. under Ar. $POCl_3$ (500 mg, 3.26 mmol) and triethylamine (450 μL, 3.26 mmol) were added, and the mixture was stirred for 30 minutes then warmed to room temperature and stirred for 1 hour. The mixture was cooled back to −78° C. L-alanine methyl ester hydrochloride (455 mg, 3.26 mmol) and triethylamine (910 μL, 6.52 mmol) were added, and the mixture was warmed to room temperature and stirred for 16 hours. The mixture was cooled to 0° C. Pentafluorophenol (600 mg, 3.26 mmol) and triethylamine (450 μL, 3.26 mmol) in dichloromethane (2 mL) were added, and the mixture was stirred for 30 minutes then warmed to room temperature and stirred for 4 hours. The mixture was partitioned between dichloromethane and water, and the aqueous layer was extracted with dichloromethane. The combined organics were washed with brine, dried over magnesium sulfate, filtered and concentrated in vacuo. The resulting material was purified by flash column chromatography on silica gel (3:1 heptane-ethyl acetate) to give a white solid.

[0647] The solid was stirred in heptane (20 mL), then filtered rinsing the cake with heptane (20 mL). The solid was dried under vacuum then by rotary evaporation to give methyl ((4-bromophenoxy)(perfluorophenoxy)phosphoryl)-L-alaninate (805 mg, 49%) as a 60:40 mixture of diastereomers. The filtrate was concentrated to give methyl ((4-bromophenoxy) (perfluorophenoxy)phosphoryl)-L-alaninate (74 mg, 4.5%) as a 30:70 mixture of diastereomers. The batches were recombined and stirred in 4:1 heptane-ethyl acetate (15 mL) at 45° C. for 30 min then at room temperature for 30 min. The solid was collected by filtration and washed with 4:1 heptane-ethyl acetate (8 mL). The solid was dried by rotary evaporation to give P-diastereomer 1 of methyl ((4-bromophenoxy)(perfluorophenoxy)phosphoryl)-L-alaninate (385 mg, 23%) as a white solid. ^{1}H-NMR (400 MHz, $CDCl_3$) δ 7.45 (d, 2H), 7.15 (d, 2H), 4.17 (dq, 1H), 3.91 (dd, 1H), 3.75 (s, 3H), 1.47 (d, 3H). ^{19}F-NMR (376 MHz, $CDCl_3$) δ −153.2 (d, 2F), −158.9 (dt, 1F), −161.7 (dd, 2F). ^{31}P-NMR (162 MHz, $CDCl_3$) δ −1.1.

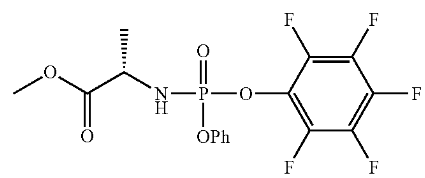
[0648] The filtrate was concentrated to a white solid. Attempts to resolve this diastereomer 2 were unsuccessful, so all filtrates were combined to give methyl ((4-bromophenoxy) (perfluorophenoxy)phosphoryl)-L-alaninate (374 mg, 21%) as a white solid, and as a 80:20 ratio of P-diastereomer 2 to P-diastereomer 1 based on integration of the Me signals in ^{1}H NMR.

Synthesis of Neopentyl ((Naphthalen-1-yloxy)(perfluorophenoxy)phosphoryl)-L-alaninate



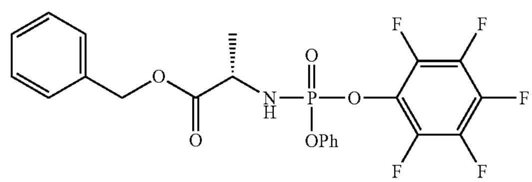
[0649] 1-Naphthol (375 mg, 2.6 mmol) was dissolved in dichloromethane (10 mL) and cooled to −78° C. under Ar. $POCl_3$ (242 μL, 2.6 mmol) and triethylamine (362 μL, 2.6 mmol) were added, and the mixture was stirred for 30 minutes then warmed to room temperature and stirred for 1 hour. The mixture was cooled back to −78° C. Neopentyl L-alaninate hydrochloride (500 mg, 2.6 mmol) and triethylamine (725 μL, 5.2 mmol) were added, and the mixture was warmed to room temperature and stirred for 16 hours. The mixture was cooled to 0° C. Pentafluorophenol (479 mg, 2.6 mmol) and triethylamine (362 μL, 2.6 mmol) were added, and the mixture was warmed to room temperature and stirred for 16 hours. Pentafluorophenol (479 mg, 2.6 mmol) and triethylamine (362 μL, 2.6 mmol) were added and the mixture was stirred for a further 2 hours. The mixture was partitioned between ethyl acetate and water, and the aqueous was extracted with ethyl acetate. The combined organics were washed with brine, dried over magnesium sulfate, filtered and concentrated in vacuo. The resulting material was purified by normal phase purification (Biotage Isolera, 40 g cartridge, eluent: 0-30% ethyl acetate in heptane) to give neopentyl ((naphthalen-1-yloxy)(perfluorophenoxy)phosphoryl)-L-alaninate as a mixture of two diastereomers (552 mg, 55%). ^{1}H-NMR (400 MHz, $CDCl_3$) δ 8.12-8.10 (m, 2H), 7.87 (d, 2H), 7.71 (d, 2H), 7.61-7.51 (m, 6H), 7.42 (t, 2H), 4.33-4.25 (m, 2H), 4.12-4.00 (m, 2H), 3.90-3.72 (m, 4H), 1.48 (s, 3H), 1.46 (s, 3H), 0.93 (s, 9H), 0.90 (s, 9H). 23H for each diastereomer. ^{31}P-NMR (162 MHz, $CDCl_3$) δ −0.6.

Synthesis of Methyl ((Perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate



[0650] L-Alanine methyl ester hydrochloride (1.0 g, 7.16 mmol) was dissolved in dichloromethane (10 mL) and the mixture was cooled to −78° C. under Ar. Phenyl dichlorophosphate (1.07 mL, 7.17 mmol) and triethylamine (2.0 mL, 14.3 mmol) were added, and the mixture was warmed to room temperature and stirred for 16 hours. Pentafluorophenol (1.32 g, 7.16 mmol) and triethylamine (1.0 mL, 7.16 mmol) were added, and the mixture was stirred for 16 hours. The mixture was partitioned between ethyl acetate and water, and the aqueous was extracted with ethyl acetate. The combined organics were washed with brine, dried over magnesium sulfate, filtered and concentrated in vacuo. The resulting material was purified by normal phase purification (Biotage Isolera, 80 g cartridge, eluent: 0-30% ethyl acetate in heptane) to give 2-ethylbutyl ((perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate as a mixture of two diastereomers (1.38 g, 45%). ^{1}H-NMR (400 MHz, $CDCl_3$) δ 7.38-7.34 (m, 2H), 7.28-7.21 (m, 3H), 4.14-4.27 (1H), 3.85-4.00 (1H), 3.75 (d, 3H), 1.46 (t, 3H). ^{31}P-NMR (162 MHz, $CDCl_3$) δ −1.1.

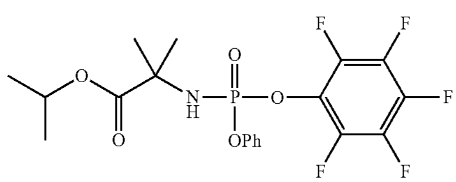
Synthesis of Benzyl ((Perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate



[0651] L-Alanine benzyl ester hydrochloride (1 g, 4.64 mmol) was dissolved in dichloromethane (10 mL), and the mixture was cooled to −78° C. under Ar. Phenyl dichlorophosphate (691 μL, 4.64 mmol) and triethylamine (1.29 mL, 9.27 mmol) were added, and the mixture was warmed to room temperature and stirred for 3 hours. Pentafluorophenol (853 mg, 4.64 mmol) and triethylamine (646 μL, 4.64 mmol) were added, and the mixture was stirred for 16 hours. A further portion of triethylamine (646 μL, 4.64 mmol) was added, and the mixture was stirred for a further 16 hours. The mixture was partitioned between ethyl acetate and water, and the aqueous layer was extracted with ethyl acetate. The combined organics were washed with brine, dried over magnesium sulfate, filtered and concentrated in vacuo. The resulting material was purified by normal phase purification (Biotage Isolera, 80 g cartridge, eluent: 0-30%

ethyl acetate in heptane) to give benzyl ((perfluorophenoxy) (phenoxy) phosphoryl)-L-alaninate as a mixture of two diastereomers (771 mg, 33%). 1H-NMR (400 MHz, $CDCl_3$) δ 7.35-7.31 (m, 7H), 7.25-7.17 (m, 3H), 5.16 (s, 2H), 4.23 (q, 1H), 3.89 (dd, 1H), 1.47 (d, 3H).

Synthesis of Isopropyl 2-Methyl-2-((Perfluorophenoxy(phenoxy)phosphoryl)amino) propanoate

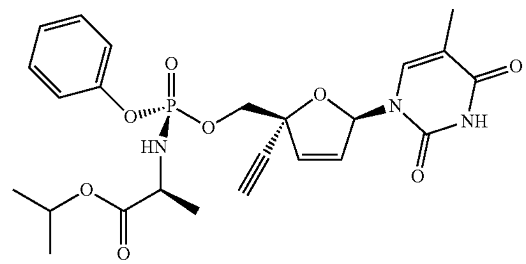


[0652] To a stirred solution of phenyl dichlorophosphate (5 g, 23.7 mmol) in DCM (50 mL) at −78° C. under nitrogen atmosphere was added pentafluorophenol (4.4 g, 23.7 mmol) and TEA (3.6 mL, 26.1 mmol). The resulting mixture was stirred for 30 min at −78° C. To the above mixture was added isopropyl 2-amino-2-methylpropanoate hydrochloride (4.3 g, 23.7 mmol) and TEA (3.6 mL, 26.7 mmol) at −78° C. under nitrogen atmosphere. The resulting mixture was stirred for additional 3 h at −20° C. After completion, the reaction was quenched with aq. NH_4Cl and extracted with EtOAc. The organic layer was dried over anhydrous sodium sulfate. After filtration and concentration in vacuo, the residue was purified by reverse phase flash (column, C18 silica gel; Mobile Phase, water ((0.1% NH4HCO3) and ACN (0% ACN up to 100% in 25 min; UV detection at 254/220 nm) to afford isopropyl 2-methyl-2-((perfluorophenoxy(phenoxy) phosphoryl)amino) propanoate (4.4 g, 9.4 mmol, 39.7%) as a white solid. LC-MS (ES, m/z): 468 $(M+H^+)$.

Example 2—Synthesis of Isopropyl ((R)-(((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (I-1) and Related Compounds

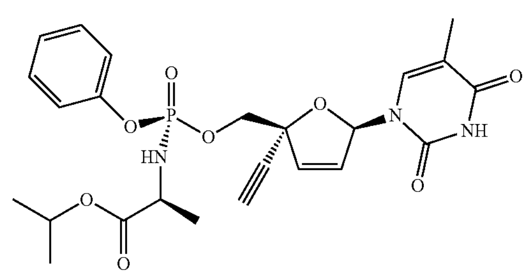
[0653] The phosphoramidate compounds below were prepared according to the following general procedure. (2R, 5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol was dissolved in THF and cooled to 0° C. under argon. tert-Butylmagnesium chloride (1M solution in THF, 2.1 eq.) was added and the mixture was stirred at 0° C. for 30 minutes. Phosphoramidate (1.2 eq.) was dissolved in THF under argon and added dropwise. The mixture was warmed to room temperature and stirred for 3 hours. The mixture was quenched with aqueous ammonium chloride and partitioned between ethyl acetate and a saturated aqueous solution of $NaHCO_3$. The aqueous layer was extracted with ethyl acetate. The combined organics were washed with brine, dried over magnesium sulfate, filtered and concentrated in vacuo. The resulting material was purified as specified.

Synthesis of Isopropyl ((R)-(((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (I-1)



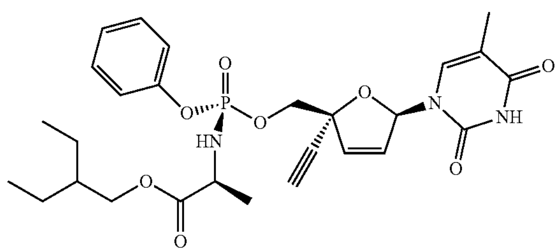
[0654] To a solution of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol (102 mg, 0.41 mmol) in 2 mL THF at 0° C. was added t-butylmagnesium chloride (860 μL of a 1M solution in THF, 0.86 mmol), and the mixture was stirred for 30 min. A solution of a 4:1 mixture of diastereomers of isopropyl ((perfluorophenoxy) (phenoxy)phosphoryl)-L-alaninate (220 mg, 0.49 mmol; prepared as described in Example 1, above) in 2 mL THF was added dropwise, and the reaction mixture was warmed to room temperature. After quenching and extraction (as described in the general procedure above), the resulting material was purified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane), and then the Rp diastereomer was isolated by chiral prep HPLC (Chiralpak IA, 20% EtOH in heptane over 30 minutes, 18 mL/min) giving 22 mg isopropyl ((R)-(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy) (phenoxy)phosphoryl)-L-alaninate (10%). Assignment of the Rp stereochemistry at phosphorous was made by comparison to compound I-2, which was synthesized from the commercially available Sp phosphoramidate reagent. ^{1}H-NMR (400 MHz, DMSO-D6) δ 7.31 (t, 2H), 7.21 (s, 1H), 7.12 (q, 3H), 6.88 (s, 1H), 6.43 (d, 1H), 6.14 (d, 1H), 6.01 (dd, 1H), 4.87-4.80 (m, 1H), 4.31 (q, 1H), 4.12 (q, 1H), 3.84 (s, 1H), 3.67 (dd, 1H), 1.59 (s, 3H), 1.14-1.11 (m, 9H). ^{31}P-NMR (162 MHz, DMSO-D6) δ 4.3. UPLC (Basic method, C18) 97.81% purity (M−H=516.01).

Synthesis of Isopropyl ((S)-(((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (I-2)



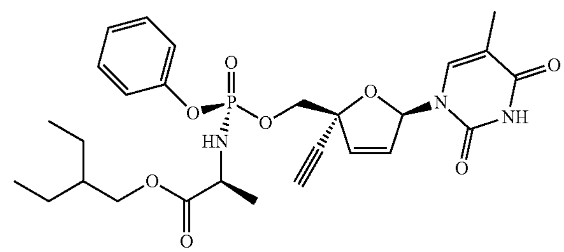
[0655] To a solution of 100 mg of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol in 2 mL THF at 0° C. was added t-butylmagnesium chloride (840 μL, 1M in THF, 0.84 mmol), and the mixture was stirred for 30 min. Commercial Sp phosphoramidate (CAS 1334513-02-8) dissolved in 2 mL THF was added dropwise, and the reaction mixture was warmed to room temperature and stirred for 2 h. The reaction mixture was quenched with aqueous ammonium chloride, partitioned between ethyl acetate and aqueous sodium bicarbonate, and the aqueous layer was extracted with ethyl acetate. The combined organic layers were washed with brined, dried over magnesium sulfate, filtered, and concentrated. The resulting material was purified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane), then repurified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-100 ethyl acetate:heptane) and then by reverse phase purification (Biotage Isolera, 12 g cartridge, eluent: 5% to 100% acetonitrile in water) to afford 73 mg of isopropyl ((S)-(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (35%). ^{1}H-NMR (400 MHz, DMSO-D6) δ 7.33 (t, 2H), 7.20-7.13 (m, 4H), 6.88 (t, 1H), 6.35 (q, 1H), 6.14 (dd, 1H), 6.02 (dd, 1H), 4.84-4.78 (m, 1H), 4.16 (d, 2H), 3.81 (s, 1H), 3.70 (td, 1H), 1.64 (s, 3H), 1.14-1.10 (m, 9H). 1×NH not observed. ^{31}P-NMR (162 MHz, DMSO-D6) δ 3.5. UPLC (Basic method, C18) 99.03% purity (M−H=516.02).

Synthesis of 2-Ethylbutyl ((R)-(((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (I-5)



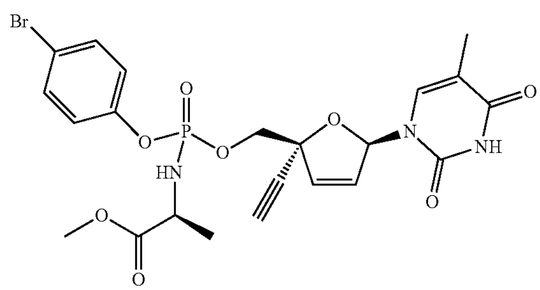
[0656] Reaction carried out on 169 mg (0.68 mmol) of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol in 4 mL THF at 0° C. was was added t-butylmagnesium chloride (1.4 mL, 1M in THF, 1.4 mmol), and the mixture was stirred for 30 min. A solution of 2-ethylbutyl ((perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate as a mixture of two diastereomers at phosphorus (prepared as described in Example 1, above) dissolved in 4 mL THF was added dropwise, and the mixture was warmed to room temperature and stirred overnight. The reaction mixture was quenched with aqueous ammonium chloride, partitioned between ethyl acetate and sodium carbonate, and the aqueous layer extracted with ethyl acetate. The combined organic layers were washed with brine, dried over magnesium sulfate, filtered, and concentrated. The resulting material was purified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane), and then the Rp diastereomer was isolated by chiral prep HPLC (Chiralpak IA, 20% EtOH in heptane over 30 minutes, 18 mL/min) to provide 49 mg 2-ethylbutyl ((R)-(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy) phosphoryl)-L-alaninate (13%). Assignment of the Rp stereochemistry at phosphorous was made by comparison to compound I-6, which was synthesized from the commercially available Sp phosphoramidate reagent. ^{1}H-NMR (400 MHz, DMSO-D6) δ 7.31 (t, 2H), 7.21 (s, 1H), 7.12 (q, 3H), 6.88 (s, 1H), 6.42 (d, 1H), 6.14 (d, 1H), 6.04 (t, 1H), 4.31 (q, 1H), 4.11 (dd, 1H), 3.93 (td, 2H), 3.84 (s, 1H), 3.75 (d, 1H), 1.59 (s, 3H), 1.43 (t, 1H), 1.29-1.22 (m, 4H), 1.14 (d, 3H), 0.80 (t, 6H). ^{31}P-NMR (162 MHz, DMSO-D6) δ 4.2. UPLC (Basic method, C18) 96.87% purity (M−H=558.11).

Synthesis of 2-Ethylbutyl ((S)-(((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (I-6)



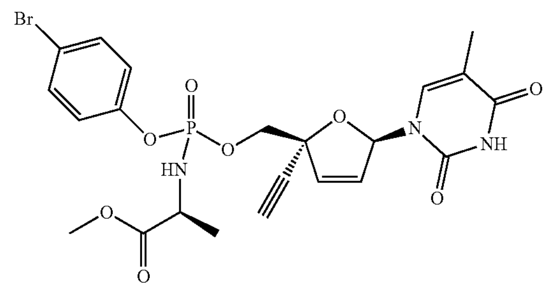
[0657] To a solution of 100 mg (0.4 mmol) of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol in 2 mL THF at 0° C. was added a solution of t-butylmagnesium chloride (0.84 mL of a 1M solution in THF, 0.84 mmol). A solution of commercial Sp phosphoramidate (CAS No. 1911578-98-7, 238 mg, 0.48 mmol) dissolved in 2 mL THF was added dropwise. The solution was warmed to room temperature and stirred for 2 hours. The reaction mixture was quenched with aqueous ammonium chloride, partitioned between ethyl acetate and aqueous sodium bicarbonate, and the aqueous layer was extracted with ethyl acetate. The combined organic layers were washed with brine, dried over magnesium sulfate, filtered, and concentrated. The resulting material was purified by reverse phase purification (Biotage Isolera, 12 g cartridge, eluent: 50% to 100% acetonitrile in water) then repurified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane) giving 124 mg product (55%). ^{1}H-NMR (400 MHz, DMSO-D6) 8 11.37 (s, 1H), 7.32 (t, 2H), 7.20-7.13 (m, 4H), 6.88 (s, 1H), 6.34 (q, 1H), 6.14 (d, 1H), 6.05 (dd, 1H), 4.16 (d, 2H), 3.90 (dq, 2H), 3.81-3.71 (m, 2H), 1.64 (s, 3H), 1.44-1.38 (m, 1H), 1.28-1.20 (m, 4H), 1.16 (d, 3H), 0.79 (t, 6H). ^{31}P-NMR (162 MHz, DMSO-D6) δ 3.4. UPLC (Basic method, C18) 99.75% purity (M−H=558.16).

Synthesis of P-diastereomer 1 of Methyl ((4-Bromophenoxy)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)-L-alaninate (I-7 or I-8)



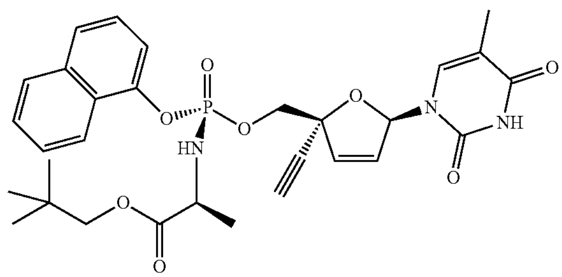
[0658] To a solution of 100 mg (0.4 mmol) of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol in 2 mL THF at 0° C. was added a solution of t-butylmagnesium chloride (0.84 mL of a 1M solution in THF, 0.84 mmol), and the mixture was stirred for 30 min. A solution of methyl ((4-bromophenoxy)(perfluorophenoxy) phosphoryl)-L-alaninate (242 mg, 0.48 mmole, diastereomerically pure; P-diastereomer 1 described in Example 1 above) in 2 mL THF was added dropwise. The solution was warmed to room temperature and stirred for 2 hours. The reaction mixture was quenched with aqueous ammonium chloride, partitioned between ethyl acetate and aqueous sodium bicarbonate, and the aqueous layer was extracted with ethyl acetate. The combined organic layers were washed with brine, dried over magnesium sulfate, filtered, and concentrated. The resulting material was purified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane) then repurified by reverse phase purification (Biotage Isolera, 12 g cartridge, eluent: 50% to 100% acetonitrile in water) giving 112 mg of P-diastereomer 1 of methyl ((4-bromophenoxy)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)-L-alaninate (49%). ^{1}H-NMR (400 MHz, DMSO-D6) δ 7.53 (d, 2H), 7.19-7.13 (m, 3H), 6.87 (t, 1H), 6.37 (dd, 1H), 6.15-6.09 (m, 2H), 4.16 (d, 2H), 3.81-3.73 (m, 2H), 3.53 (s, 3H), 1.64 (d, 3H), 1.16 (d, 3H). 1×NH not observed. ^{31}P-NMR (162 MHz, DMSO-D6) δ 3.6. UPLC (Basic method, C18) 98.17% purity (M+H=567.96).

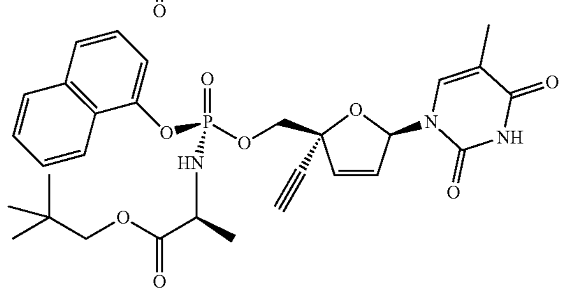
Synthesis of P-diastereomer 2 of Methyl ((4-Bromophenoxy)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)-L-alaninate (I-7 or I-8)



[0659] To a solution of 100 mg (0.4 mmol) of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol in 2 mL THF at 0° C. was added a solution of t-butylmagnesium chloride (0.84 mL of a 1M solution in THF, 0.84 mmol), and the mixture was stirred for 30 min. A solution of methyl ((4-bromophenoxy)(perfluorophenoxy) phosphoryl)-L-alaninate (242 mg, 0.48 mmole, 80:20 ratio of P-diastereomers; P-diastereomer 2 described in Example 1 above) in 2 mL THF was added dropwise. The solution was warmed to room temperature and stirred for 3 hours. The reaction mixture was quenched with aqueous ammonium chloride, partitioned between ethyl acetate and aqueous sodium bicarbonate, and the aqueous layer was extracted with ethyl acetate. The combined organic layers were washed with brine, dried over magnesium sulfate, filtered, and concentrated. The resulting material was purified by reverse phase purification (Biotage Isolera, 12 g cartridge, eluent: 50% to 100% acetonitrile in water) then the Rp diastereomer was isolated by chiral prep HPLC (Chiralpak IA, 20% EtOH in heptane over 30 minutes, 18 mL/min) giving 29 mg of P-diastereomer 2 of methyl ((4-bromophenoxy)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)-L-alaninate (13%). ^{1}H-NMR (400 MHz, DMSO-D6) δ 11.37 (s, 1H), 7.51 (d, 2H), 7.18 (s, 1H), 7.08 (d, 2H), 6.88 (s, 1H), 6.43 (d, 1H), 6.17-6.12 (m, 2H), 4.31 (t, 1H), 4.13 (s, 1H), 3.84 (s, 1H), 3.75 (d, 1H), 3.56 (s, 3H), 1.59 (s, 3H), 1.16 (d, 3H). ^{31}P-NMR (162 MHz, DMSO-D6) δ 4.0. UPLC (Basic method, C18) 99.89% purity (M−H=567.88).

Synthesis of Neopentyl ((R)-(((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(naphthalen-1-yloxy)phosphoryl)-L-alaninate (I-16) and Neopentyl ((S)-(((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(naphthalen-1-yloxy)phosphoryl)-L-alaninate (III-1)

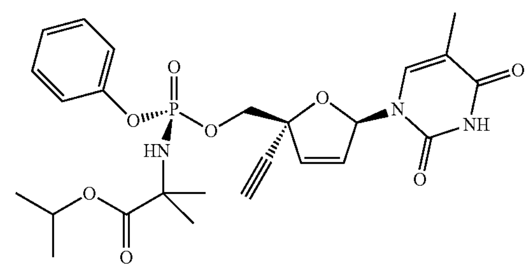


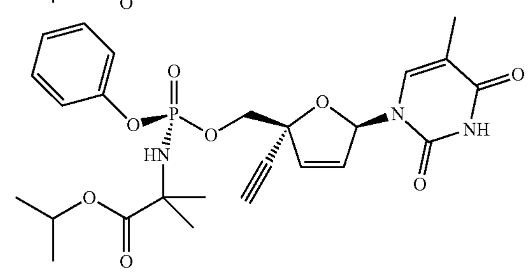


[0660] To a solution of 243 mg (0.98 mmol) of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol in 4 mL THF at 0° C. was added a solution of t-butylmagnesium chloride (2.06 mL of a 1M solution in THF, 2.06 mmol), and the mixture was stirred for 30 min. A solution of neopentyl ((naphthalen-1-yloxy) (perfluorophenoxy)phosphoryl)-L-alaninate (627 mg, 1.18 mmol, mixture of P-diastereomers; prepared as described in Example 1 above) in 4 mL THF was added dropwise. The solution was warmed to room temperature and stirred for 2 hours. The reaction mixture was quenched with aqueous ammonium chloride, partitioned between ethyl acetate and aqueous sodium bicarbonate, and the aqueous layer was extracted with ethyl acetate. The combined organic layers were washed with brine, dried over magnesium sulfate, filtered, and concentrated. The resulting material was purified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane) then the diastereomers were separated by chiral prep HPLC (Chiralpak IA, 20% EtOH in heptane over 30 minutes, 18 mL/min) giving 41 mg of P-diastereomer 1 of neopentyl ((((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl) methoxy)(naphthalen-1-yloxy) phosphoryl)-L-alaninate (7%). ^{1}H-NMR (400 MHz, DMSO-D6) δ 11.37 (s, 1H), 8.09 (s, 1H), 7.93 (s, 1H), 7.71 (d, 1H), 7.55 (q, 2H), 7.41 (t, 2H), 7.22 (s, 1H), 6.89 (s, 1H), 6.35 (d, 1H), 6.26 (t, 1H), 6.15 (d, 1H), 4.24 (s, 2H), 3.86-3.80 (m, 2H), 3.68 (dd, 2H), 1.56 (s, 3H), 1.21 (d, 3H), 0.82 (s, 9H). ^{31}P-NMR (162 MHz, DMSO-D6) δ 4.0. UPLC (Basic method, C18) 99.44% purity (M−H=594.10).

[0661] The other diastereomer was repurified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane) and then repurified by reverse phase purification (Biotage Isolera, 12 g cartridge, eluent: 50% to 100% acetonitrile in water) to give 35 mg of P-diastereomer 2 of neopentyl ((R)-(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(naphthalen-1-yloxy) phosphoryl)-L-alaninate (6%). ^{1}H-NMR (400 MHz, DMSO-D6) δ 8.03 (d, 1H), 7.90 (d, 1H), 7.70 (d, 1H), 7.55-7.48 (m, 2H), 7.39 (q, 2H), 7.22 (s, 1H), 6.90 (s, 1H), 6.46 (dd, 1H), 6.26 (dd, 1H), 6.17 (d, 1H), 4.40 (q, 1H), 4.19 (q, 1H), 3.87 (dd, 2H), 3.79 (d, 1H), 3.65 (d, 1H), 1.41 (s, 3H), 1.22 (d, 3H), 0.84 (s, 9H). ^{31}P-NMR (162 MHz, DMSO-D6) δ 4.1. UPLC (Basic method, C18) 98.65% purity (M−H=594.11).

Synthesis of Isopropyl 2-{[(R)-[(2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3H-pyrimidin-1-yl)-5H-furan-2-yl]methoxy(phenoxy)phosphoryl]amino}-2-methylpropanoate (II-22) and Isopropyl 2-{[(S)-[(2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3H-pyrimidin-1-yl)-5H-furan-2-yl]methoxy(phenoxy) phosphoryl]amino}-2-methylpropanoate (II-23)





[0662] To a stirred solution of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol (70 mg, 0.28 mmol) in anhydrous THF (5 mL) was added t-BuMgCl (0.5 mL, 0.5 mmol, 1.0 M in TIF) dropwise at 0° C. under nitrogen atmosphere. The resulting mixture was stirred for 30 min at 0° C., and then a solution of isopropyl 2-methyl-2-{[perfluorophenoxy(phenoxy)phosphoryl]amino}propanoate (130 mg, 0.28 mmol) in THF (5 mL) was added dropwise at 0° C. The resulting mixture was warmed to room temperature and stirred for 2 h. The reaction was quenched with MeOH and concentrated. The residue was purified by Prep-TLC (CH_2Cl_2/MeOH=10/1) to afford isopropyl 2-{[(2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3H-pyrimidin-1-yl)-5H-furan-2-yl]methoxy(phenoxy) phosphoryl]amino}-2-methylpropanoate (110 mg, 0.21 mmol, 73%) as a white solid. LC-MS (ES, m/z): 532 $(M+H^{+})$.

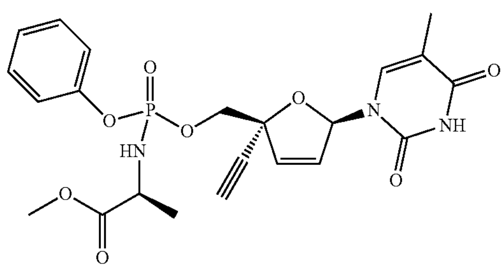
[0663] The mixture of P-diastereomers of isopropyl 2-{[(2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3H-pyrimidin-

1-yl)-5H-furan-2-yl]methoxy(phenoxy)phosphoryl]amino}-2-methylpropanoate (80 mg, 0.15 mmol) was separated by Chiral-Prep-HPLC with the following conditions (Column: CHIRALPAK IE, 2*25 cm, 5 m; Mobile Phase A: Hex (0.5% 2M NH_3-MeOH), Mobile Phase B: EtOH; Flow rate: 20 mL/min; Gradient: 50% B to 50% B in 19 min; Wave Length: 220/254 nm; RT1(min): 12.667; RT2(min): 16.654) to afford P-diastereomer 1 of isopropyl 2-{[(2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3H-pyrimidin-1-yl)-5H-furan-2-yl]methoxy(phenoxy)phosphoryl]amino}-2-methylpropanoate (9.0 mg, 11.3%, first peak) as a white solid and P-diastereomer 2 of isopropyl 2-{[(2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3H-pyrimidin-1-yl)-5H-furan-2-yl]methoxy(phenoxy)phosphoryl]amino}-2-methylpropanoate (10.0 mg, 12%, second peak) as a white solid. Analysis of the compounds by LC/MS was conducted under the following conditions: Column: Kinetex EVO C18, 30*3.0 mm, 2.6 μm; Mobile Phase A: Water with 5 mM NH_4HCO_3, Mobile Phase B: ACN; Flow rate: 1.50 mL/min; Gradient: 10% B to 95% B in 1.2 min, 95% B to 95% B in 1.78 min, 95% B to 10% B in 1.83 min; Wave Length: 254/220 nm; RT(min): 0.980.

[0664] P-diastereomer 1 LC-MS (ES, m/z): 532 ($M+H^+$); 98.7% purity. ^{1}H NMR (400 MHz, DMSO-d_6) δ 11.41 (s, 1H), 7.39-7.30 (m, 2H), 7.22 (q, J=1.1 Hz, 1H), 7.19-7.11 (m, 3H), 6.91 (dd, J=2.1, 1.4 Hz, 1H), 6.47 (dd, J=5.8, 2.1 Hz, 1H), 6.18 (dd, J=5.8, 1.4 Hz, 1H), 5.87 (d, J=10.4 Hz, 1H), 4.85 (hept, J=6.2 Hz, 1H), 4.35 (dd, J=11.2, 6.1 Hz, 1H), 4.17 (dd, J=11.2, 4.9 Hz, 1H), 3.87 (s, 1H), 1.59 (d, J=1.2 Hz, 3H), 1.36-1.31 (m, 3H), 1.27 (s, 3H), 1.16 (t, J=6.6 Hz, 6H).

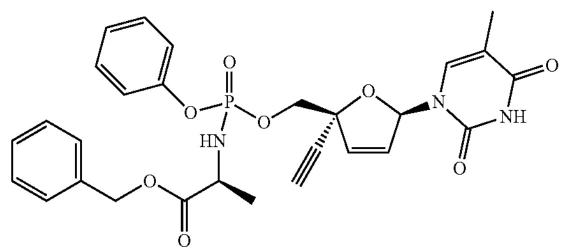
[0665] P-diastereomer 2 LC-MS (ES, m/z): 532 ($M+H^+$); 98.3% purity (RT=0.984 min). ^{1}H NMR (400 MHz, DMSO-d_6) δ 11.41 (s, 1H), 7.41-7.32 (m, 2H), 7.26 (t, J=1.2 Hz, 1H), 7.24-7.13 (m, 3H), 6.91 (dd, J=2.1, 1.4 Hz, 1H), 6.39 (dd, J=5.8, 2.1 Hz, 1H), 6.17 (dd, J=5.8, 1.4 Hz, 1H), 5.82 (d, J=10.2 Hz, 1H), 4.84 (hept, J=6.2 Hz, 1H), 4.28 (dd, J=11.1, 5.1 Hz, 1H), 4.20 (dd, J=11.1, 4.9 Hz, 1H), 3.85 (s, 1H), 2.53 (s, 3H), 1.65 (d, J=1.2 Hz, 3H), 1.34-1.29 (m, 3H), 1.26 (s, 3H), 1.15 (dd, J=6.2, 2.3 Hz, 6H).

Synthesis of Methyl ((((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (Mixture of III-2 and III-3)



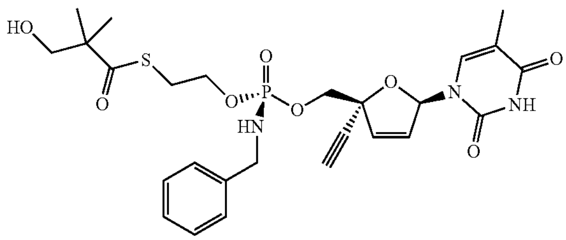
[0666] Reaction of 200 mg of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol with methyl ((perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate (mixture of P-diastereomers; prepared as described in Example 1 above) was conducted according to the general procedure described above. The resulting material was purified by normal phase purification (Biotage Isolera, 12 g cartridge, eluent: 0-10% methanol:dichloromethane) giving 311 mg of methyl ((((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (77%) as a mixture of P diastereomers. ^{1}H-NMR (400 MHz, DMSO-D6) δ 7.32 (q, 4H), 7.20-7.09 (m, 8H), 6.89-6.87 (m, 2H), 6.43 (dd, 1H), 6.35 (dd, 1H), 6.14 (dd, 2H), 6.07 (s, 2H), 4.27-4.36 (1H), 4.15 (dd, 3H), 3.82 (d, 2H), 3.69-3.79 (2H), 3.55 (d, 6H), 3.25 (s, 2H), 1.64 (d, 3H), 1.59 (d, 3H), 1.15 (dd, 6H). 24H for each diastereomer. ^{31}P-NMR (162 MHz, DMSO-D6) δ 4.1, 3.5. UPLC (Basic method, C18): 44.55% faster eluting diastereomer (M−H=488.0) and 53.58% slower eluting diastereomer (M−H=488.0).

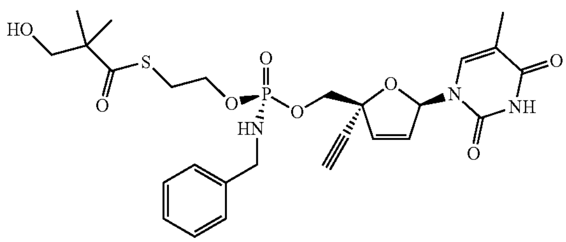
Synthesis of Benzyl ((((2R,5R)-2-Ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (Mixture of III-4 and III-5)



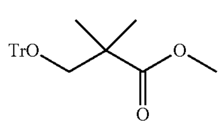
[0667] To a solution of 53 mg (0.21 mmol) of (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol in 1 mL THF at 0° C. was added a solution of t-butylmagnesium chloride (0.45 mL of a 1M solution in THF, 0.45 mmol), and the mixture was stirred for 30 min. A solution of benzyl ((perfluorophenoxy)(phenoxy)phosphoryl)-L-alaninate (107 mg, 0.213 mmol, mixture of P-diastereomers; prepared as described in Example 1 above) in 0.5 mL THF was added dropwise. The solution was warmed to room temperature and stirred for 2 hours. The reaction mixture was quenched with aqueous ammonium chloride, partitioned between ethyl acetate and aqueous sodium bicarbonate, and the aqueous layer was extracted with ethyl acetate. The combined organic layers were washed with brine, dried over sodium sulfate, filtered, and concentrated. The resulting material was purified by reverse phase purification (Biotage Isolera, 22 g cartridge, eluent: 10% to 80% acetonitrile in water) giving 58 mg of benzyl ((((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)(phenoxy)phosphoryl)-L-alaninate (48%) as a mixture of P diastereomers. ^{1}H NMR (400 MHz, MeOD-d4): δ 1.32 (dd, 3H), 1.73 (d, 3H), 3.24 (s, 1H), 3.90-3.95 (m, 1H), 4.23-4.32 (m, 2H), 5.12 (s, 2H), 6.05 (dd, 1H), 6.24 (dd, 1H), 7.02-7.03 (m, 1H), 7.17-7.19 (m, 3H), 7.30-7.36 (m, 8H). 2H exchangeable not observed. UPLC: (Acidic method, C18) 97.5% purity (M+H=566.2).

Example 3—Synthesis of S-(2-(((R)-(Benzylamino)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)oxy) ethyl) 3-hydroxy-2,2-dimethylpropanethioate (II-1) and S-(2-(((S)-(Benzylamino)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)oxy)ethyl) 3-hydroxy-2,2-dimethylpropanethioate (II-2)



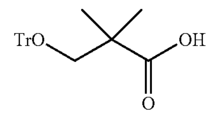


Part I—Synthesis of Methyl 2,2-Dimethyl-3-(trityloxy)propanoate



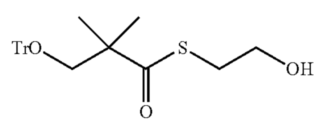
[0668] Triethylamine (3.75 mL, 26.9 mmol) and 4-(dimethylamino)pyridine (219 mg, 1.79 mmol) were added to a stirred solution of methyl 2,2-dimethylhydroxypropionate (2.4 mL, 18.8 mmol) and trityl chloride (5 g, 17.9 mmol) in dichloromethane (25 mL) at room temperature under argon. The resulting solution was stirred at room temperature for 3 days. A further portion of methyl 2,2-dimethylhydroxypropionate (0.57 mL, 4.48 mmol) and triethylamine (1.25 mL, 8.97 mmol) was added, and the resulting mixture was stirred overnight at room temperature. The resulting mixture was diluted with dichloromethane (20 mL) and water (50 mL). The layers were separated, and the dichloromethane layer was washed with water (25 mL), dried over magnesium sulfate, filtered and concentrated under reduced pressure to afford a golden oil (8.1 g). Purification by flash column chromatography on silica with heptane-ethyl acetate (4:1) as eluent gave methyl 2,2-dimethyl-3-(trityloxy)propanoate (6.23 g, 93%) as a pale straw coloured oil. ^{1}H NMR ($CDCl_3$, 400 MHz): 7.42-7.20 (m, 15H), 3.66 (s, 3H), 3.11 (s, 2H), 1.17 (s, 6H). UPLC (Basic method, C18): 84.98% purity, Major ion observed: CPh_3^+ 243.13.

Part II—Synthesis of 2,2-Dimethyl-3-(trityloxy)propanoic Acid



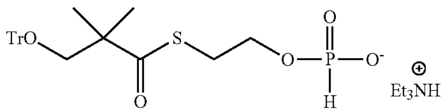
[0669] To a stirred solution of methyl 2,2-dimethyl-3-(trityloxy)propanoate (6.23 g, 16.6 mmol) in 1,2-dimethoxyethane (25 mL) at room temperature was added a solution of sodium hydroxide (1.33 g) in water (20 mL). The resulting mixture was heated at 80° C. for 18 h then cooled to room temperature. The mixture was diluted with dichloromethane (100 mL) and citric acid (100 mL of a 10% aqueous solution), and the layers were separated. The dichloromethane layer was dried over magnesium sulfate, filtered, and concentrated under reduced pressure. The resulting oil was suspended in heptane-ethyl acetate (50 mL of a 4:1 mixture) and concentrated at 50° C. to ~1/4 volume. The resulting slurry was diluted with heptane (50 mL) and stirred at room temperature for 30 min. The solid was collected by filtration, washed with heptane (25 mL), and dried by rotary evaporation to give 2,2-dimethyl-3-(trityloxy)propanoic acid (4.63 g, 72% over two steps) as a white solid. ^{1}H NMR ($CDCl_3$, 400 MHz): 7.42 (d, 6H), 7.30-7.20 (m, 9H), 3.16 (s, 2H), 1.21 (s, 6H).

Part III—Synthesis of S-(2-Hydroxyethyl) 2,2-Dimethyl-3-(trityloxy)propanethioate



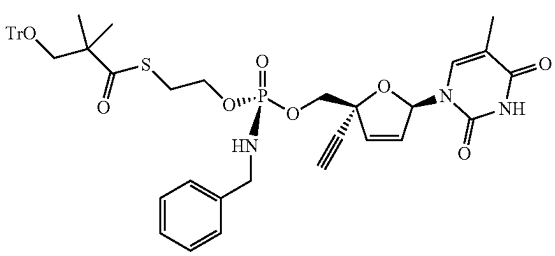
[0670] 1,1'-Carbonyldimidazole (2.29 g, 14.1 mmol) was added to a stirred solution of 2,2-dimethyl-3-(trityloxy)propanoic acid (4.63 g, 12.8 mmol) in dichloromethane (30 mL) at room temperature under argon. The resulting solution was stirred at room temperature for 90 min, and then 2-mercaptoethanol (1.08 mL, 15.4 mmol) was added. The resulting solution was stirred at room temperature for 2 h. then quenched by addition of water (100 mL). The mixture was extracted with dichloromethane (70 mL). The dichloromethane layer was washed with sodium carbonate (50 mL of a saturated aqueous solution), dried over magnesium sulfate, filtered and concentrated under reduced pressure to give S-(2-hydroxyethyl) 2,2-dimethyl-3-(trityloxy)propanethioate (5.44 g, 100%) as a colourless oil. ^{1}H NMR ($CDCl_3$, 400 MHz): 7.41 (d, 6H), 7.31-7.20 (m, 9H), 3.73 (t, 2H), 3.17 (s, 2H), 3.08 (t, 2H), 1.80 (br s, 1H), 1.23 (s, 6H). UPLC (Basic method, C18): 97.56% purity, molecular ion not observed.

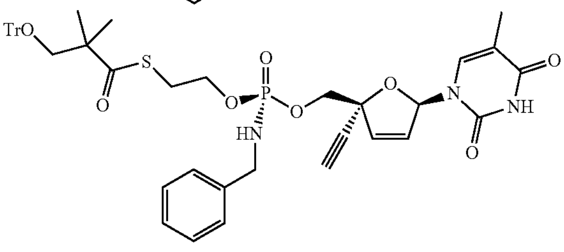
Part IV—Synthesis of Triethylammonium 2-((2,2-Dimethyl-3-(trityloxy)propanoyl)thio) ethyl phosphonate



[0671] Phosphonic acid (390 mg, 4.76 mmol) was added to a stirred solution of S-(2-hydroxyethyl) 2,2-dimethyl-3-(trityloxy)propanethioate (1 g, 2.38 mmol) in dichloromethane at room temperature under argon. Triethylamine (0.71 mL, 5.11 mmol) was added and the resulting mixture stirred at room temperature for 10 min, then trimethylacetyl chloride (0.44 mL, 3.57 mmol) was added. An exotherm was noted but not measured. The mixture was cooled with an ice-water bath, then stirred at room temperature for 1 h. Triethylamine (0.66 mL, 4.76 mmol) was added, and the resulting suspension was stirred at room temperature for 10 min. Water (10 mL) and dichloromethane (10 mL) were added, and the layers were separated. The dichloromethane layer was washed with water (10 mL). The aqueous layer was back-extracted with dichloromethane (5 mL). The dichloromethane layers were combined and concentrated at <50° C., azeotroping with acetonitrile (2×20 mL), to give triethylammonium 2-((2,2-dimethyl-3-(trityloxy)propanoyl) thio)ethyl phosphonate (1.29 g, 92%) as a colourless oil. ^{1}H NMR ($CDCl_3$, 400 MHz): 7.41 (d, 6H), 7.30-7.25 (m, 6H), 7.21 (t, 3H), 6.83 (d, 1H, P—H, J=636 Hz), 3.96 (q, 2H), 3.16 (t, 2H), 3.14 (s, 2H), 3.07-3.00 (m, 6H), 1.29 (t, 9H), 1.18 (s, 6H). ^{31}P NMR ($CDCl_3$, 162 MHz): 4.91 (s). UPLC (Basic method, C18): 88.42% purity (M^-=483.0).

Part V—Synthesis of (R) and (S) Diastereomers at Phosphorous of S-(2-(((Benzylamino)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)oxy)ethyl) 2,2-dimethyl-3-(trityloxy) propanethioate





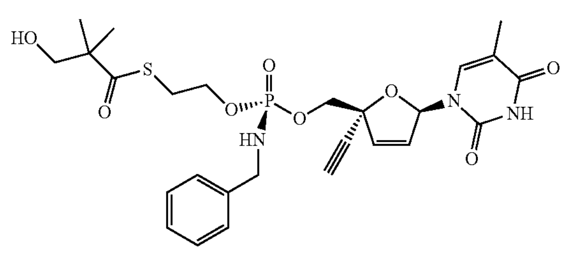
[0672] N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride (1.18 g, 6.16 mmol) was added to a stirred solution of triethylammonium 2-((2,2-dimethyl-3-(trityloxy)propanoyl)thio)ethyl phosphonate (1.27 g, 2.18 mmol), (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol (300 mg, 1.21 mmol), and pyridine (0.98 mL, 12.1 mmol) in acetonitrile (6.5 mL) at room temperature under argon. The resulting solution was stirred at room temperature for 2 h.

[0673] Benzylamine (0.82 mL, 7.49 mmol) and carbon tetrachloride (0.54 mL, 5.56 mmol) were added and the resulting suspension was stirred at room temperature for 1 h. The mixture was diluted with ethyl acetate (50 mL) and water (50 mL), and the layers were separated. The ethyl acetate layer was washed with citric acid (25 mL of a 10% aqueous solution), sodium hydrogen carbonate (25 mL of a saturated aqueous solution), and brine (25 mL), dried over magnesium sulfate, filtered, and concentrated under reduced pressure to give the crude product (1.3 g) as a yellow oil. Purification by flash column chromatography on silica with dichloromethane-methanol (2.5%) as eluent gave S-(2-(((benzylamino)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)oxy)ethyl) 2,2-dimethyl-3-(trityloxy)propanethioate P-diastereomer 1 (faster eluting diastereomer, 213 mg, 21%) as a pale yellow residue and P-diastereomer 2 (slower eluting diastereomer, 183 mg, 18%) as a pale yellow residue. 34 mg (3%) of mixed fractions were also obtained.

[0674] P-diastereomer 1 (faster eluting diastereomer): ^{1}H NMR ($CDCl_3$, 400 MHz): 7.88 (br s, 1H), 7.39 (d, 6H), 7.30-7.25 (m, 14H), 7.11 (dd, 1H), 6.17 (dd, 1H), 5.82 (dd, 1H), 4.30 (dd, 1H), 4.11-4.00 (m, 6H), 3.20-3.05 (m, 2H), 3.14 (s, 2H), 2.99-2.91 (m, 1H), 2.61 (s, 1H), 1.87 (d, 3H), 1.19 (s, 6H). ^{31}P NMR ($CDCl_3$, 162 MHz): 9.23 (s). UPLC (Basic method, C18): 98.39% purity (M^-=818.11).

[0675] P-diastereomer 2 (slower eluting diastereomer): ^{1}H NMR ($CDCl_3$, 400 MHz): 7.88 (br s, 1H), 7.39 (d, 6H), 7.30-7.25 (m, 14H), 7.10 (dd, 1H), 6.20 (dd, 1H), 5.88 (dd, 1H), 4.23 (dd, 1H), 4.16-4.02 (m, 6H), 3.17-3.12 (m, 2H), 3.15 (s, 2H), 2.99-2.91 (m, 1H), 2.59 (s, 1H), 1.84 (d, 3H), 1.19 (s, 6H). ^{31}P NMR ($CDCl_3$, 162 MHz): 8.50 (s). UPLC (Basic method, C18): 99.62% purity (M^-=818.13).

Part VI—Synthesis of S-(2-(((R)-(Benzylamino)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)oxy) ethyl) 3-hydroxy-2,2-dimethylpropanethioate (II-1) and S-(2-(((S)-(Benzylamino)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)oxy)ethyl) 3-hydroxy-2,2-dimethylpropanethioate (II-2)



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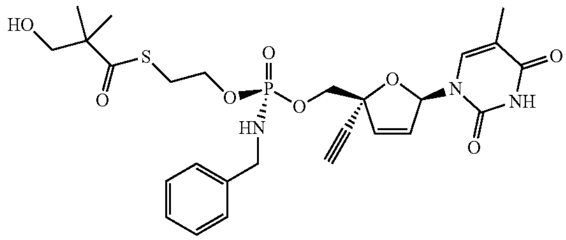
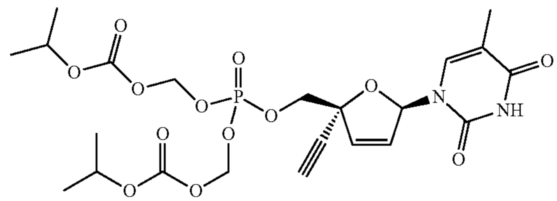


[0676] Each phosphorous diastereomer from Part V above was individually treated as follows. Hydrochloric acid (0.13 mL of a 4.0 M solution in 1,4-dioxane, 0.52 mmol) was added to a stirred solution of S-(2-(((benzylamino)(((2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-yl)methoxy)phosphoryl)oxy)ethyl) 2,2-dimethyl-3-(trityloxy)propanethioate (P-diastereomer 1: 213 mg, 0.26 mmol; P-diastereomer 2: 183 mg, 0.22 mmol) in industrial ethanol (4.5 mL) at 0° C. The mixture was stirred at 0° C. for 1 h, then at room temperature over the weekend. The mixture was diluted with dichloromethane (25 mL), then poured into a stirred mixture of dichloromethane (12 mL) and sodium hydrogen carbonate (25 mL of a saturated aqueous solution) and stirred at room temperature for 10 min. The layers were separated, and the aqueous layer was extracted with dichloromethane (2×12 mL). The combined dichloromethane layers were dried over magnesium sulfate, filtered, and concentrated under reduced pressure to give the crude product (250 mg) as a pale yellow residue.

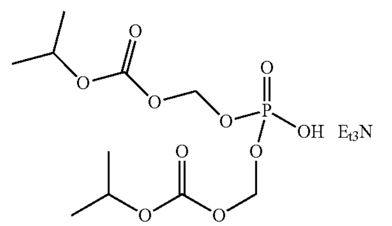
[0677] The crude product from the reaction of P-diastereomer 1 was purified by flash column chromatography on silica with dichloromethane-methanol (5-7%) as eluent to afford a pale yellow gum, which was further purified by Biotage Isolera reverse phase (12 g Silicycle C18, eluting with acetonitrile-water: 15-45%). Product-containing fractions were combined and freeze dried to give P-diastereomer 1 (85 mg, 57%) as a white solid. ^{1}H NMR (DMSO-d$_6$, 400 MHz): 11.37 (br s, 1H), 7.30-7.18 (m, 6H), 6.86 (br s, 1H), 6.33 (dd, 1H), 6.10 (dd, 1H), 5.71 (dt, 1H), 4.88 (t, 1H), 4.07 (dd, 1H), 3.92-3.80 (m, 5H), 3.78 (s, 1H), 3.39 (s, 2H), 2.99 (t, 2H), 1.69 (s, 3H), 1.07 (s, 6H). ^{31}P NMR (CDCl$_3$, 162 MHz): 10.05 (s). UPLC (Acidic method, C18): 100% purity ([M−$C_7H_{14}O_2S$]$^-$=416.05).

[0678] The crude product from the reaction of P-diastereomer 2 was purified by Biotage Isolera reverse phase (12 g Silicycle C18, eluting with acetonitrile-water: 15-45%). Product-containing fractions were combined and freeze dried to give P-diastereomer 2 (68 mg, 53%) as a white solid. ^{1}H NMR (DMSO-d$_6$, 400 MHz): 11.36 (br s, 1H), 7.28-7.17 (m, 6H), 6.86 (br s, 1H), 6.34 (dd, 1H), 6.11 (dd, 1H), 5.67 (dt, 1H), 4.90 (t, 1H), 4.04-3.82 (m, 6H), 3.78 (s, 1H), 3.40 (s, 2H), 3.01 (t, 2H), 1.68 (s, 3H), 1.08 (s, 6H). ^{31}P NMR (CDCl$_3$, 162 MHz): 9.78 (s). UPLC (Acidic method, C18): 99.02% purity ([M−$C_7H_{14}O_2S$]$^-$=416.05).

Example 4—Synthesis of 1-[(2R,5R)-5-[({di[(isopropoxycarbonyl)oxy]methoxyphosphoryl}oxy)methyl]-5-ethynyl-2H-furan-2-yl]-5-methyl-3H-pyrimidine-2,4-dione (IV-1)

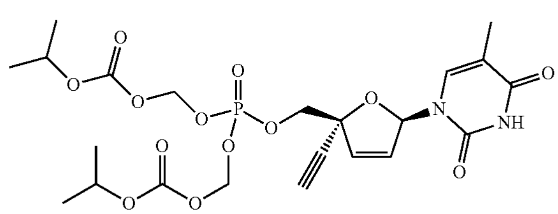


Part I—Synthesis of Di[(isopropoxycarbonyl)oxy]methoxyphosphinic Acid Triethylamine Salt



[0679] Trimethyl phosphate (3 g, 21.4 mmol) and chloromethyl isopropyl carbonate (9.8 g, 64.2 mmol) were dissolved in acetone (100 mL), and sodium iodide (10.6 g, 70.7 mmol) was added with stirring. The reaction was fitted with a reflux condenser and heated to 55° C. for 4 h. When the reaction was complete, the solid which formed (NaCl) during the reaction was filtered off and discarded. Acetone was then removed by rotary evaporation, and the residue was dissolved in 150 mL water. The water layer was washed with ethyl acetate (3×50 mL). The organic washes were discarded and the aqueous layer collected. The aqueous phase was acidified using hydrochloric acid (1N aq). After acidification the desired product was extracted with ethyl acetate (3×100 mL). The organic extracts were combined and evaporated to obtained the di[(isopropoxycarbonyl)oxy]methoxyphosphinic acid (2.0 g, 6.06 mmol, 28%). The free acid was re-dissolved in water and TEA was added until basic to form TEA salt. The solution was freeze dried for storage at −4° C. LC-MS (ES, m/z): 331 (M+H$^+$).

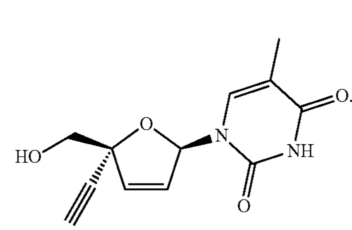
Part II—Synthesis of 1-[(2R,5R)-5-[({di[(isopropoxycarbonyl)oxy]methoxyphosphoryl}oxy)methyl]-5-ethynyl-2H-furan-2-yl]-5-methyl-3H-pyrimidine-2,4-dione (IV-1)



[0680] Into a round-bottom vial, di[(isopropoxycarbonyl)oxy]methoxyphosphinic acid (239 mg, 0.7 mmol) and (2R,5R)-2-ethynyl-5-(5-methyl-2,4-dioxo-3,4-dihydropyrimidin-1(2H)-yl)-2,5-dihydrofuran-2-methanol (60 mg, 0.3 mmol) were suspended in pyridine (2 mL). 1-methyl-1H-imidazole (298 mg, 3.6 mmol) was added, and the mixture was sonicated for 30 s. The mixture was then evaporated to a residue by rotary evaporation for 20 minutes. The residue was then re-suspended in acetonitrile (3 mL) and sonicated for 30 s. The mixture was fitted with a stir bar and stirring was commenced. Bis(2-oxo-1,3-oxazolidin-3-yl)phosphinoyl chloride (338 mg, 1.3 mmol) was then added, and the mixture was stirred at room temperature for 3 h. The reaction was monitored by LCMS. When the reaction was complete, water (3 mL) was added, and the resulting mixture was extracted with ethyl acetate (3×5 mL). The combined organic layers were washed with saturated NaCl (3×5 mL), dried over anhydrous Na_2SO_4. After filtration, the filtrate was concentrated under reduced pressure. The residue was purified by Prep-TLC with petroleum ether/ethyl acetate (1/5) to afford 90 mg crude product. The crude product was purified by Prep-HPLC with the following conditions (Column: XBridge Prep OBD C18 Column, 30*150 mm, 5 μm; Mobile Phase A: Water (10 mmol/L NH_4HCO_3), Mobile Phase B: ACN; Flow rate: 60 mL/min; Gradient: 15% B to 60% B in 10 min, 60% B; Wave Length: 254/220 nm). The product-containing fractions were partially evaporated under reduced pressure and lyophilized overnight to afford 1-[(2R,5R)-5-[({di[(isopropoxycarbonyl)oxy]methoxyphosphoryl}oxy)methyl]-5-ethynyl-2H-furan-2-yl]-5-methyl-3H-pyrimidine-2,4-dione (33.3 mg, 0.06 mmol, 24.34%) as an off-white solid. LC-MS (ES, m/z): 561 ($M+H^+$). 99.1% purity. LCMS conditions: Column: Shim-pack ScepterC18 Column, 33*3.0 mm, 3 μm; Mobile Phase A: Water/5 mM NH_4HCO_3, Mobile Phase B: ACN; Flow rate: 1.5000 mL/min; Gradient: 10% B to 95% B in 1.2 min, 95% B in 1.78 min, 95% B to 10% B in 1.83 min; Wave Length: 254/220 nm). ^1H NMR (400 MHz, DMSO-d_6) δ 11.42 (s, 1H), 7.13 (d, J=1.4 Hz, 1H), 6.93-6.88 (m, 1H), 6.43 (dd, J=5.8, 2.1 Hz, 1H), 6.18 (d, J=5.8, 1.4 Hz, 1H), 5.66-5.55 (m, 4H), 4.82 (m, 2H), 4.30 (dd, J=11.0, 5.7 Hz, 1H), 4.17 (dd, J=11.0, 5.8 Hz, 1H), 3.88 (s, 1H), 1.73 (d, J=1.3 Hz, 3H), 1.25 (ddd, J=6.2, 2.4, 1.2 Hz, 12H).

Example 5—Transient Cellular Assay for Inhibiting LINE1 Reverse Transcriptase

[0681] The following compound (hereinafter "Test Compound") was tested for ability to inhibit LINE1 reverse transcriptase using a transient artificial-intron Cis LINE1 reporter assay:



Additional exemplary compounds were also tested for ability to inhibit LINE1 reverse transcriptase using a transient artificial-intron Cis LINE1 reporter assay. Assay procedures and results are described below.

Part I—Procedure for Transient Artificial-Intron Cis LINE1 Reporter Assay

[0682] Intron-disrupted Firefly luciferase (FLuc) expression cassettes were generated as described by Xie, Y. et al. "Characterization of L1 retrotransposition with high-throughput dual-luciferase assays," *Nucleic Acid Res.* (2011) Vol. 39, No. 3, e16. In addition, the plasmid contained an intact Renilla luciferase (RLuc) expression cassette on the vector backbone, in order to normalize transfection efficiency and measure potential cell toxicity.

[0683] HEK 293 cells were seeded in 96-well plates at 1,000 cells/well in 55 μL and grown for 24 hours. Cells were transfected with FuGeneHD (Promega) following the manufacturer's protocol. Each well received 0.133 ng plasmid, 0.4 μL FuGeneHD reagent, and 4.5 μL GlutaMAX-I-supplemented Opti-MEM I reduced-serum medium (Invitrogen). Cells were simultaneously treated with test compound serially diluted starting at 100 μM in a 3-fold dilution dose response.

[0684] Luminescence was measured using the Dual-Glo Luciferase Assay System (Promega) following the manufacturer's instructions. The ratio between FLuc and RLuc gene expression was used to report LINE1 activity.

Part II—Results

[0685] Experimental results are provided in FIG. **1**, where the Test Compound was found to inhibit FLuc (labeled "FF/RR" in FIG. **1**) with an IC_{50} of 0.06 μM.

[0686] Additional experimental results are provided in Table 6, below. The symbol "***" indicates an IC_{50} less than or equal to 0.05 μM. The symbol "**" indicates an IC_{50} in the range of greater than 0.05 μM to less than or equal to 0.5 μM. The symbol "*" indicates an IC_{50} greater than 0.5 μM.

TABLE 6

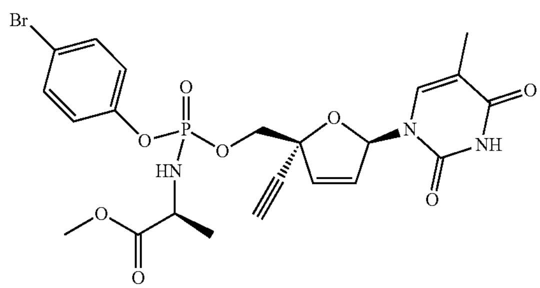
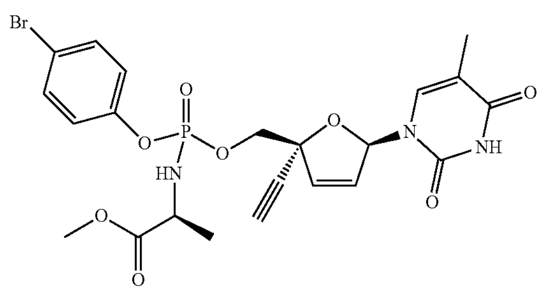
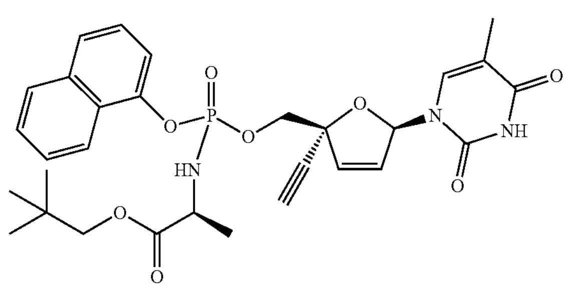
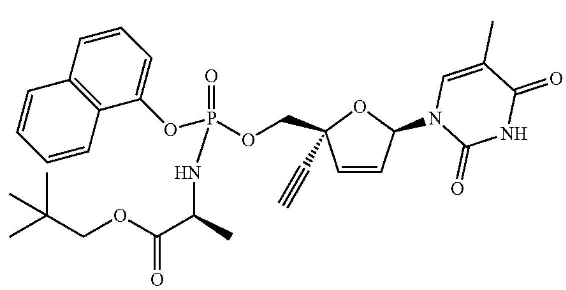
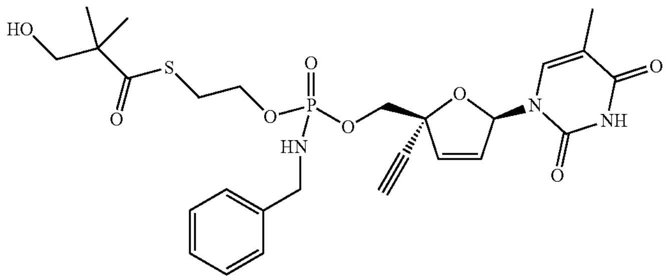
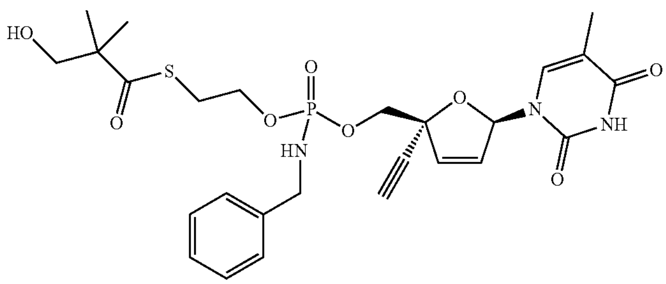
Compound	IC_{50}
I-1	**
I-2	**
I-5	**
I-6	**
 (P-Diastereomer 1 from Example 2; I-7 or I-8)	**
 (P-Diastereomer 2 from Example 2; I-7 or I-8)	**
 (P-Diastereomer 1 from Example 2; I-16 or III-1)	**
 (P-Diastereomer 2 from Example 2; I-16 or III-1)	***

TABLE 6-continued

Compound	IC_{50}
 (P-Diastereomer 1 from Example 3; II-1 or II-2)	**
 (P-Diastereomer 2 from Example 3; II-1 or II-2)	**

Example 6—Stable Cellular Assay for Inhibiting LINE1 Reverse Transcriptase

[0687] Exemplary compounds were tested for ability to inhibit LINE1 reverse transcriptase using a stable artificial-intron Cis LINE1 reporter assay. Assay procedures and results are described below.

Part I—Procedure for Stable Artificial-Intron Cis LINE1 Reporter Assay

[0688] A stable HeLa Tet-On 3G (Takara, cat no 631183) cell line expressing a bi-directional inducible LINE1 construct was generated as described in Xie, Y. et al. "Cell division promotes efficient retrotransposition in a stable L1 reporter cell line," Mobile DNA (2013) 4:10. Single cell clones were screened for high Luciferase expression and the highest expression Firefly expressing clone was chosen for compound testing.

[0689] Test compounds were serially diluted in DMSO and spotted in 96-well plates. Subsequently the HeLa L1 artificial-intron reporter cells were plated into the compound-containing wells (8,000 cells/well), and the cells were induced for reporter expression with doxycycline (Sigma cat no D9891) at a final concentration of 500 ng/mL. Luminescence was measured 72 h after plating using the Dual-Glo Luciferase Assay System (Promega cat no E2940) following the manufacturer's instructions. The Firefly Luciferase activity was used to report LINE1 activity.

Part II—Results

[0690] Experimental results are provided in Table 7, below. The symbol "***" indicates an IC_{50} less than or equal to 0.05 μM. The symbol "**" indicates an IC_{50} in the range of greater than 0.05 μM to less than or equal to 0.5 μM. The symbol "*" indicates an IC_{50} greater than 0.5 μM.

TABLE 7

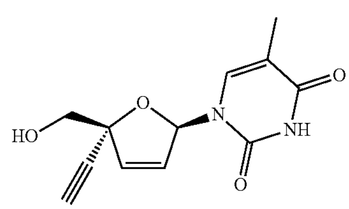
Compound	IC_{50}
	**
I-1	*
I-2	*

TABLE 7-continued

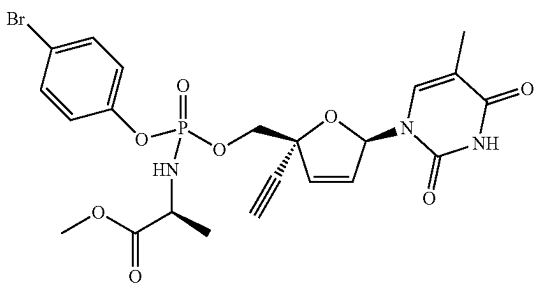
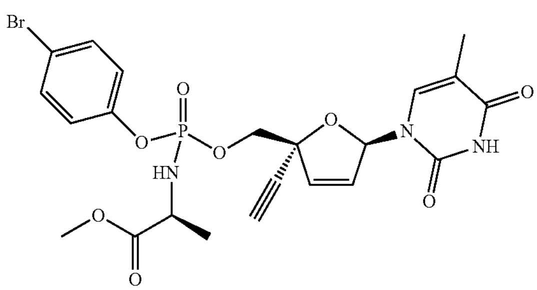
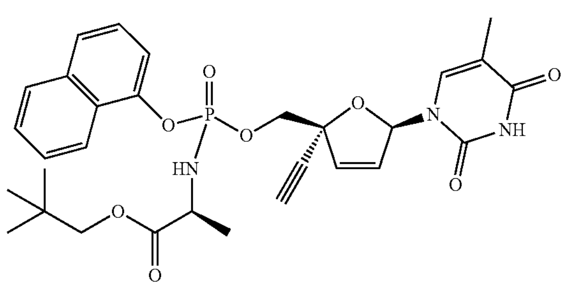
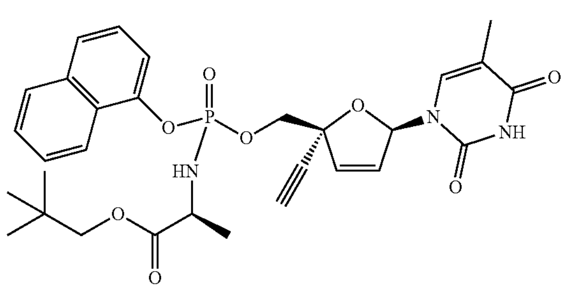
Compound	IC_{50}
I-5	**
I-6	**
 (P-Diastereomer 1 from Example 2; I-7 or I-8)	**
 (P-Diastereomer 2 from Example 2; I-7 or I-8)	**
 (P-Diastereomer 1 from Example 2; I-16 or III-1)	**
 (P-Diastereomer 2 from Example 2; I-16 or III-1)	**

TABLE 7-continued

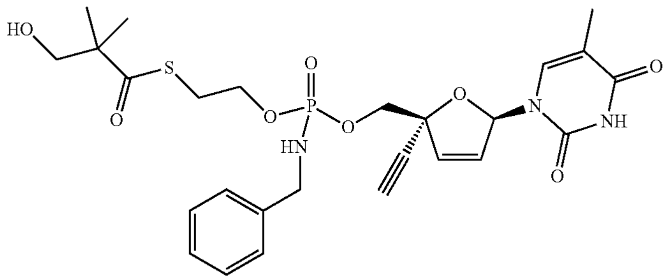
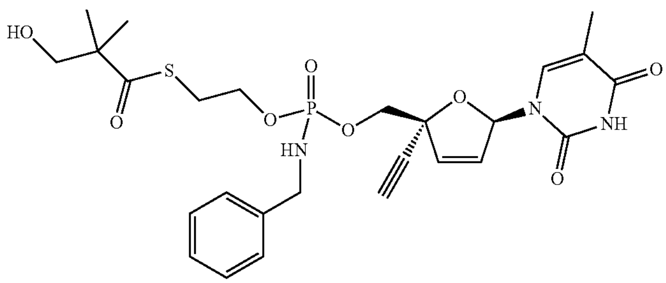
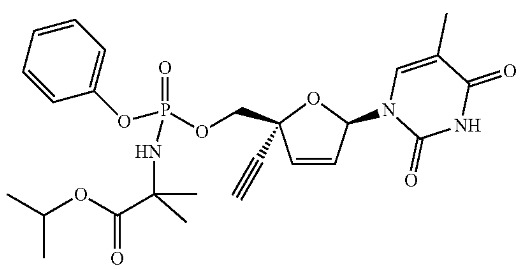
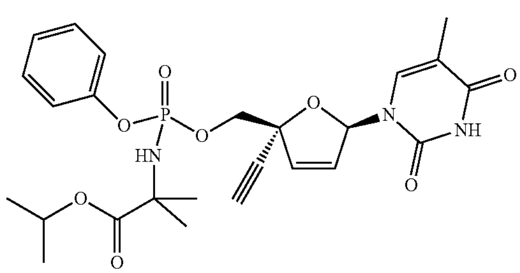
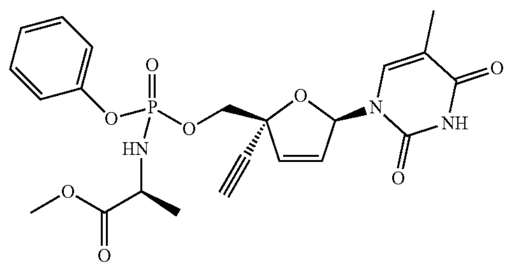
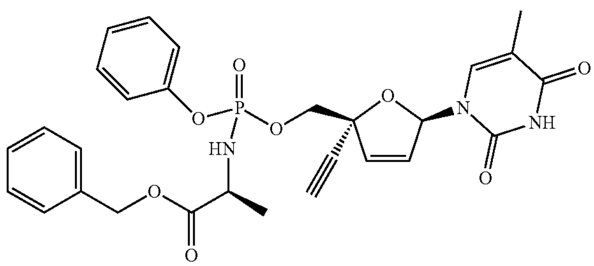
Compound	IC50
 (P-Diastereomer 1 from Example 3; II-1 or II-2)	**
 (P-Diastereomer 2 from Example 3; II-1 or II-2)	**
 (P-Diastereomer 1 from Example 2; II-22 or II-23)	*
 (P-Diastereomer 2 from Example 2; II-22 or II-23)	*

TABLE 7-continued

Compound	IC_{50}
 (Mixture of III-2 and III-3 from Example 3)	*
 (Mixture of III-4 and III-5 from Example 3)	**
IV-1	***

Example 7—Biochemical Assay for Inhibiting LINE1 Reverse Transcriptase

[0691] Exemplary compounds were tested for ability to inhibit LINE1 reverse transcriptase using a homogeneous time-resolved fluorescence (HTRF) assay. Assay procedures and results are described below.

Part I—Procedure for Homogeneous Time-Resolved Fluorescence LINE1 RT Assay

[0692] The LINE1 reverse transcriptase homogeneous time-resolved fluorescence (HTRF) assay was performed with recombinant MBP-tagged LINE1 protein (238-1061) (generated and purified according to procedures in Dai L. et al. BMC Biochemistry 2011; 12:18) in a 384-well format. Test compound was serially diluted in DMSO and further diluted in the assay buffer (50 mM Tris-HCl, 50 mM KCl, 10 mM $MgCl_2$, 10 mM DTT, pH 8.1) to achieve a final DMSO concentration of 1%. The serially diluted compound was mixed with 64 ng/well of LINE1 enzyme, 5 nM of pre-annealed template/biotin-primer pair (synthesized at Generay Biotechnology), 10 nM of Fluorescein-12-dATP fluorescent probe (Perkin Elmer), and 1 μM dGTP/dCTP/dTTP (Thermo Fisher Scientific) in the assay buffer. The template/biotin-primer sequences were as follows:

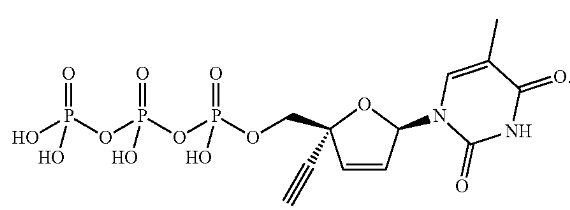
```
                                     (SEQ ID NO: 1)
5' to 3' GUAACUAGAGAUCCCUCAGACCCUUUUAGUCAGAAU

                                     (SEQ ID NO: 2)
5' to 3' Biotin-TTCTGACTAAAAGGGTCTGAGGGAT.
```

[0693] After incubating at 25° C. for 60 minutes, the detection reagent (20 mM EDTA with streptavidin-terbium cryptate, Cisbio Bioassay) in the PPI buffer (Cisbio Bioassay) was added, and the mixture was incubated at 25° C. for 30 minutes. At the end of the incubation, fluorescence was read at ex/em=337/485 nm and ex/em=337/520 nm on an Envision 2104 plate reader (Perkin Elmer). The fluorescence ratio at 520/485 nm was used for the calculation. Percent inhibition was calculated with the DMSO sample as 0% inhibition and no enzyme as 100% inhibition. The IC_{50} was calculated by fitting the compound dose inhibition curve with a 4-parameter non-linear regression equation.

Part II—Results

[0694] The tetra(triethylammonium) salt of the following compound was found to inhibit LINE1 reverse transcriptase with an IC_{50} of 0.070 uM:



Example 8—Biochemical Assay for Inhibiting HERV-K Reverse Transcriptase

[0695] Exemplary compounds were tested for ability to inhibit HERV-K reverse transcriptase using a homogeneous time-resolved fluorescence (HTRF) assay. Assay procedures and results are described below.

Part I—Procedure for Homogeneous Time-Resolved Fluorescence HERV-K RT Assay

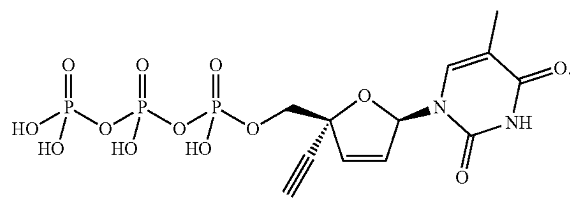
[0696] The HERV-K reverse transcriptase homogeneous time-resolved fluorescence (HTRF) assay was performed in a 384-well format with HERV-K reverse transcriptase (2-596)-8His protein. Baculoviruses were created using Bac-to-Bac technology (Invitrogen). pFastBac donor plasmids containing HERV-K reverse transcriptase sequence (NCBI GenBank number AAC63291.1, J. Virology (1999) Vol. 73, No. 3, pp. 2365-2375) were transformed into DH10 Bac cells following the manufacturer's instructions. Recombinant bacmids were then isolated clonally and transfected into SF9 cells with lipofectin. HERV-K reverse transcriptase was expressed in the SF9 insect cells and then purified using immobilized metal affinity chromatography (IMAC) followed by size-exclusion chromatography (SEC).

[0697] Test compound was serially diluted in DMSO and further diluted in the assay buffer (50 mM Tris-HCl, 50 mM KCl, 10 mM $MgCl_2$, 10 mM DTT, pH 8.1) to achieve a final DMSO concentration of 1%. The serially diluted compound was mixed with 16 ng/well of HERV-K enzyme, 5 nM of pre-annealed template/biotin-primer pair (synthesized at Geneway Biotechnology), and 10 nM of Fluorescein-12-dUTP fluorescent probe (Perkin Elmer). The template/biotin-primer sequences were poly(rA_{45}) and biotin-oligo(dT)$_{16}$.

[0698] After incubating at 25° C. for 30 minutes, the detection reagent 20 mM EDTA with streptavidin-terbium cryptate (Cisbio Bioassay) in the PPI buffer (Cisbio Bioassay) was added, and the mixture was incubated at 25° C. for 60 minutes. At the end of the incubation, fluorescence was read at ex/em=337/485 nm and ex/em=337/520 nm on an Envision 2104 plate reader (Perkin Elmer). The fluorescence ratio at 520/485 nm was used for the calculation. Percent inhibition was calculated with the DMSO sample as 0% inhibition and no enzyme as 100% inhibition. The IC_{50} was calculated by fitting the compound dose inhibition curve with a 4-parameter non-linear regression equation.

Part II—Results

[0699] The tetra(triethylammonium) salt of the following compound was found to inhibit HERV-K reverse transcriptase with an IC_{50} of 6.4 μM:



INCORPORATION BY REFERENCE

[0700] The entire disclosure of each of the patent documents and scientific articles referred to herein is incorporated by reference for all purposes.

EQUIVALENTS

[0701] The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

SEQUENCE LISTING

```
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source                 1..36
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 1
gtaactagag atccctcaga cccttttagt cagaat                  36

SEQ ID NO: 2           moltype = DNA  length = 25
FEATURE                Location/Qualifiers
source                 1..25
                       mol_type = other DNA
                       organism = synthetic construct
SEQUENCE: 2
ttctgactaa aagggtctga gggat                              25
```
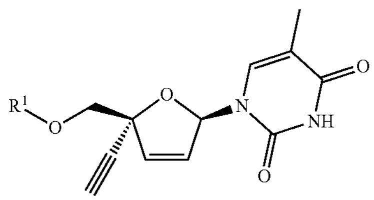

1. A compound represented by Formula I:



(I)

or a pharmaceutically acceptable salt thereof; wherein:

R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —P(O)(N(R^3)(R^4))$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^8;

R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen; or

R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

R^3 and R^9 each represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, or two instances of R^9, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

R^4 represents independently for each occurrence —C(R^5)$_2$—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

R^6 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —N(R^9)$_2$;

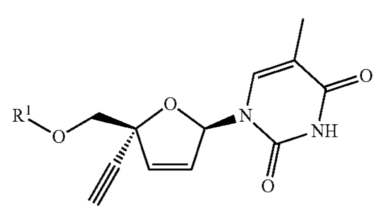
R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl;

m and p are independently for each occurrence 0, 1, 2, or 3; and

nis 1, 2, or 3.

2. The compound of claim 1, wherein the compound is a compound of Formula I.

3. A compound represented by Formula I-1:



(I-1)

or a pharmaceutically acceptable salt thereof; wherein:

R^1 is —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^8;

R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring;

R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

R^4 is —C(H)(R^5)—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

R^5 is C_{1-6} alkyl, C_{1-6} haloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur;

R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl;

R^8 is phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-7} alkyl, C_{1-20} alkyl substituted with hydroxyl, C_{1-10} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-8} alkylene)-S—(C_{1-10} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein said phenyl is substituted with n instances of R^7; wherein each of said naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-10} haloalkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl;

m is independently for each occurrence 0, 1, 2, or 3; and

n is 1, 2, or 3.

4. The compound of claim **3**, wherein the compound is a compound of Formula I-1.

5. The compound of any one of claims **1-4**, wherein R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N$(R^3)_2$.

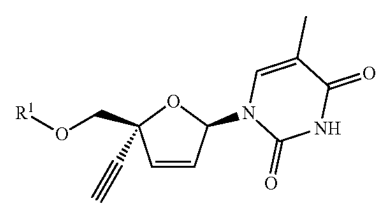
6. The compound of any one of claims **1-5**, wherein R^2 represents independently for each occurrence phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; each of which is substituted with m instances of R^7.

7. The compound of any one of claims **1-6**, wherein R^4 is —C(H)(R^5)—CO_2R^6.

8. The compound of any one of claims **1-7**, wherein R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl.

9. The compound of claim **1** represented by Formula I-A:

(I-A)



or a pharmaceutically acceptable salt thereof; wherein:

R^1 is —P(O)(OR^2)(N(R^3)(R^4)) or —C(O)—C(H)(R^5)—N$(R^3)_2$;

R^2 is phenyl or naphthyl, each of which is substituted with m instances of R^7;

R^3 represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

R^4 is —C(H)(R^5)—CO_2R^6;

R^5 is C_{1-6} alkyl or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl,

R^6 is C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom;

R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, or C_{1-4} alkoxyl; and

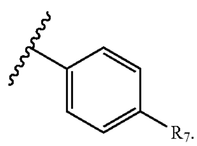
m is 0, 1, 2, or 3.

10. The compound of claim **9**, wherein the compound is a compound of Formula I-A.

11. The compound of any one of claims **1-10**, wherein R^1 is —P(O)(OR^2)(N(R^3)(R^4)).

12. The compound of any one of claims **1-11**, wherein R^2 is phenyl substituted with m instances of R^7.

13. The compound of any one of claims **1-11**, wherein R^2 is



14. The compound of any one of claims **1-12**, wherein m is 1.

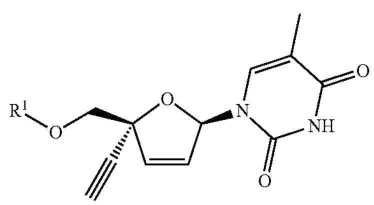
15. The compound of any one of claims **1-12**, wherein m is 0.

16. The compound of any one of claims **1-14**, wherein R^7 represents independently for each occurrence halo.

17. The compound of any one of claims **1-5** and **8-10**, wherein R^1 is —C(O)—C(H)(R^5)—N(R^3)$_2$.

18. The compound of claim **1** represented by Formula I-B:

(I-B)



or a pharmaceutically acceptable salt thereof; wherein:

R^1 is —P(O)(OR^2)$_2$ or —P(O)(N(R^3)(R^4))$_2$;

R^2 represents independently for each occurrence C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen; or two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

R^3 and R^9 each represents independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, or two instances of R^9, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

R^4 represents independently for each occurrence —C(R^5)$_2$—CO$_2$R^6;

R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), phenyl, or C_{3-7} cycloalkyl; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

R^6 represents independently for each occurrence C_{1-6} alkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one nitrogen or oxygen atom;

R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —N(R^9)$_2$; and

p is 0, 1, 2, or 3.

19. The compound of claim **18**, wherein the compound is a compound of Formula I-B.

20. The compound of any one of claims **1-2** or **18-19**, wherein R^1 is —P(O)(OR^2)$_2$.

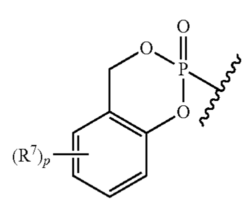
21. The compound of any one of claims **1-2** or **18-20**, wherein R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl); wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene.

22. The compound of any one of claims **1-2** or **18-20**, wherein R^2 represents independently for each occurrence —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl) or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene.

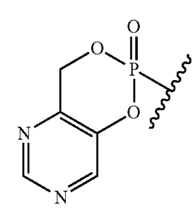
23. The compound of any one of claims **1-2** or **18-20**, wherein R^2 represents independently for each occurrence —(C_{1-10} alkylene)-O—(C_{1-20} alkyl) or —(C_{1-10} alkylene)-S—(C_{1-20} alkyl); wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); and wherein one instance of R^2 is additionally selected from hydrogen.

24. The compound of any one of claims **1-2** or **18-20**, wherein two instances of R^2 are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7.

25. The compound of any one of claims **1-2** or **18-20**, wherein R^1 is



or



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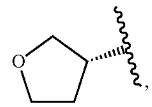
26. The compound of any one of claims **1-2** or **18-19**, wherein R^1 is —P(O)(N(R^3)(R^4))$_2$.

27. The compound of any one of claims **1-16**, **18-19**, or **26**, wherein R^6 is C_{1-6} alkyl, allyl, C_{3-5} cycloalkyl,



—CH_2-phenyl, or —CH_2—(C_{3-5} cycloalkyl).

28. The compound of any one of claims **1-16**, **18-19**, or **26**, wherein R^6 is C_{1-4} alkyl or C_{3-5} cycloalkyl.

29. The compound of any one of claims **1-19** or **26-28**, wherein R^3 is hydrogen.

30. The compound of any one of claims **1-19** or **26-29**, wherein R^5 is C_{1-6} alkyl or hydrogen.

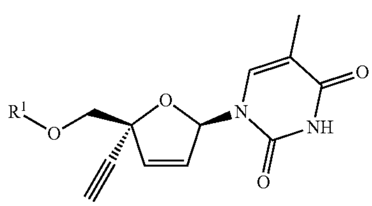
31. A compound in Table 1 or 2 herein, or a pharmaceutically acceptable salt thereof.

32. A compound in Table 1, 2, 3, 4, or 5 herein, or a pharmaceutically acceptable salt thereof.

33. A pharmaceutical composition comprising a compound of any one of claims **1-32** and a pharmaceutically acceptable carrier.

34. A method of treating a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder other than a viral infection, comprising administering to a subject in need thereof a therapeutically effective amount of a compound of Formula II to treat the disorder; wherein Formula II is represented by:

(II)



or a stereoisomer thereof; or a pharmaceutically acceptable salt of either of the foregoing; wherein:

R^1 is hydrogen, —P(O)(OR^2)(N(R^3)(R^4)), —P(O)(OR^2)$_2$, —P(O)(N(R^3)(R^4))$_2$, —C(O)—C(H)(R^5)—N(R^3)$_2$, or —C(O)R^2;

R^2 represents independently for each occurrence hydrogen, —P(O)(OH)$_2$, —P(O)(OH)—O—P(O)(OH)$_2$, phenyl, naphthyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-20} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—N(R^9)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-20} alkyl), or —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl); wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^7; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl; and wherein one methylene unit in each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally replaced with a C_{3-5} cycloalkylene; and wherein each of said C_{1-10} alkylene is optionally substituted with one —O—(C_{1-20} alkyl); or R^2 and R^4, or two instances of R^2, are taken together with their intervening atoms to form a 4-7 membered saturated heterocyclic ring or a 7-12 membered bicyclic heterocyclic ring, each of which is substituted with p instances of R^7;

R^3 and R^9 each represent independently for each occurrence hydrogen or C_{1-4} alkyl; or R^3 and R^5, or two instances of R^3, or two instances of R^9, are taken together with the atom or atoms to which they are attached to form a 4-7 membered saturated heterocyclic ring containing 1 nitrogen atom;

R^4 represents independently for each occurrence —C(R^5)$_2$—CO_2R^6, C_{1-20} alkyl, C_{1-20} haloalkyl, —(C_{1-10} alkylene)-O—(C_{1-10} alkyl), —(C_{1-10} alkylene)-OC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-S—(C_{1-10} alkyl), —(C_{1-10} alkylene)-SC(O)—(C_{1-10} alkyl), —(C_{1-10} alkylene)-phenyl, phenyl; naphthyl; a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein each of said phenyl, naphthyl, 5-6 membered monocyclic heteroaryl, and 8-10 membered bicyclic heteroaryl is substituted with m instances of R^6; and wherein each of said C_{1-20} alkyl, C_{1-20} haloalkyl, and C_{1-10} alkyl is optionally substituted with one hydroxyl;

R^5 represents independently for each occurrence C_{1-6} alkyl, C_{1-6} haloalkyl, C_{3-5} cycloalkyl, or hydrogen, wherein said C_{1-6} alkyl is optionally substituted with —S—(C_{1-4} alkyl), —SH, C_{1-4} alkoxyl, hydroxyl, phenyl, C_{3-7} cycloalkyl, a 5-6 membered monocyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur, or an 8-10 membered bicyclic heteroaryl having 1, 2, or 3 heteroatoms independently selected from nitrogen, oxygen, and sulfur; or two instances of R^5 are taken together with the carbon atom to which they are attached to form a 3-5 membered saturated carbocyclic ring;

R^6 is C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-6} alkenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur; wherein said C_{1-6} alkyl is optionally substituted with C_{1-4} alkoxyl, phenyl, C_{3-7} cycloalkyl, or a 4-7 membered saturated monocyclic heterocyclyl having one or two heteroatoms independently selected from nitrogen, oxygen, and sulfur;

R^7 represents independently for each occurrence halo, C_{1-4} alkyl, C_{1-4} haloalkyl, C_{1-4} alkoxyl, or —N(R^9)$_2$; and

m and p are independently for each occurrence 0, 1, 2, or 3.

35. A method of treating a disorder selected from the group consisting of cancer, an inflammatory disorder, a neurodegenerative disorder, and an immune disorder, comprising administering to a subject in need thereof a therapeutically effective amount of a compound of any one of claims **1-32** to treat the disorder.

36. The method of claim **35**, wherein the disorder is an immune disorder that is a viral infection.

37. The method of claim **36**, wherein the viral infection is an infection by human immunodeficiency viruses 1 or 2 (HIV-1 or HIV-2), human T-cell leukemia viruses 1 or 2 (HTLV-1 or HTLV-2), respiratory syncytial virus (RSV), human papilloma virus (HPV), adenovirus, hepatitis B virus (HBV), hepatitis C virus (HCV), Epstein-Barr virus (EBV), varicella zoster virus (VZV), cytomegalovirus (CMV), herpes simplex viruses 1 or 2 (HSV-1 and HSV-2), human herpes virus 8 (HHV-8, also known as Kaposi's sarcoma-associated virus), or a flavivirus selected from Yellow Fever virus, Dengue virus, Japanese Encephalitis, and West Nile virus.

38. The method of claim **34** or **35**, wherein the disorder is cancer.

39. The method of claim **38**, wherein the cancer is breast cancer, ovarian cancer, uterine cancer, cervical cancer, prostate cancer, testicular cancer, lung cancer, leukemia, head and neck cancer, oral cancer, esophageal cancer, stomach cancer, bile duct and gallbladder cancers, bladder cancer, urinary tract cancer, colon cancer, rectal cancer, thyroid cancer, pancreatic cancer, kidney cancer, liver cancer, brain cancer, skin cancer, or eye cancer.

40. The method of claim **34** or **35**, wherein the disorder is an inflammatory disorder.

41. The method of claim **40**, wherein the inflammatory disorder is rheumatoid arthritis, osteoarthritis, ankylosing spondylitis, inflammatory bowel disease, Crohn's disease, ulcerative colitis, nonalcoholic steatohepatitis (NASH), nonalcoholic fatty liver disease (NAFLD), cholestatic liver disease, sclerosing cholangitis, psoriasis, dermatitis, vasculitis, scleroderma, asthma, bronchitis, chronic obstructive pulmonary disease (COPD), pulmonary fibrosis, pulmonary hypertension, sarcoidosis, myocarditis, pericarditis, gout, myositis, Sjogren's syndrome, or systemic lupus erythematosus.

42. The method of claim **34** or **35**, wherein the disorder is an immune disorder other than a viral infection.

43. The method of claim **42**, wherein the immune disorder is a type 1 interferonopathy, type 1 diabetes, Aicardi-Goutieres syndrome (AGS), arthritis, psoriasis, systemic lupus erythematosus (SLE), lupus nephritis, cutaneous lupus erythematosus (CLE), familial chilblain lupus, systemic sclerosis, STING-associated vasculopathy with onset in infancy (SAVI), graft versus host disease, scleroderma, polymyositis, inflammatory bowel disease, dermatomyositis, ulcerative colitis, Crohn's disease, vasculitis, psoriatic arthritis, Reiter's syndrome, exfoliative psoriatic dermatitis, pemphigus vulgaris, Sjogren's syndrome, autoimmune uveitis, glomerulonephritis, post myocardial infarction cardiotomy syndrome, pulmonary hemosiderosis, amyloidosis, sarcoidosis, aphthous stomatitis, thyroiditis, gastritis, adrenalitis (Addison's disease), ovaritis, primary biliary cirrhosis, myasthenia gravis, gonadal failure, hypoparathyroidism, alopecia, malabsorption syndrome, pernicious anemia, hepatitis, hypopituitarism, diabetes insipidus, or sicca syndrome.

44. The method of claim **34** or **35**, wherein the disorder is a neurodegenerative disorder.

45. The method of claim **44**, wherein the neurodegenerative disorder is Alzheimer's disease, amyotrophic lateral sclerosis (ALS), multiple sclerosis, Parkinson's disease, Huntington's disease, peripheral neuropathy, age-related macular degeneration, Creutzfeldt-Jacob disease, stroke, prion disease, frontotemporal dementia, Pick's disease, progressive supranuclear palsy, spinocerebellar ataxias, Lewy body disease, dementia, multiple system atrophy, epilepsy, bipolar disorder, schizophrenia, an anxiety disorder, or major depression.

46. The method of any one of claims **34-45**, wherein the method further comprises administering an effective amount of an additional therapeutic agent.

47. The method of any one of claims **34-46**, wherein the subject has (i) expression of LINE1 RNA, LINE1 ORF1 polypeptide, and/or LINE1 ORF2 polypeptide; and/or (ii) activity of LINE1 reverse transcriptase.

48. The method of any one of claims **34-47**, wherein the subject has (i) expression of HERV-K RNA and/or (ii) activity of HERV-K reverse transcriptase.

49. The method of any one of claims **34-48**, wherein the subject is a human.

50. A method of inhibiting LINE1 reverse transcriptase activity, comprising contacting a LINE1 reverse transcriptase with an effective amount of a compound of any one of claims **1-32**, in order to inhibit the activity of said LINE1 reverse transcriptase.

51. A method of inhibiting HERV-K reverse transcriptase activity, comprising contacting a HERV-K reverse transcriptase with an effective amount of a compound of any one of claims **1-32**, in order to inhibit the activity of said HERV-K reverse transcriptase.

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